Privileged Access Manager v. 15.0

Administration

Exercise Guide

CYBERARK®

The Identity Security Company™

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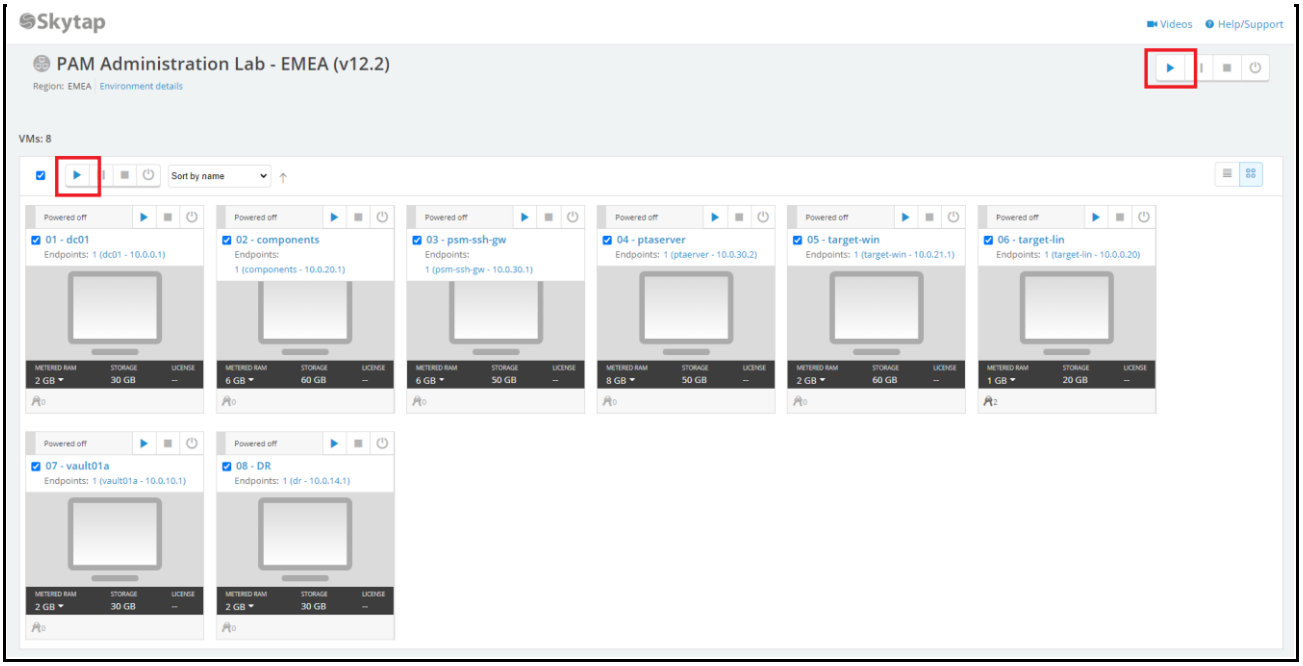
Introduction

Using Kyndryl Cloud Uplift

Before beginning the exercises, here are a few tips to help you navigate the labs more effectively. You can refer to the section for ***International Users*** for instructions on changing the keyboard.

The virtual machines need to be running for you to be able to do the exercises. You can start all the virtual machines with one click by pressing the start button (there are two). The buttons are highlighted in red in the image below.



Note:	The number and names of virtual machines vary by course. The image above is given as an example and might not match exactly what you see.

The environments have been configured to start up gradually beginning with the domain controller and the **Vault**. It will take a few minutes for them to get up and running. Also note that some machines are designed *not* to start automatically. These servers are not needed until later in the course, so you can start them when instructed in the manual or by the **CyberArk** trainer.

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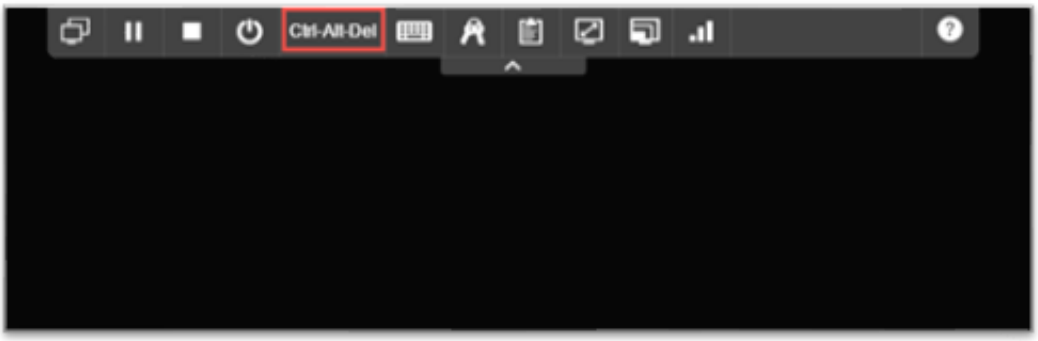
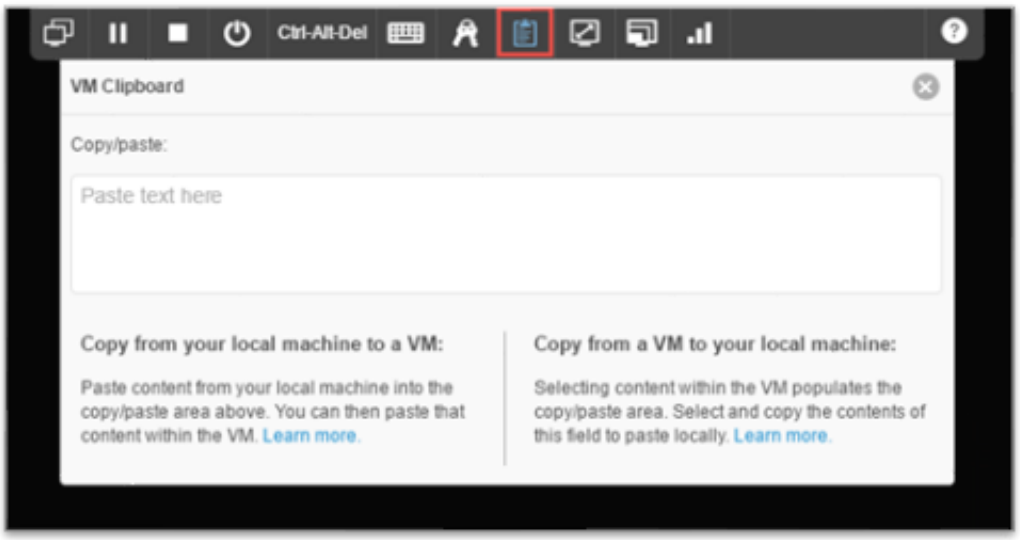
Occasionally, for reasons outside our control, one or more machines may fail to start up when requested or may shut itself down. If you notice that a particular machine is not responding to a ping or if you cannot log in using Active Directory, you should check your virtual machines to make sure they are all running properly.

1. Click on the large monitor icon to connect to a virtual machine with the HTML 5 client.

2. Use the **Ctrl-Alt-Del** button on the tool bar to send a Ctrl-Alt-Del to the machine.



3. The clipboard icon will allow you to copy and paste text between your computer and your lab machine. Do **NOT** copy and paste from this PDF into the **CyberArk PAM** tool. It will not work properly.



4. The full screen icon will resize your virtual screen to adapt to your computer's screen settings to avoid scrolling.

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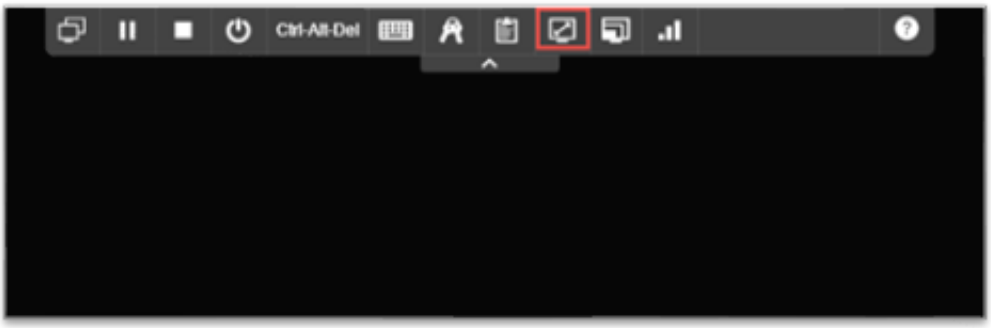
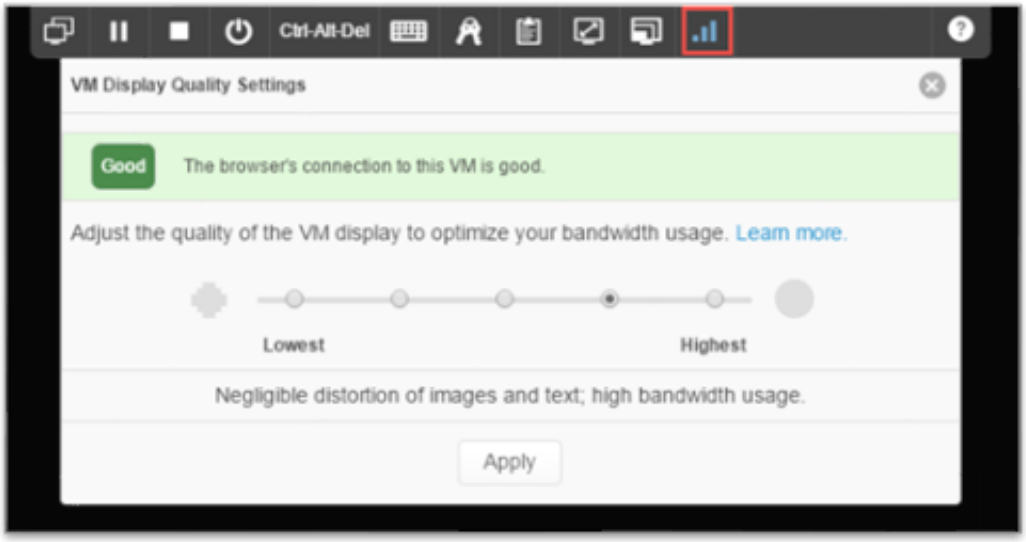


5. You may need to adjust your bandwidth setting on slower connections.



International Users

By default, the lab machines are configured to use a US English keyboard layout. If you use a machine from a country other than the US, you may experience odd behavior in your lab machines. The solution is to install the keyboard layout for your keyboard on our lab machines. Follow the process below to find and configure the correct keyboard layout for your keyboard.

1. From the **Start Menu**, go to **Settings | Time & Language | Language | Add a language**.

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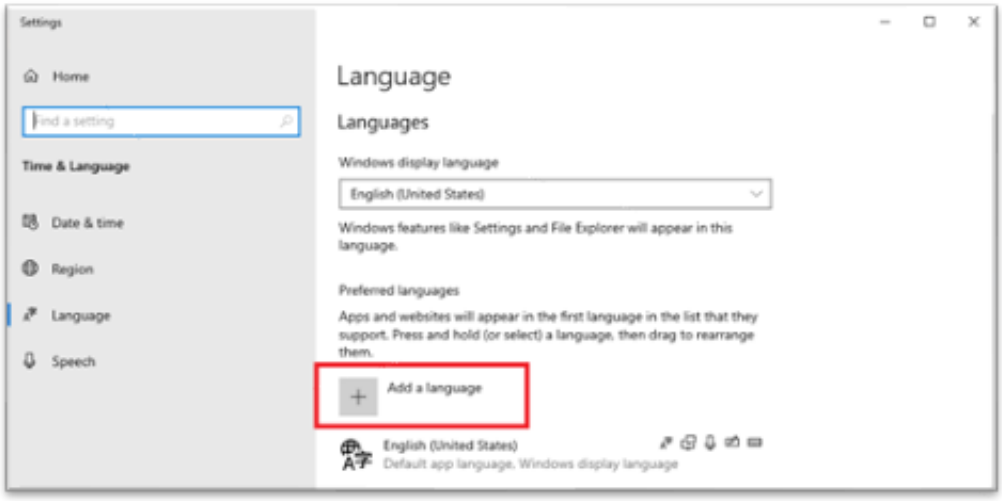

2. Select your language. Click **Next**. You can uncheck the options for voice and handwriting and then click **Install**.

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Note:	If you use an alternate keyboard layout (e.g., AZERTY, Dvorak), you can click the options next to your language to install that. Otherwise, close the **Language** window.

3. In the system tray, click **ENG**, then choose your keyboard layout. You may switch back and forth between keyboard layouts. Your instructor may need to switch back to **ENG** to help you with exercises, so do not uninstall any language options.

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Introduction to CyberArk Privileged Access Manager

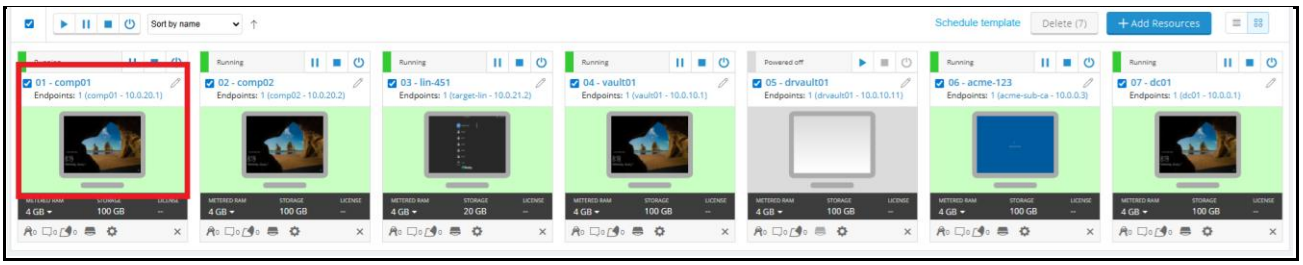
Welcome to **CyberArk Privileged Access Manager Administration** training. The purpose of this training is to introduce the **Privileged Access Manager** (or **PAM**) solution. Specifically, this training focuses on the **CyberArk PAM Self-Hosted** solution.

As the focus of this training is on the administration of the solution, the necessary software has already been installed. The scenario is that the implementation team has done its job and it is now up to the system administrators (that is, us) to take ownership of **CyberArk PAM** and configure it according to the organization's requirements.

Getting to Know the Acme Corp Environment

For the purposes of the training, we have created an IT environment for the fictitious **Acme Corporation**. The domain name is **acme.corp**.

Our environment consists of a total of seven virtual servers. Some host **CyberArk** components, such as the **Vault**; some are IT infrastructure, such as the Acme domain controller (**dc01**); and finally, others have nothing to do with either **CyberArk** or the IT environment and are what we call the *target servers*, such as machines hosting Acme human resources applications or financial information.



The goal is to provide trainees with an environment that resembles as closely as possible an actual production environment. As such, there is a domain with Active Directory, an email server, and so on. Our goal is to integrate **CyberArk PAM** in this corporate environment and to bring the principal privileged accounts under **CyberArk** control.

Acme Servers

The table below lists the various servers, their roles, and configuration. The lines shaded blue represent servers hosting **CyberArk** services.

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VM no.	Host name	IP Address	Operating system	Role
01	*comp01*	*10.0.20.1*	*Windows 2022 Server*	*Component server hosting:* • *PVWA* • *CPM* *Admin workstation*
02	*comp02*	*10.0.20.2*	*Windows 2022 Server*	*Component server hosting:* • *PSM*
03	*lin-451*	*10.0.21.2*	*Linux Rocky 9.6*	*Component server hosting:* • *PSM for SSH* • *HTML5 Gateway*
04	*vault01*	*10.0.10.1*	*Windows 2022 Server*	*Primary Vault*
*05 **	*drvault01*	*10.0.10.5*	*Windows 2022 Server*	*Disaster Recovery Vault*
06	*acme-123*	*10.0.0.3*	*Windows 2022 Server*	*Intermediate certificate authority* *Email server* *Oracle DB*
07	*dc01*	*10.0.0.1*	*Windows 2022 Server*	*Domain controller* *Active Directory* *Email server*

Note * This VM is configured not to start up automatically. We will start it manually later in the course.

We will do most of our work on **comp01**, also referred to as the **Component** server. As indicated above, the **Component** server runs several **CyberArk** component services. For the purposes of this training, it also serves as the workstation for the **Vault** administrator.

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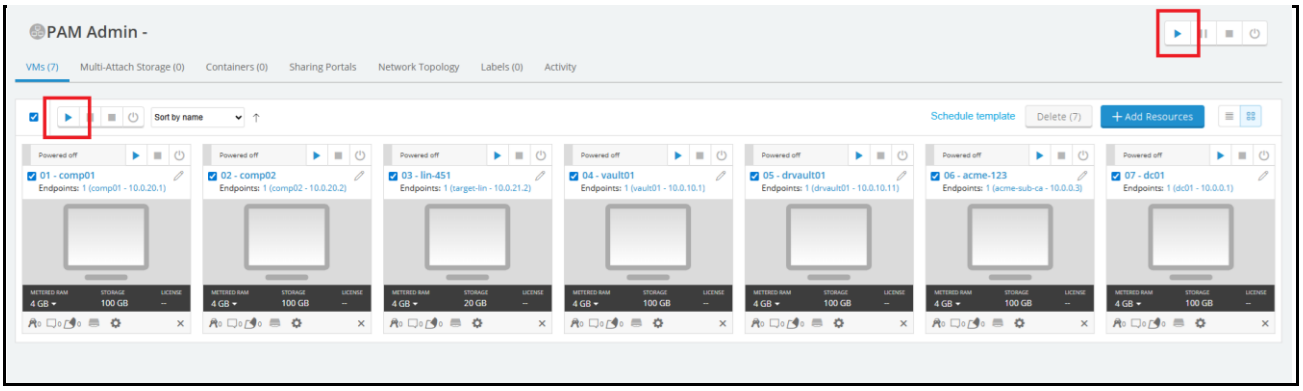
Starting the Virtual Machines

→ IMPORTANT! ←

Because of changes to certificate lifetimes, there are a number of scripts that run every time the virtual machines are started to generate new certificates. These scripts have interdependencies among several machines, so for them to work, they need to be started at the same time using the general **Start** button in Kyndryl Cloud Uplift.

All the servers (except for the **drvault01**) are configured to start automatically when the general power-on button is clicked in Uplift. For **CyberArk PAM** to work properly, all those servers need to be running. So, if you run into problems connecting to the **PVWA** or opening a **PSM** session to a Linux machine, the first thing to do is to check that all the machines and corresponding services are running.

Reminder: Usually, the VMs should be started from the LMS. However if more than 2 hours have passed without activity, the environment will shut down automatically and you will need to restart it using the start buttons, as shown in the image below.



Because we won't need it immediately, we will start up the disaster recovery Vault server virtual machine manually later in the course.

Getting to know CyberArk PAM

In this first exercise, we will perform a few basic tasks to start to familiarize ourselves with the various **CyberArk PAM** tools and interfaces. We will:

- Log into the **Comp01** server, which will also serve as our workstation.
- Log into **Password Vault Web Access** (**PVWA**)
- Connect via **PrivateArk Client**

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- Connect via **Remote Control Client**
- View the **Vault Server Central Administration**

All actions should be performed on the **Comp01** server unless otherwise indicated.

Note:	For the sake of convenience, we use the **Comp01** server in this training as our workstation. It is important to note, however, that this is something you would never do in the real world. As the host of sensitive **CyberArk** services, your component servers must be placed under **CyberArk** control, with their passwords stored securely in the **Vault** and accessed only through the **Privileged Session Manager**.

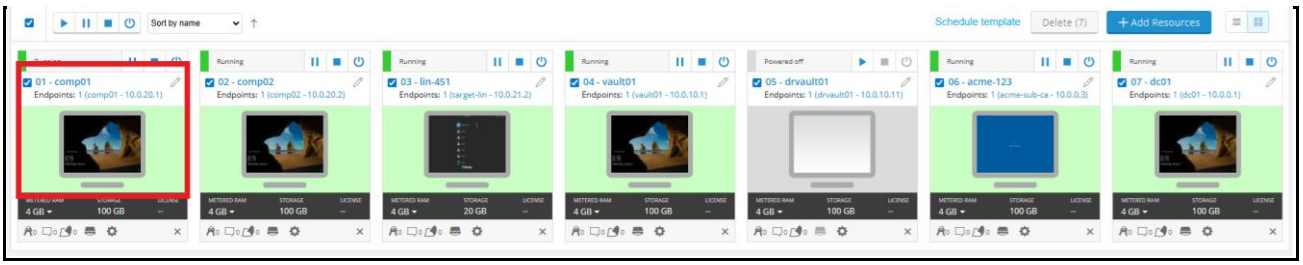
Log into the Comp01 server

First, we need to log into Windows. As already mentioned, we are going to use the **Comp01** server as our workstation. The account we will use is ***Mike***, an Active Directory user who has been given the responsibility for configuring and maintaining the **CyberArk PAM** solution in Acme. He is a member of the AD group **CyberArk Vault Admins**.

1. Make sure that you have started the virtual machines using one of the global start buttons, as shown in the image below.

Note:	Make sure all the VMs (except for the **DR**) are running before proceeding to the next step.

2. In Uplift, click on the screen for the **01 – COMP01** virtual machine. This will open a browser window with the machine's login screen.



3. Click the **Ctrl-Alt-Del** button in the Uplift toolbar at the top of the window to bring up the login dialog. You can also use the keyboard combination **Ctrl+Alt-End** to send **Ctrl+Alt+Delete**.

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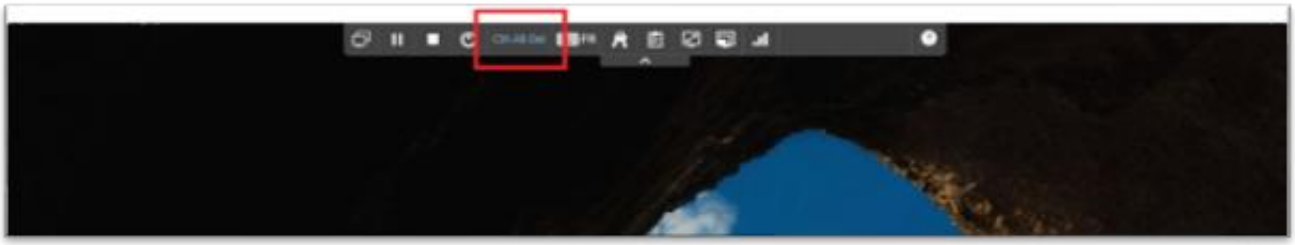

4. Enter *Mike* as the username and *Cyberark1* as the password. Remember, the machines use the US English keyboard as the default, so you may have to adjust the keys you use. When you are finished, hit **Enter** to log in.

5. And you should find yourself logged into the **Comp01** server with the Active Directory credentials of *Mike*, the CyberArk Vault Administrator.

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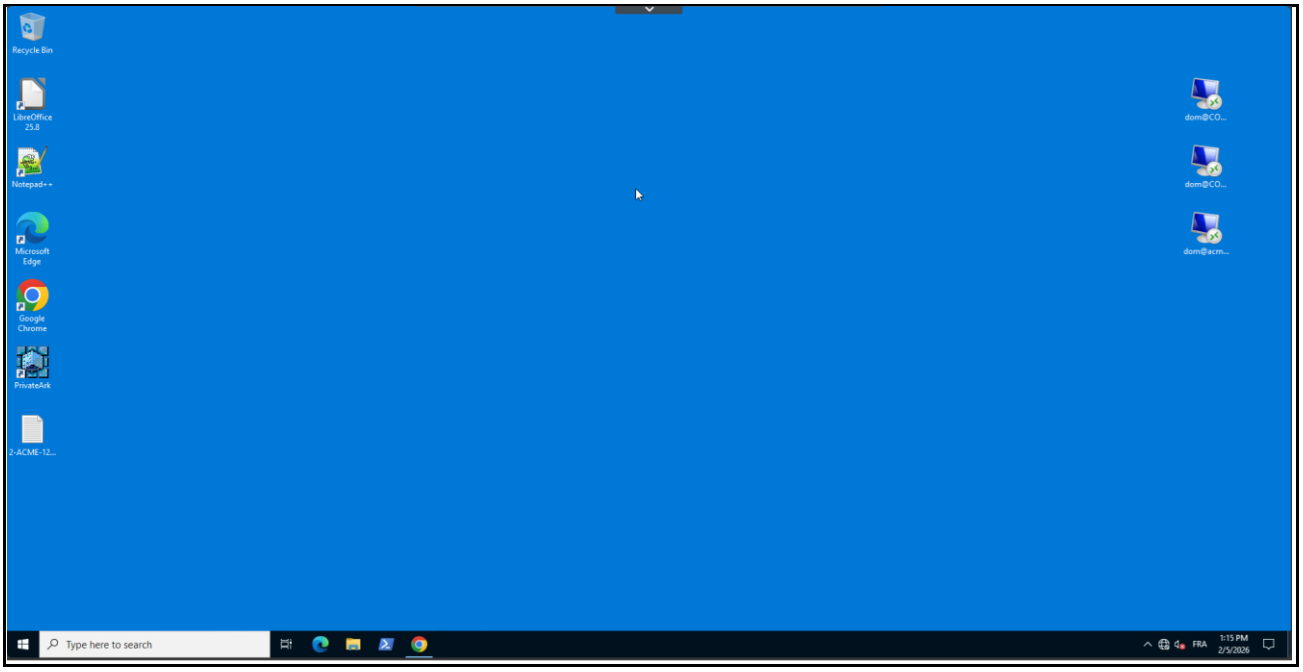

PVWA

In this section, we will perform some basic operations using the **Password Vault Web Access**, or **PVWA**. We will:

- Log in to the **PVWA** as *Mike*, our CyberArk Vault Administrator
- Activate the **PSM**
- Deactivate "Reason for Access"
- Launch a **PSM** connection in the New UI
- Retrieve a password in the Classic UI

Log in as Mike

1. On the **Comp01** server, launch **Google Chrome** using the shortcut in the taskbar at the bottom of the screen. You should arrive directly on the login screen for the **PVWA**.

2. There are currently two authentication methods available to us: **CyberArk** and **LDAP**. LDAP integration has already been performed by the installation team, so we can connect with the Active Directory credentials of our CyberArk Vault Administrator *Mike*. Click on the **LDAP** icon.

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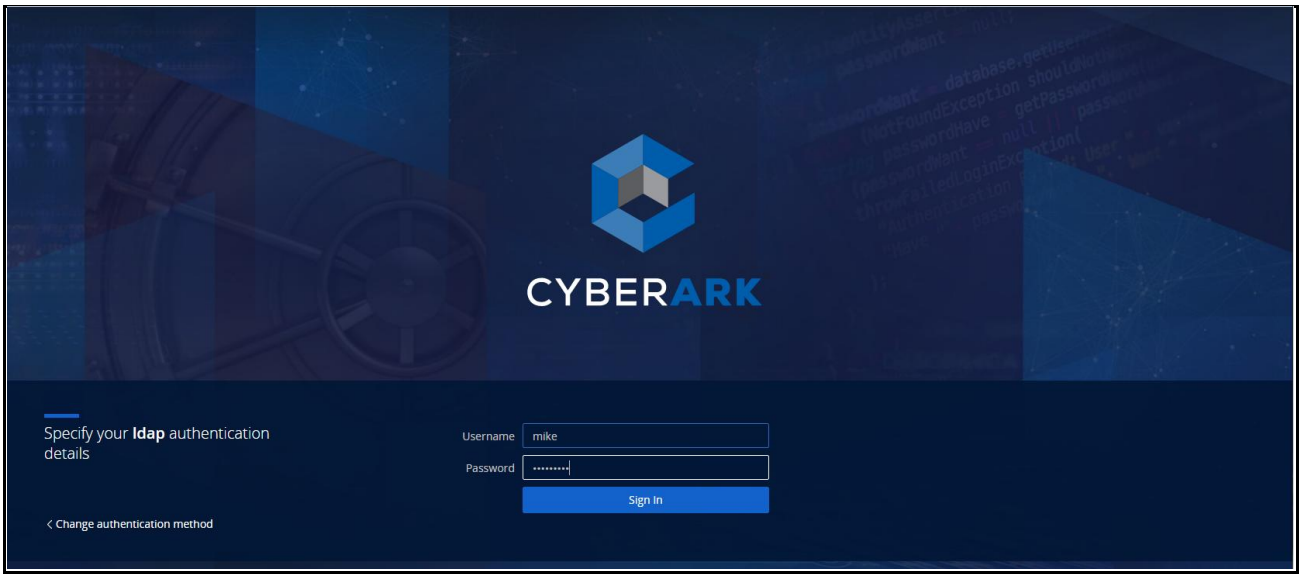
3. Enter the username *Mike* and password *Cyberark1* and then hit **Enter** or click **Sign In**.



4. By default, you will be in the **Accounts View**, which provides access to all the privileged accounts in the *Safes* of which *Mike* is a member. There are not many accounts at the moment. It will be our job to add more.

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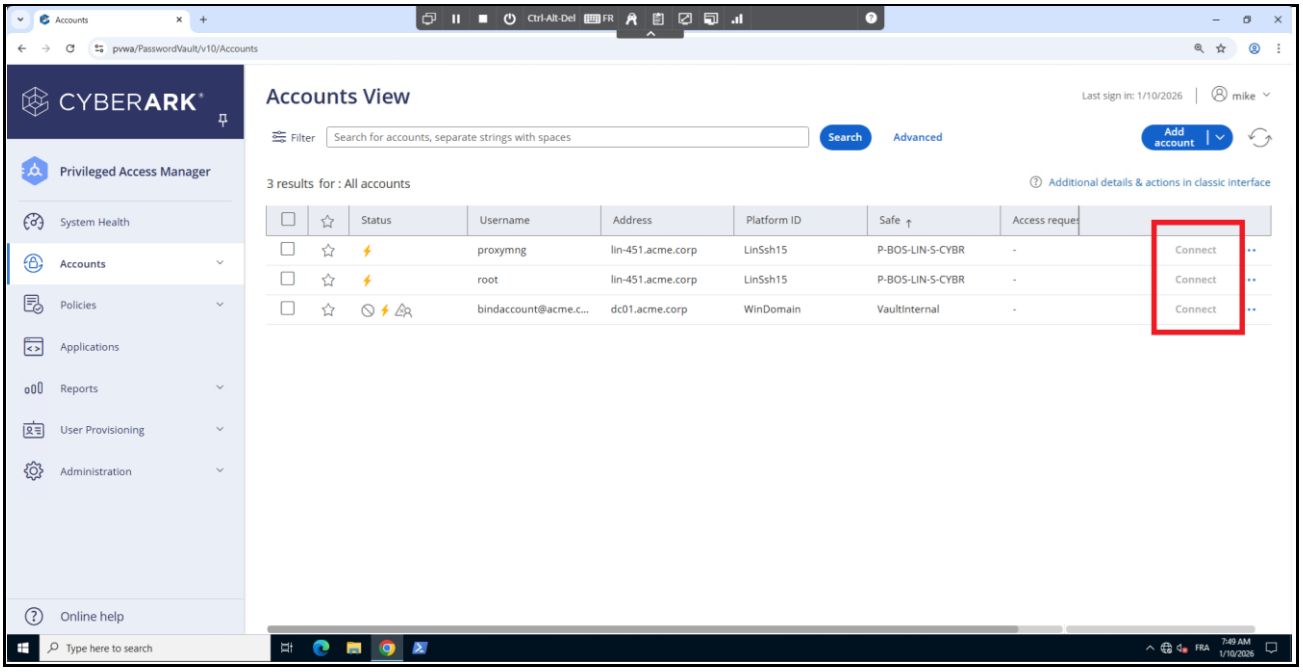
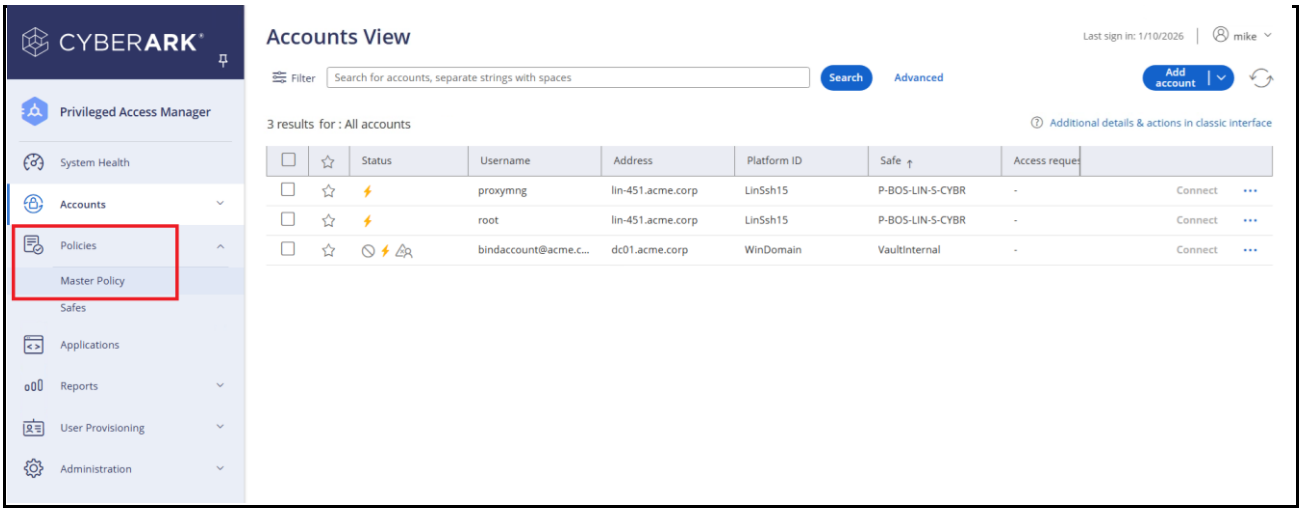


Activate the PSM

As you can see in the image above, the **Connect** button is greyed out. The reason for this is that the **PSM** has not yet been activated (this is the default value), we will activate it now.

1. To activate the **PSM**, we will need to modify the **Master Policy**. Click on the **Policies** tab.



2. In the **Master Policy**, expand the **Session Management** section, select *Require privileged session monitoring and isolation*, and then click the pencil icon in the upper right-hand corner.

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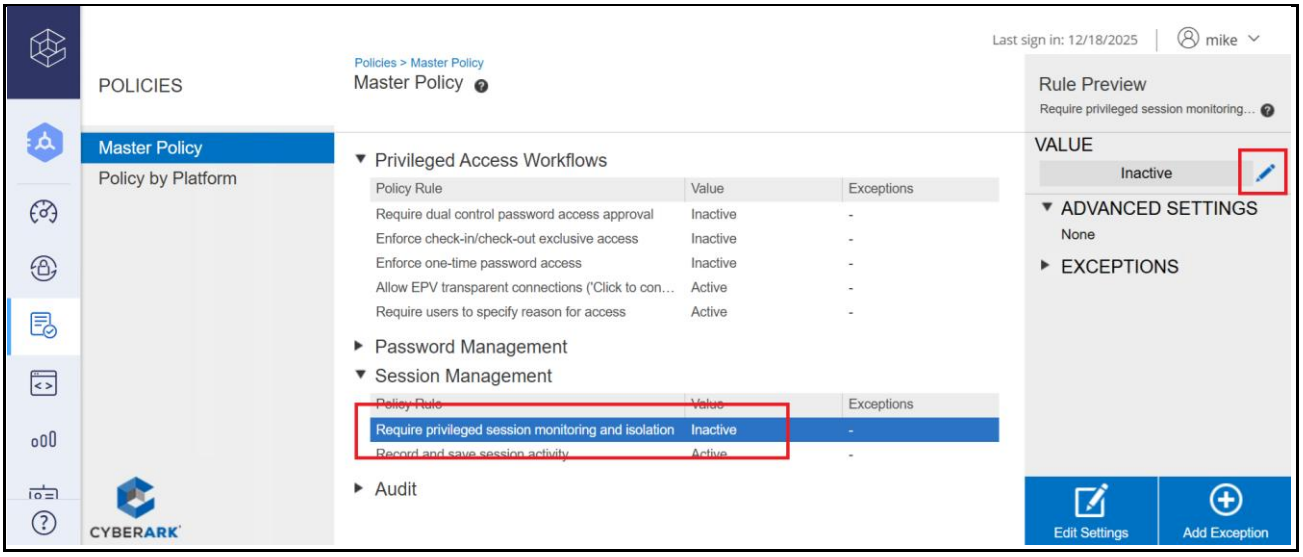
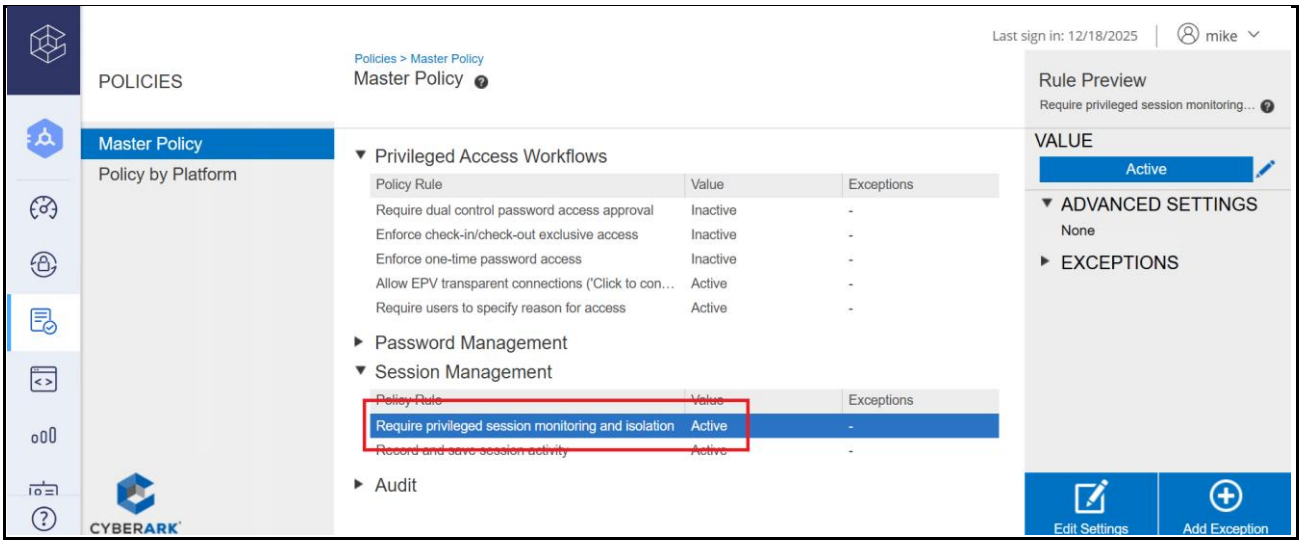


3. Toggle the value from *Inactive* to *Active* and then click the diskette icon to save the change.



Deactivate "Reason for Access"

While we are here, we can make our lives easier by deactivating the option **Require users to specify a reason for access**, which we won't need immediately in our training environment. That way we will not be required to provide a reason every time we want to test a newly created account. It can be found in the **Privileged Access Workflows** section at the top.

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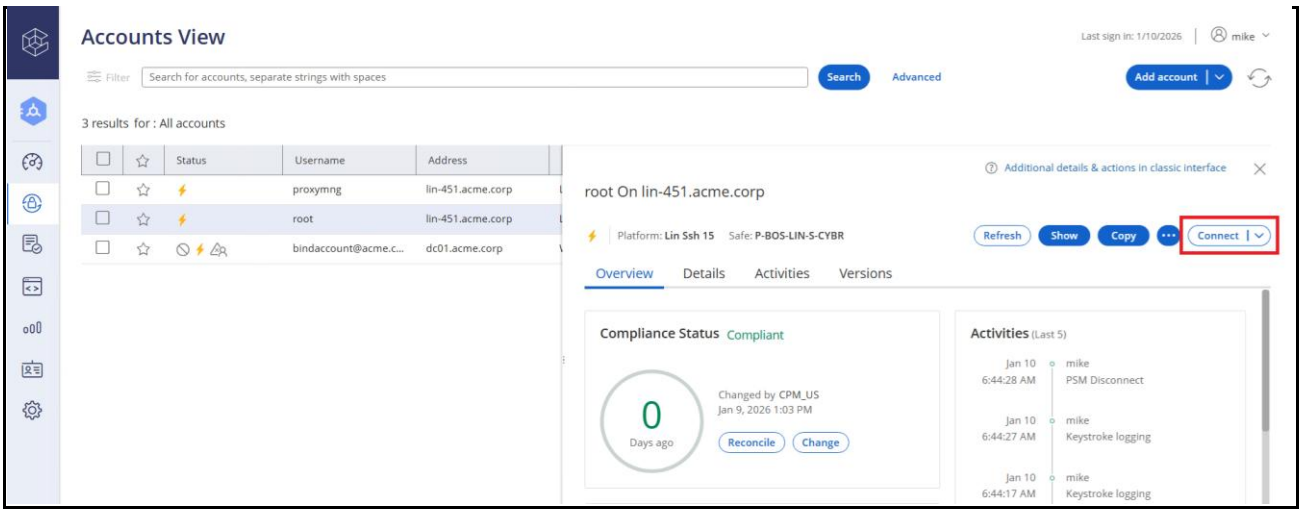
Select the policy, click the pencil icon, toggle the value from *Active* to *Inactive*, and then click the diskette icon to save your changes.

Connect to an Account in the New UI

Now we will test using a password from the **Vault** to connect to a target device using a test account.

1. Go back to **Accounts View** by clicking on the tab along the left-hand side of the screen (second from the top) and then click on the *root* account. You should now see that the **Connect** button is enabled. Click on **Connect**.

Note:	The account ***proxymng*** is a maintenance account created in the PSM for SSH Server. We will discuss the PSM for SSH later in the course.



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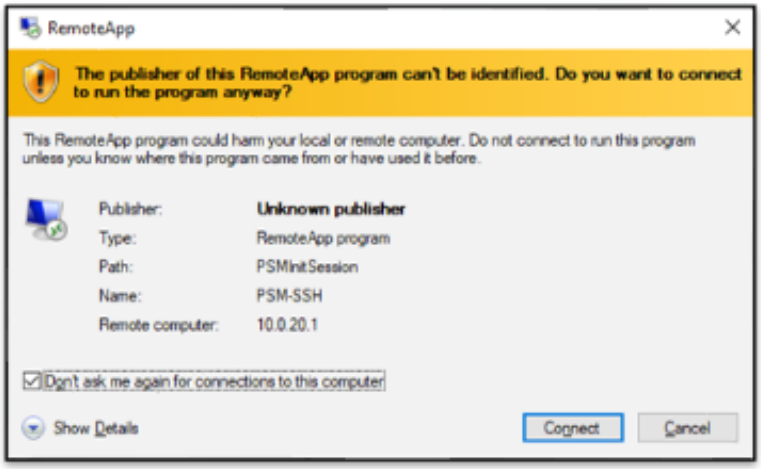
2. The **PVWA** server will send an RDP file. In Chrome, it is downloaded to the local machine and appears in the upper right-hand corner of the screen. There is a slight delay after clicking the **Connect** button.

3. Click on the RDP file to launch the connection.

4. You may be prompted to allow the RemoteApp program to run. If you are, you can check the box "**Don't ask me again…**" and click **Connect**.



Note:	If it is the first time the currently logged in user (in this case *Mike*) connects to the target server, you will be prompted to accept the server's key. You must accept the key.

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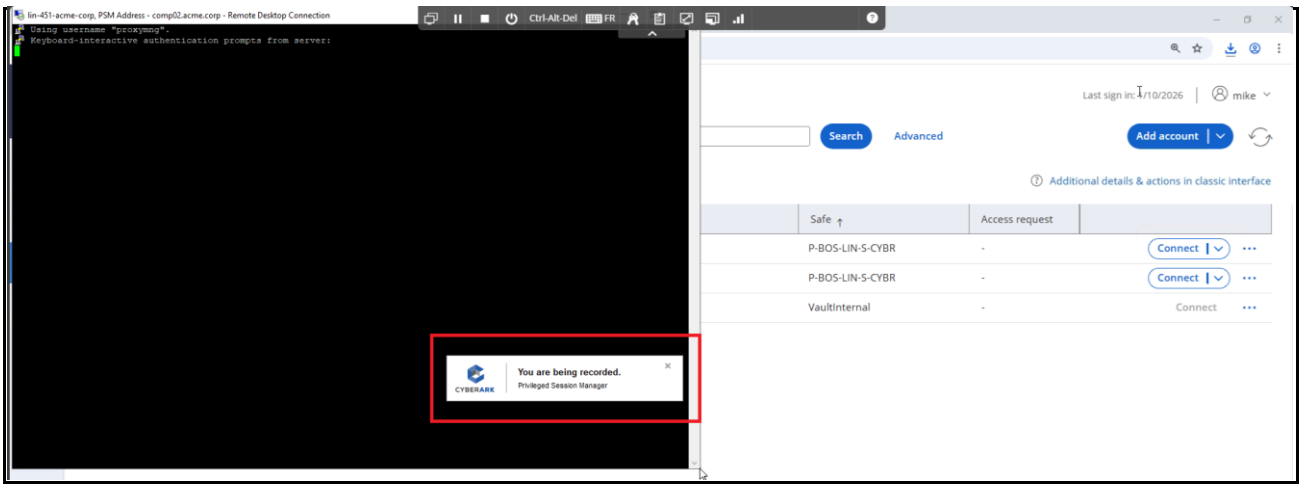
5. You will see a banner telling you that your session is being recorded by the **Privileged Session Manager** (this will eventually disappear), and then you will see a **PuTTY** window with the SSH connection to the machine **lin-451** with the username *proxymng*. This usually pops up ***behind*** the browser window, so you will have to click on the



Note:	The password for this account was retrieved by the **PSM** from the **Vault** and inserted into the PuTTY session running on the **PSM** server at the time of connection. At no time did the password appear on the user machine.

6. Now take a closer look at the Putty session. Notice that although we clicked to connect to the *root* account, the connection was established by the account *proxymng* and then a switch user to *root* occurred. This is an example of what we call a ***logon account.*** We will examine logon accounts a little later.

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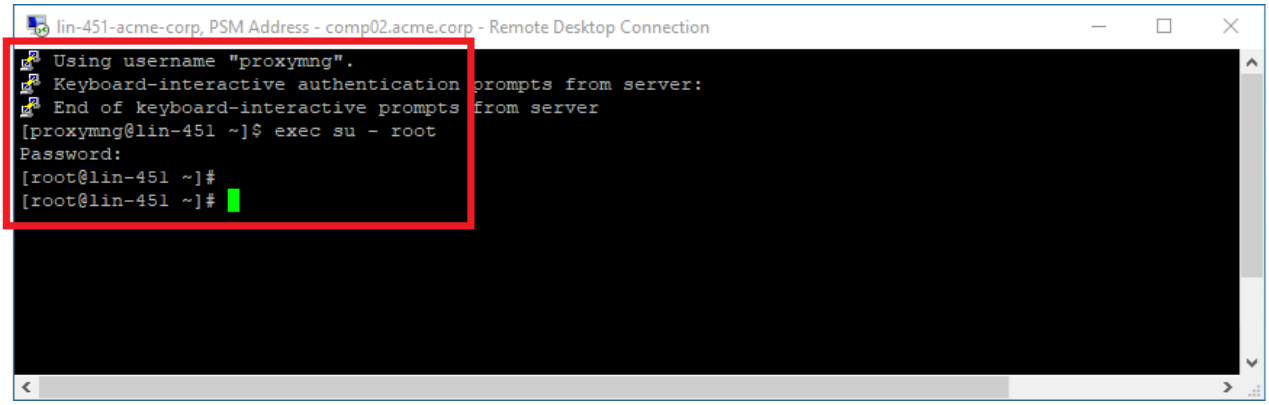


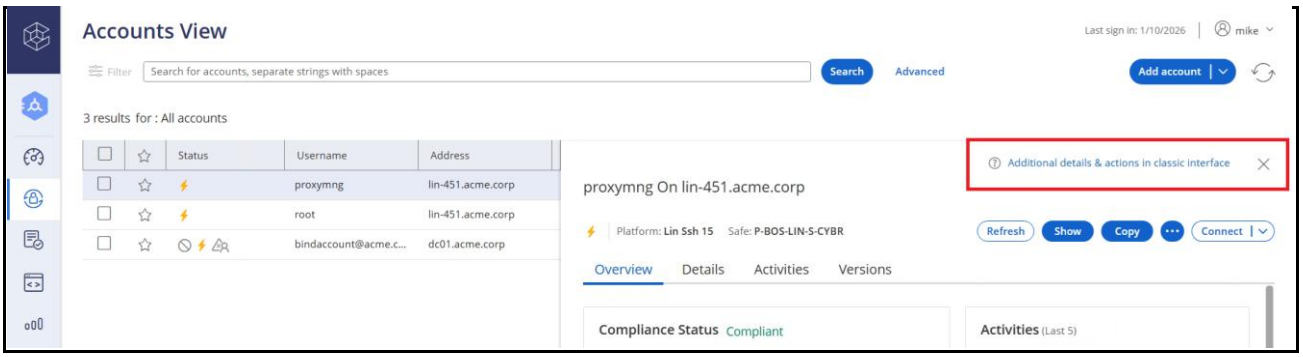
7. Enter “exit” (without the quotes) into the SSH session and hit **Enter** to close the session. This closes the SSH session and the RDP connection.

Retrieve a Password in the Classic UI

CyberArk introduced a new user interface beginning with version 10. There is, however, still some functionality that can only be accessed through the old interface, or **Classic UI**, so we will now look at how to access this user interface.

In this section, we will use another method to retrieve a password by using the **Show** button in the classic interface, although there are buttons for both **Show** and **Copy** in the new interface.

1. Back in the **PVWA**, click on the account *proxymng*. In the upper right-hand corner of the **Accounts View** you will see a link labelled **Additional details & actions in classic interface**. Click the link.



2. Here, you are looking at the **Account Details** for *proxymng* in the Classic interface. Notice that we are still in the new interface: You still have access to the tabs along the left-hand side. Now click the **Show** button.

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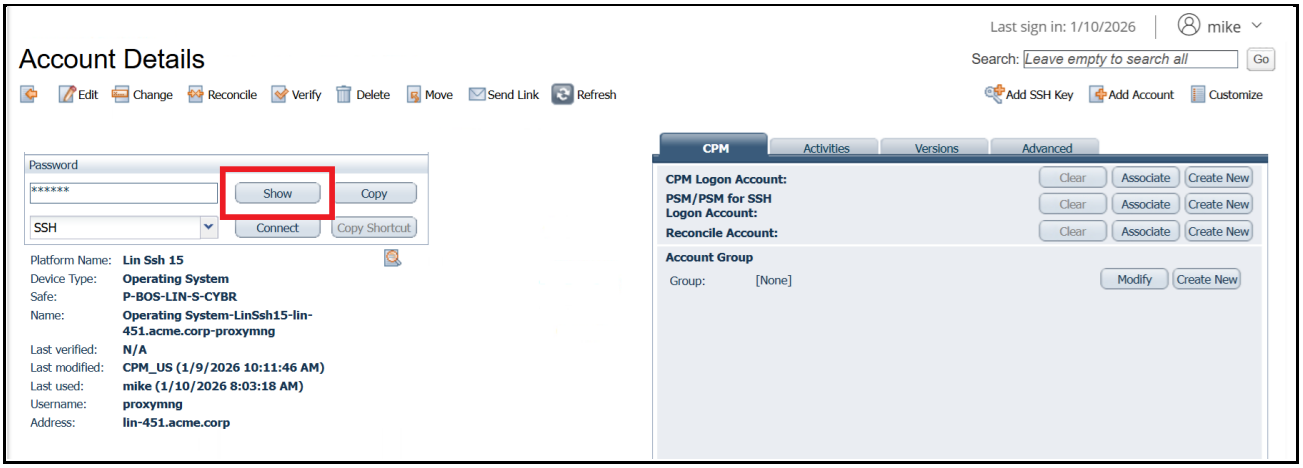

3. We can now see the password that is currently stored in the **Vault** for the account *proxymng*.

4. As a last step, click the **Change** button at the top of the **Account Details** view. You are presented with three options. The first option – *Change the password immediately (by the CPM)* – is available in both the Classic and the new interface. The third option – *Change the password only in the Vault* – is also available in the new UI. The second option – *Specify the password for the next CPM change* – is for the moment only available in the Classic interface.

   Select **Change the password immediately** and click **OK** to change the password.

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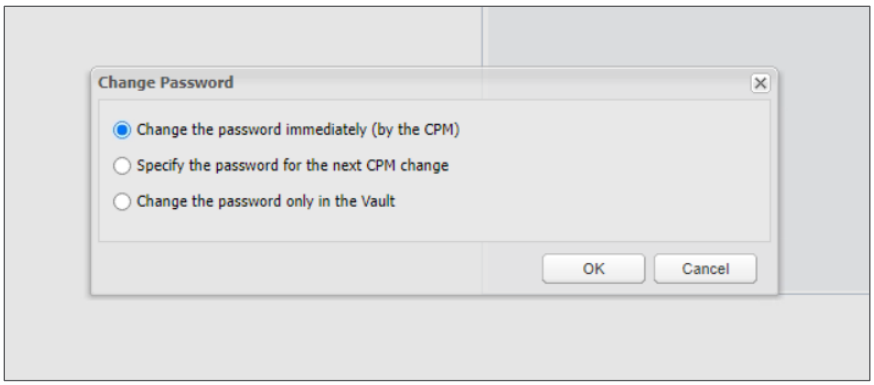
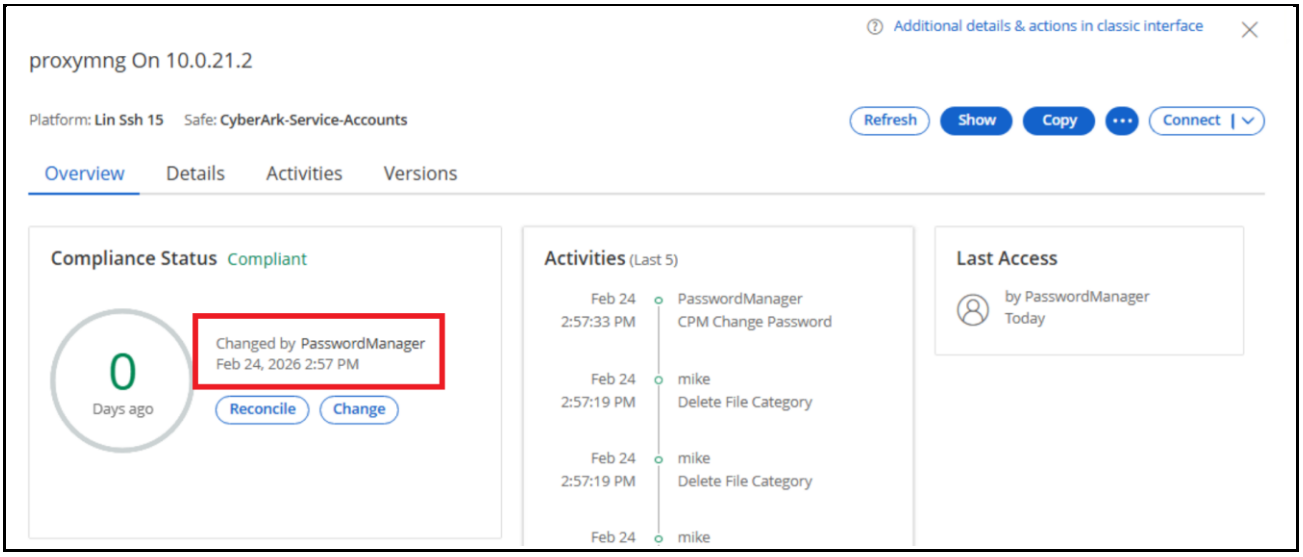


5. Now hover the mouse over the **Accounts** tab on the left-hand side and select **Accounts**. This will bring us back to the new interface. Click on the *proxymng* account again. You can see that the password has been scheduled for change. Press refresh until you see the password has been changed. After a minute or two, you will see that the password has indeed been changed by the user *PasswordManager* (in other words, the **CPM**).



6. Close the **PVWA**.

PrivateArk Client

In this section, we will see how to perform a basic file retrieval using the **PrivateArk Client**. The file we are going to retrieve is *italog.log*, the **Vault**'s main log file.

Connecting

1. In the Windows taskbar, click on the shortcut to launch the **PrivateArk Client**.

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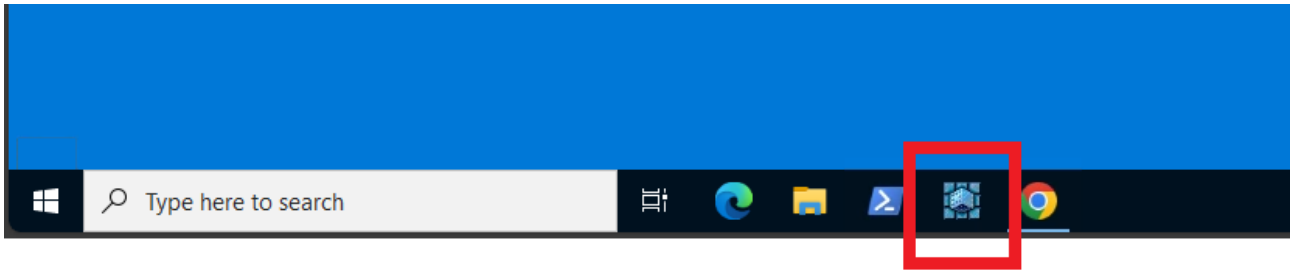
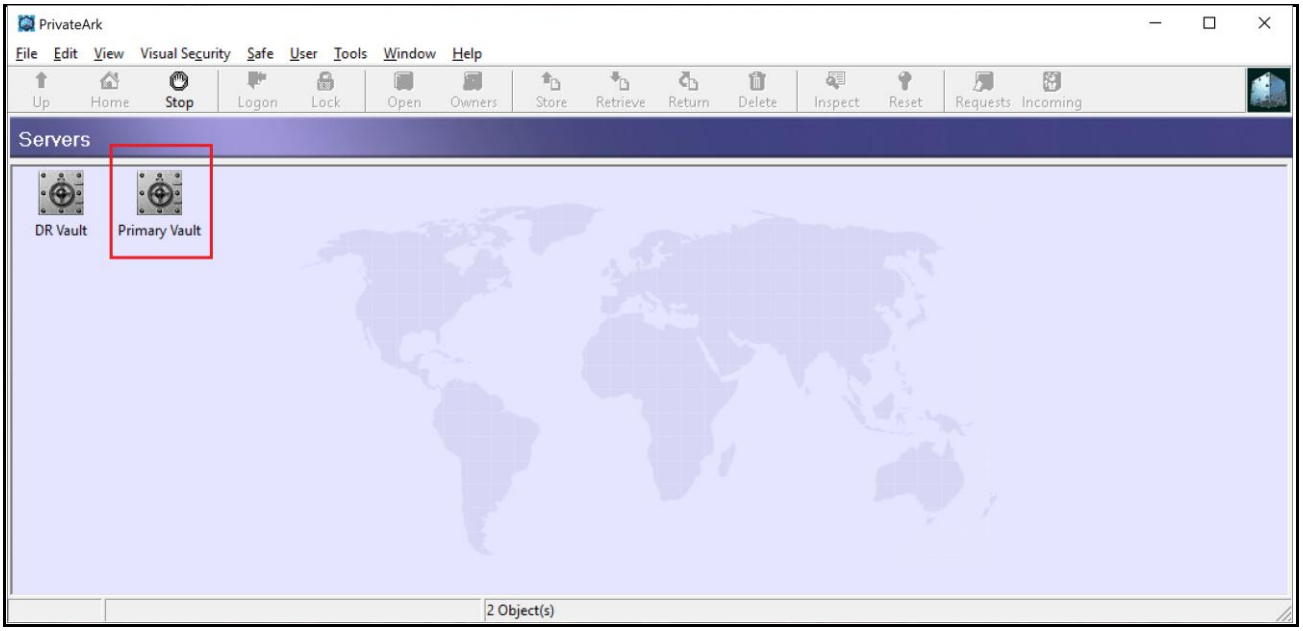


2. Now double-click on the link ***Primary Vault***. You can configure multiple **Vault** connections here: Primary, Disaster Recovery, etc.

Note:	You will notice you have two servers configured: ***Primary Vault*** and ***DR Vault***. When you are requested in this guide to connect using the **PrivateArk Client**, always use the ***Primary Vault*** unless stated otherwise.



3. Enter the *Cyberark1* as the built-in **CyberArk** administrator's password and click **Logon** (or hit **Enter**).

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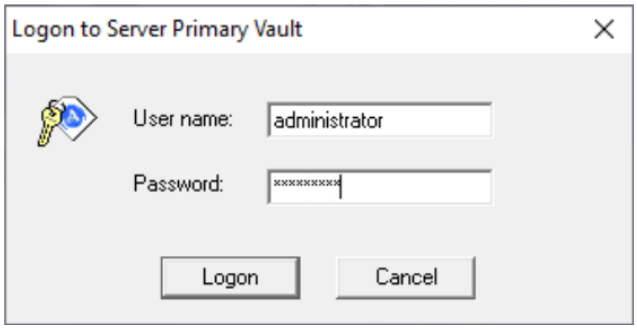
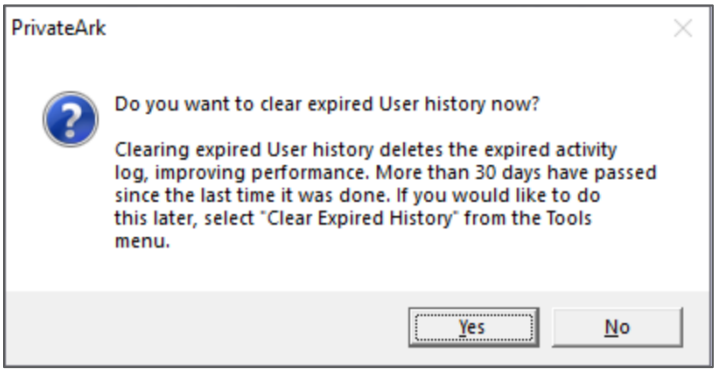


4. You may receive a message asking you whether you want to clear expired user history. Click **Yes**.



Note: It is not possible to be connected to the **CyberArk PAM** solution via two interfaces at the same time using the same user account. If you have not logged out of your session on the **PVWA**, logging into the **PrivateArk Client** with the same user will terminate your **PVWA** session. The reverse, however, is not true: if you leave your **PrivateArk Client** session open and try to log into the **PVWA** with the same user, you will not be able to connect until the **PrivateArk Client** session is terminated.

Accessing a File in a Safe

1. Now we are in the main window looking at the **Safes** to which the current user has access. The **Safe** we are interested in is named *System*. Double-click on it to open it and “step into” the **Safe**.

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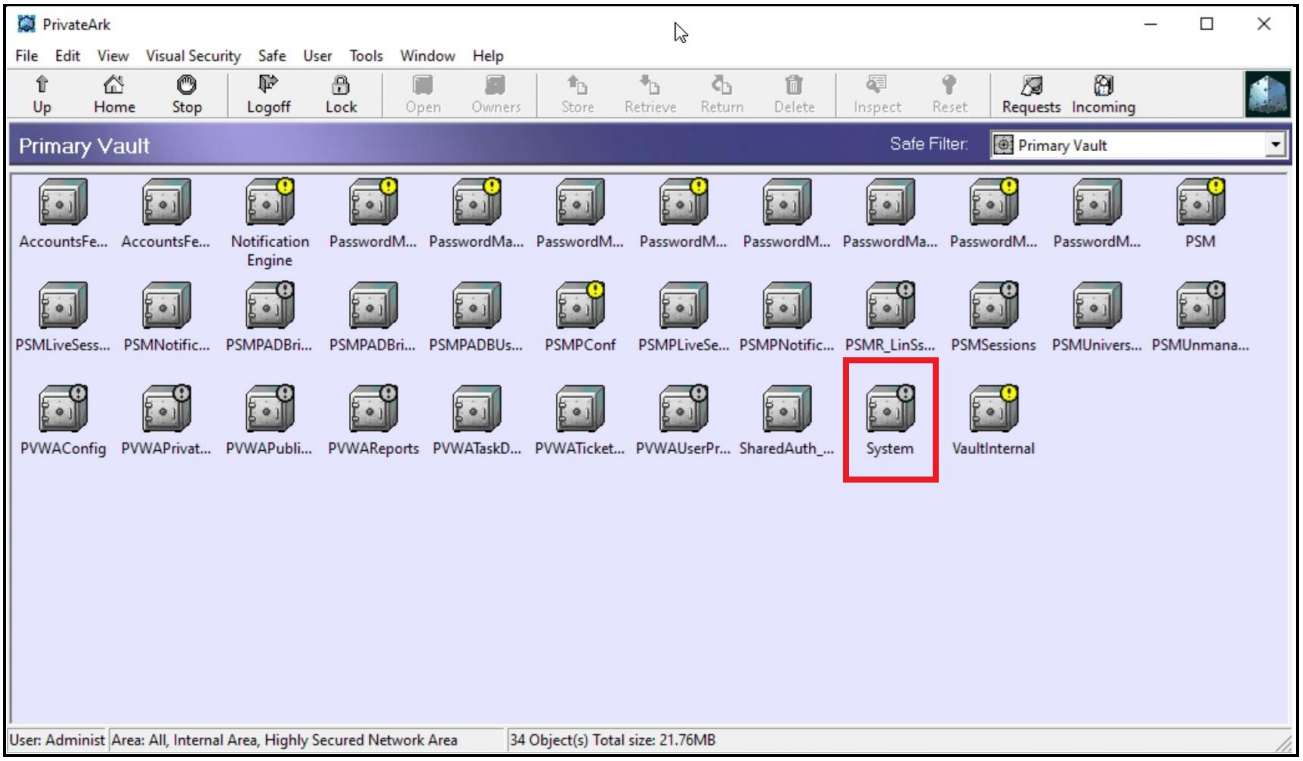
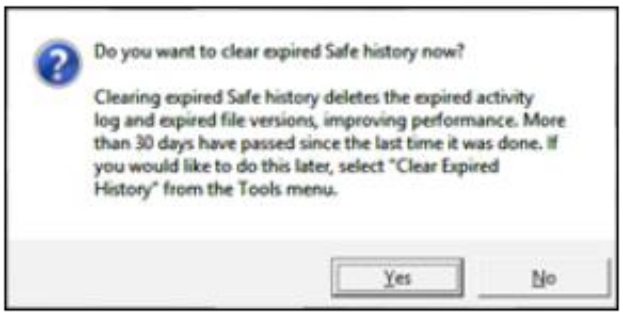


You may receive a message asking if you want to clear expired **Safe** history. Click **Yes**.



2. The file we want to view is *italog.log*. We are not going to modify the file, so right-click on it and select *Retrieve for Read-Only*.

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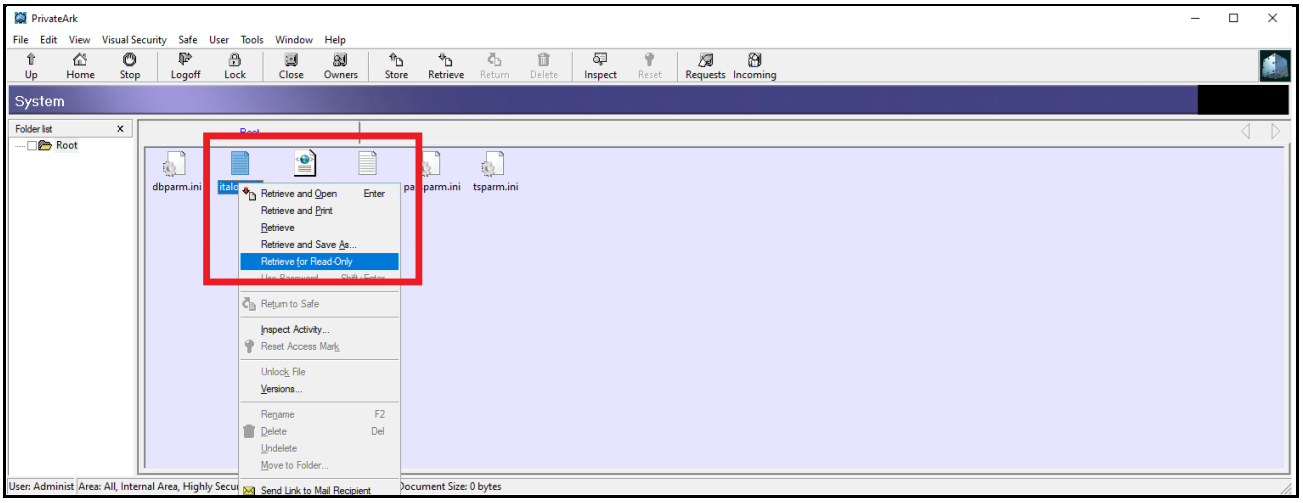
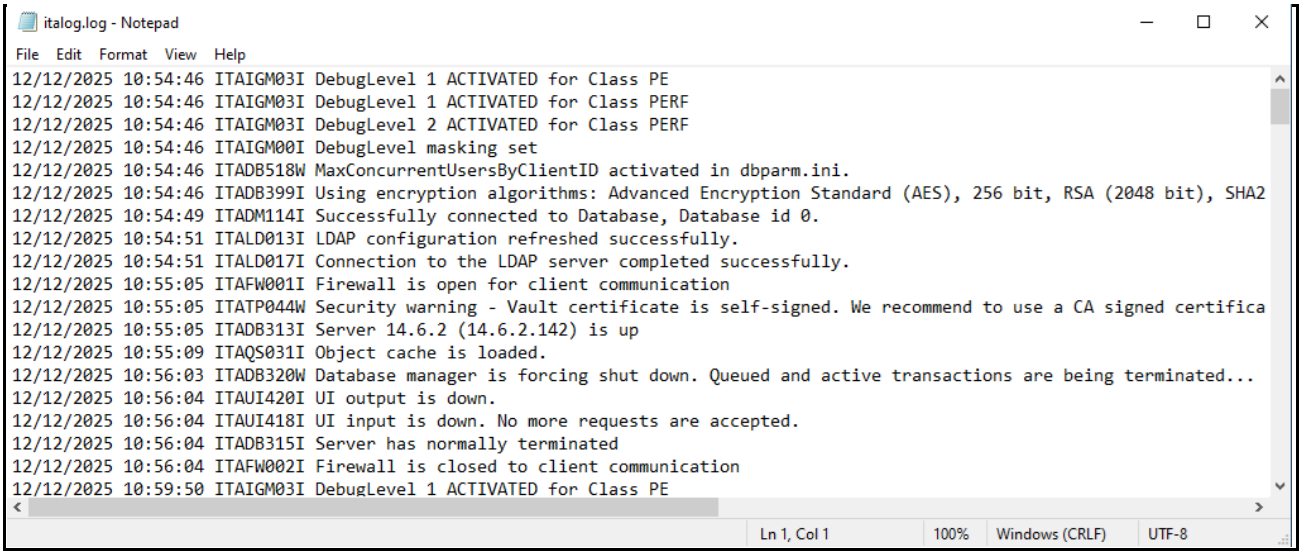


3. The file is extracted from the **Safe** and displayed. Take a moment to view some of the log messages and then close the file.



4. To indicate to the **Vault** that we are finished with the file, right-click on it again in the **PrivateArk Client** and select *Return to Safe*.

Modifying the View

You can change how you view the **Safes** by going to the **View** menu.

1. Click the **Up** button (in the upper lefthand corner of the window) to navigate back to the top of the **Safes** list and click **View** and then **Details** (or whatever view you prefer).

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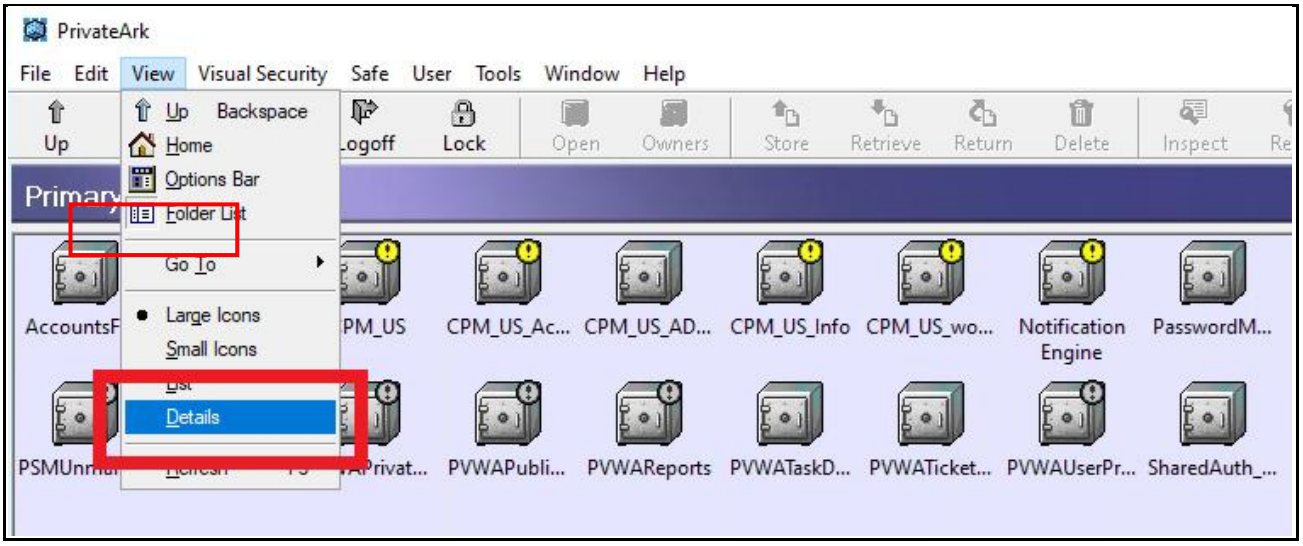
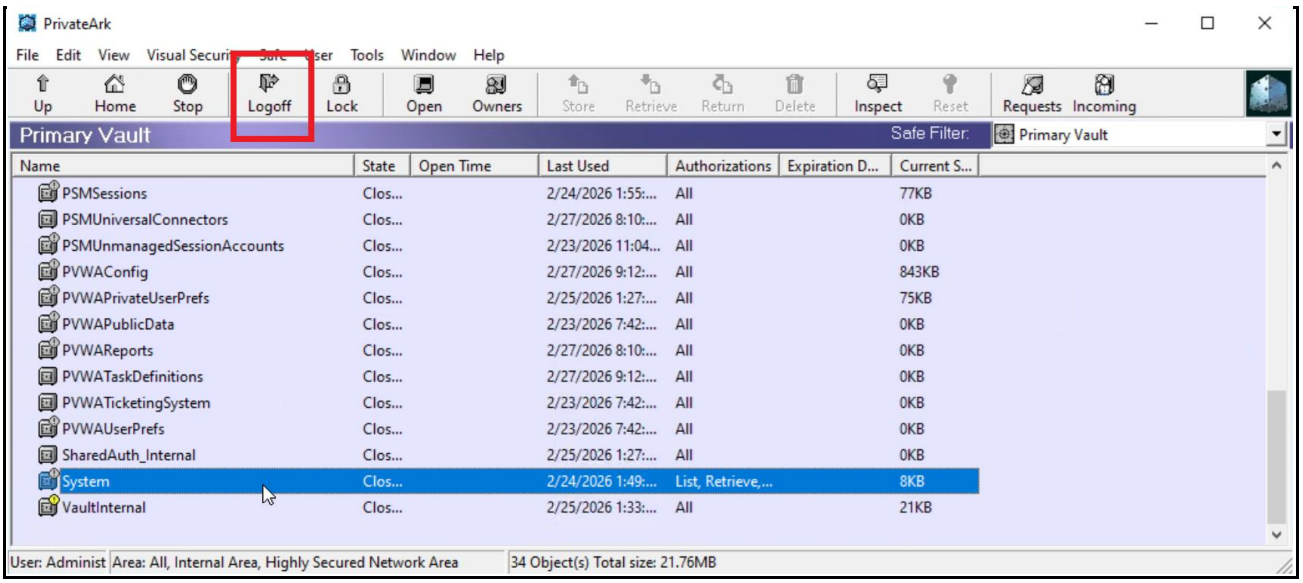


2. Notice that the *System* **Safe** is still open. You can either use the **Logoff** button or simply close the **PrivateArk Client**. Both will close the **Safe** and terminate your session.



Remote Control Client

We are now going to execute a few simple commands using the **Remote Control Client**, a command-line tool for performing remote administration on the **Vault**.

This command line application can only be run as an administrator, so in order to use it we will connect to a Windows machine (actually, the same one we are on: **Comp01**) using an account with administrative privileges on the local machine.

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Note:	You can run the **Remote Control Client** using either Windows terminal or Powershell. We will use Powershell.

1. On the desktop of the **Comp01** server, there are three RDP shortcuts in the upper-right corner:

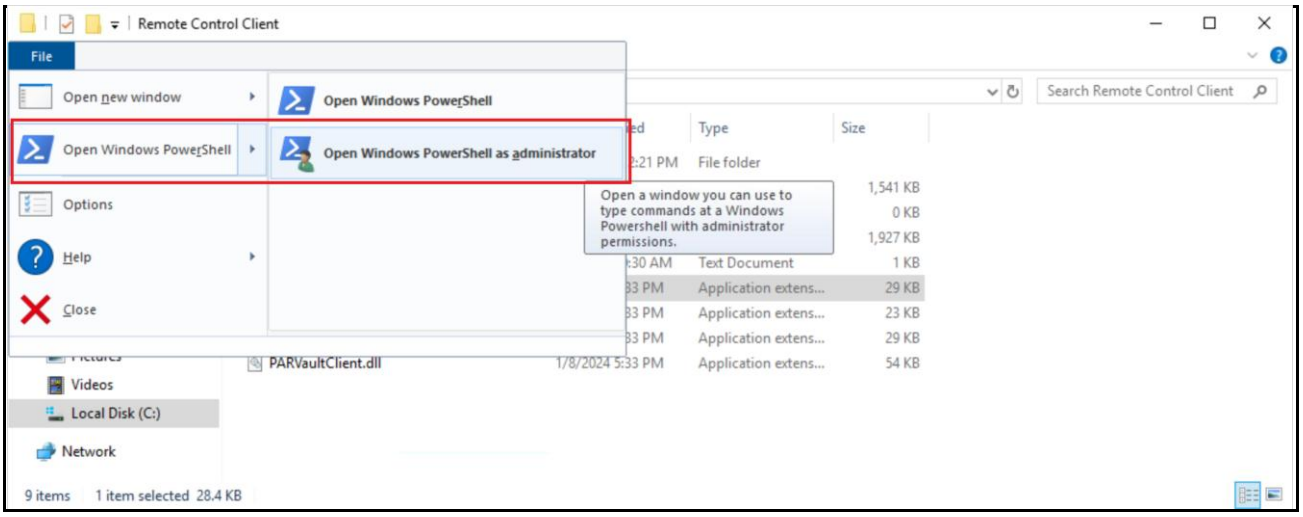
   - ***dom@COMP01***
   - ***dom@COMP02***
   - ***dom@acme-123***

   Our Vault Admin *Mike* uses these shortcuts to get administrative access to the servers running **CyberArk** components, but they represent a security threat, so we will have to deal with this problem at some point.

   Double-click on the first one - ***dom@COMP01*** and when you are prompted, enter *Dom*'s password: *Cyberark1*.

2. In *Dom*'s session, launch Windows file explorer and browse to: *c:\Remote Control Client*.

3. Click on **File | Open Windows Powershell | Open Windows Powershell as administrator**.



4. This will open a Powershell terminal in administrative mode. Execute the command shown highlighted in yellow below. Note the dot + back slash at the beginning of the command.

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```
.\PARClient.exe 10.0.10.1

Cyber-Ark Remote Administration Client (15.0.XX.x)

Working with agent on: 10.0.10.1

Loaded component from [C:\Remote Control Client\PARClusterVaultClient.dll]
Loaded component from [C:\Remoteexit Control Client\PARDRClient.dll]
Loaded component from [C:\Remote Control Client\PARENEClient.dll]
Loaded component from [C:\ Remote Control Client\PARVaultClient.dll]
PARCLIENT>
```

Note:	The connection string is made up of the path to the executable and the **Vault** IP address. You will be prompted for the password the first time you run a command. The password for the **Remote Control Client** is set during installation and cannot be managed by the **Vault**.

5. Once you have the **PARCLIENT** prompt, get the current **Vault** status by running **status vault**. You will be prompted for your password: *Cyberark1*

```
PARCLIENT> status vault
Password:
Vault is running.
```

Note:	Depending on how long after logging in you wait to run the command, you may be prompted to re-enter the password.

6. To stop the **Vault**, run the following:

```
PARCLIENT> stop vault
Are you sure you want to stop the remote Vault (Y/N)? y
Vault was stopped successfully
```

7. To restart the **Vault**, run the following:

```
PARCLIENT> start vault
Vault was started, pending service running. Use status command for further
details.
```

8. When you stop the **Vault**, the ***Event Notification Engine***, or ENE, service is also stopped because it is dependent on the **Vault** service. However, when you start the **Vault**, the ENE is not automatically restarted. You must restart it manually by running:

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```
PARCLIENT> start ene
ENE was started, pending service running. Use status command for further details
```

9. As a final step, check the status on these two **Vault** services by running:

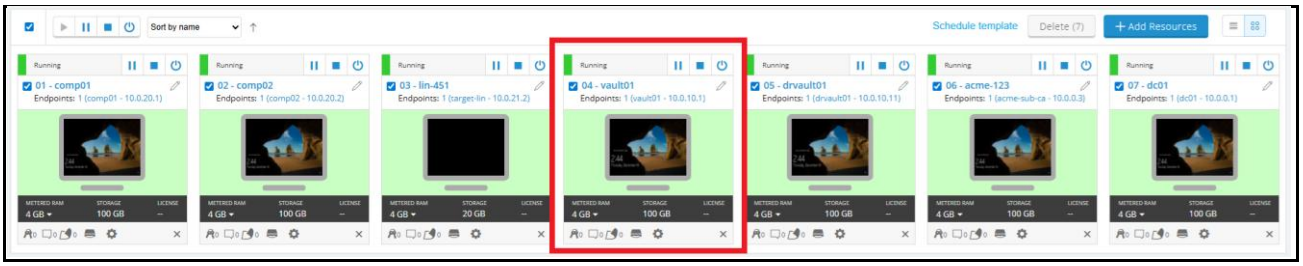
```
PARCLIENT> status ene
ENE is running.
PARCLIENT> status vault
Vault is running.
```

10. Type **exit** and hit enter to exit the **PrivateArk Remote Control Client**.

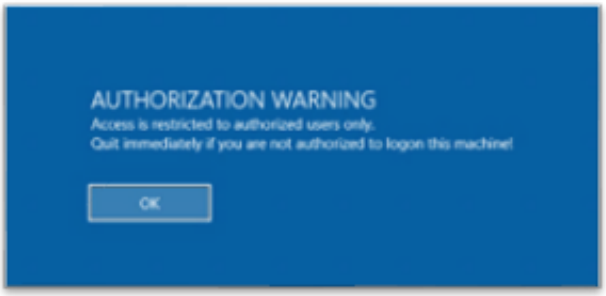
Note:	Make sure that you do not shut down the server when exiting the RDP session.

The Vault Server

In the last section for this first chapter, we will see how to stop and restart the **Vault** service directly on the **Vault** Windows server itself. To do this, we will need to switch in Uplift from the **Component** server to the **Vault** server.



1. Log into Windows with the local administrator account *administrator*, password *Cyberark1*. You will receive an authorization warning message.



2. On the desktop of the **Vault** server, you will find two **CyberArk** icons:

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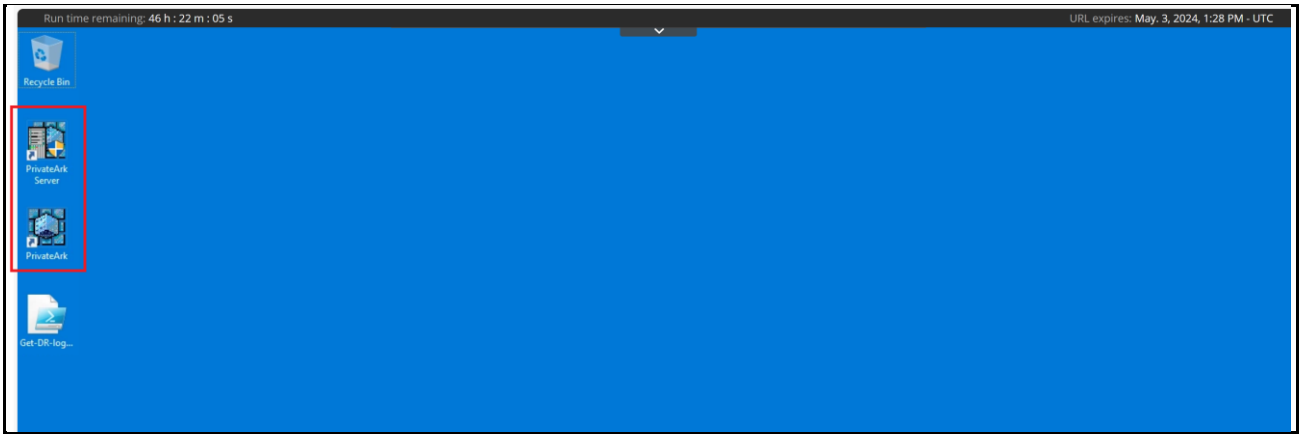
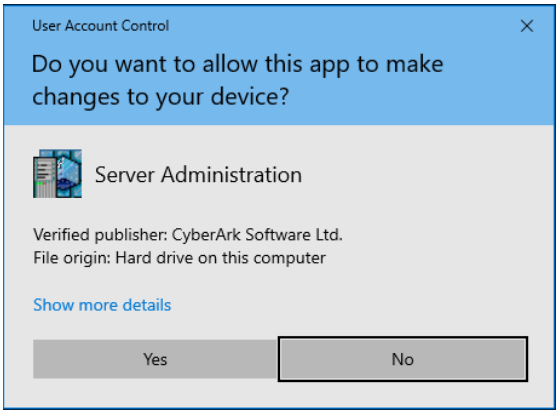
- **PrivateArk Server**
- **PrivateArk Client**

Double-click on the **PrivateArk Server** shortcut.



3. You will receive a **User Account Control** alert. Click **Yes** to allow the action.



4. The main function of the **Server Central Administration** tool is to view the *italog.log* file and to stop and restart the **Vault**. Click on the red traffic light icon to stop the **Vault** service.

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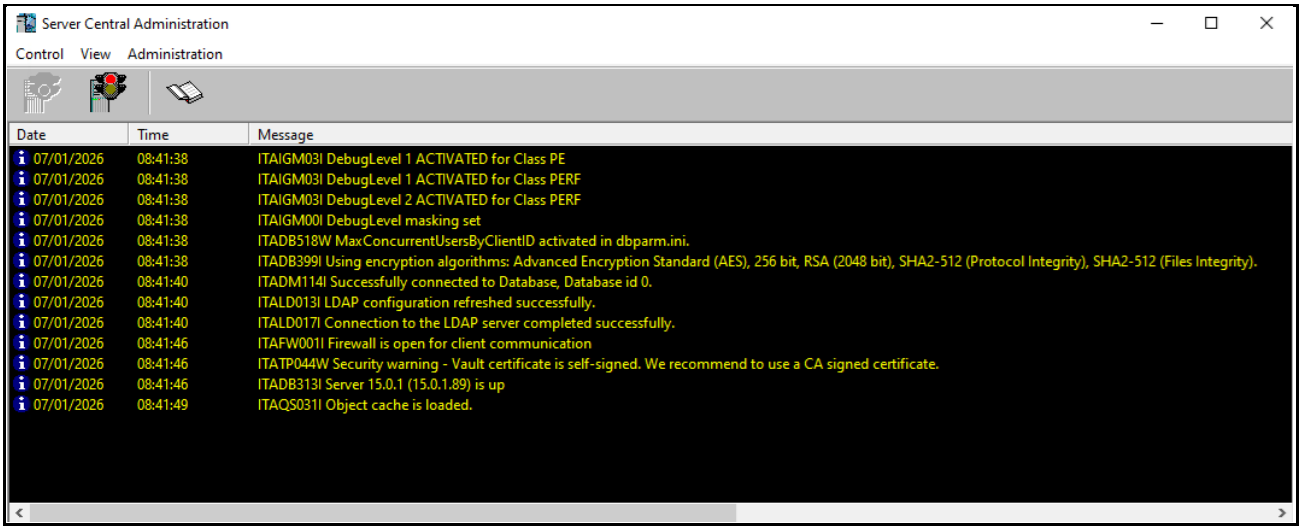
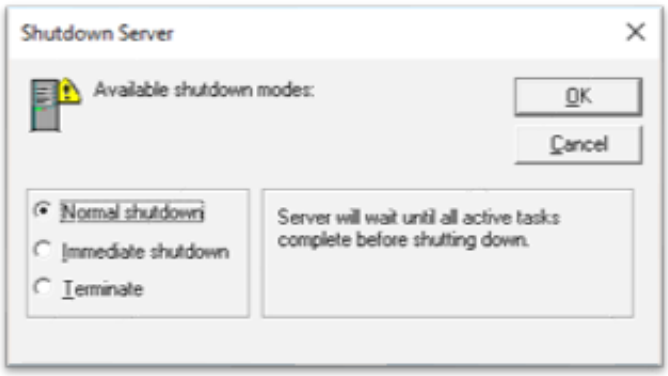


Note:	If you see an error message related to LDAP, it can be safely ignored. It is linked to our Uplift environment and simply means that when the **Vault** started, LDAP (Active Directory in our case) was not ready to communicate.

5. You will be prompted for the type of shutdown. Choose *Normal shutdown* and click **OK**.



6. You will be asked to confirm **Vault** shutdown. Click **Yes** to shut down the **Vault**.

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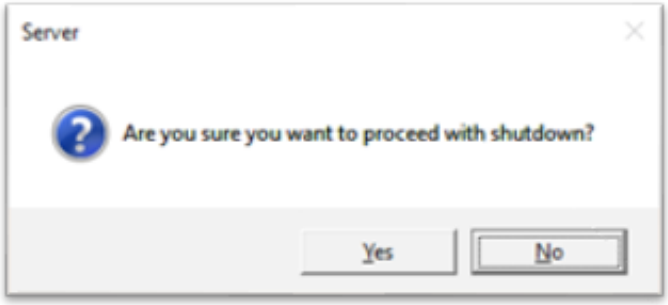
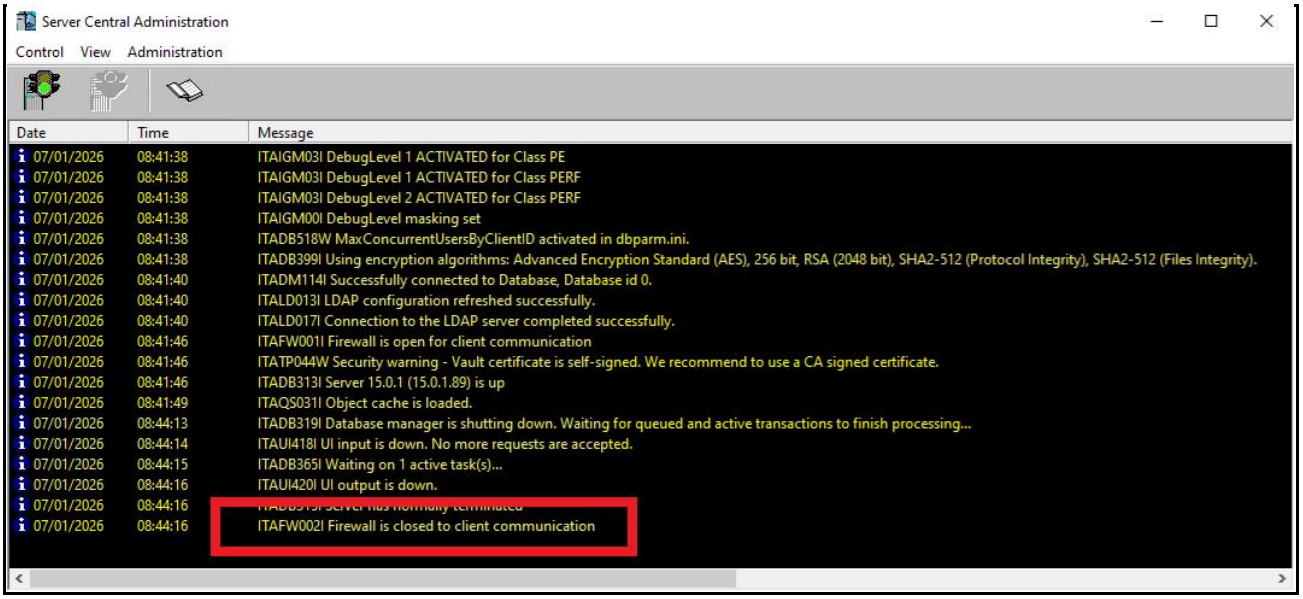


7. You will see the messages indicating the shutdown procedure, ending with the message: **`ITAFW002I Firewall is closed to client communication.`** To restart the **Vault** service, click on the green traffic light icon.



8. You will see several messages indicating that the **Vault** is starting up.

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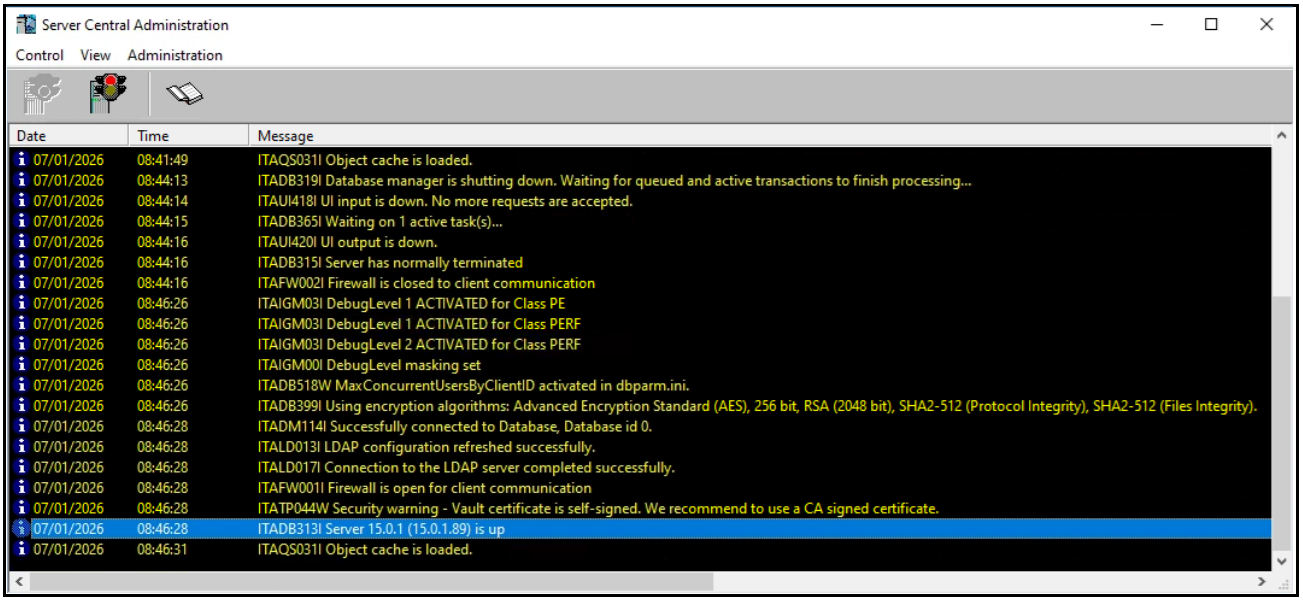
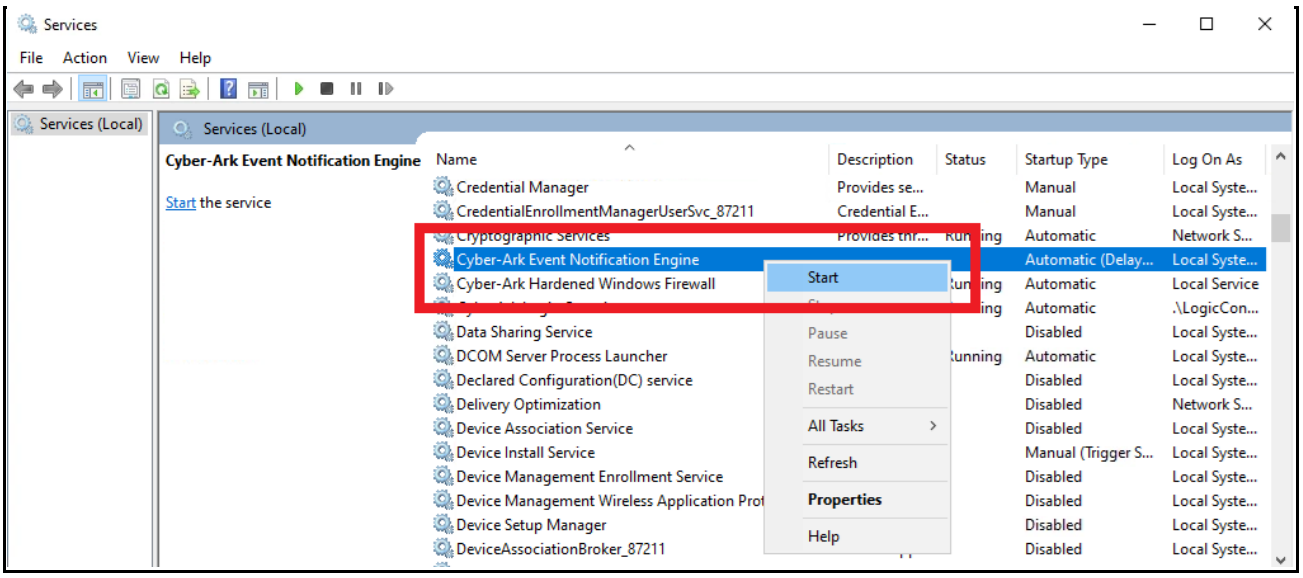


9. As was the case with the **Remote Control Client**, starting the **Vault** in the **Server Central Administration** tool does not restart the **Cyber-Ark Event Notification Engine** (as it is listed in the local services). The **ENE** is essential for the **Vault** to send emails and alerts, so you will have to start it by going into the **Services** tool on the **Vault** server and starting the service there. You will find a shortcut in the taskbar on the **Vault** desktop.



10. When you have finished, you can sign out of Windows on the **Vault**. Make sure you don't shut down the **Vault** server by accident.

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Note:	If you need to perform a restart of the **PrivateArk Server** service (for example, after making a change to the **Vault** configuration), using the **Services** interface is probably a better option. You can simply select the service, right-click on it, and select **Restart**. Because the **Event Notification Engine** is dependent on the **PrivateArk Server**, you will be prompted to restart that as well.

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User Management

Know the Players

Before we begin, let's first get to know the different users we will be using throughout this lab and their roles. The password for all these users is *Cyberark1*.

Username	Authentication Method	CyberArk Role	LDAP Group
Administrator	***CYBERARK***	***Vault Admin***	**-**
Master	***CYBERARK***	***Master User***	**-**
CyberArk Team (AD)			
Mike	***LDAP / PKI***	***Vault Admin***	***CyberArk Vault Admins***
Cindy	***LDAP***	***Auditor***	***CyberArk Auditors***
Dexter	***LDAP***	***Help Desk (custom)***	***CyberArk Help Desk***
Linux Team			
Paul	***LDAP***	***Safe Manager***	***CyberArk Safe Managers***
Carlos	***LDAP***	***User***	***LinuxA dmins***
Windows Team			
Tom	***LDAP***	***Safe Manager***	***CyberArk Safe Managers***
John	***LDAP***	***User***	***Windows Admins***
Oracle Team			
Robert	***LDAP***	***Safe Manager***	***CyberArk Safe Managers***

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LDAP Integration and Directory Mapping

In this section we will review LDAP integration with **CyberArk PAM** and the predefined directory mapping to four common CyberArk roles.

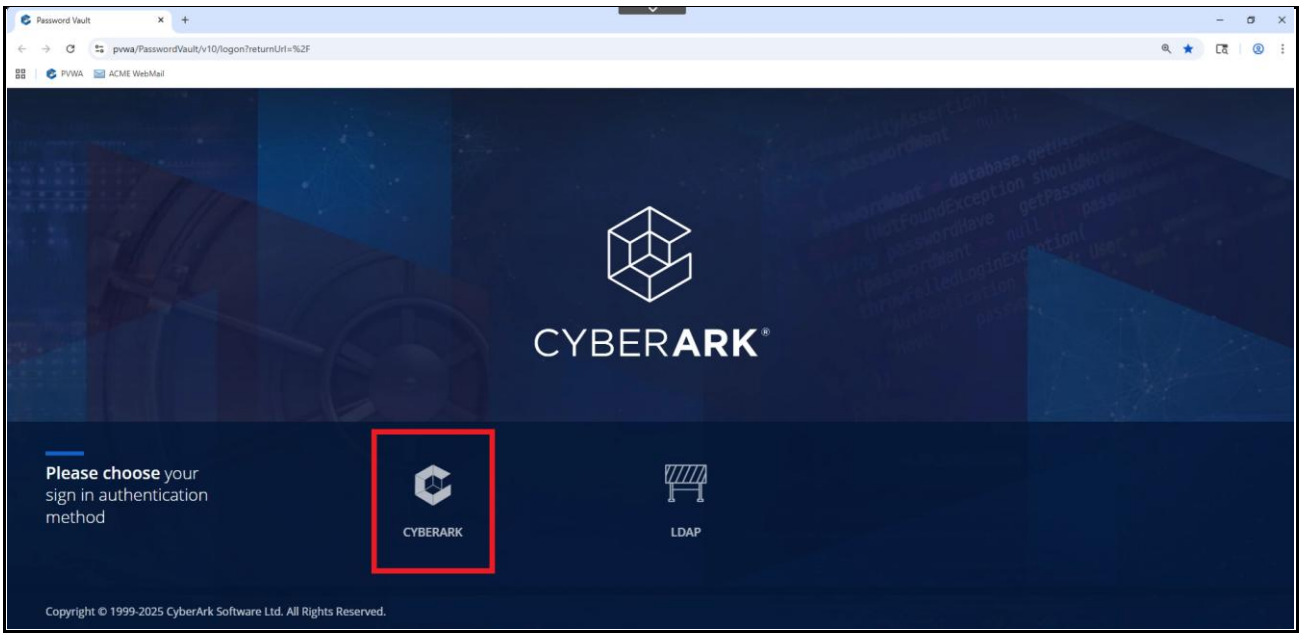
LDAP integration is a two-step process:

1. Create the connection to the LDAP server, which in our case is Active Directory.
2. Create the directory mappings between the AD groups and the built-in CyberArk roles.

The above steps have already been completed by the implementation team. We will simply review the predefined directory mappings and examine the authorizations assigned to four common **PAM** roles. Because this defines how **CyberArk PAM** interacts with LDAP, an LDAP user cannot perform this task, so we will be using the built-in **CyberArk** *administrator* account.

Review LDAP Integration and pre-defined Directory Mappings

1. Launch Chrome to open the **PVWA** page and click on **CYBERARK** as the authentication method.



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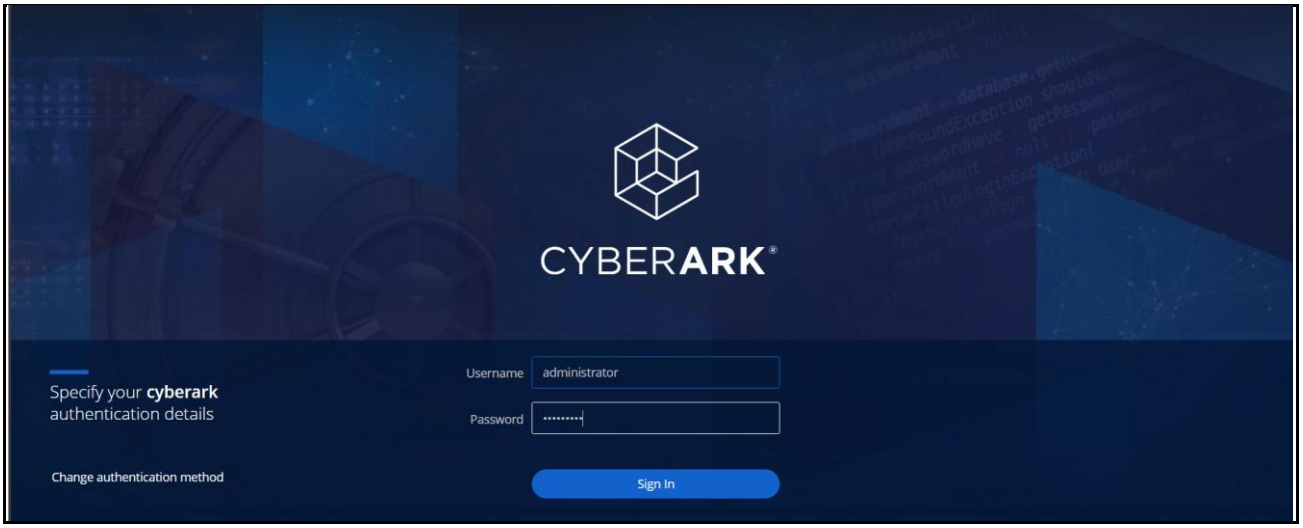
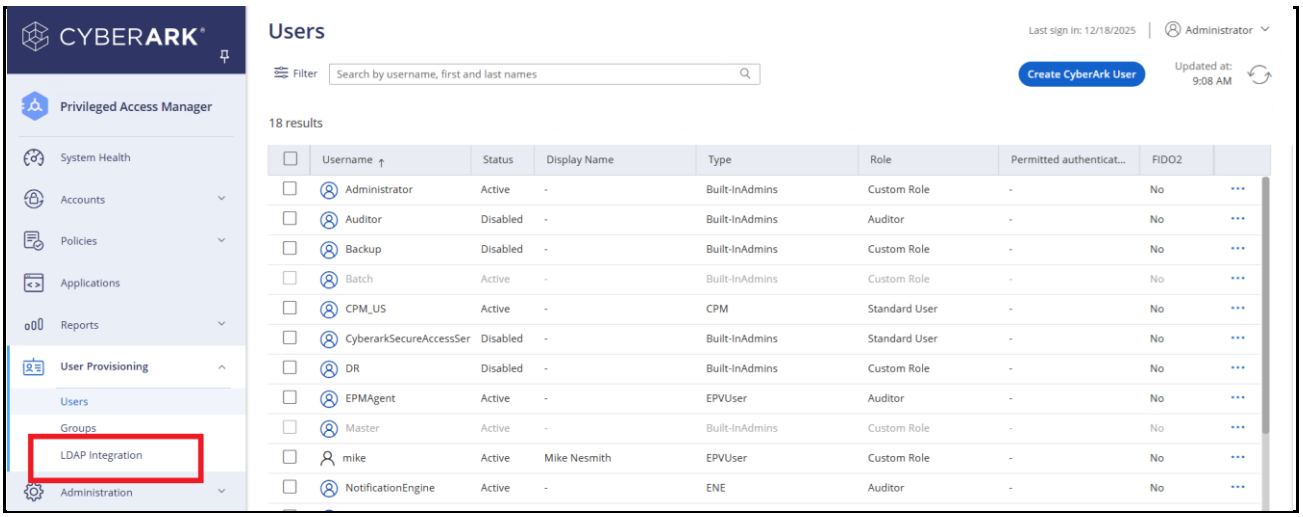
2. Enter the credentials: Username: *administrator,* password: *Cyberark1.*



3. Along the left side of the window, you will find the navigation tabs. The **User Provisioning** tab is the next to last one. Click the down arrow and select **LDAP Integration**.



Note: You can pin the left-hand menu to the screen using the thumbtack icon just below the name **CYBERARK** at the top.

4. Click on *acme.corp* and note that **CyberArk PAM** has been integrated with the domain and that four directory mappings have been defined.

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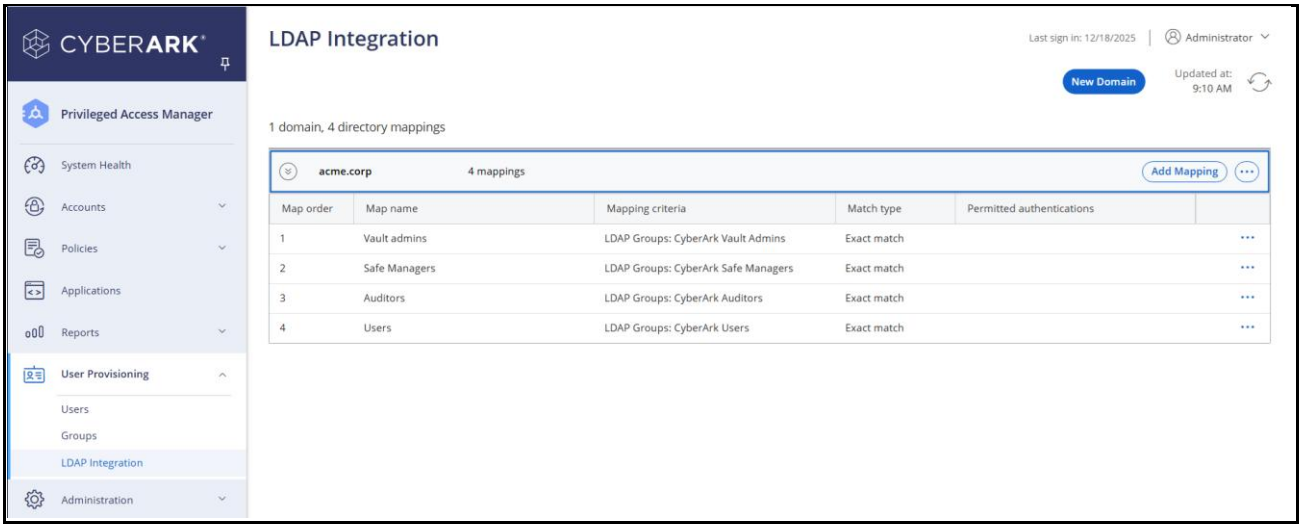

5. Each LDAP group is mapped to a **CyberArk PAM** role as shown in the table below.

LDAP Group	CyberArk Role
CyberArk Vault Admins	Vault Admins
CyberArk Safe Managers	Safe Managers
CyberArk Auditors	Auditors
CyberArk Users	Users

6. Click on the **Vault admins**. Clicking on the mapping brings you to the **Details** tab, where you can see the mapping criteria, the mapping destination group in the **Vault**, the authentication method the mapped users will use to authenticate to **CyberArk PAM**, and how many days user activity logs are kept.

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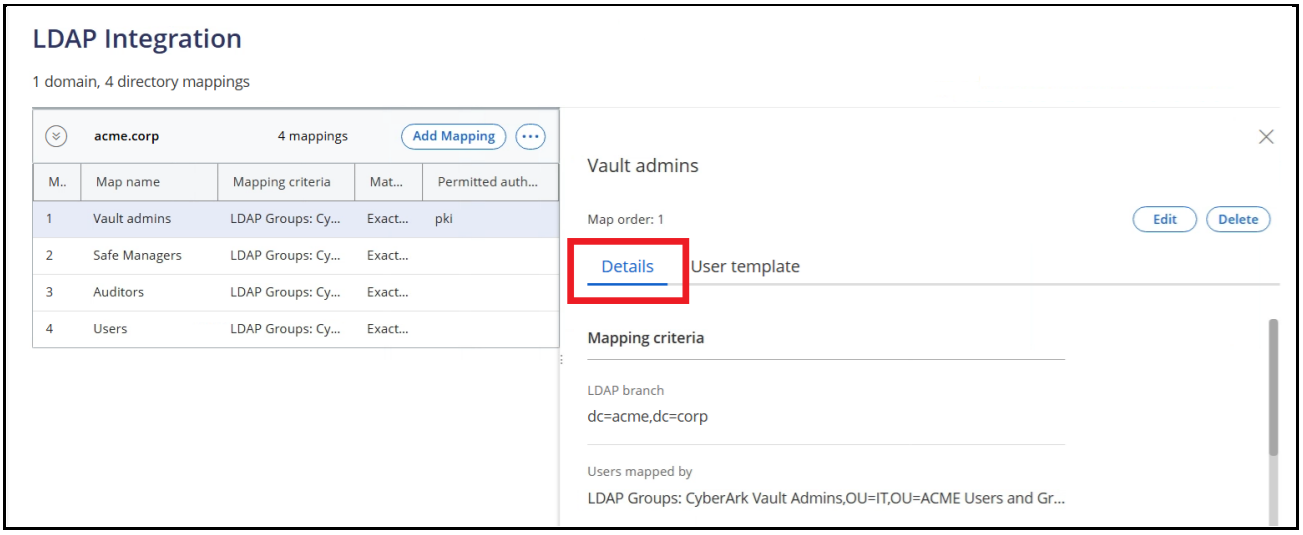
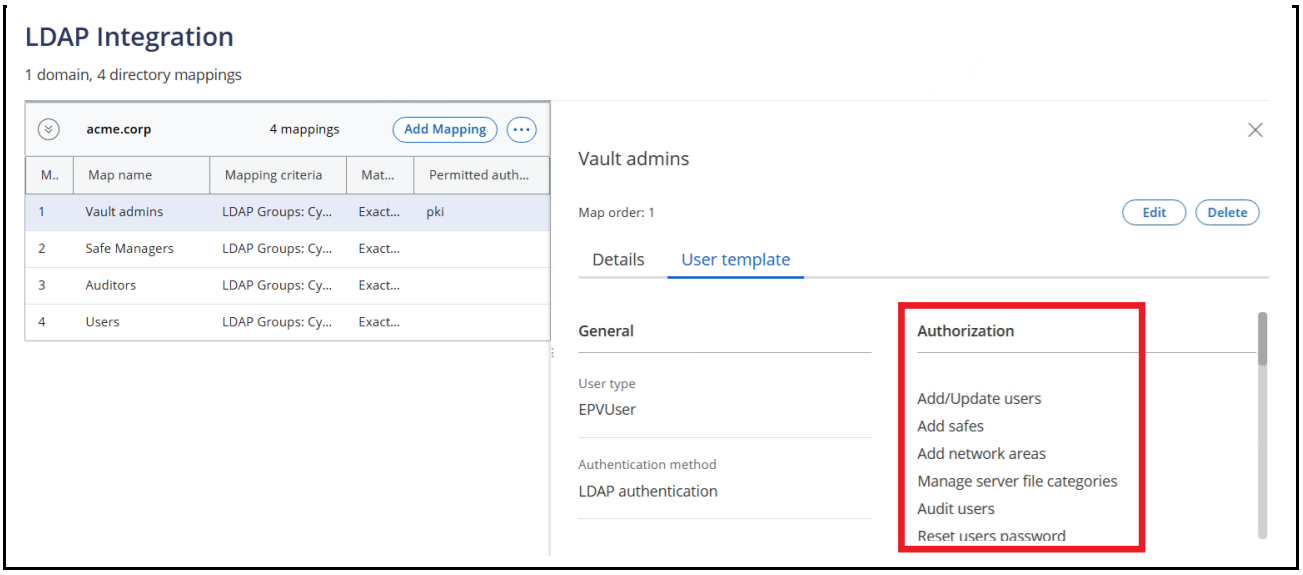


Note: In the above example, we can see that users who belong to the AD group *CyberArk Vault Admins* are mapped to this role, and that the authentication method they will use is LDAP.

7. To know what **Vault** authorizations are assigned to the mapped users, click on the **User template** tab. Here we can see the **Vault** authorizations assigned to this mapping. The members of the AD group *CyberArk Vault Admins* will be assigned these **Vault** authorizations when they authenticate to **CyberArk PAM** for the first time. When you are ready, click on the **Edit** button.



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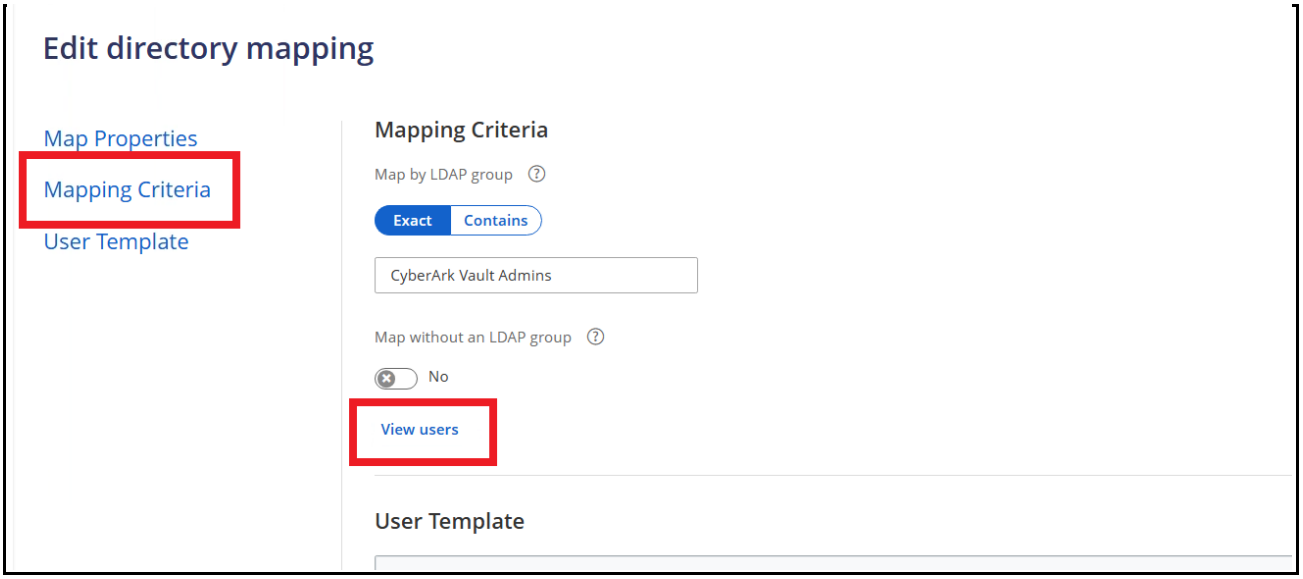
8. Note you can now edit all the settings we reviewed in the **Details** page, as well as edit the **Vault** authorizations that are assigned to users who meet the search criteria. Click on the Mapping Criteria link to jump to that section, and then click **View users**.



9. Here we can review which LDAP users are currently members of the AD group **CyberArk Vault Admins** and who are eligible for this role.

10. Repeat the above steps to review the details of the other three pre-defined mappings: *Safe Managers, Auditors and Users*. Note the following for each mapping:

    - What are the mapping criteria for this mapping?
    - Which users currently meet the mapping criteria?
    - What **Vault** authorizations are assigned to users who meet the criteria?

Test the LDAP Integration and Pre-defined Mappings

Test the integration by logging in with the following users (all have the password *Cyberark1*).

- Cindy
- Paul
- Carlos

Take note of the differences in access to different **PVWA** panes and buttons.

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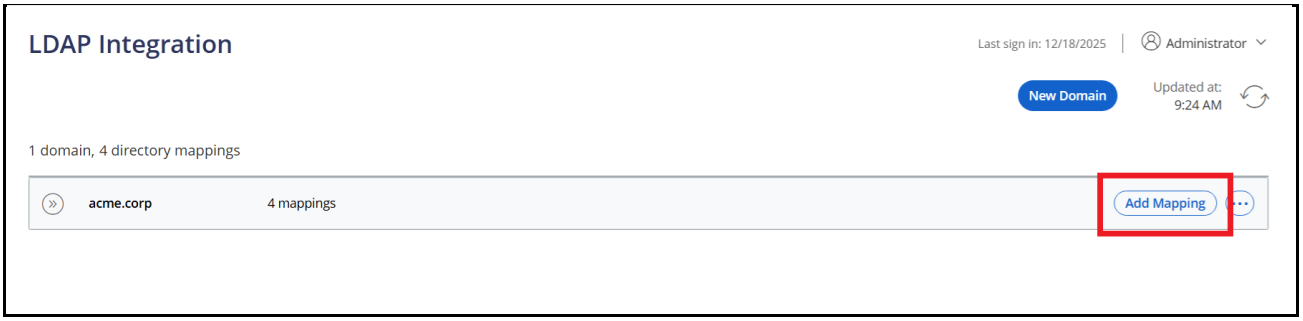
Configure Custom Directory Mapping

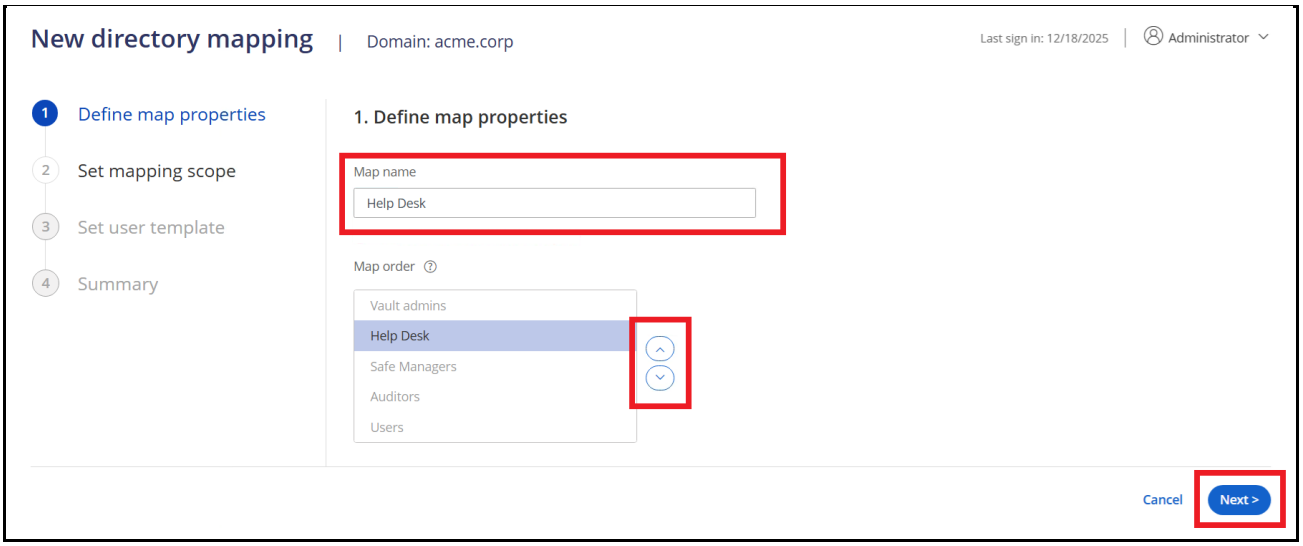
In this section we will create a custom directory mapping for ***CyberArk Help Desk*** – a group that needs unique **Vault** authorizations to manage users in **CyberArk PAM**.

1. Login to the **PVWA** as *administrator* using **CYBERARK** authentication with the password *Cyberark1*.

2. Navigate to **User Provisioning** | **LDAP Integration**. This time, select **Add Mapping**.



In the **Map name** field, enter *Help Desk*. Click in the **Map order** section to update the display and move *Help Desk* to the second position using the up and down arrows. Then click on **Next**.



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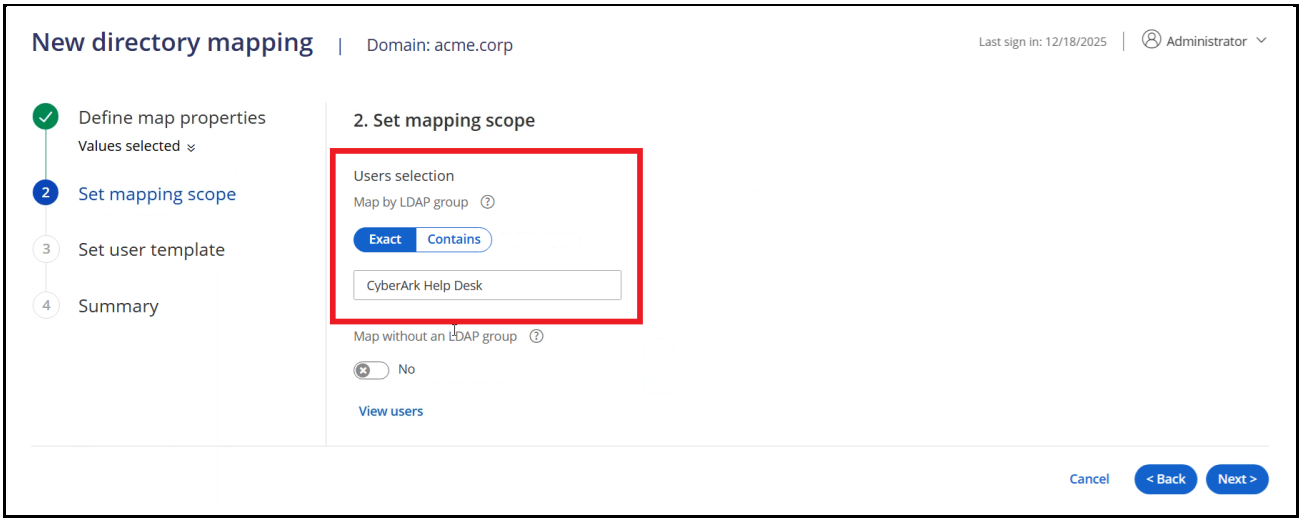
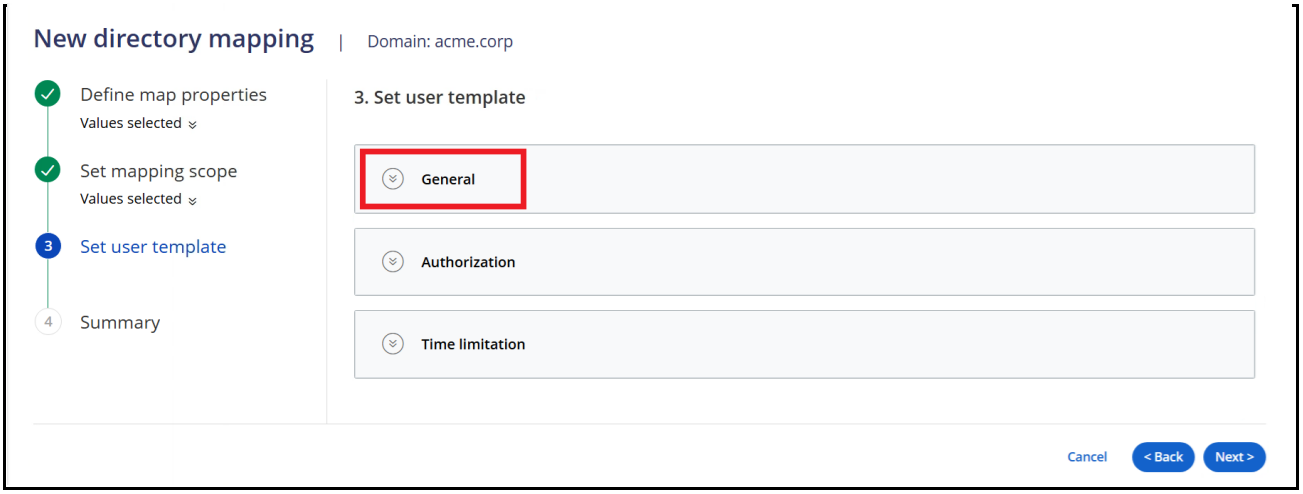
Note:	The mapping order is important for users who belong to multiple groups/mappings. For example, if a user belongs to both *Help Desk* and *Vault Admins* mappings, the user will receive the privileges for the first mapping listed. If *Help Desk* was listed first, a user who is also a help desk user would only receive the help desk subset of Vault authorizations, instead of the full set provided by the *Vault Admins* mapping.

3. In the field **LDAP group**, type 'cyber' (once you begin typing the name should autocomplete with all the groups that begin with that string). Select the Active Directory group **CyberArk Help Desk**. You may click on **View users** to view the users to which the directory mapping will be applied. Then click on **Next**.



4. In step 3, we will configure the user template that is used to create users when they log in for the first time. Click on **General** to expand that section.



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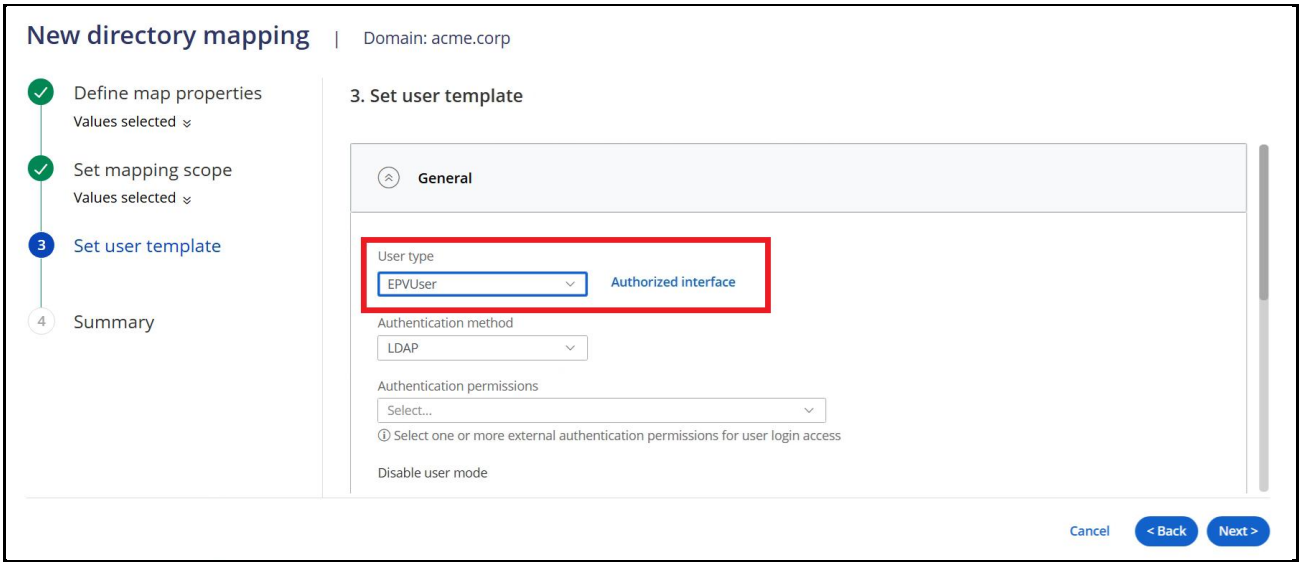
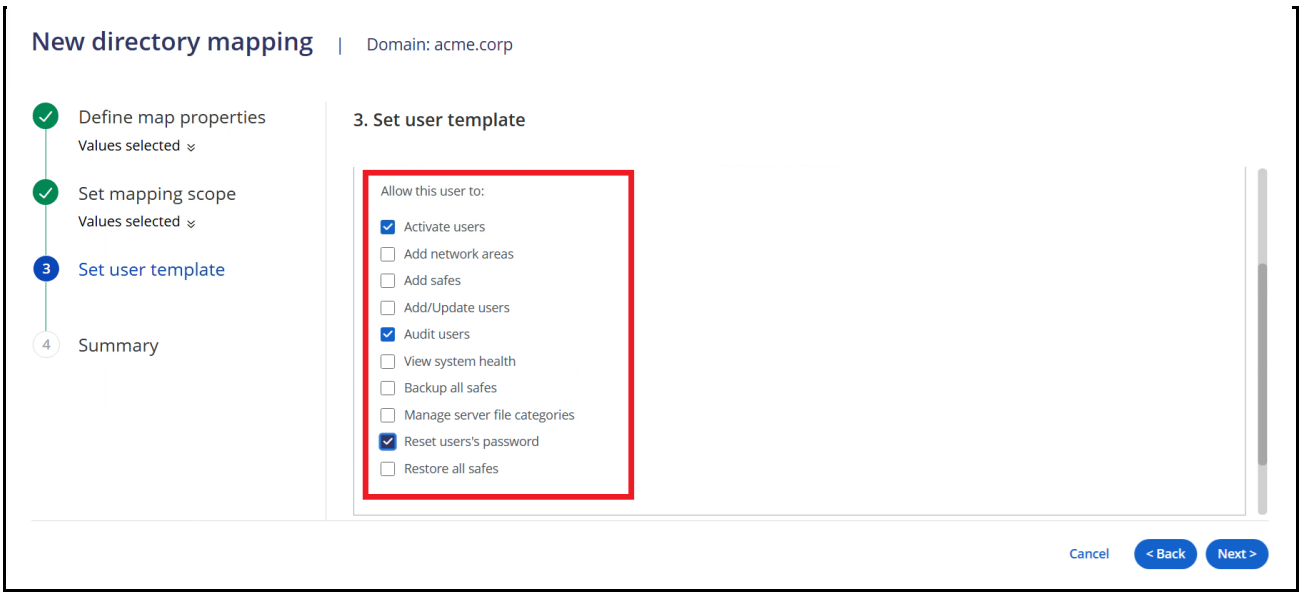
5. Drop down the **User type** list and select **EPVUser**. If we wanted to increase the security of our environment, we could limit the clients to which Help Desk users have access by clicking on **Authorized interface** and unchecking any we did not want to grant.
   Scroll down to the **Authorization** section.



6. We will grant Help Desk users the right to:

   - **Activate users**
   - **Audit users**
   - **Reset users' passwords**



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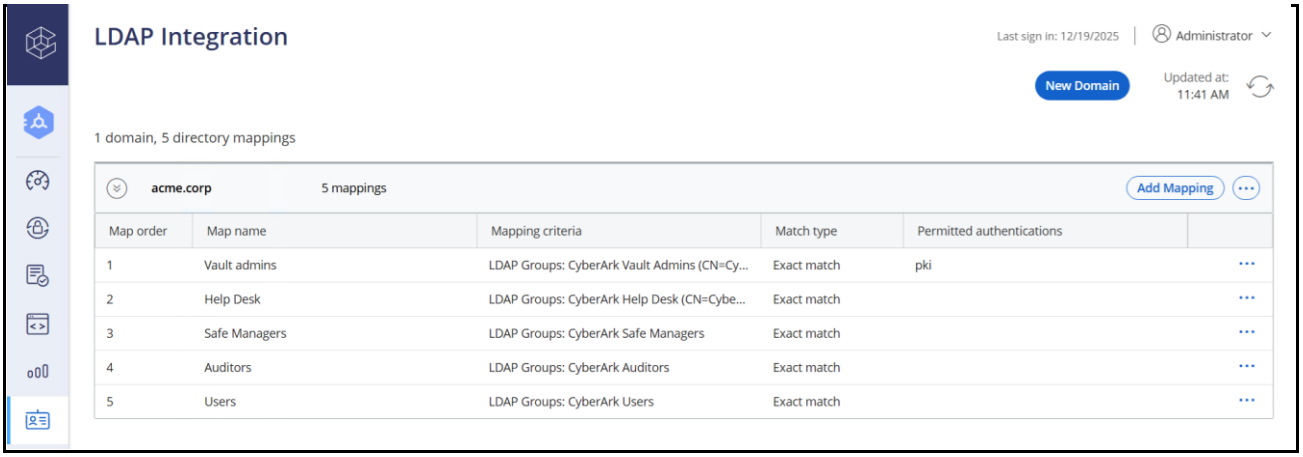
7. We are not going to set any time limitations on these users, so click **Next**, review the **Summary** page, and then click **Save**. When you return to the main **LDAP Integration** page, you will see that you now have five mappings.



8. Log off the **PVWA**.

Test Custom Directory Mapping

To test this custom mapping, we will log in to the **PrivateArk Client** as *Dexter*, who works in the **CyberArk Help Desk** for **Acme**. In this exercise we will also see how to change the authentication method used in the **PrivateArk Client**.

Note:	User management capabilities have now been introduced also in the **PVWA** interface. The new interface enables you to create and edit **CyberArk** users, create groups, disable users, reset user passwords, and much more. In the following steps we will go through **PrivateArk Client** user management, and we will unsuspend a suspended user using the new web interface.

1. Open the **PrivateArk Client** using the shortcut on the desktop of **comp01**.

2. Right-click on the **Primary Vault** and select **Properties**.

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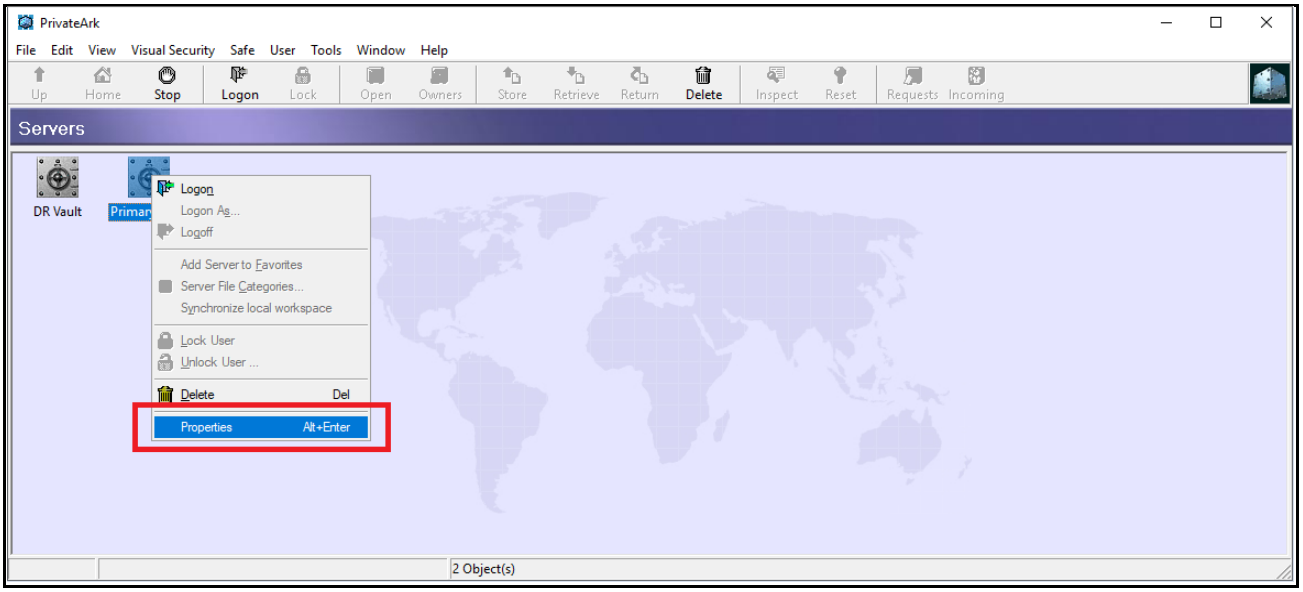
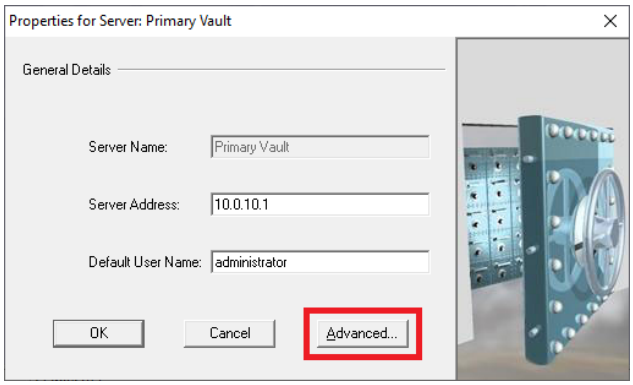


3. Click on **Advanced**.



4. Select *LDAP authentication* and then click on **OK**.

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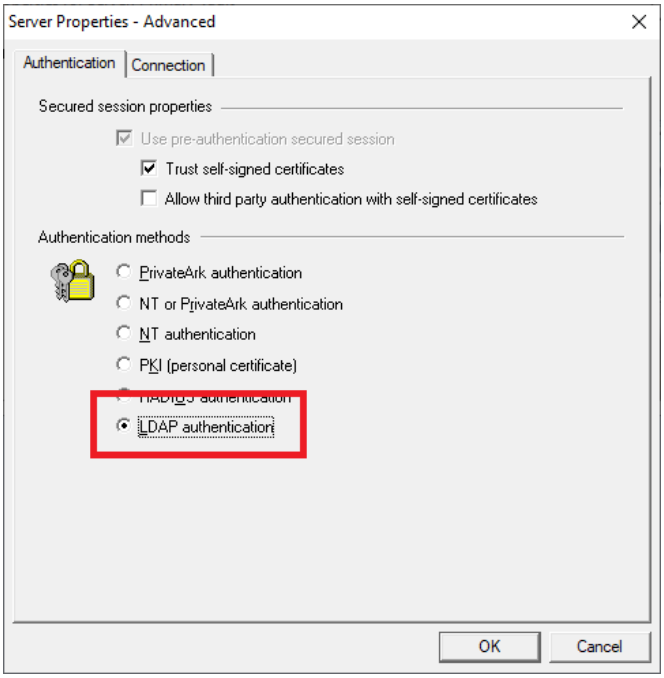
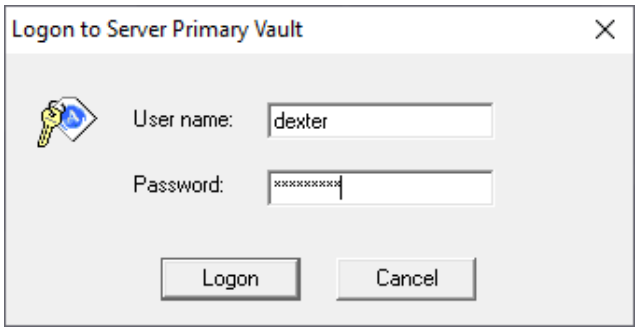


5. Click on **OK** again (no need to change the default username).

6. Double-click on the **Primary Vault** icon login. Enter *Dexter* as the username and *Cyberark1* as the password.



Note: You should not see any Safes when logged in to the **PrivateArk Client** as *Dexter*.

7. Navigate to **Tools** | **Administrative Tools** | **Users and groups**.

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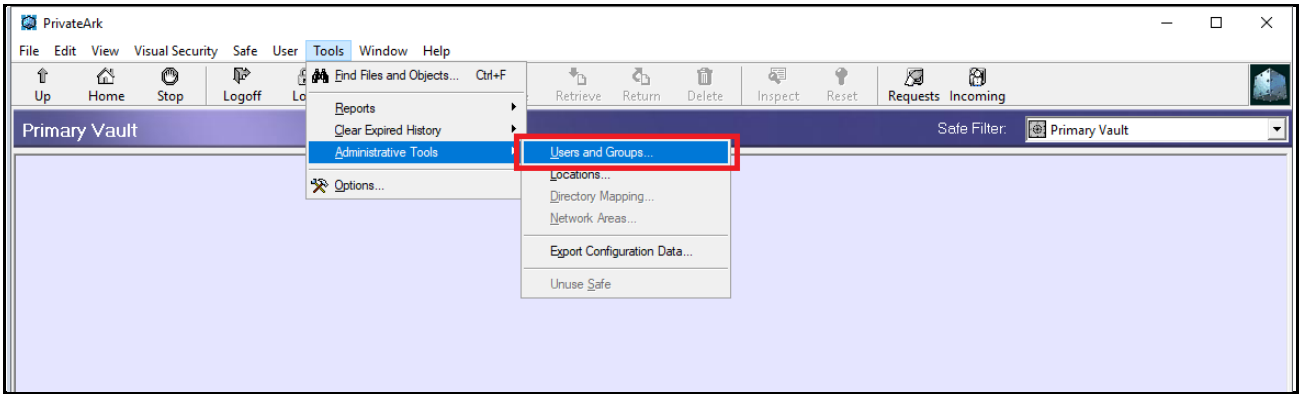
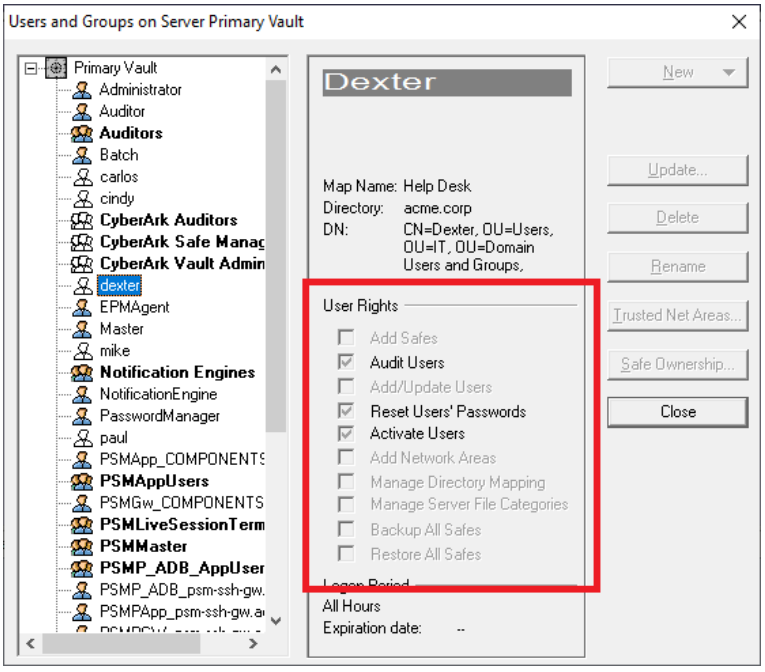


8. You should be able to see all the users provisioned in the **Vault**, both internal users and transparent users. You should also be able to see the newly added *Dexter* transparent user.

9. Select *Dexter* to view the **Vault** authorizations granted to the user.



10. Select another user, for example, *Mike*, and review the his **Vault** authorizations. Then click on **Trusted Net Areas**…

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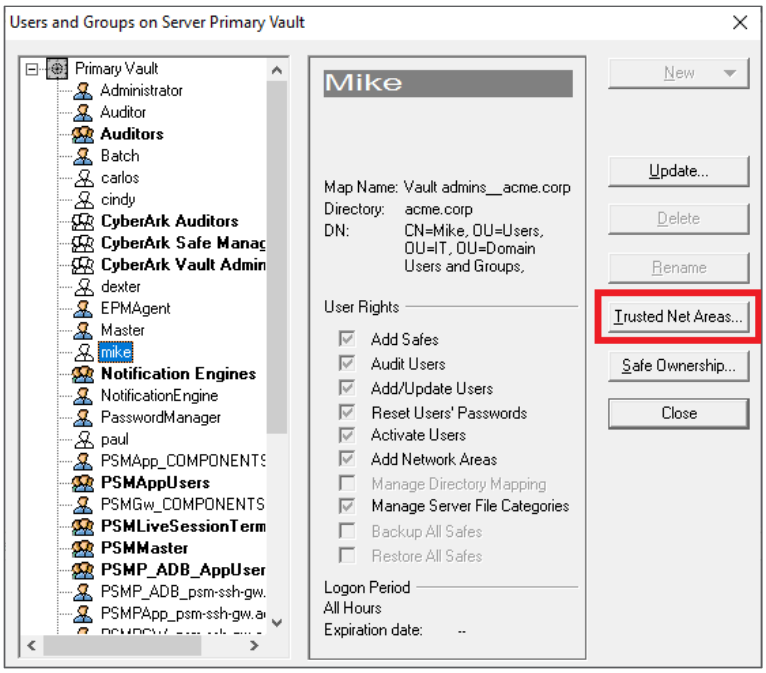
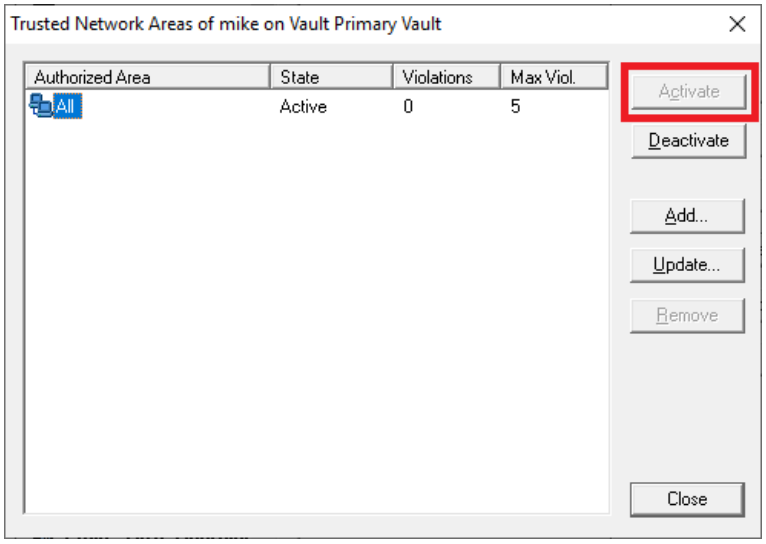


11. As you can see, the user is now active (there is no need to deactivate it). In the event that *Mike* or any other user gets suspended, *Dexter* (or any other user of the *CyberArk Help Desk* group) will now be able to re-activate the user by clicking on **Activate**.



12. **Close** the dialog boxes and then log-off the **PrivateArk Client**.

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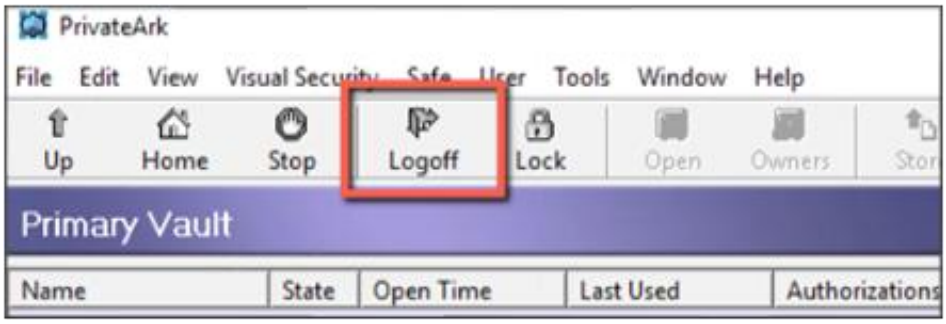
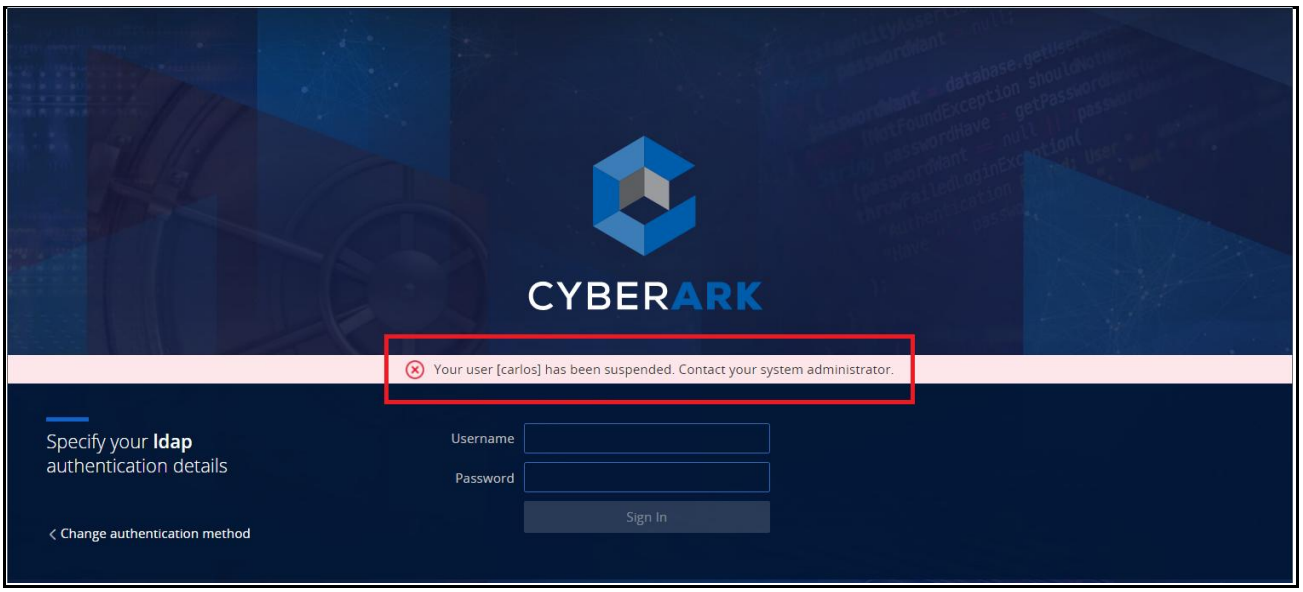


Unsuspend a suspended user

In this exercise, you will provoke a user suspension by entering the incorrect password for a user and then see how an administrator or a help desk user can unsuspend the user.

1. From the **Comp01** server, perform some incorrect logins as *Carlos* via the **PVWA**. After 5 unsuccessful attempts the user will be suspended and you will receive the message below on the 6th attempt.



2. On the **Comp01** server launch Chrome to open the **PVWA** page.
3. Login as *Dexter* (using LDAP authentication). Enter the username *Dexter* and password *Cyberark1* and then hit **Enter** or click **Sign In**.
4. Navigate to **User Provisioning** | **Users**
5. Locate the *Carlos* user. Click on the ellipsis on the right side and **Activate**.

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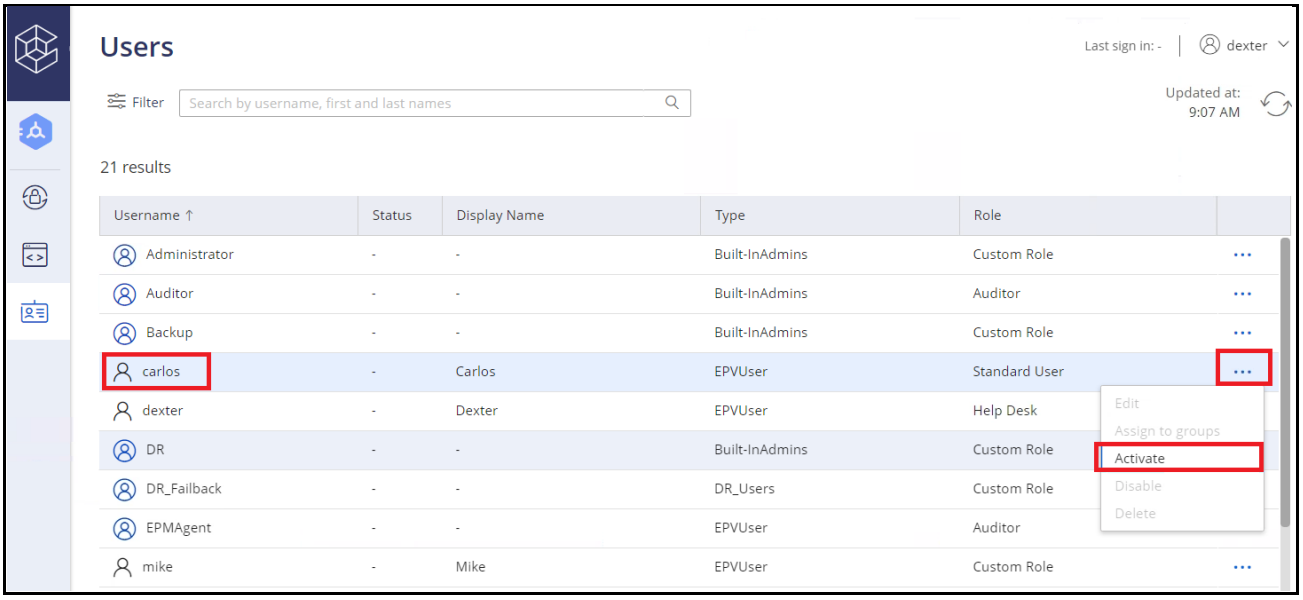
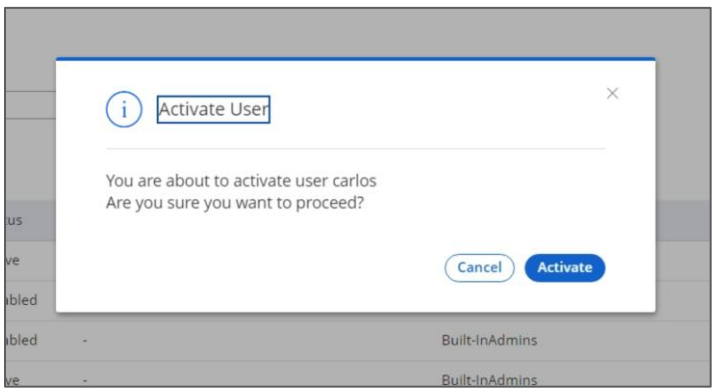


6. On the popup message select **Activate** to unsuspend Carlos.



7. Sign out from **PVWA**.

8. Try to login as *Carlos*, this time using the correct password (*Cyberark1*). Verify you can now login as *Carlos*.

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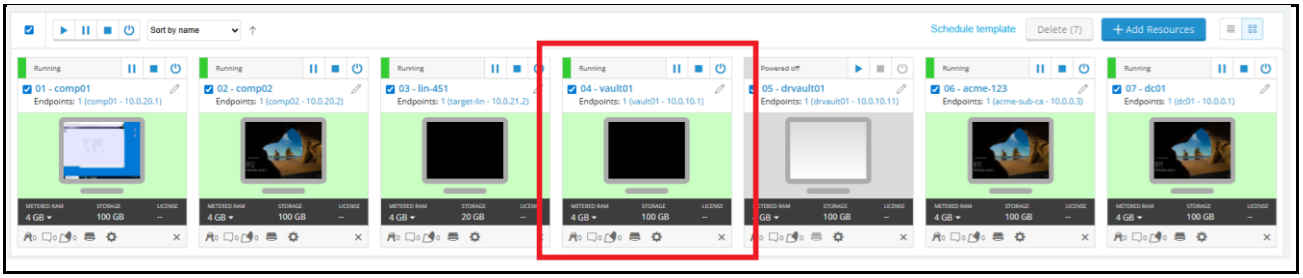
Log in with Master

There are some cases where you will need to log in to the **Vault** with the *Master* user. This can be in the event of an emergency or to give permissions to a user for a Safe when there are no active users with the necessary permissions.

To use the Master user, the *dbparm.ini* file, which is the **Vault**'s main configuration file, must point to the location of the *Recovery Private Key*. By default, this is the CD-ROM drive of the **Vault** Windows server.

1. Go back to the main Uplift view and click on the icon for **04 - vault01**.



2. Connect to the **Vault** server as *Administrator*/*Cyberark1*.

3. Navigate to *C:\Program Files (x86)\PrivateArk\Server* using the shortcut on the desktop.

4. Then navigate to: *Conf\dbparm.ini*.

This is the **Vault**'s main configuration file. In it are all the parameters for modifying the **Vault**'s behavior. For the moment, we are only interested in two parameters:

- **RecoveryPrvKey**
- **EmergencyStationIP**

The first parameter – **RecoveryPrvKey** – gives the path to the location of the recovery private key, sometimes called the **Master Key**. This key is the private half of a public/private key-pair and is required to log in with the *Master* user account. It can also be used to access account passwords in the event of a failure of the **Vault Server** application.

The second parameter – **EmergencyStationIP** – lists the IP address of the only machine that is allowed to connect to the **Vault** with the *Master* account using the **PrivateArk Client**.

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Important:	In earlier versions of **CyberArk PAM**, you could connect with the *Master* account using the **PrivateArk Client** from the **Vault** server itself without specifying an **EmergencyStationIP**. That is no longer the case.

5. The **RecoveryPrvKey** parameter has been changed in the training environment to point to the location of the file called ***recprv.key*** in the ***Master CD*** folder:

```
RecoveryPrvKey="C:\CYBR_Files\02 -License and Operator Keys\Master
CD\recprv.key"
```

Important:	You don't need to do anything here, but in a real environment, you need a way for the **Vault** to retrieve the recovery private key. Only then would you be able to connect to the **Vault** as *Master* with the appropriate password. We will discuss Master key management later in the course.

6. Open the **PrivateArk Client** from the **Vault** server machine.

7. Enter the username *Master*. The password is *Cyberark1*. These values were set during installation.

Question:	How many **Safes** are listed?

8. Log off the **PrivateArk Client** session and log in as *Administrator*.

Question:	How many **Safes** are listed?

9. You should notice that there are more **Safes** displayed when you are logged in as the *Master* user.

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Securing Windows Domain Accounts

In this section, we will look at how to secure Windows domain accounts. We will begin with accounts that are owned by the CyberArk Vault Administrators and that are used by **CyberArk PAM** to perform **CPM** operations. We will look at:

- A reconciliation account – ***s-reconcile***
- A discovery account – ***cpm-scan***

To achieve this, we will perform the following tasks:

- Import a **Platform**
- Duplicate a **Platform**
- Add an exception to the **Master Policy**
- Create a **Safe**
- Add the accounts to the **Safe**

Configure a Platform

In this section, we will duplicate a platform for Windows domain accounts that has been downloaded from the **CyberArk Marketplace**. If you look in the list of platforms that we have out of the box, you will see that we already have one: **Windows Domain Accounts via LDAP**. However, this platform won't work for one of our use cases (specifically the discovery account, but more on that later).

Import the new platform

1. Log into the **PVWA** as either *administrator* or *Mike*: either user can perform this action.

1. Browse to **Administration | Platform Management** by opening the left-hand menu, scrolling down, clicking on the *down icon* of the **Administration** menu to expand it, then click on **Platform Management**.

   In the upper right-hand corner, click on the **Import platform** button.

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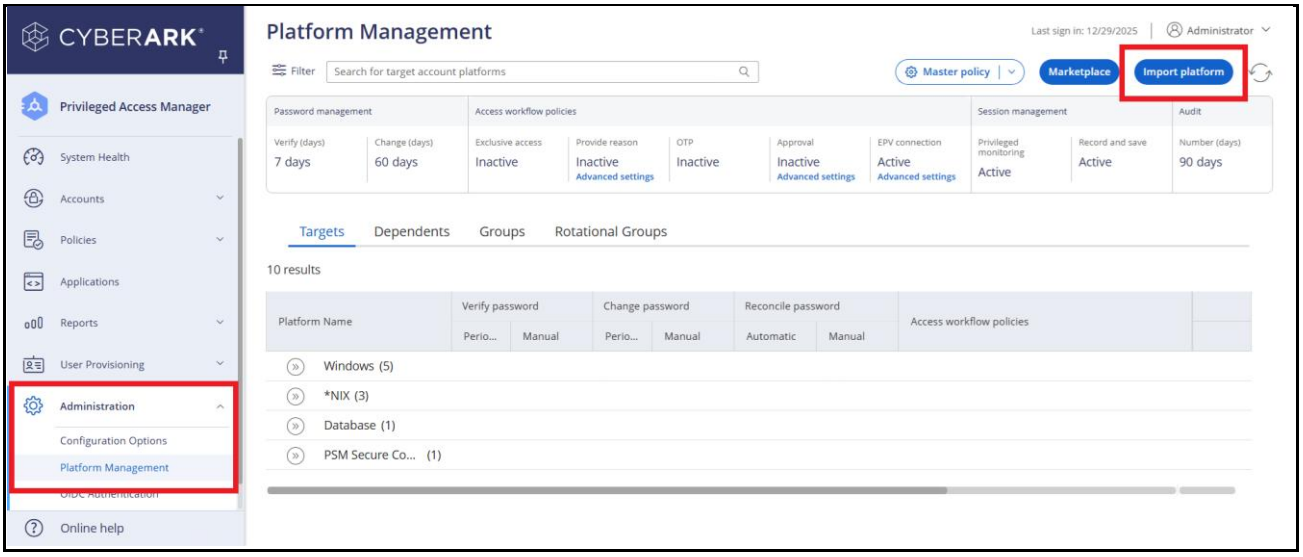
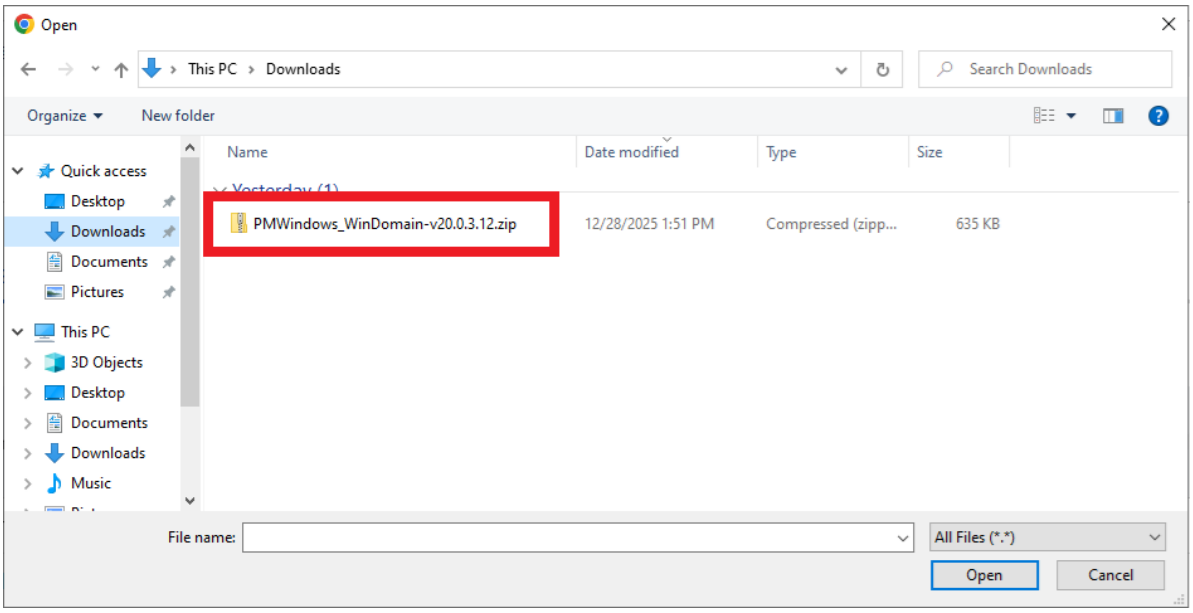


2. Browse to the directory ***c:\CYBR_Files\Platforms***, select the ***PMWindows_WinDomain-vXXyy.zip*** file, and click **Open**.



3. And that's it. When you expand the list of Windows platforms, you will see there is a new one there: **Windows Domain Account**.

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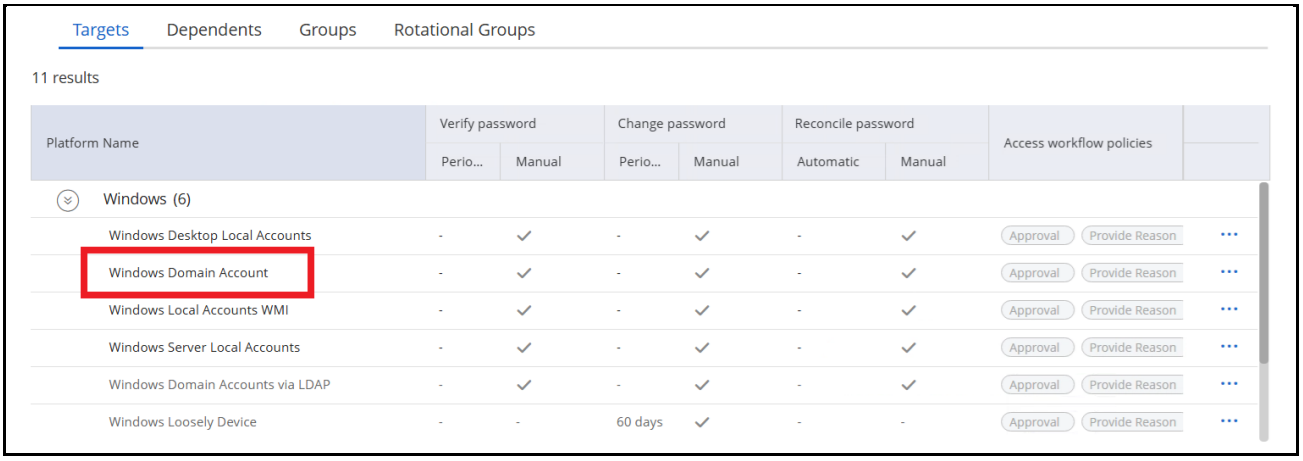
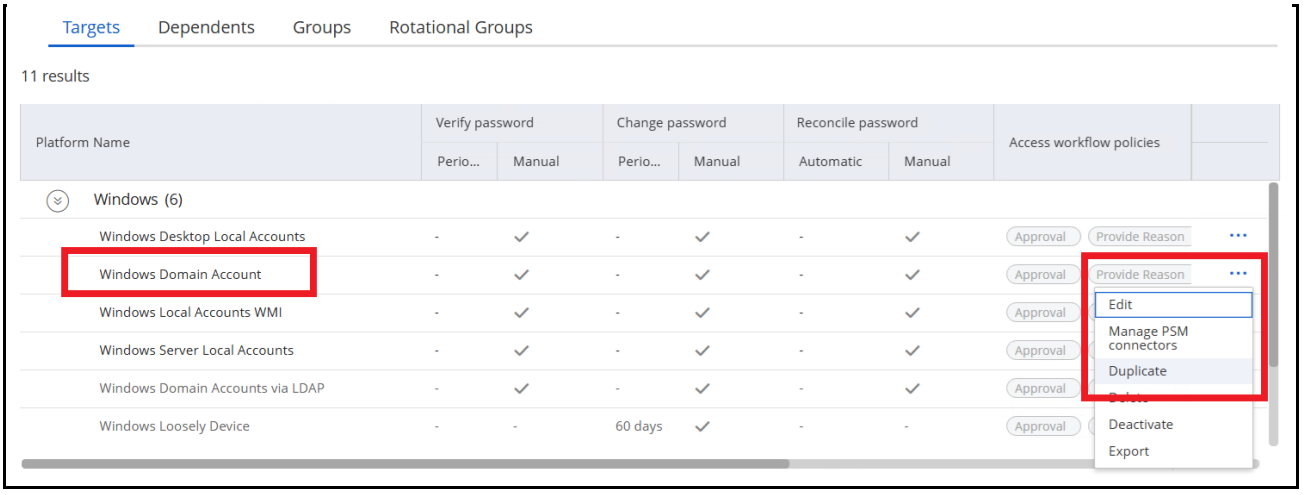


4. Stay in the **Platform Management** section.

Duplicating a Platform

As a general rule, CyberArk recommends that our customers not modify the platforms we provide. Rather, you should duplicate the base platform, providing it with a name relevant to your organization, and then modifying that. In this way, if ever you have a problem with the platform you modify, you can always refer to the base platform or simply delete the problem platform and being again. That is what we will be doing in this course.

1. Go to the end of the line for the new Window Domain Account platform, click on the three dots to drop down the menu of options, and select **Duplicate**.



2. Enter as the name *Dom Cybr 15* (you can also give it a meaningful description) and then press **Create**.

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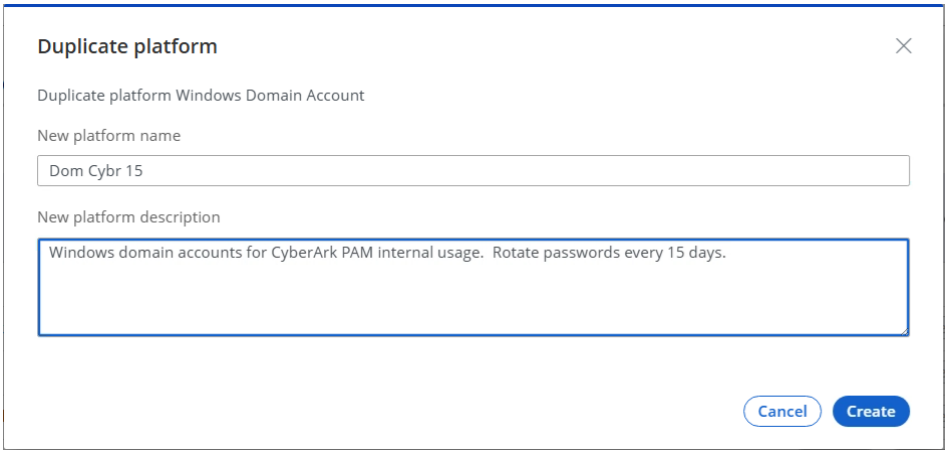
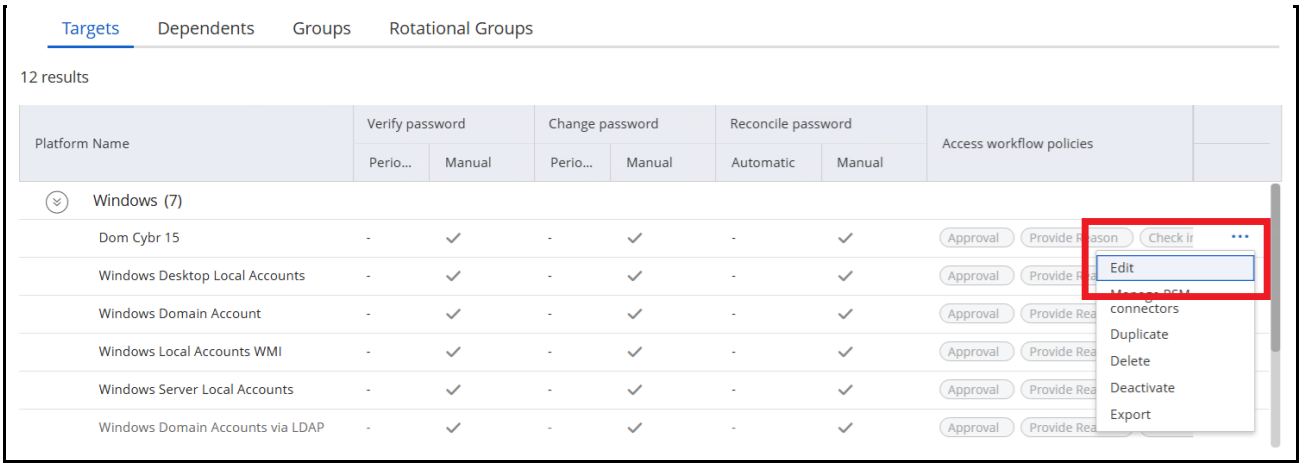


Edit the new Platform

In this section, we will make a few changes to our new platform so that it meets our requirements.

1. Select the new platform ***Dom Cybr 15*** and as we did previously, click on the three dots at the end of the line, but this time click on **Edit**.



2. Click on **UI & Workflows** and change *AutoChangeOnAdd* from ***No*** to **Yes**.

Note:	While not required, it is always a good idea to press the **Apply** button to make sure your changes are saved (bottom right of the screen).

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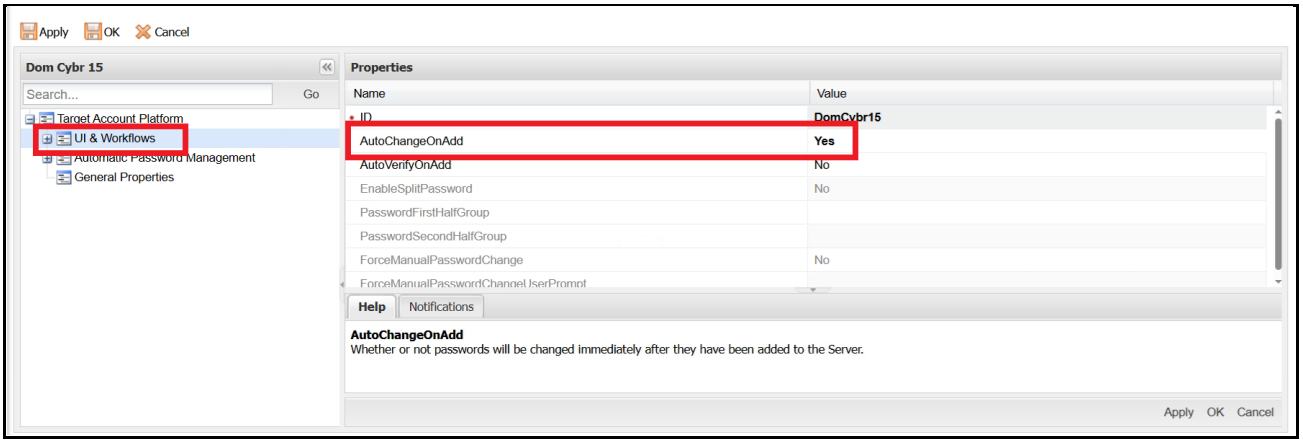
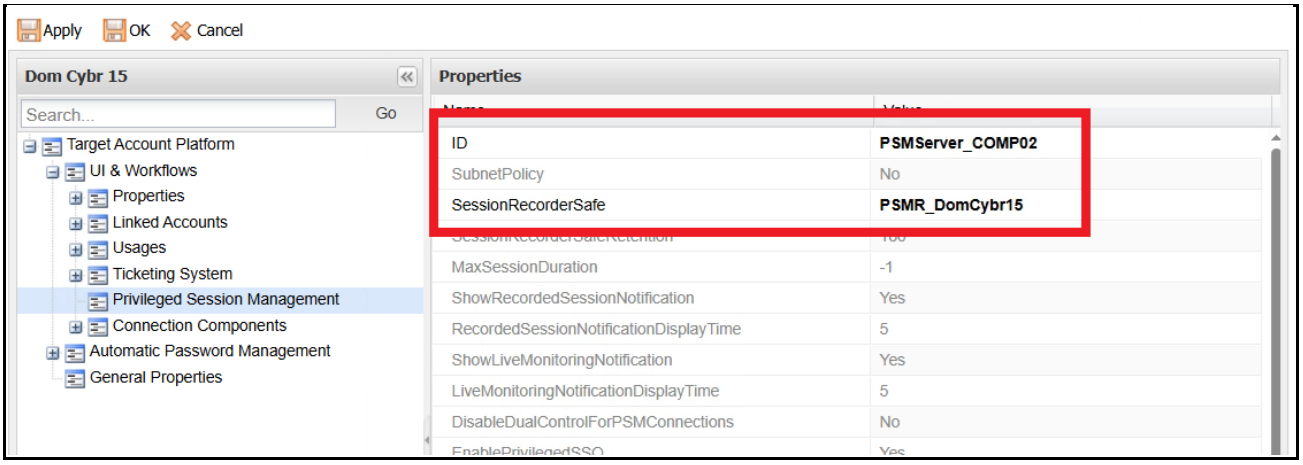


Note: This setting will prompt the **CPM** to automatically change the password whenever a new account assigned to this platform is added.

3. Browse to **UI & Workflows | Privileged Session Management** and modify the two following parameters:

ID	***PSMServer_COMP02***
SessionRecorderSafe	***PSMR_DomCybr15***



Note: Because we imported this platform, it is not yet aware of the local PSM server, so we need to set that value. The second parameter will ensure that recordings for all accounts associated with this platform will be stored in a dedicated safe.

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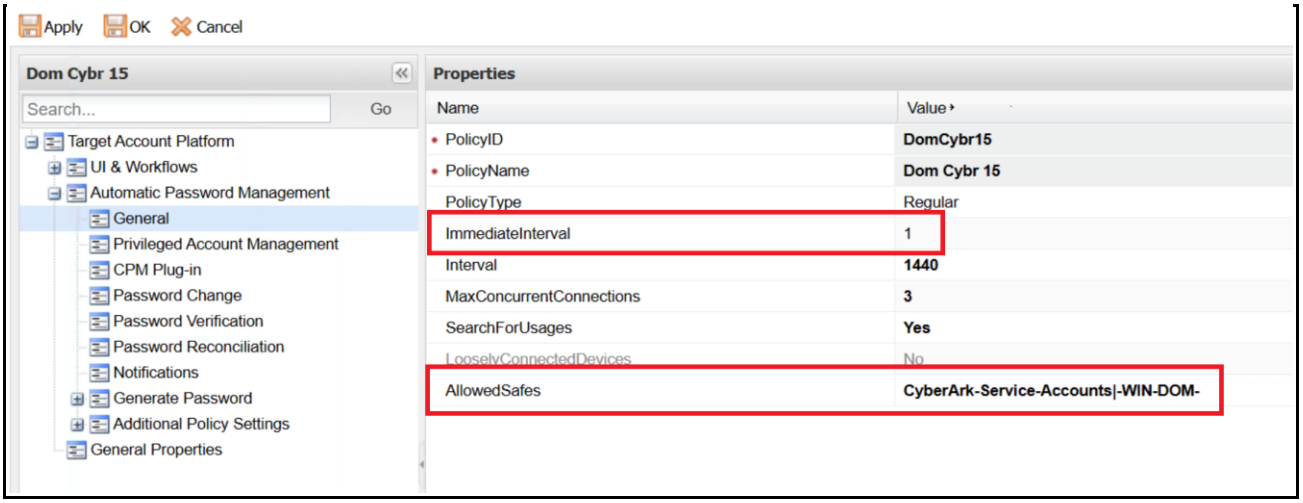
4. Now expand the section **Automatic Password Management | General** and change the value of *ImmediateInterval* to ***1***.

Note:	Changing the *ImmediateInterval* to **1** is only suitable for testing and training. In a production environment, this should be left to its default value.

5. Still in **Automatic Password Management | General**, enter the following into the *AllowedSafes* parameter:

```
CyberArk-Service-Accounts|-WIN-DOM-
```

Warning!	Do **NOT** copy and paste from the PDF file. It will not work because of special characters in PDF files. Make sure there is no space in front of or behind the pipe symbol |.



Note:	This regular expression restricts the **Accounts** to which this **Platform** can be applied to only those stored in **Safes** that start with the string "-WIN-DOM-" or the **Safe** named "CyberArk-Service-Accounts". This field is case sensitive. Using the ***AllowedSafes*** parameter is one of the best ways of improving CPM performance.

6. Go to **Password Change** and set *PerformPeriodicChange* to **Yes**.

7. Go to **Password Verification** and set *VFPerformPeriodicVerification* to **Yes**.

8. Finally, go to **Generate Password**. Here, we are going to modify the password length to give us more secure passwords for our domain admin accounts. Set the values as follows:

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PasswordLength	*17*
MinUpperCase	*2*
MinLowerCase	*2*
MinDigit	*1*
MinSpecial	*1*

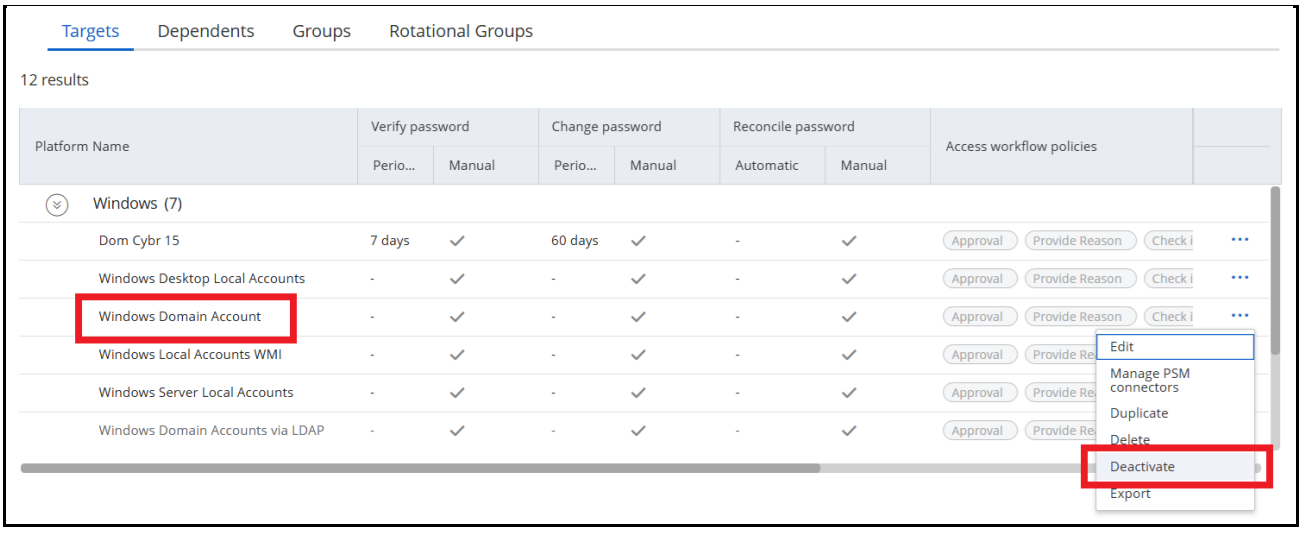
Note: The sum of the various complexity parameters must be less than or equal to *PasswordLength* for password change to function. However, the system does not check the values for you. The value of 17 here was chosen at random. You can choose any (reasonable) length you want.

9. Press **Apply** and **OK** to save all your changes and close the **Platform**.

Deactivate the base Platform

Notice that some of the **Platforms** are *Active* while others are *Inactive* (greyed out). You can significantly improve the performance of **CyberArk PAM** by deactivating all **Platforms** that are not being actively used. Now that we have created our own **Platform** for Windows domain accounts, we can deactivate the base platform.

1. Click on the three dots at the end of the line for **Windows Domain Account** platform and click **Deactivate**. You will be prompted to confirm the action. Click on **Deactivate**.



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2. Remain for a moment in the **Platform Management** section.

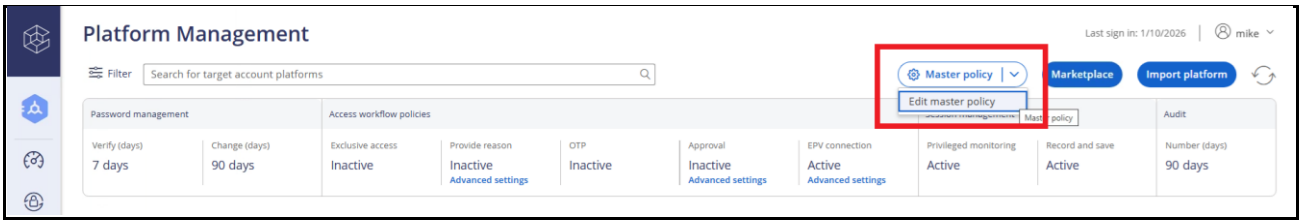
Editing the Master Policy

In this section, you will modify the ***Master Policy*** to:

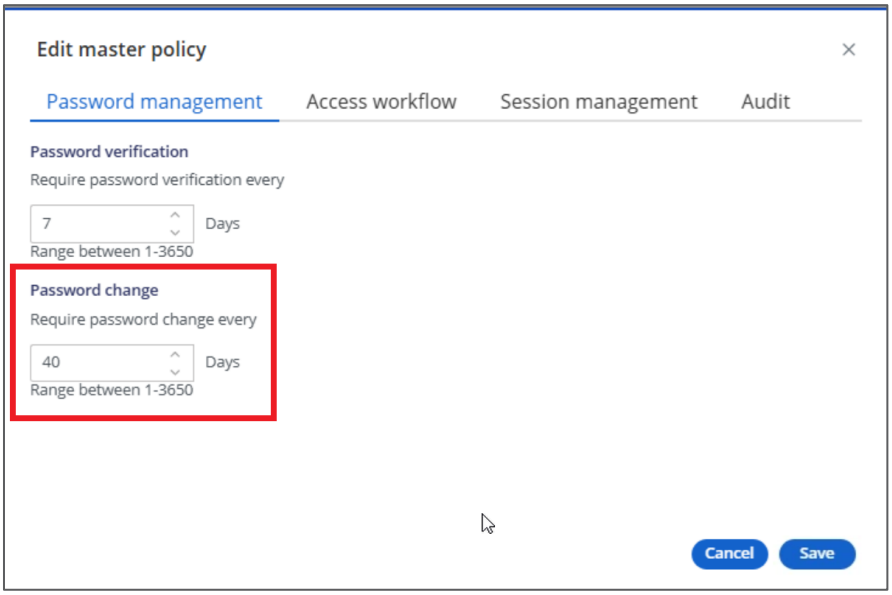
- Change passwords for all accounts every 40 days
- Create an exception for the Platform ***Dom Cybr 15*** to rotate passwords every 15 days

Password Change Policy

1. A recent feature added to **CyberArk PAM** is a shortcut from the **Platform Management** page to a new interface for editing the global **Master Policy** settings. Click on the down arrow and then click **Edit master policy**.



2. Change the value of *Require password change every…* to ***40 days*** and then click **Save**. This is a global change and will impact every Platform, and therefore every account, in **CyberArk PAM**.



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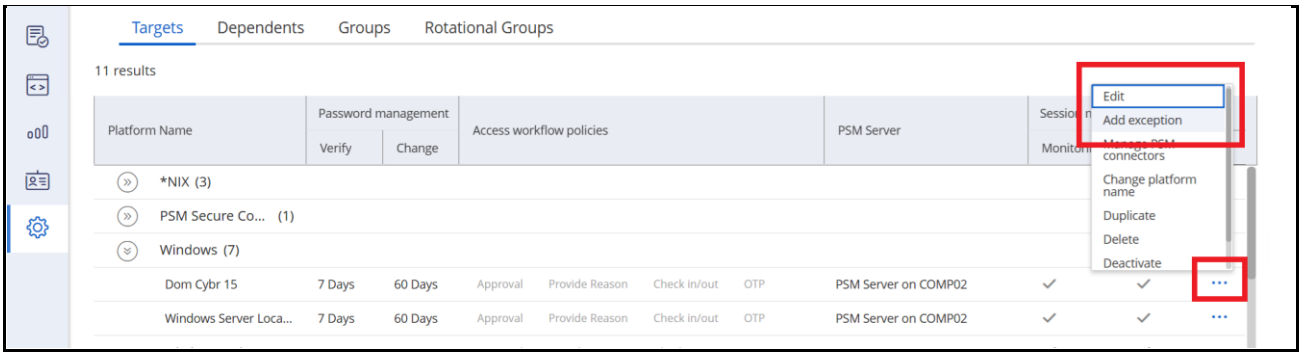
Note:	We have set the maximum password age to 40 days so that our **CPM** policy will not conflict with the default maximum age of 42 days on the Windows 2022 servers.

Add Exceptions

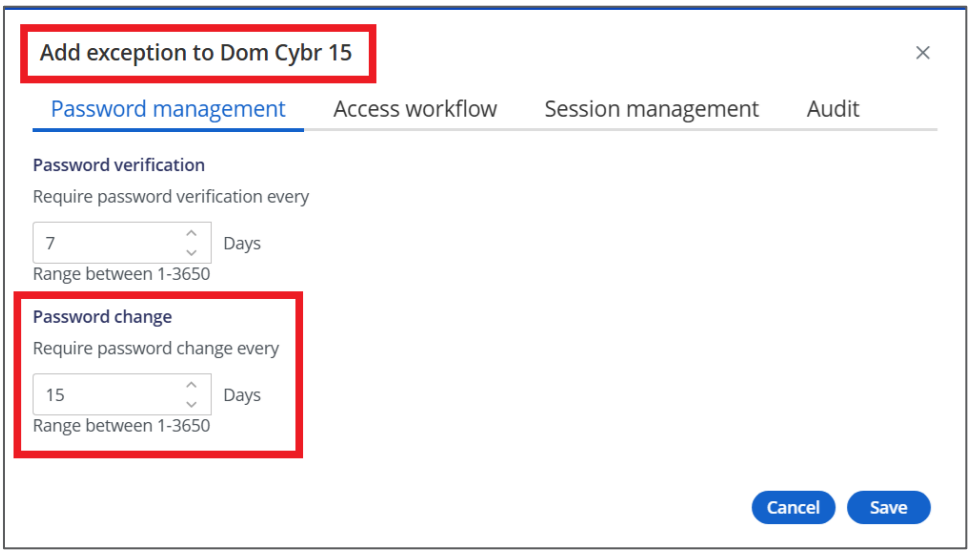
We said that we wanted to rotate the passwords for the domain accounts used by **CyberArk PAM** every 15 days, not every 40 days, so we need to add an exception to the global rule.

We can now do that within the **Platform Management** section.

1. Go to the newly created Platform and click on the three dots (or ellipsis) at the end of the line. Now click on **Add exception**.



2. Here we can create an exception to the global rule that will apply only to the selected Platform: **Dom Cybr 15**. Set the parameter *Require password change every…* and to ***15 days*** and click **Save**.



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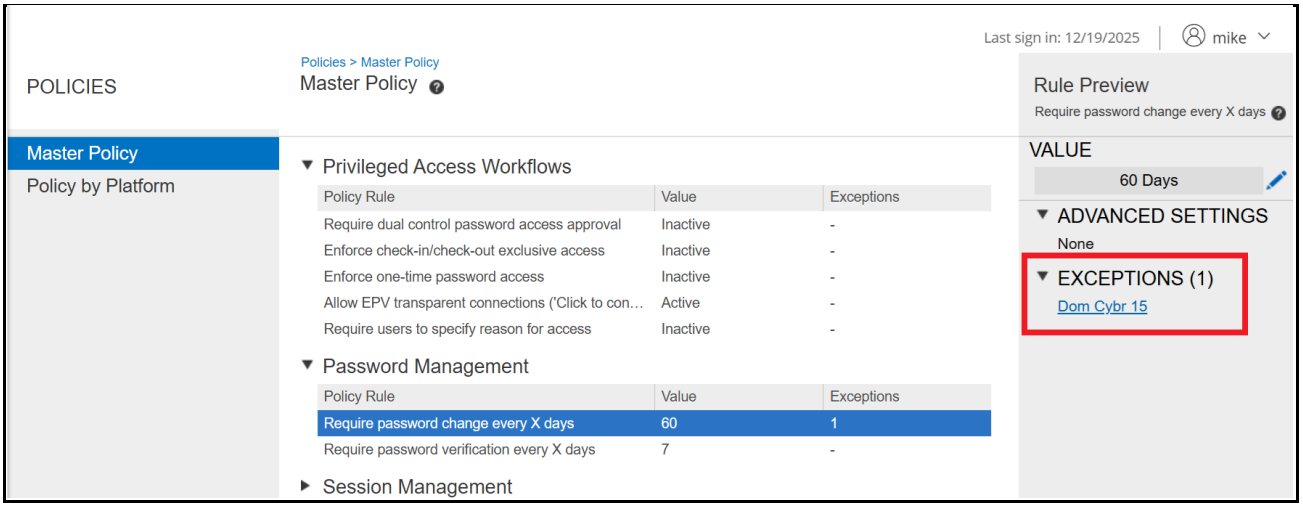
3. For the moment, you can still see your exceptions to the **Master Policy** in the old interface, but that will eventually be phased out.



Add an Account

In this section, we are going to add two accounts from Active Directory to **CyberArk PAM**, beginning with our ***reconcile*** account.

The reconcile account is a domain account named ***s-reconcile***. It has been added to the local *Administrators* group on all Acme Windows servers (except the domain controller) through a GPO.

The safe for this type of account was created by the implementation team so we will use that. It is safe we listed in the ***AllowedSafes*** parameter for the platform: **CyberArk-Service-Accounts**.

Add the reconcile account

We don't need to create a Safe just yet because one already exists that suits our requirements: **CyberArk-Service-Accounts**. This Safe was created by the implementation team initially to store the *root* and *proxymng* accounts.

1. Log into the **PVWA** as *mike*. Go to the **ACCOUNTS** tab and press the **Add account** button.

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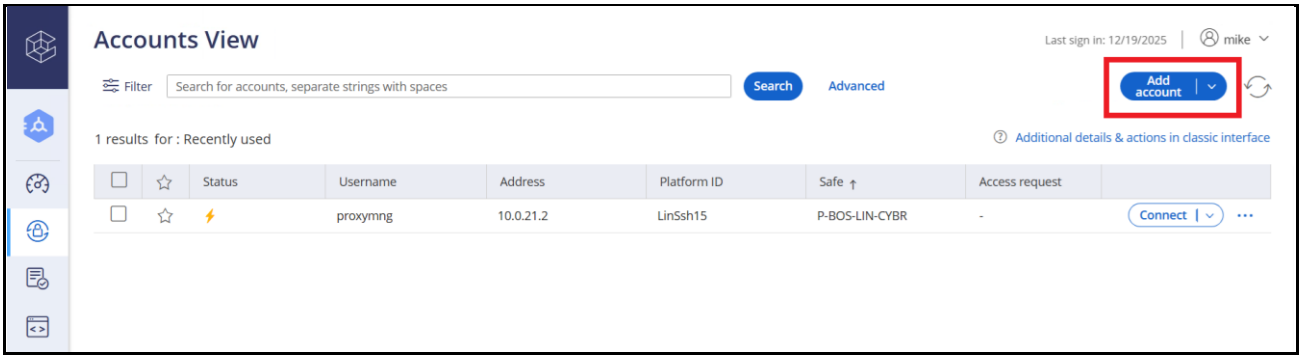
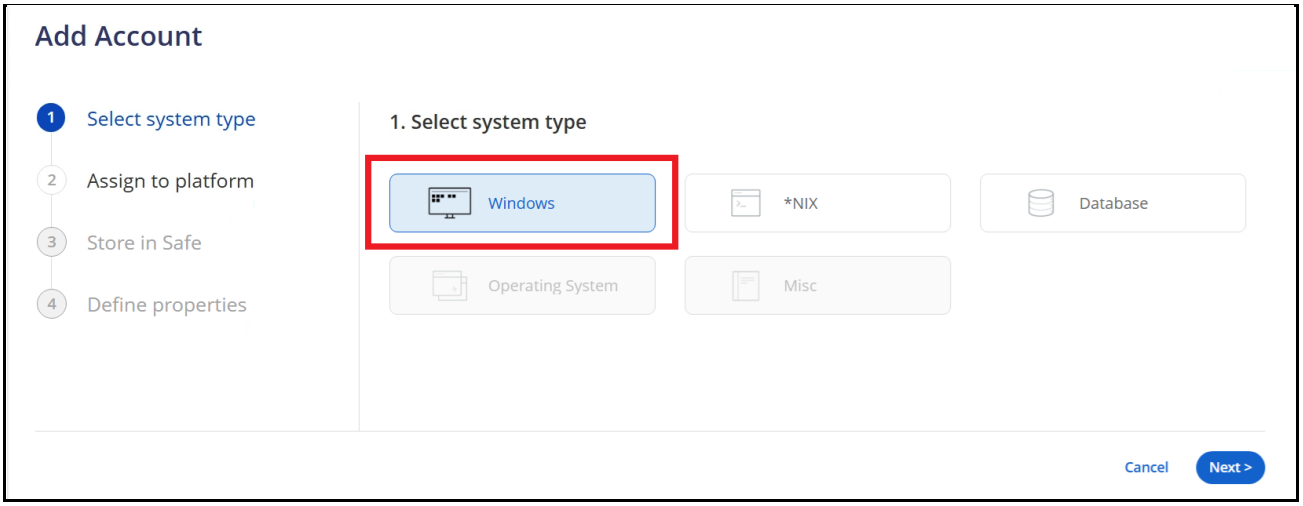
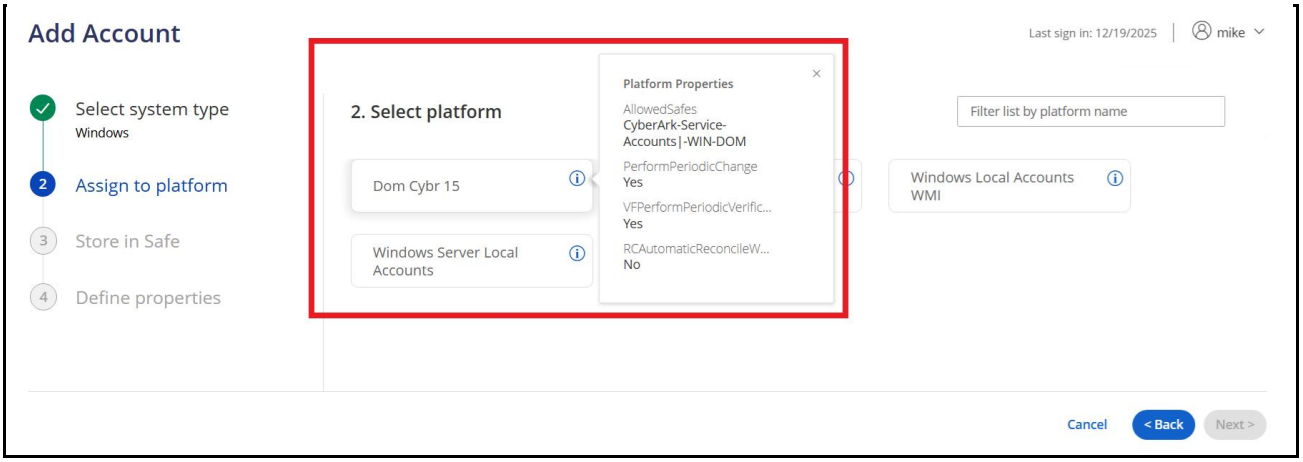


2. First select the *System Type*. Click on **Windows**



3. Next, select the **Platform** we created for domain accounts: **Dom Cybr 15**.



Note:	If you are uncertain about which **Platform** you want to choose, you can click on the **Information** icon to view some additional information.

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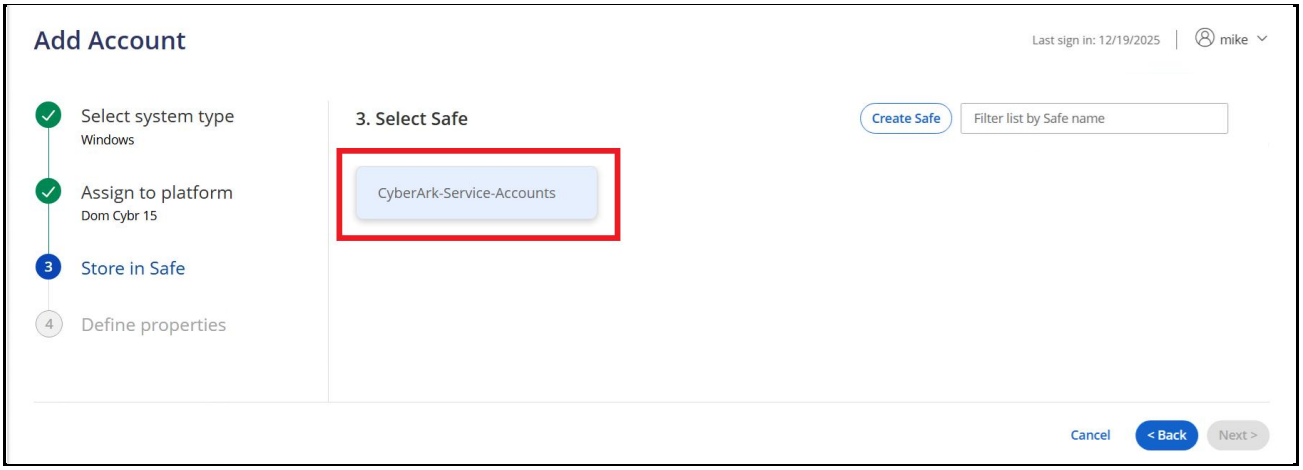
4. Next, select the **Safe** we created: **CyberArk-Service-Accounts**.



Note:	Notice that in the image above, we cannot see any other **Safes**. Why do you think that is? There is a hint in the previous image.

5. Enter the following and then press **Add**:

Address:	***acme.corp***
Username:	***s-reconcile***
Password (optional)	***Cyberark1***
Confirm Password	***Cyberark1***
Customize account name *	***s-reconcile***
Log On To (optional) **	***ACME***

Note:	***** Giving the reconcile account a name short and easy to remember will make our jobs a vault administrators easier, as we will see later. ****** Scroll down to find the **Log On To** parameter. The interface says this parameter is optional, but that does not seem to be the case, at least in this environment.

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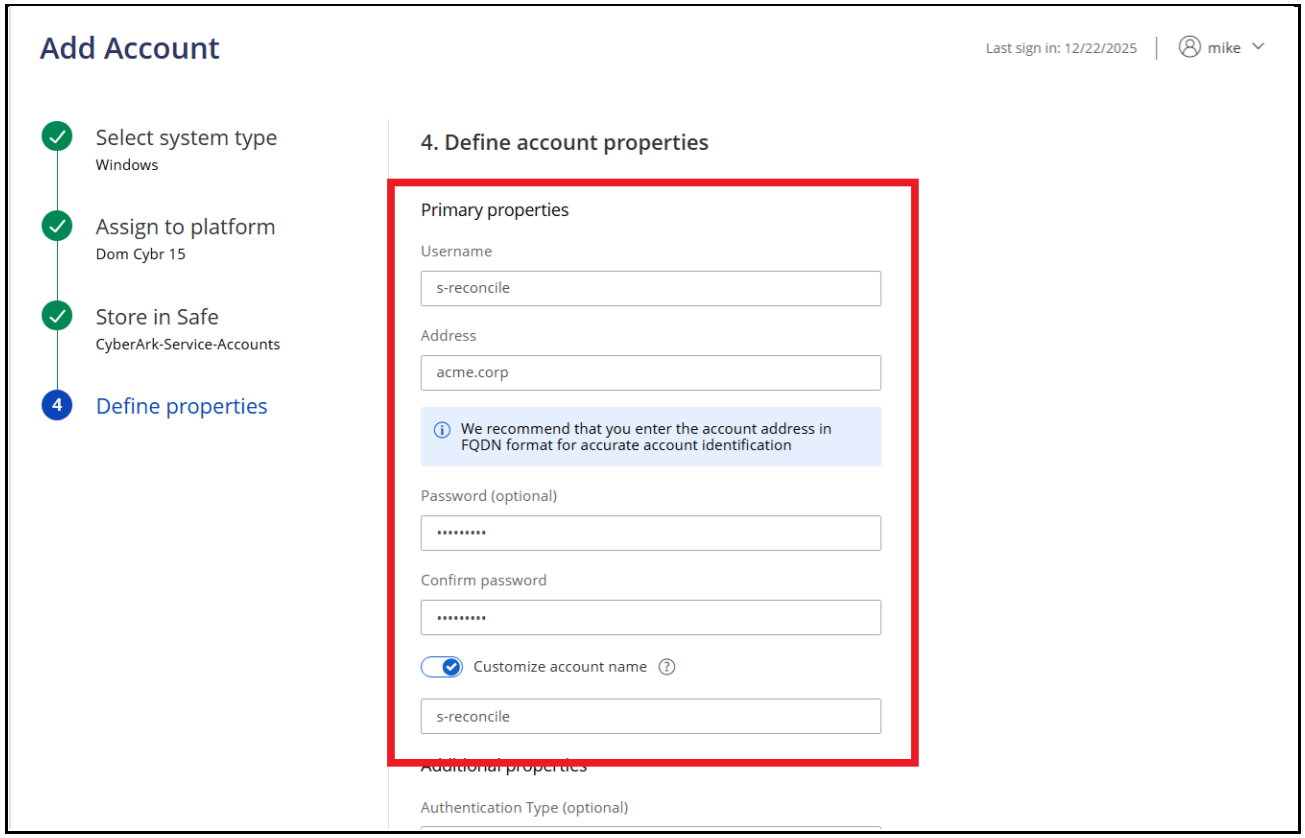
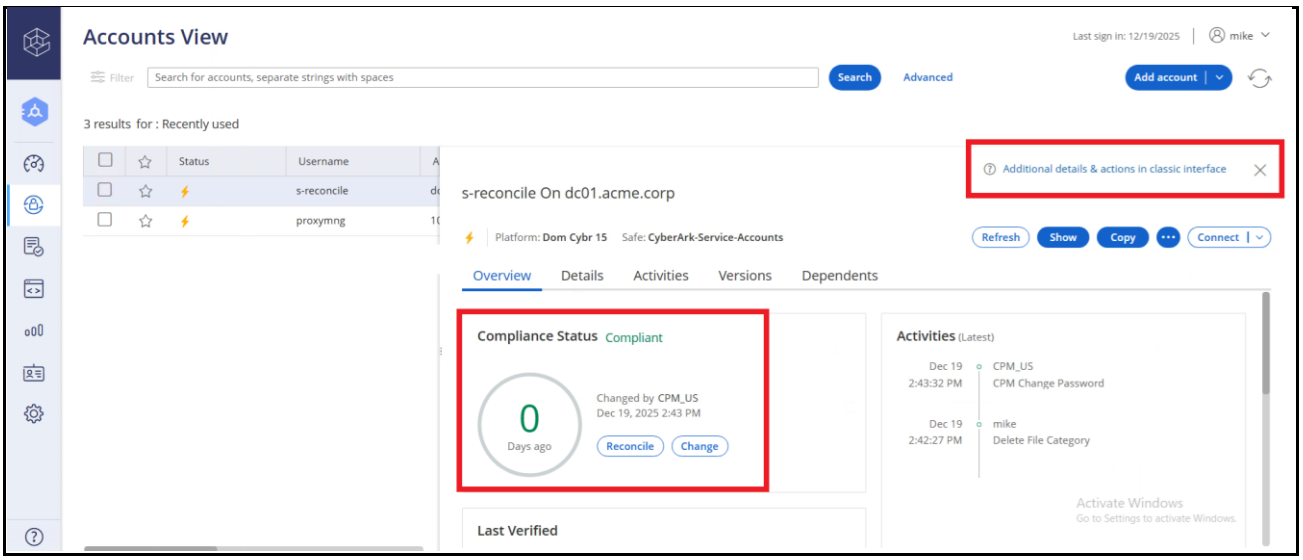


6. Click on the new account *s-reconcile* in the **Accounts** view. Because *AutoChangeOnAdd* was set to **Yes** for the **Platform** managing this account, it is scheduled for immediate password change. In a minute or two, you should see that the account was changed by the **CPM** user *PasswordManager*.



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7. Select the newly created account from the list and then click on the link **Additional details & actions in classic interface** (see above).

Note:	You can find this information under the **Details** tab in the new interface, but it is slightly more convenient to get it here.

Add the discovery account

We will need another Windows account for a later exercise – *cpm-scan*, so we will now add a second Windows domain account using the information below.

System Type:	*Windows*
Platform Name:	*Dom Cybr 15*
Store in Safe:	*CyberArk-Service-Accounts*
Address:	*acme.corp*
User Name:	*cpm-scan*
Password:	*Cyberark1*
Confirm Password:	*Cyberark1*
Log On To	***(leave blank)***

The password for this account will also be automatically changed. We will use these accounts in later exercises.

Note:	We do not need to customize the name for the **CPM** discovery account. We are leaving the **Log On To** parameter blank for accounts discovery.

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Securing Unix SSH Accounts

In this section, we will be managing Unix SSH accounts or, to be more precise, Linux accounts that connect using SSH.

In the previous section, we were managing what we could call "meta-accounts": accounts that are owned by the Vault Administrators and that are used by **CyberArk PAM** to manage other accounts (which we will see later). Here, we are dealing with typical accounts, the kind of accounts that Unix administrators might use on a daily basis.

We'll start with an account that is owned by the IT team who manage Linux servers. Members of this team are members of the Active Directory group *Linux Admins*. Our Vault Administrators do not need to know the Linux passwords or have access to them. They need to enable the Linux team to manage their accounts securely and efficiently. We will, however, allow the Vault Admins to see the Safes (and the accounts in them).

To achieve these goals, we are going to divide the tasks of configuring **CyberArk PAM** to manage these accounts into separate phases and perform the actions by "changing hats"; that is, logging into **CyberArk PAM** with different user accounts according to the table below:

Role	Action	User
Vault Administrator	Configuring **Platforms** and setting **Policies**.	*Mike*
Safe Manager	Creating **Safes**, adding members, adding accounts.	*Paul*
Auditor	Verifying that accounts are being used according to corporate policy.	*Cindy*

Vault Administrator Tasks – Mike

Vault administrator tasks are handled by *Mike*, so use this account to login to the **PVWA**.

Duplicating a Unix Platform

Here you will duplicate and configure a **Platform** to manage Linux accounts that connect to their targets with SSH.

1. Navigate to **ADMINISTRATION** | **Platform Management**, expand the section ***NIX** then click **UNIX via SSH**. Click on the three dots (or ellipsis) at the end of the line and select **Duplicate**.

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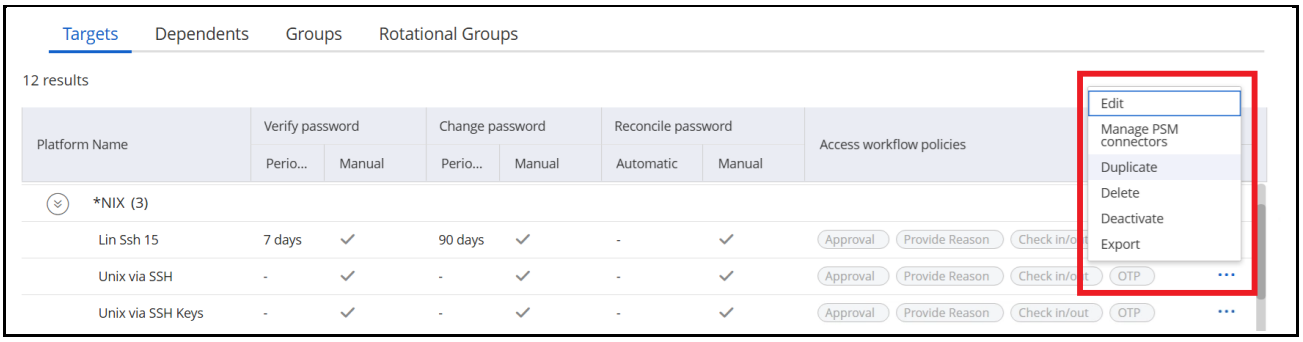
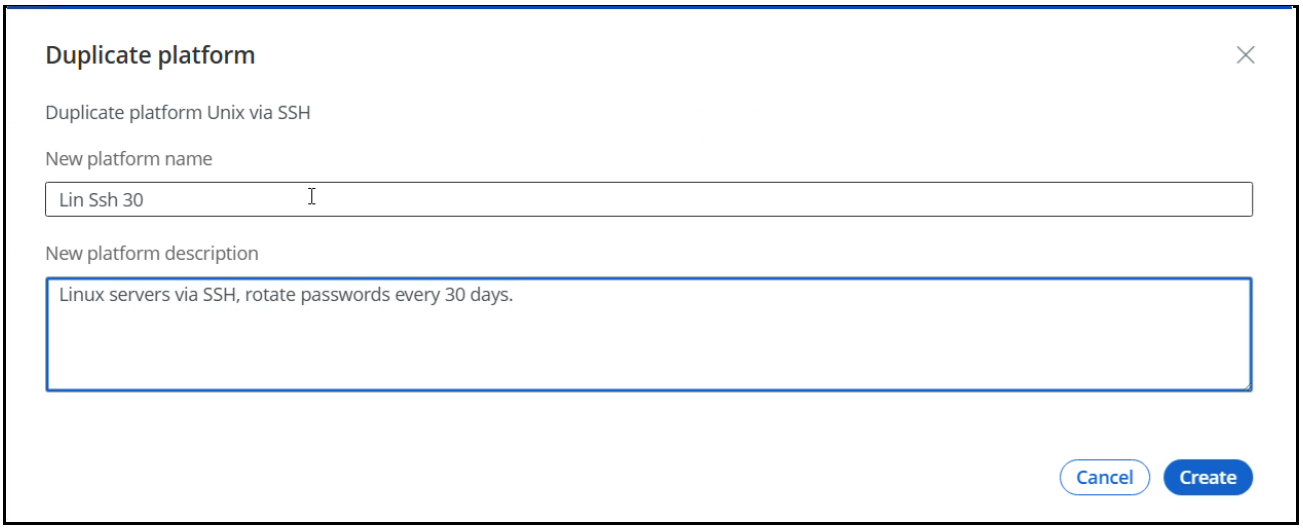
Note:	Notice that there is a *Nix platform that has already been created. This was created by the implementation to test that the basic functionality of the system was working correctly.



2. Enter *Lin Ssh 30* in the **Name** field and optionally something like *Linux servers via SSH, rotate passwords every 30 days* for a description and then press **Create**.



Important!	Although you are free (and encouraged) to apply your own naming conventions for **Platforms** and **Safes** in your own environments, please note that we will be referring to the names provided here in later exercises. If you choose to give your **Platforms** and **Safes** different names, it may complicate the exercises. We therefore recommend that you use the names suggested in this guide.

3. Highlight the newly created **Platform**, click the ellipsis at the end of the line, and press **Edit**.

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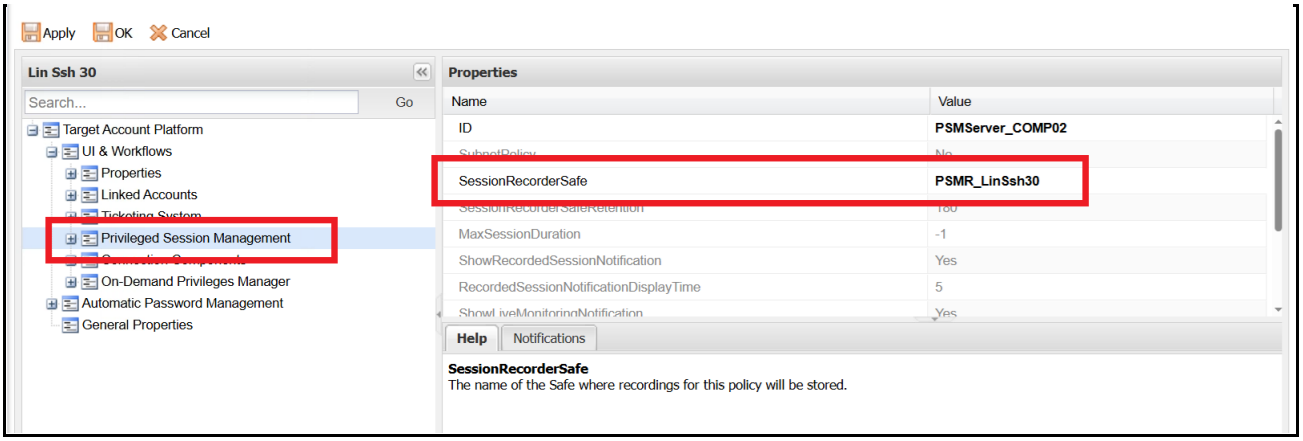
4. Click on **UI & Workflows | Privileged Session Management**. We are going to modify the parameter **SessionRecorderSafe** so that this Platform will have a dedicated Safe in which to store the recordings of all PSM session activities. Modify the value so that is reads ***PSMR_LinSsh30***.



Note: We will do this systematically for all our production platforms: Modify the **SessionRecorderSafe** parameter so that its name includes the **PlatformID**. This will make management of those Safes easier as our **CyberArk PAM** environment grows.

5. Go **Automatic Password Management | General**.

6. Change *ImmediateInterval* to **1**

Note: Changing the *ImmediateInterval* field to *1* is only suitable for testing and should set to **5** or higher in a real environment.

7. In the **AllowedSafes** parameter, add ***-LIN-S-*** (case sensitive). This determines which **Safes** can use this platform. Only **Safes** that contain the string '-LIN-S-' (for "Linux Servers") will be able to use this **Platform**.

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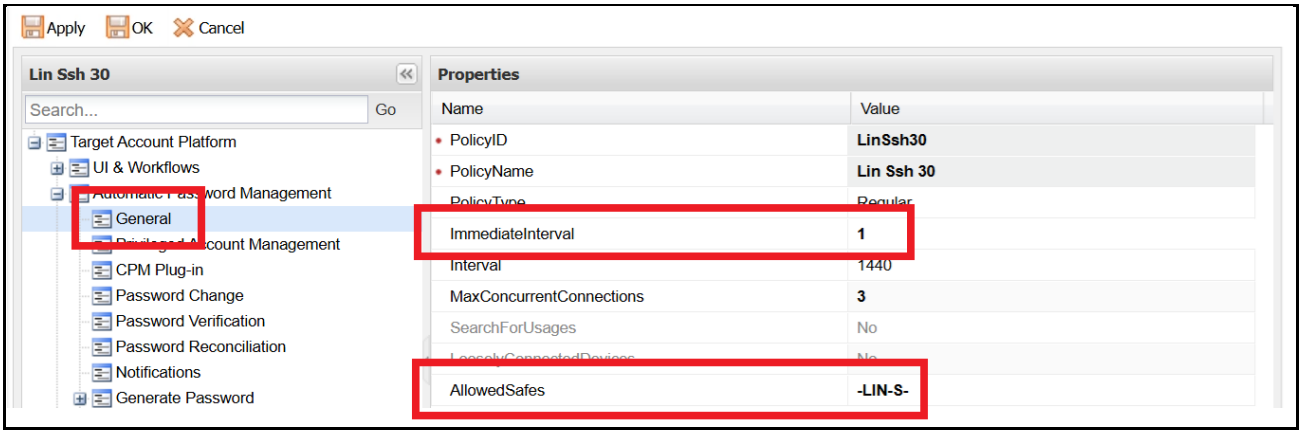

8. Now go to **Password Change** and change the value of the parameter *PerformPeriodicChange* from **No** to **Yes**. This will enable the application of the **Master Policy** rule *Require password change every X days* to accounts managed by this platform.

9. Within the same window, go to **Password Verification** and change *VFPerformPeriodicVerification* from **No** to **Yes**. This will allow the password to be verified by the **CPM** automatically and without user intervention.

10. Finally, in **Generate Password**, note that the default password length for *Nix machines is 12 characters. This value can be changed to reflect your organization's requirements.

Note:	Until recently, the default password length for *nix accounts in **CyberArk PAM** was 8. It has been increased to **12**.

11. Click **OK** to save your changes and close the **Platform**.

12. As a final step, locate the baseline **Platform** *Unix via SSH* and deactivate it.

Add an Exception to the Master Policy

We have already seen how to create a **Master Policy** exception. Create an exception for our new **Platform** *Lin Ssh 30* that rotates passwords every 30 days. While we are here, we can also create an exception for the Platform *Lin Ssh 15* so that its policy is in line with its name.

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Safe Manager Tasks – Paul

For this section, we will need to "change hats"; that is, we need to imagine that we are a different user. We are no longer *Mike* the Vault Administrator. We are now the team leader of the Linux system administrators named *Paul*. We have been instructed to place all our privileged Linux accounts into **CyberArk PAM** so that their passwords (and SSH keys) will be stored in the **Vault**.

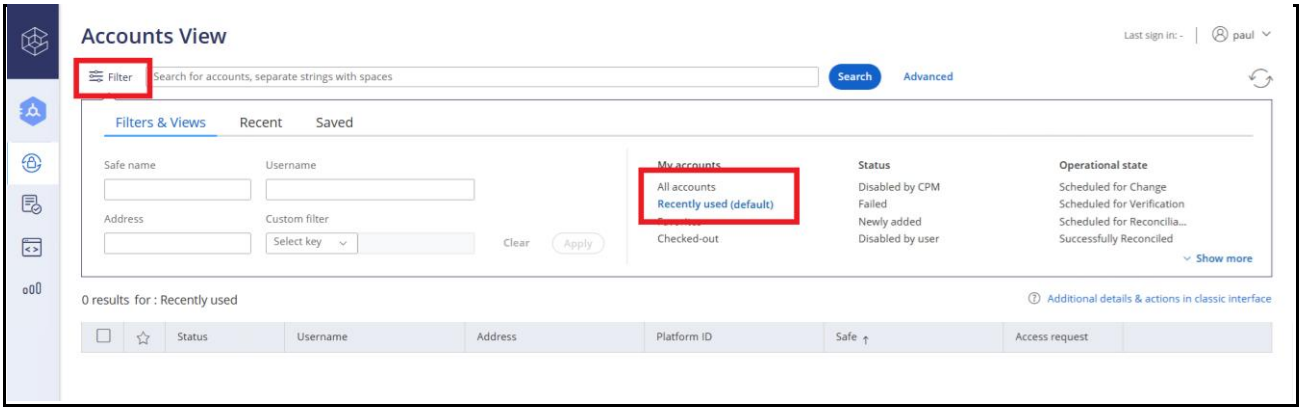
Paul is a member of the Active Directory group **CyberArk Safe Managers**. This means that when he logs in to **CyberArk PAM**, he will have the right to create **Safes**, add users to the **Safes** he creates, and to add new accounts to those **Safes**, which is what we shall do.

We will perform the basic tasks required to manage privileged accounts on a Linux server to which we connect using SSH. We will create **Safes** to store our accounts securely and add AD groups of users who are authorized to use those accounts. We will then add the new accounts, verify that we can connect with them, and see how an auditor can monitor the account activity.

Creating a Safe

1. Log in to the **PVWA** as *Paul* with the password *Cyberark1* using LDAP authentication.

   At first glance, it might look like Paul cannot see any accounts, but we said Paul was a *Safe Manager* and we added the Safe Manager group to the **Safe** *CyberArk-Service-Accounts*, so theoretically he should be able to see those accounts. And he can. His view, however, is filtered to show only "Recently used" accounts, and as Paul is just beginning his **CyberArk PAM** journey, he has not used any accounts yet.



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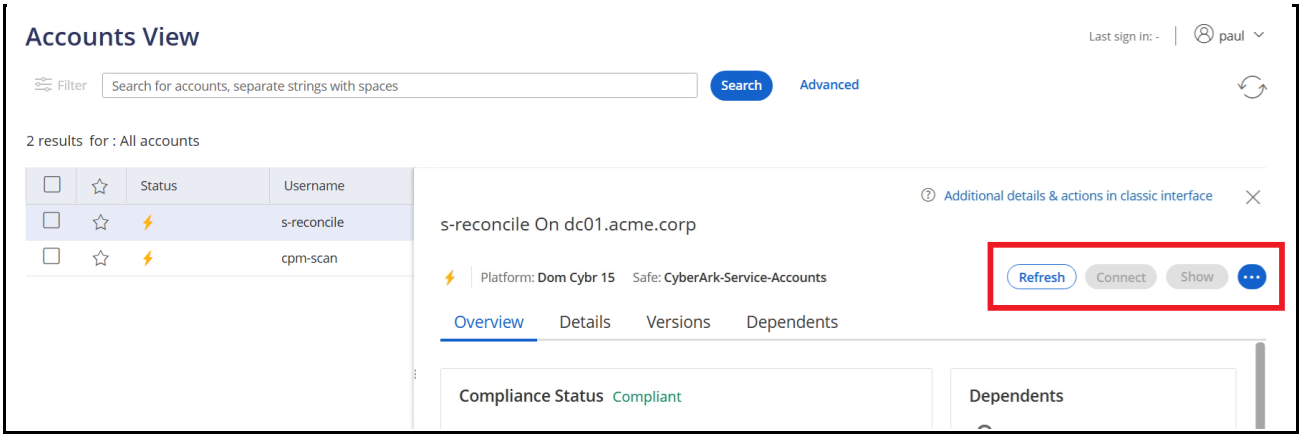
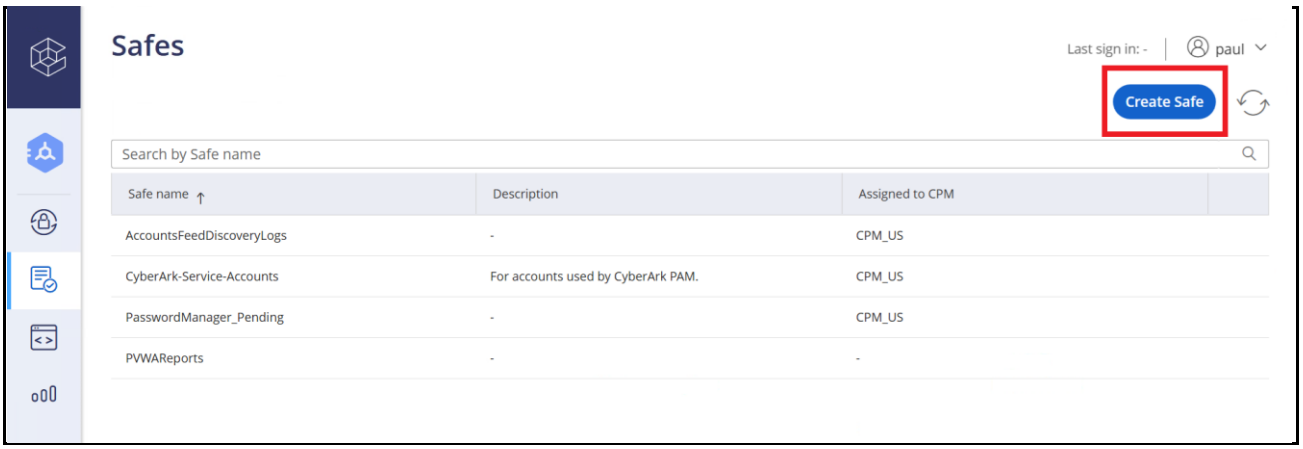
2. Click on **All accounts** to view the accounts. If you select one of the accounts, you will see that you can neither connect nor show the password. Remember, Safe Managers only have *List* privileges on that **Safe**.



3. Now go to **POLICIES | Safes** and click **Create Safe**.



4. Enter *P-BOS-LIN-S-FIN* as the **Safe Name**. This is the **Safe** where the Acme Corporation will store the privileged accounts for its Linux servers that hold financial data for its US division (which happens to be located in Boston). Click on **Next**

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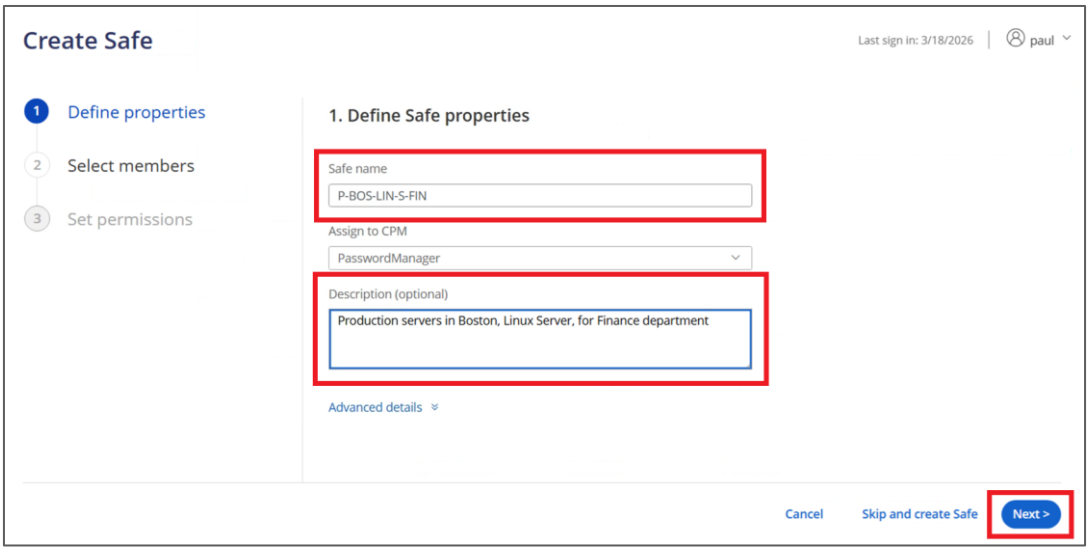


Add Safe Members

Now we will add a group to the **Safe**.

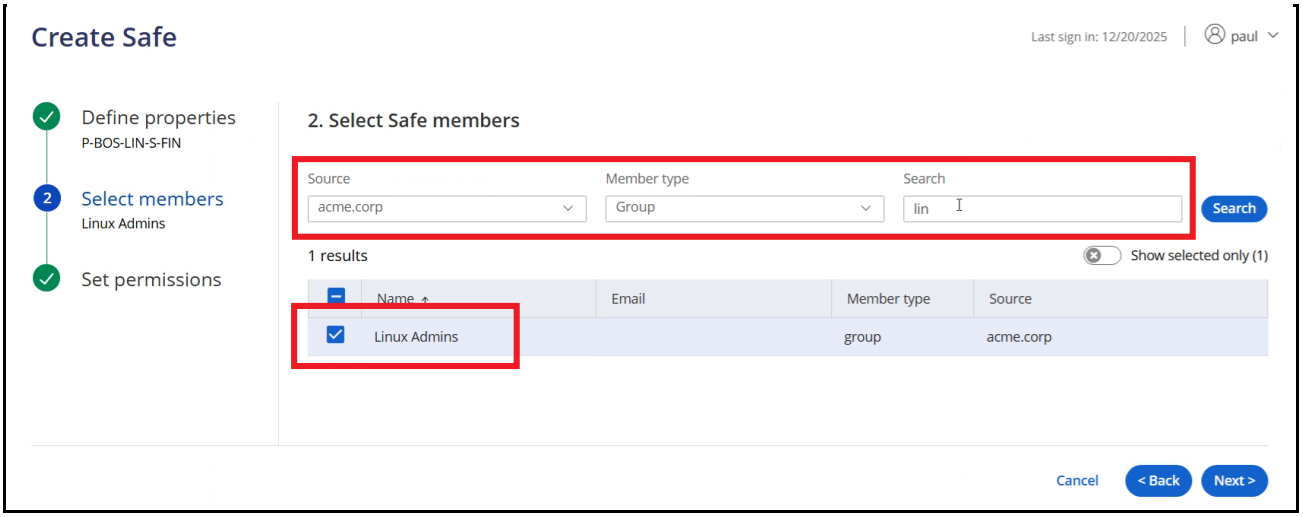
1. Enter the following search parameters:

   Source: *acme.corp*

   Member type: *Group*

   Search: *lin*

   And then click **Search**. Select the group *Linux Admins* and click **Next**.



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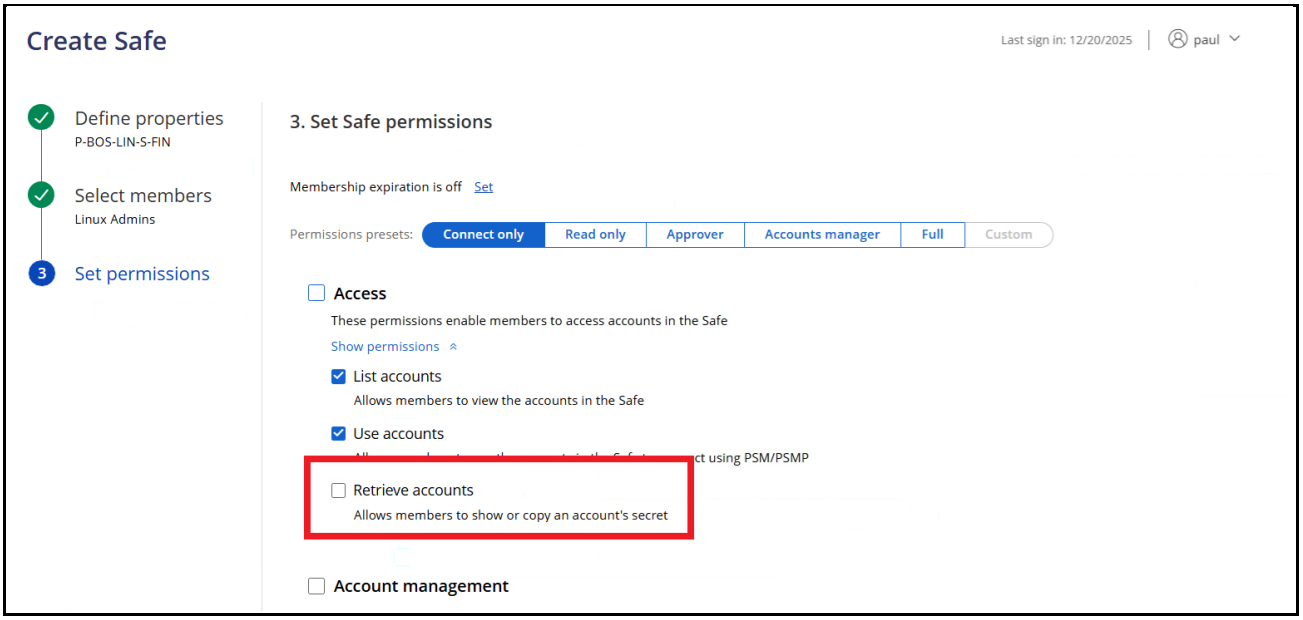
2. The Linux Administrators are the users who will be accessing the accounts in this **Safe** to connect to target servers to do their jobs. They need to be able to see the accounts and to use (but not see) the passwords. To achieve this, we can use the **Permission preset** option **Connect only**. Click on it and then click on the link **Show permissions** to review what this option gives them.



We will give them an additional permission. Scroll down to the section **Safe management and monitoring** and expand it. Tick the box for *View Safe members*. We will need this for a later exercise when we look at **Privileged Access Workflows**. Click on **Create Safe** when you are ready.

3. We want to give the group **CyberArk Vault Admins** the same permissions they have on the **CyberArk-Service-Accounts** safe. In the **Safes** view, click on the new Safe **P-BOS-LIN-S-FIN**, select the **Members** tab, and then click **Add members**.

4. Search for the acme.corp AD group *CyberArk Vault Admins* as we did before and then select **Next**.

5. Once again, we don't have a permission preset for this, so leave **Read only** as the selected option, click on the **Show permissions** link under **Access**, and uncheck the options for **Use accounts** and **Retrieve accounts**. Notice that when you make this change, the **Permissions presets** label changes to **Custom**.

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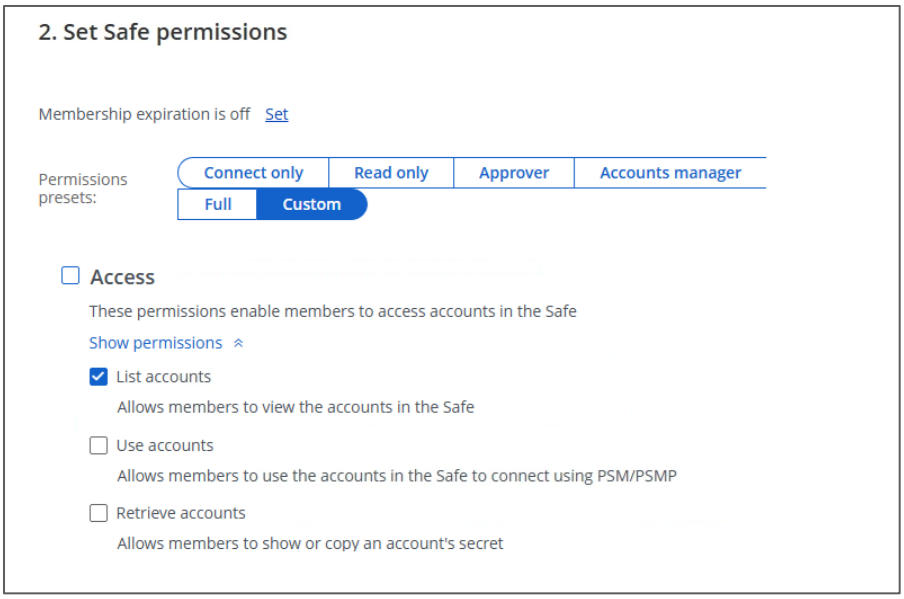
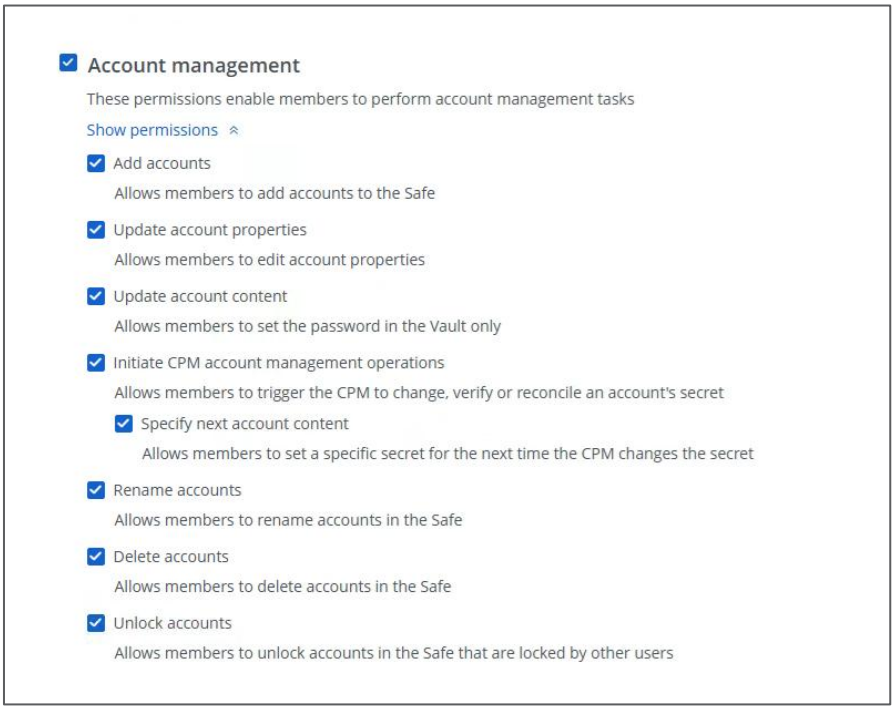


6. Now scroll down and tick the box next to **Account management**. This will give the group all the permissions associated with account management.



7. Scroll down to the **Safe management and monitoring** section and check the boxes for **View Safe members** and **View audit log**. Click **Add** to complete the process and close the current window.

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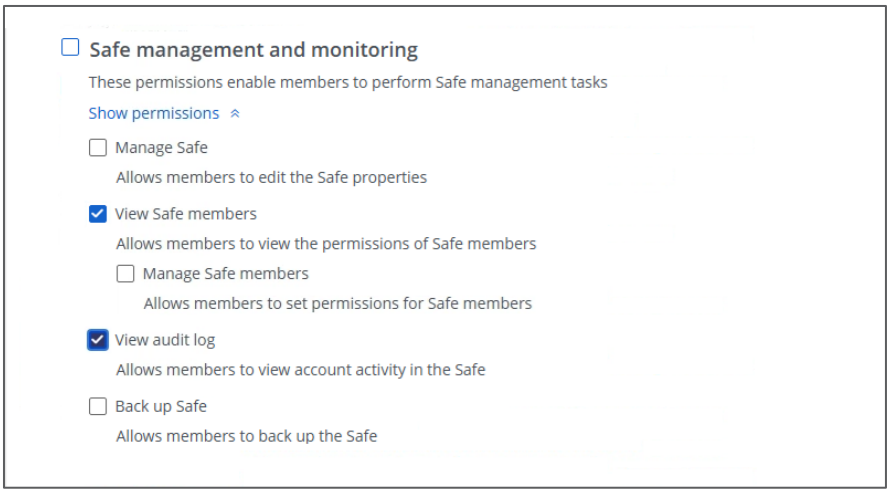

Note:	We added the *CyberArk Vault Admins* group so that they will be able to perform account onboarding, which we will see later, but they will not be able to view the passwords or use the accounts to connect.

A first Linux account

Now that we have created **Platforms** and **Safes**, we are ready to add our first Linux account. We will do this as the user *Paul*. If you are not still logged in to the **PVWA**, log in now.

Add the account

1. Navigate to **ACCOUNTS** and click **Add Account**. On the **Add Account** page, select the system type **NIX*.

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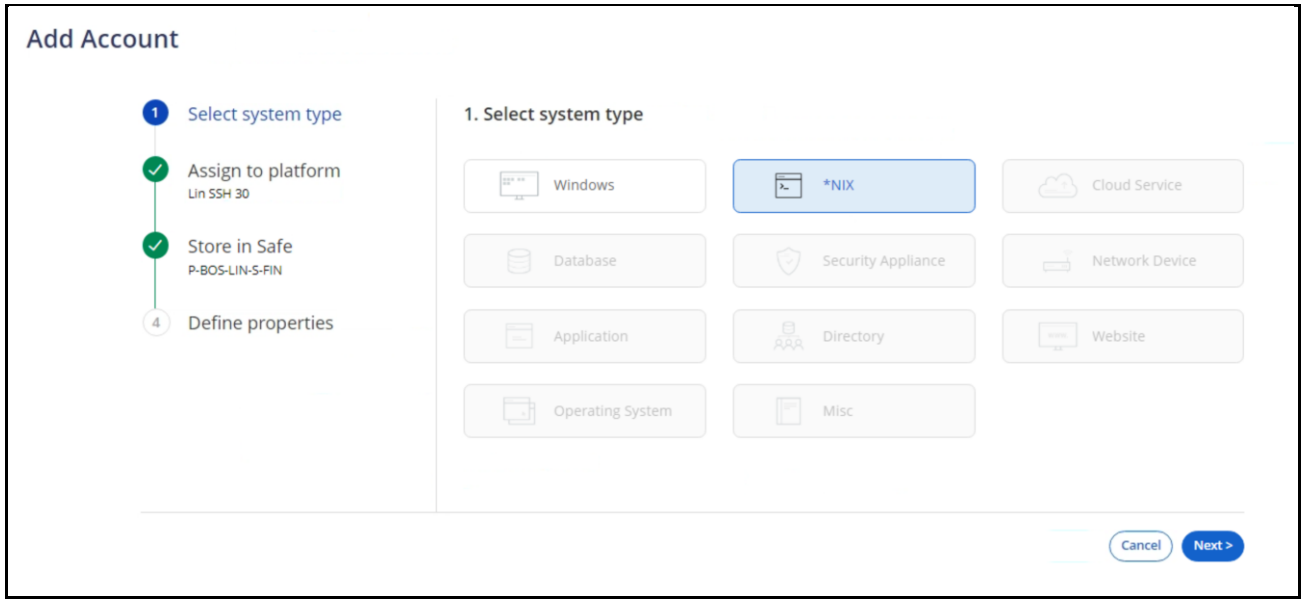

2. Select the *Lin Ssh 30* platform and click **Next**:

3. Select the **Safe** we created earlier: *P-BOS-LIN-S-FIN*.

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Note:	In the image above, only one **Safe** appears. Why is that?

4. Enter the account details as shown below and click on **Add**:

Address:	*lin-451.acme.corp*
Username:	*root01b*
Password:	*Cyberark1*
Confirm Password:	*Cyberark1*

Note:	The account ***root01b*** is one of the many test accounts we use in this training environment as a replacement for the *root* accounts on Linux servers. There are several dozen ***rootXXb*** accounts, as well as several dozen ***rootXX*** accounts. All of the ***'b'*** accounts can log in via SSH. All of the ***rootXX*** accounts (*root01*, *root02*, etc.) are blocked from logging in via SSH, which is a common configuration in the *nix world. We will see how to overcome this challenge in a later exercise.

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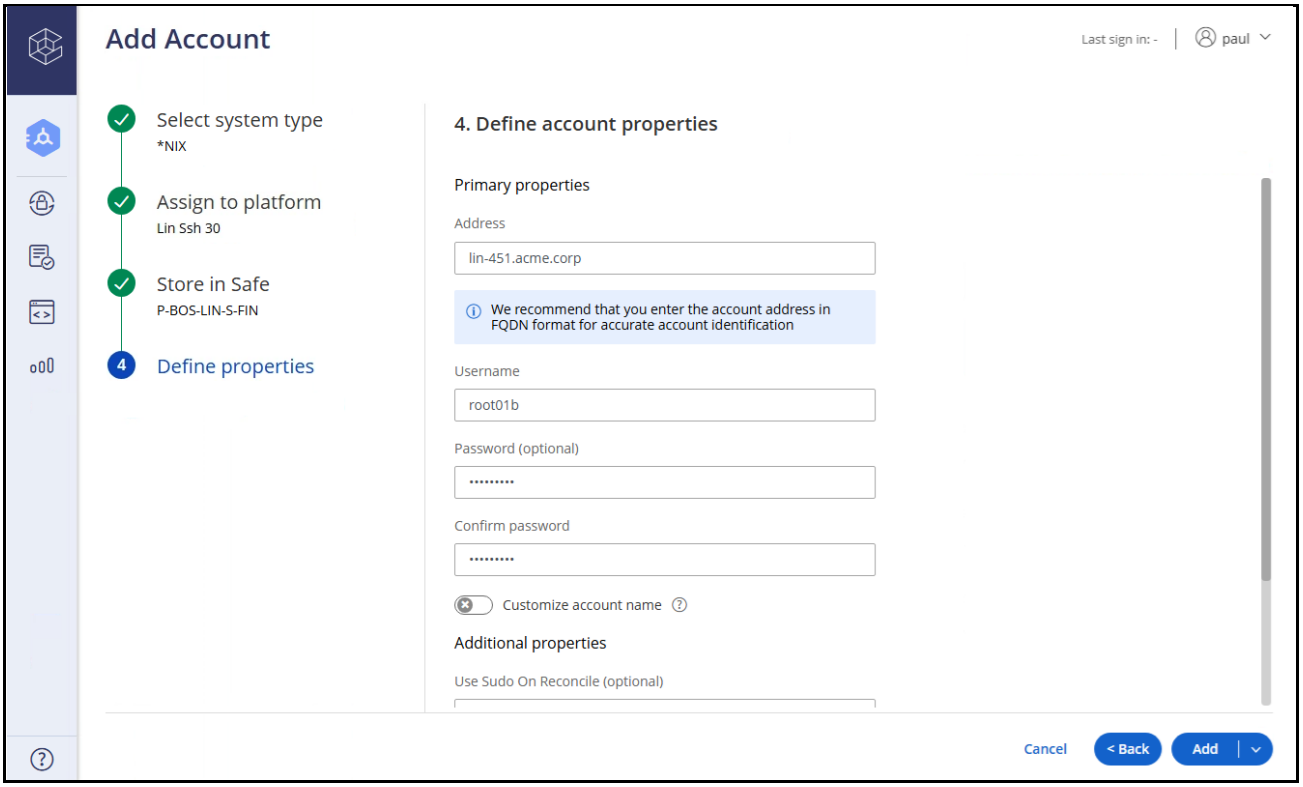

5. On the **Accounts** page, select the newly created account. In **Account Details**, press the **Change** button to confirm that you have created the account correctly and to change the password to a value known only to **CyberArk PAM**.

6. You will be asked to confirm the password change. Click **Change**.

7. You will see a brief message in a green bar at the top of the screen and you will see that the account has been scheduled for change.

8. After a minute or two, you will see that the value for **Compliance Status** is updated to ***Changed by PasswordManager***; that is, by the **CPM**.

Test the New Account as an Ordinary User

Paul wants to make sure that his new account is working correctly, so he asks his colleague *Carlos* to log in to **CyberArk PAM** to verify that he can see the newly created account and that he can connect with it to the target server via the **PSM**. *Carlos* is a member of the AD group **Linux Admins**, which gives him access to the Safe *Paul* created. So, we will have to log out and log back in as *Carlos* (or open a separate browser instance).

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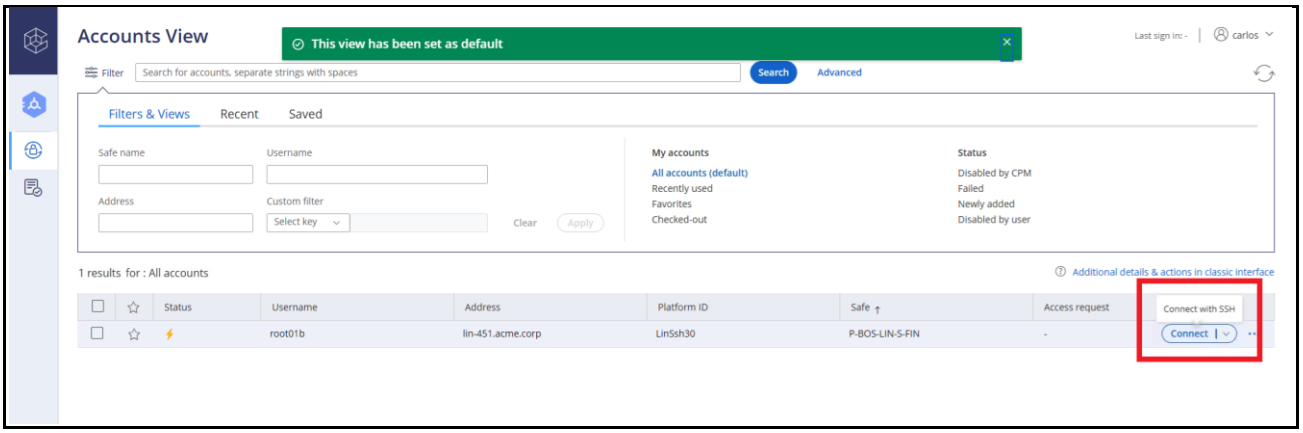
Note:	Sometimes, if you do this exercise too soon after the previous one, you will see a connection error. This simply means that the **PSM** has not yet had time to detect the new platform. Performing a restart on the **PSM** service on **COMP02** will speed up this process. You can log into **COMP02** as an administrator using the RDP short-cut on the desktop. Once connected, open Windows Services, locate the service **CyberArk Privileged Session Manager**, and restart it.

1. When you log in as *Carlos*, you will not see any accounts. That is because the default view is **Recently used**. Switch to **All accounts**. You can set that as the default view.

2. Next, click on the **Connect** button next to the account *root01b* in the **Account view**.



Note:	The behavior of RDP files will depend on the browser you use. The examples shown in this guide come from Google Chrome. You may be asked to save the RDP file before launching a connection.

3. The downloaded RDP file appears in the upper right-hand corner of the screen. Click on the file to open it and launch the RDP connection.

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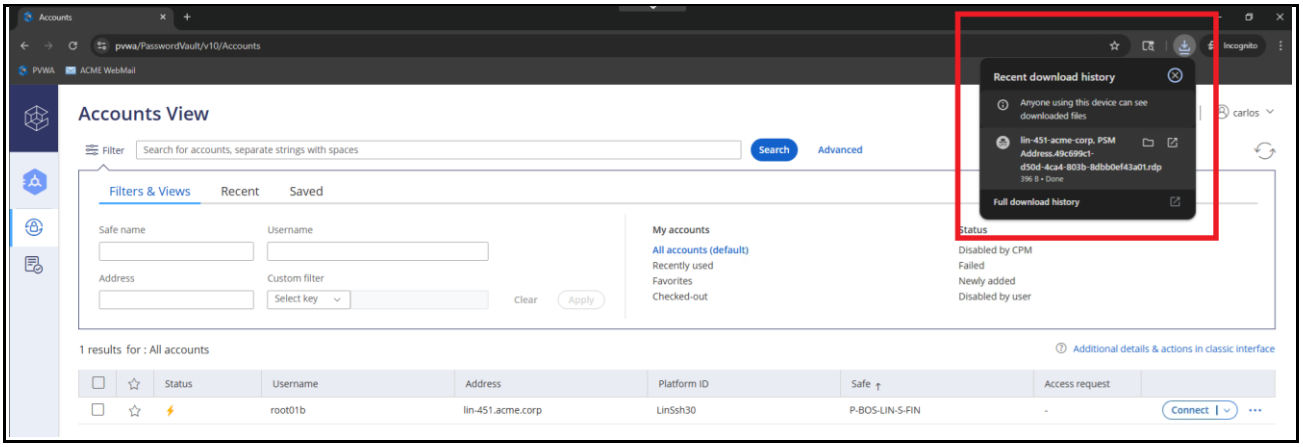
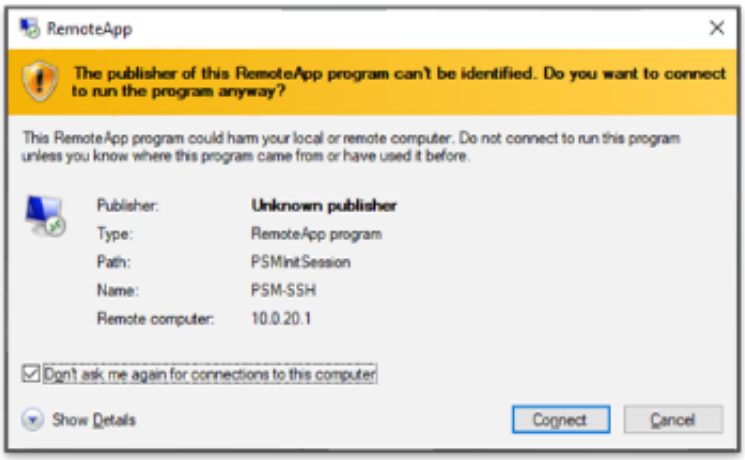


4. If this is the first time you are launching an RDP connection to this server, you will receive a pop-up warning about the publisher of the RemoteApp program. You can tick the box **Don't ask me again for connections to this computer** and then click **Connect** to continue.



5. The first time you connect to a particular target machine, you will receive an alert about the server's host key. Click **Yes** to accept the server's key.

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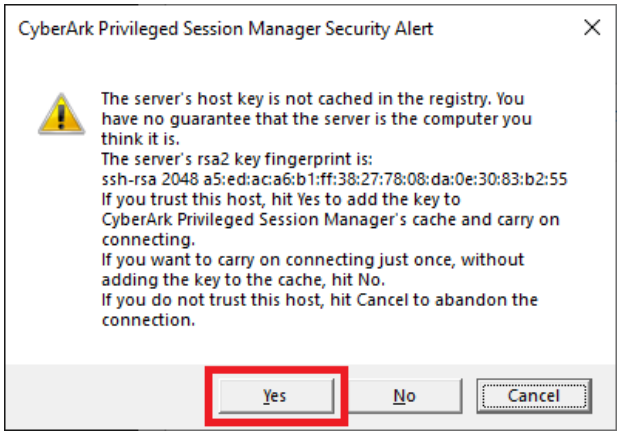
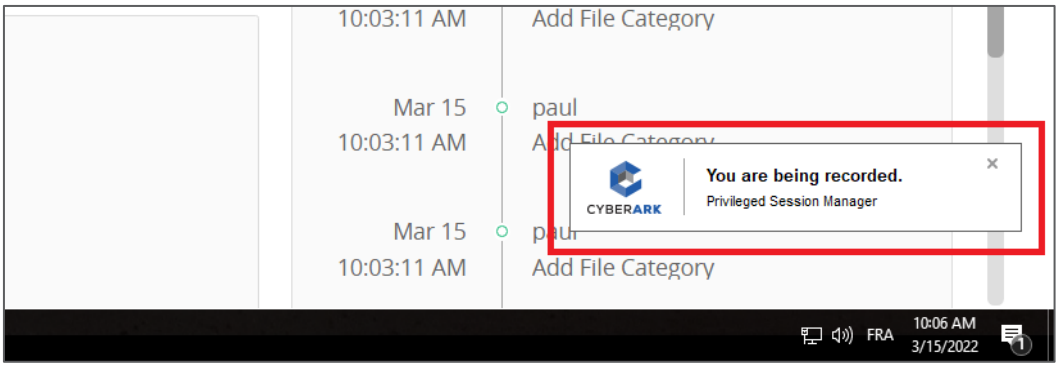
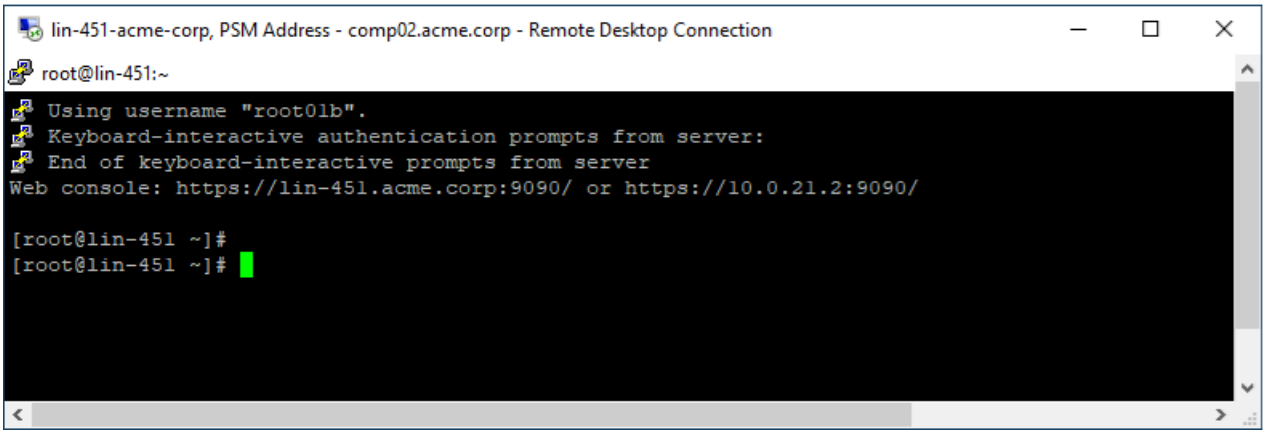


6. In the lower right-hand corner of the screen, you will see a pop-up informing you that the session is being recorded. It will disappear automatically after a few seconds.



Note: Starting with **PAM** version 14, the message displayed in the pop-up can be customized (for example, if you want to put it in your local language).

7. And then a PuTTY window will appear in your taskbar with your SSH connection to the machine *target-lin* as *root01b*.



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8. Execute a few simple, non-destructive commands (remember, you are a privileged user) such as **`pwd`** and **`ls -al`** to generate some session activity. When you are done, enter **`exit`** and hit **Enter** to close the session.

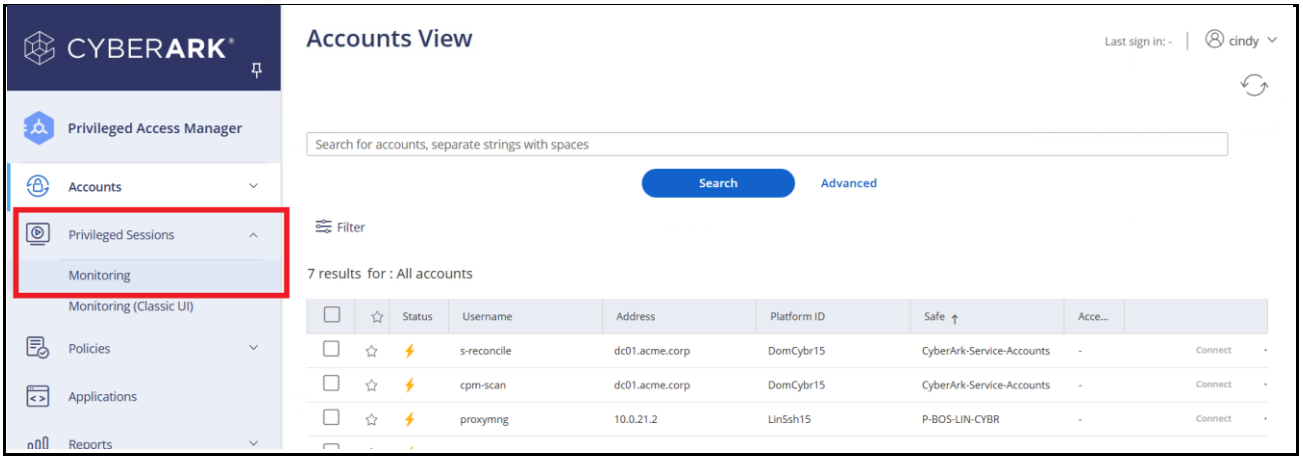
9. Log out of the **PVWA**.

Auditor Tasks

In this section, we will review the activity related to the *root01b* account by putting on our auditor's hat.

1. Sign out of the **PVWA** and log back in using **LDAP Authentication** as *Cindy*. Enter the username *Cindy* and password *Cyberark1* and then hit **Enter** or click **Sign In**.

2. In the left-hand toolbar, click on the **Privileged Sessions | Monitoring** tab.



3. Depending on your activity, you will see fewer or more recordings. Click on the most recent recording associated with *Carlos* in the list of **Recordings**.

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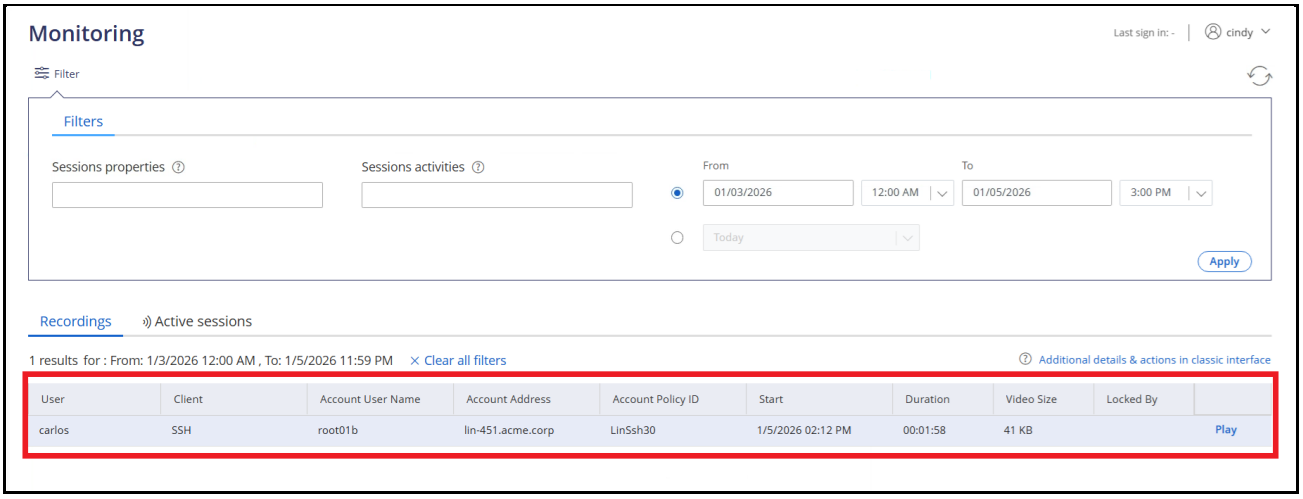
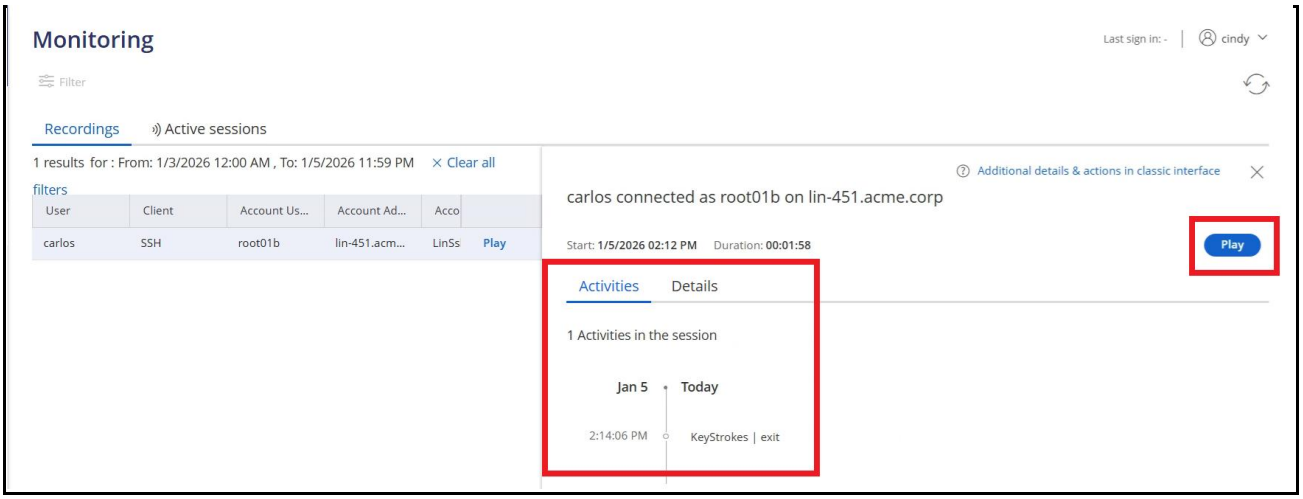


4. Notice that you have the details of what happened during the session under **Activities**, including the commands *Carlos* executed. Click on the **Play** button to view the recording.



5. The recording begins playing. You can pause, rewind, fast-forward, or jump to a specific place in the recording by clicking on a command in the time-line on the left.

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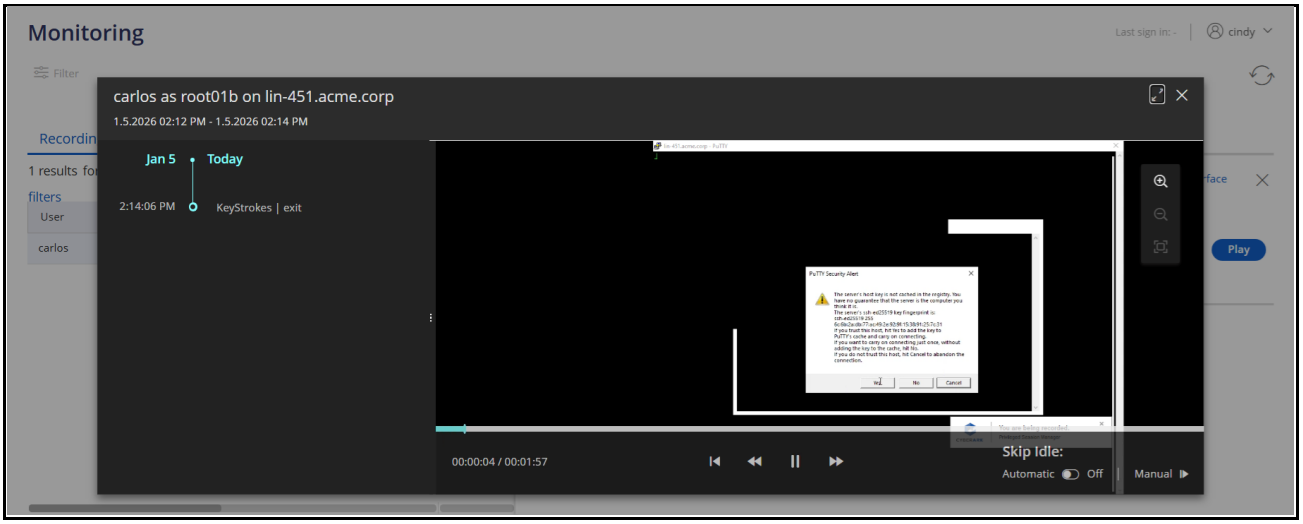

6. You can close the recording window by clicking on the **X** in the upper right-hand corner.

7. Sign out of the **PVWA**.

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Securing Oracle Database Accounts

In this section, we will configure **CyberArk PAM** to manage an Oracle DBA account. As in previous exercises, we will duplicate a **Platform**, create a **Safe**, and then add the account.

Vault Administrator Tasks

Duplicating a Platform

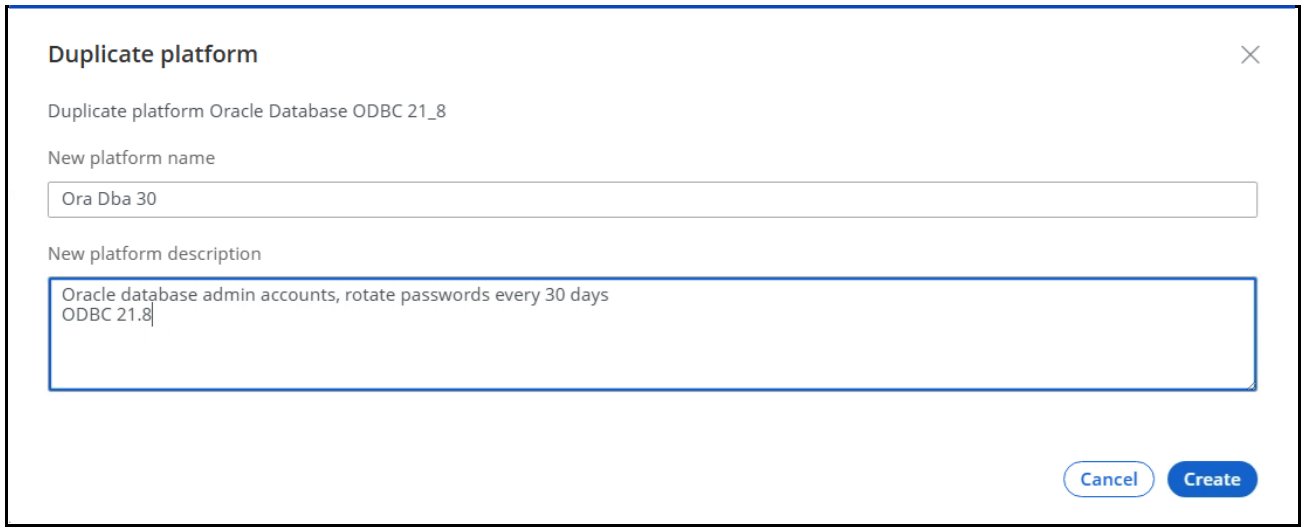
In this section, we are going to create a **Platform** dedicated to managing accounts used to access Oracle databases, such as a DBA account.

1. Log in to the **PVWA** as *Mike* and go to **ADMINISTRATION | Platform Management**.

2. Choose **Database | Oracle Database ODBC 21_8** and select **Duplicate**.

Note:	This platform was imported from the **CyberArk Marketplace** by the team responsible for implementing CyberArk PAM at the Acme Corporation. More information about importing and managing Platforms is available in the CyberArk documentation.

3. Enter *Ora Dba 30*, give the new **Platform** a meaningful description, and press **Create**.



4. The original Platform – **Oracle Database ODBC 21_8** – was already active, so our new Platform is also active. Because it is always a good idea to leave the base Platforms unchanged, and it is also recommended to deactivate any unused Platforms, you can go ahead and deactivate **Oracle Database ODBC_21_8**.

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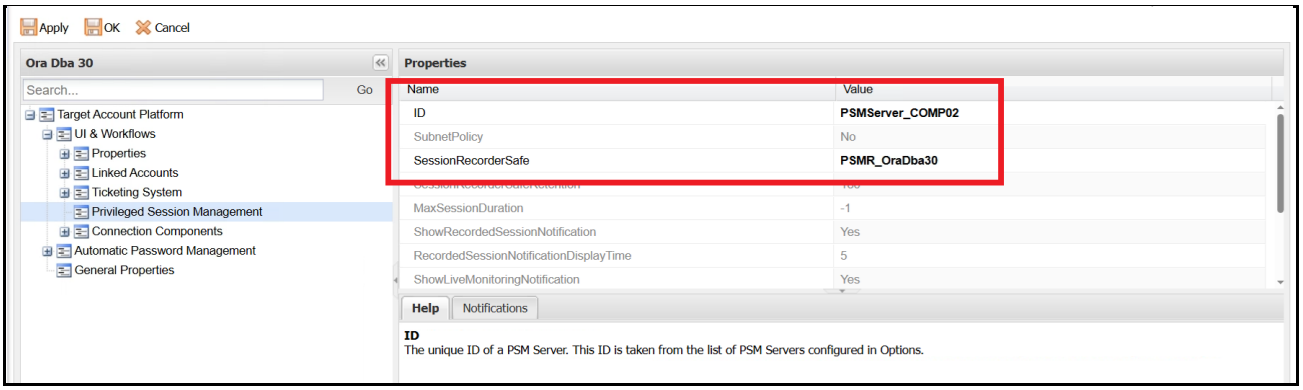
5. Now let's make a few changes to the new platform *Ora Dba 30*. Select it and click **Edit.**

6. Go to **UI & Workflows | Privileged Session Management**. We need to make two changes here.

   - First, we need to set the correct PSM Server ID, as we did with the Windows Domain platform. Change the **ID** parameter to ***PSMServer_COMP02***.
   - Then set the **SessionRecorderSafe** parameter to PSMR_<platform_id>, which in the current case would be ***PSMR_OraDba30***.

Note:	Changing the name of the recording Safe is not a requirement, but it is probably a good idea. Depending on the size of your organization, you and your colleagues might be generating a large number of recordings every day. Putting them all into a single Safe could impact performance, as well as make retrieving a recording for forensic analysis a more complicated task



7. Moving on, we will make similar changes to the ones made previously. Go to **Automatic Password Management | General**.

   - Set *ImmediateInterval* to **1**
   - Set *AllowedSafes* to **-ORA-**

8. Now go to **Automatic Password Management** | **Password Change** and change the value of the parameter *PerformPeriodicChange* from **No** to **Yes**. This will enable the application of the **Master Policy** rule *Require password change every X days* to accounts managed by this platform.

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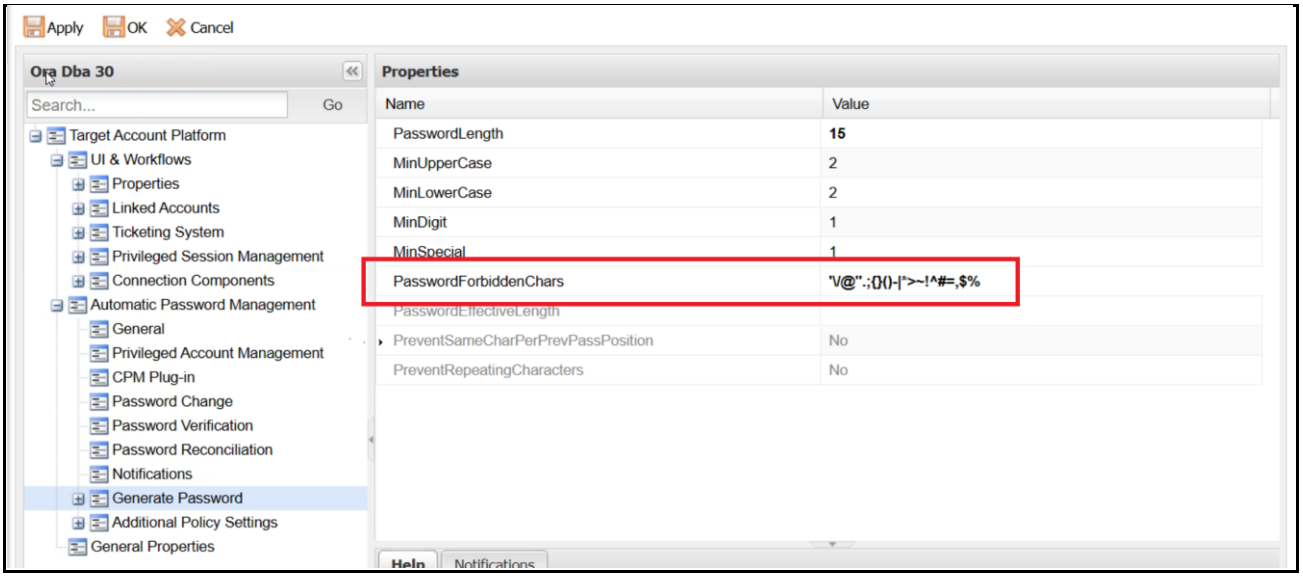
9. Within the same window, go to **Password Verification** and change *VFPerformPeriodicVerification* from **No** to **Yes**. This will allow the password to be verified by the **CPM** automatically and without user intervention. Press **Apply**.

10. In the **Generate Password** section, let's add a few characters to the *PasswordForbiddenChars* field. Recent experience has shown that some additional forbidden characters might be useful here, so add the following four characters:

Equals	**=**
Comma	**,**
Dollar	**$**
Percent	**%**

11. Make sure you add the new characters without deleting any of the existing characters.



12. Lastly, in the section **Additional Policy Settings**, locate the parameter ***ConnectionCommand*** and modify the version of the Oracle Instant Client to match the version installed on the **CPM** server. The (etc.) in the line below just indicates that there is more text in the parameter, but that it does not concern us.

```
Driver={Oracle in instantclient_21_20} (etc)
```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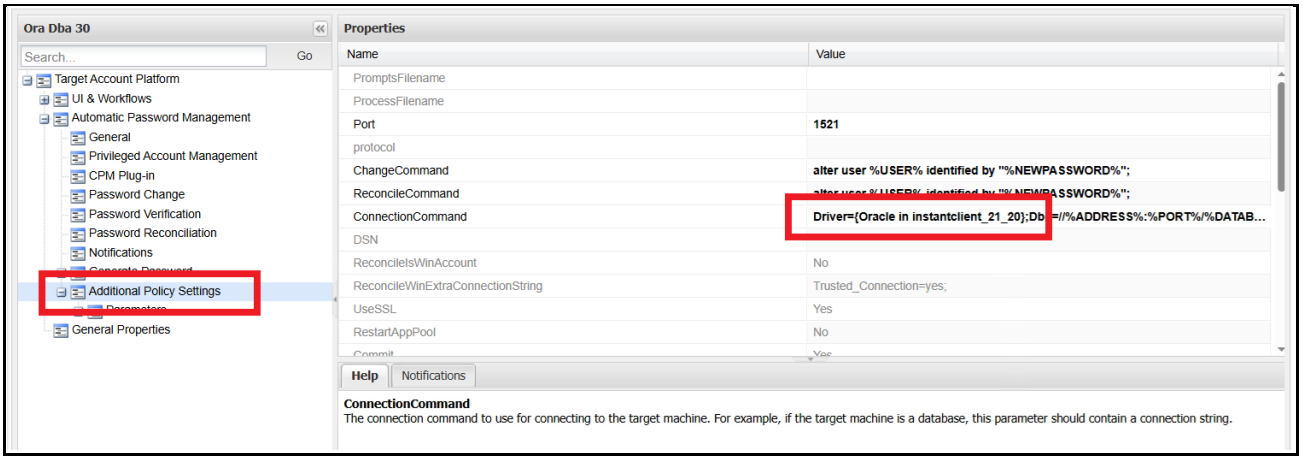

13. Click **OK** to save the changes and close the **Platform**.

Note:	Don't forget to add an exception to the Master Policy to rotate the oracle DBA passwords every 30 days.

Safe Manager Tasks

Because we are dealing with a different technology – Oracle in this case – the person responsible for managing Oracle **Safes** is different. Our Safe Manager for this exercise is named *Robert*.

Creating a Safe

1. Log in to the **PVWA** as LDAP user *Robert* (username *Robert* and password *Cyberark1*) and go to **POLICIES | Safes**.

2. Press the **Create Safe** button.

3. Enter *P-BOS-ORA-HR* as the **Safe** name and press **Next**.

4. Add the Active Directory group *Oracle Admins* to the **Safe**, using the **Connect only** permission preset (which removes the *Retrieve* permission). Click on **Create Safe** when you are ready.

5. Now on the same **Safe** add the LDAP group *CyberArk Vault Admins*. Remove the permissions: *Use accounts* and *Retrieve accounts*. Add *Account Management* (which will add all the permissions under it). We will need this for a later exercise. You can also give them *View safe members*, which is not strictly necessary for this training, but is probably a good idea.

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Adding an Account

In this next step, we will add an Oracle account. Note that database accounts have some additional settings that we need to set: **Port number** and **Database**.

In this exercise, we will add a single account, but there are 5 Oracle accounts you can use: ***dba01*** – ***dba05***.

1. Still connected as *Robert*, go the **ACCOUNTS** tab, click **Add Account** and create an account with the following parameters:

System type	***Database***
Platform	***Ora Dba 30***
Safe	***P-BOS-ORA-HR***
User Name	***dba01***
Address	***acme-123.acme.corp***
Password	***Cyberark1***
Confirm Password	***Cyberark1***
Port	***1521***
Database	***xepdb1***

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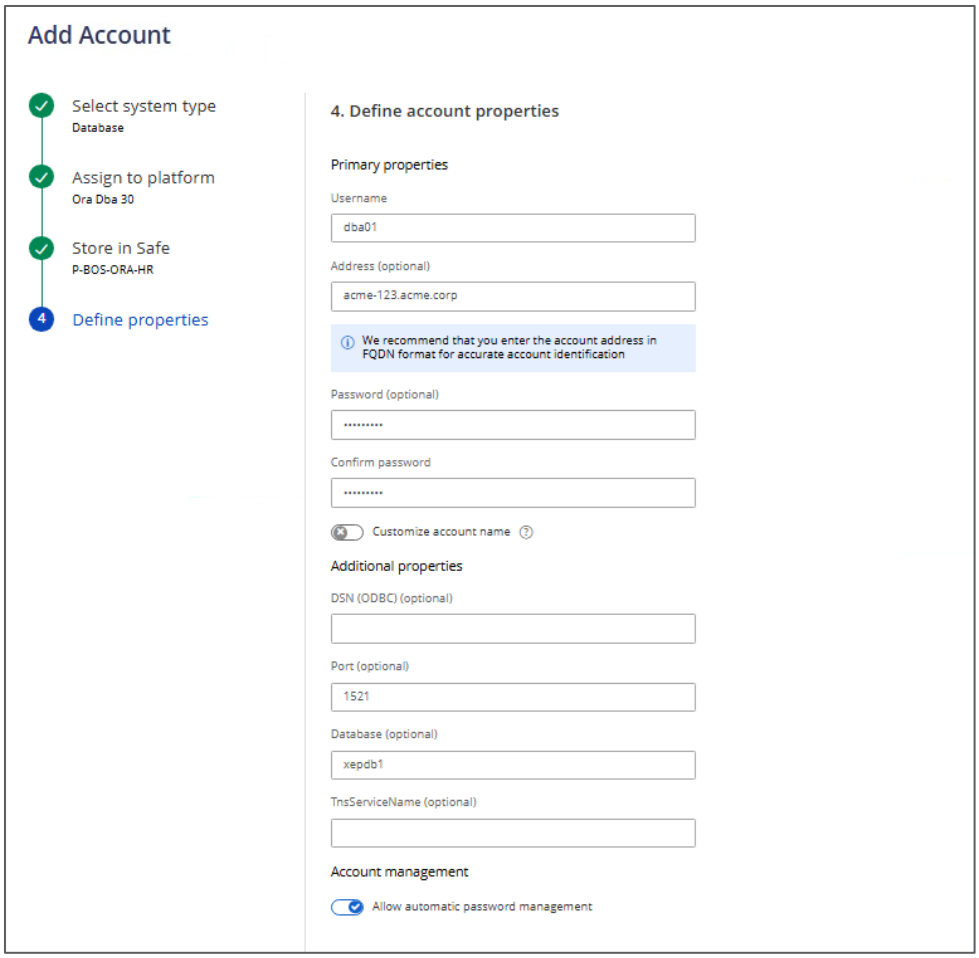

2. Press **Add**.

3. Click on the new account and then click on the **Change** button to rotate the password. Once the password is changed, you can launch a connection the database.

Note:	If you try to change the password before launching a connection, the password change will probably fail. The reason this occurs is that the existing Oracle password is expired, so when the **CPM** logs into the target server to change the password, rather than receiving the expected prompt, it sees the message about the expired password.

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Linked Accounts

In this chapter we will look at linked accounts, specifically we will be creating:

- Logon accounts
- Reconcile accounts

Securing SSH Accounts Using a Logon account

In this section, you will perform the tasks necessary to bring under **CyberArk PAM** control a Linux privileged account that is prevented by the target machine's security policy from accessing the server directly via SSH, a very common scenario for *root* accounts.

To achieve our goal, we will:

- Add a new account, *root01*, and test that we cannot connect with it via SSH
- Create a new **Safe** in which to store our logon accounts
- Add a logon account
- Associate the logon account with *root01*
- Test that we can now connect to the *root01* account using the logon account

Add the root01 account

First, we will add a new account to our existing Linux Safe – **P-BOS-LIN-S-FIN** – and manage it with the **Platform** we created earlier: **Lin Ssh 30**. This is a task for the Safe Manager.

1. Log into the **PVWA** as *Paul* and add a new account with the following details:

System Type:	**NIX*
Platform Name:	*Lin Ssh 30*
Store in Safe:	*P-BOS-LIN-S-FIN*
Address:	*lin-451.acme.corp*
Username:	*root01*
Password:	*Cyberark1*
Confirm Password:	*Cyberark1*

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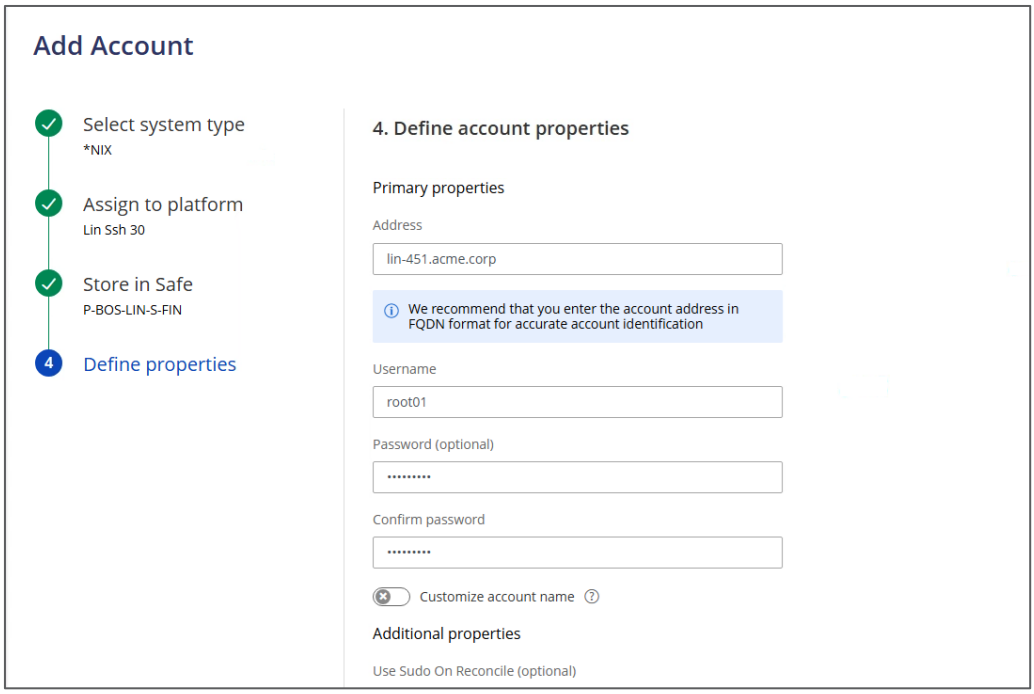

2. As a test, try either changing the password for *root01* or connecting to the target machine with the **PSM**. Either action should fail because *root01* cannot connect via SSH.

Create a New Platform for Logon Accounts

Next, we will duplicate our first Linux platform. It is already largely configured the way we want, so we will only need to make a small change in order to use it for our logon accounts.

1. Log in to the **PVWA** as *Mike* and browse to **Administration | Platforms | *NIX**. Locate the platform *Lin Ssh 30*, click on the three dots on the extreme right of the line, and select **Duplicate**.

2. Name the Platform **Lin Logon 30**, give it an appropriate description such as “Linux logon accounts, rotate passwords every 30 days” and click **Create**.

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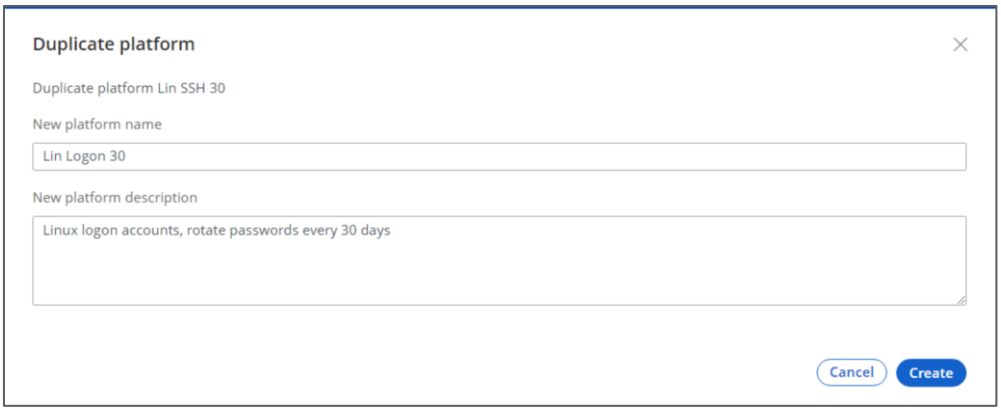
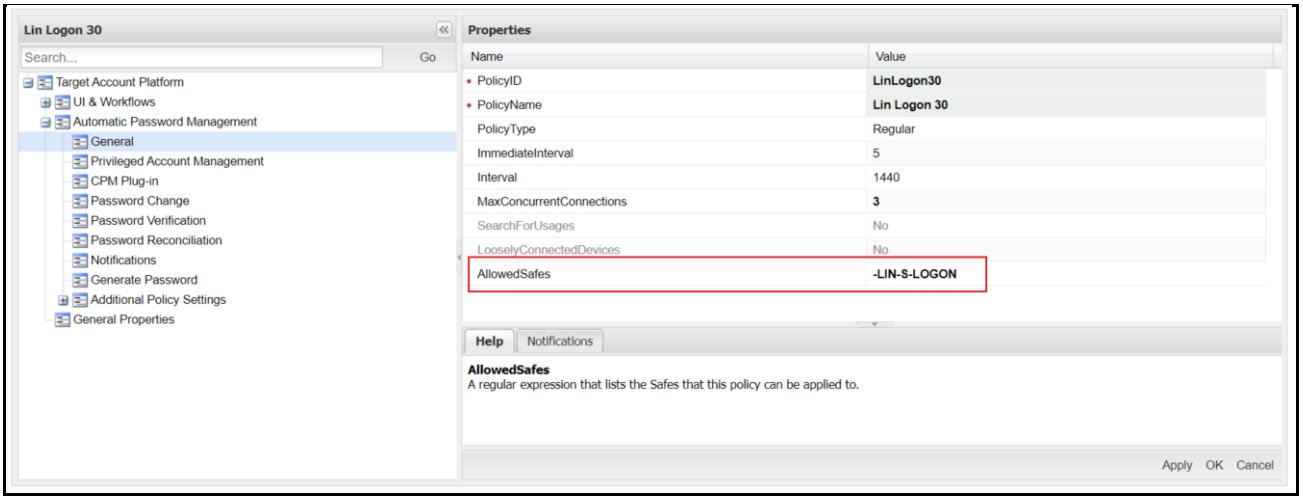


3. We need to make two small changes to this new **Platform**. Select it and click **Edit**.

4. Go to **UI & Workflows | Privileged Session Management** and change the parameter **SessionRecorderSafe** so that it matches the ID of our new Platform: ***PSMR_LinLogon30***.

5. Now go to the section **Automatic Password Management | General** and modify the **AllowedSafes** parameter so that is reads "*-LIN-S-LOGON*" (without the quotes). This will restrict this **Platform** for use only with accounts in **Safes** that contain that string.



6. When you are ready, click **OK** to save your changes.

Note: The **Master Policy** exception for 30 days was copied from the platform **Lin Ssh 30**.

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Create a Safe – P-BOS-LIN-S-LOGON

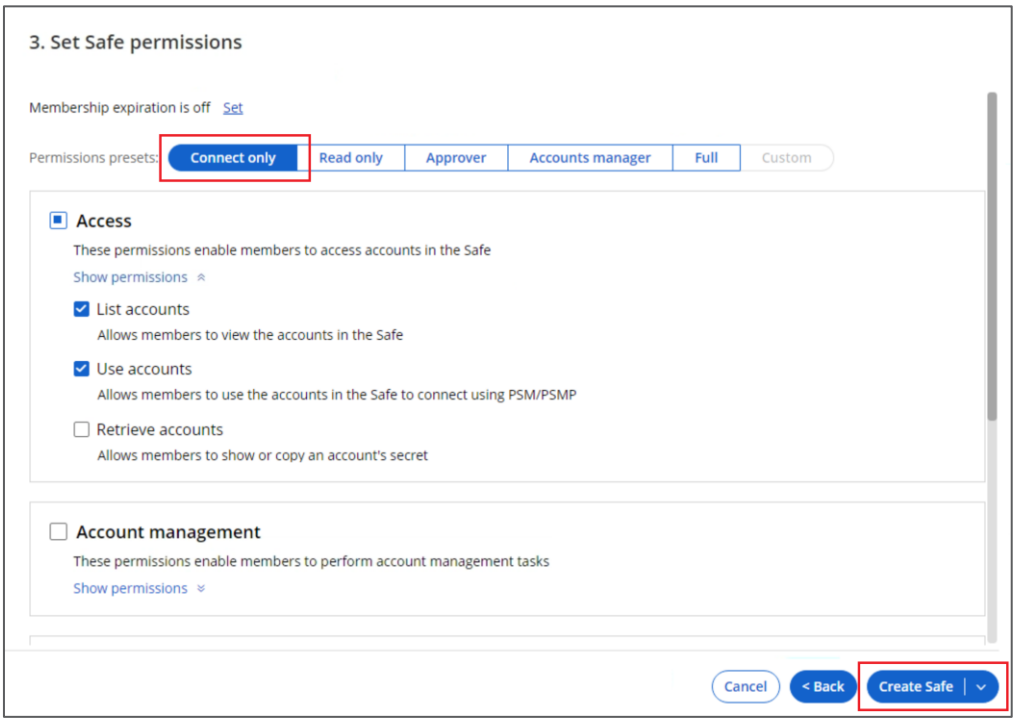
Now we will create a **Safe** for our logon accounts. By keeping the logon accounts separate from the linked accounts, we will be better able to manage who can access our target systems.

1. Log back into the **PVWA** as *Paul*. Browse to **Policies | Safes** and click on click on **Create Safe**.

2. Name the Safe **P-BOS-LIN-S-LOGON**. Give it a pertinent description, such as "Linux logon accounts for US servers with financial data" and click **Next**.

3. Search for the AD group **Linux Admins**. Click **Next**.

4. Assign them the **Connect only** permission preset and then click **Create Safe**. This step will allow *Carlos*, who is a member of the Linux Admins AD group, to use the accounts in the **Safe**, but not view the passwords.



5. As a final step, we will add another group to this **Safe** with different permissions. In the **Safes** view, select the newly created Safe **P-BOS-LIN-S-LOGON** and click on **Add Members**.

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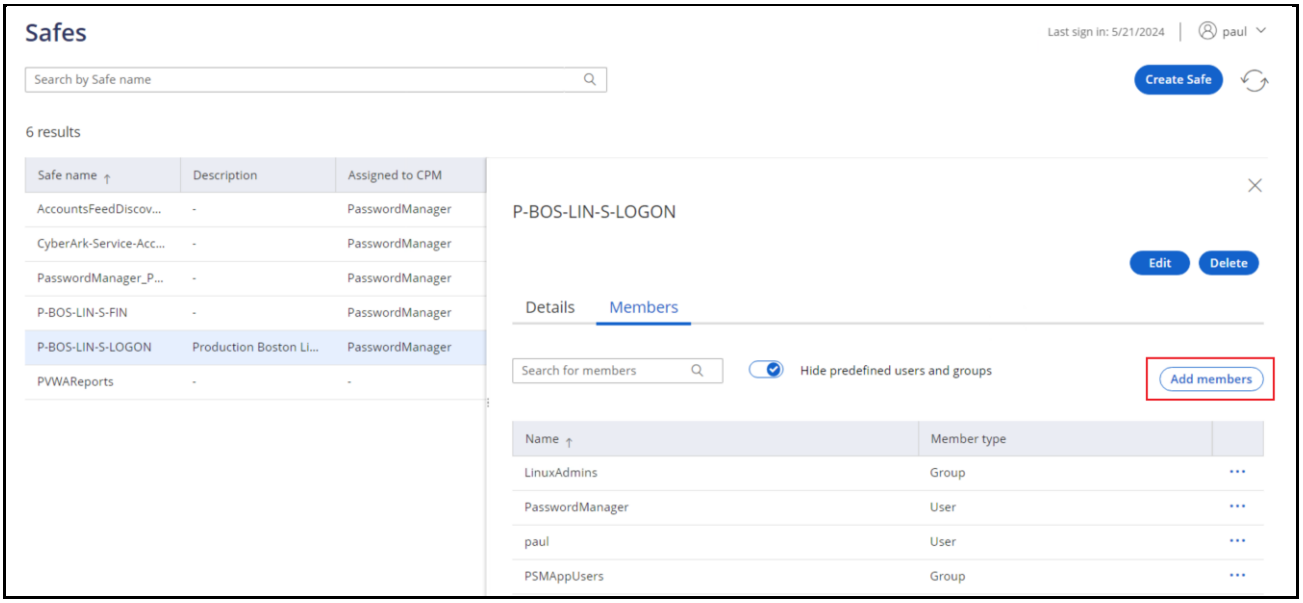

6. Add the **CyberArk Vault Admins** AD group to the **Safe** with the permissions below.

Access	*List*
Account management	*All permissions*
Safe management and monitoring	*View Safe members* and *View audit log*

Add a logon account

1. Still logged into the **PVWA** as *Paul*, add a new account with the following details:

System Type: **NIX*
Platform Name: *Lin Logon 30*
Store in Safe: *P-BOS-LIN-S-LOGON*
Address: *lin-451.acme.corp*
Username: *logon01*
Password: *Cyberark1*
Confirm Password: *Cyberark1*

Associate the logon account with root01

1. As a final step before testing our linked accounts, go to the **Accounts** view, select the *root01* account, and click on the **Details** tab. Under **Linked Accounts** and **Logon Account**, click on the ellipsis and select **Link**.

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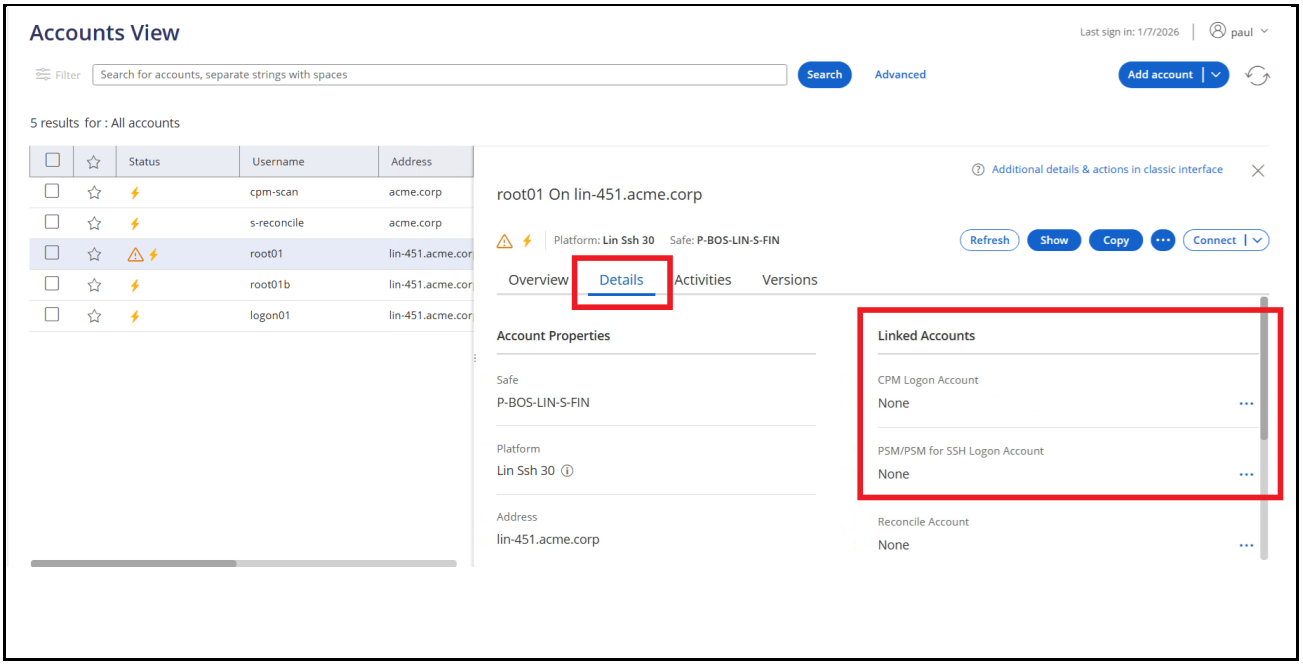
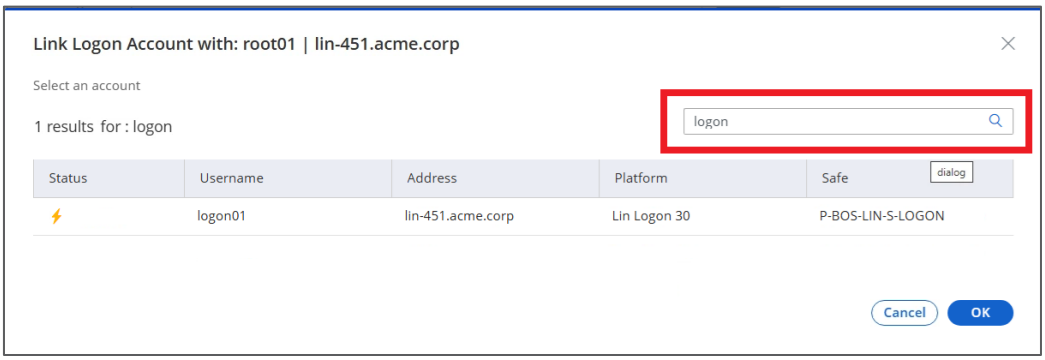


2. You can search for the account if necessary. Select the account **logon01** and click on **OK**.



Test the linked accounts

In order to test that everything is configured correctly, we are going to log in again as *Carlos*.

1. Log into the **PVWA** as Carlos. Locate the *root01* account and click the **Connect** button.

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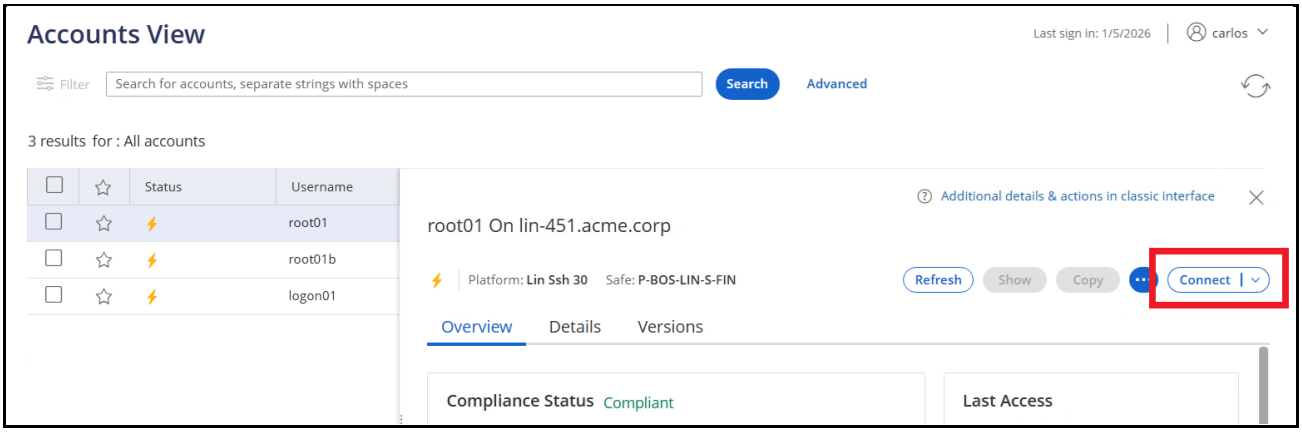
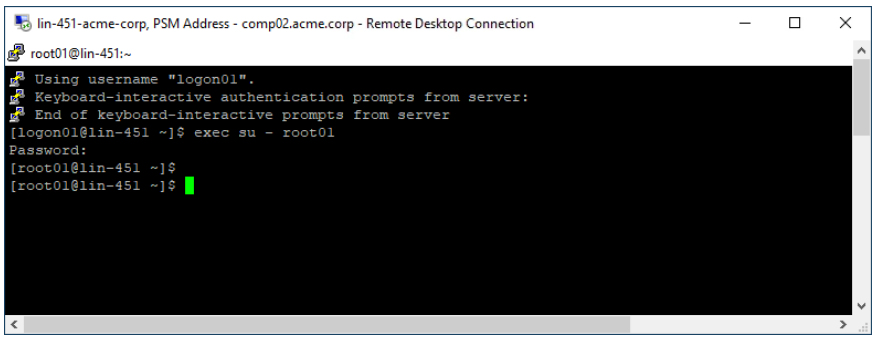


2. You should be connected to the target machine. Notice the presence in the Putty window of the connection with the *logon01* account and the switch user command.



Note:	If you receive an error that **Privileged Session Management** is not enabled or defined for the policy, restart the **CyberArk Privileged Session Manager** service in Windows Services on **COMP02** and try again.

3. Type **exit** and hit **Enter** to close the Putty window. You can log out of *Carlos*' session.

Securing Windows Server Local Accounts via a Reconcile Account

In this exercise you will add to **CyberArk PAM** a Windows server local account for which the current password is unknown. To bring this account under management, you will associate it with a domain account (*s-reconcile*) that can perform a password reset, or ***reconciliation***.

Vault Administrator Tasks - Duplicating a Platform

1. Log in to the **PVWA** as *Mike*.

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2. Go to **ADMINISTRATION | Platform Management**.

3. Select the *Windows Server Local Accounts* and click **Duplicate**.

4. Enter ***Svr Adm 20*** as the **Platform** name, you may optionally add a description like "Rotate password every 20 days", and press **Create**.

Note:	You may have noticed that there is no indication in the Platform names we give that this is a Windows platform. Inasmuch as the Platforms are now organized by type, the indication that this is a Windows platform is largely redundant. For *Nix platforms, it may be useful to know whether a particular Platform targets Linux machines as opposed to Unix machines.

5. Highlight the newly created platform and select **Edit**.

6. Go to **UI & Workflows**. Change *AutoVerifyOnAdd* from **No** to ***Yes***. This causes the **CPM** to check that the password in the **Vault** is the same as the password on the targe system. Select **Apply** to save your change.

7. Under **UI & Workflows | Privileged Session Management**, update the **SessionRecorderSafe** parameter to match the ID of the current Platform. The parameter should be: ***PSMR_SvrAdm20***.

8. Go to **Automatic Password Management | General** and set the *ImmediateInterval* to ***1***.

Note:	Once again, we are modifying this value for training purposes only, enabling us to move a little faster. A one-minute immediate interval is suitable for testing but should be set to five in a production environment.

9. Enter **-WIN-S-LA-** in the *AllowedSafes* field to limit the accounts with which this platform can be used. Click **Apply** to save your change.

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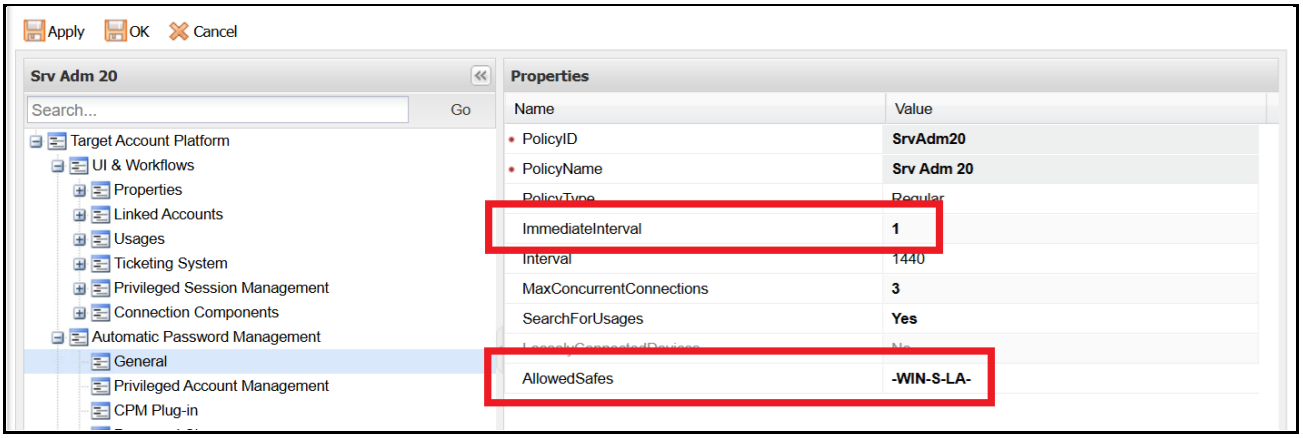

10. Now go to **Automatic Password Management** | **Password Change** and change the value of the parameter *PerformPeriodicChange* from **No** to **Yes**. This will enable the application of the **Master Policy** rule *Require password change every X days* to accounts managed by this **Platform**.

11. Within the same window, go to **Password Verification** and change *VFPerformPeriodicVerification* from **No** to **Yes**. This will allow the password to be verified by the **CPM** automatically and without user intervention.

12. Go to **Password Reconcilation** and enter the following:

RCAutomaticReconcileWhenUnsynced:	*Yes*
ReconcileAccountSafe:	*CyberArk-Service-Accounts*
ReconcileAccountName: *	*s-reconcile*

Note: This is where using a customized name for our reconcile account comes in handy. Rather than needing to go a look up the account name in the account parameters, we can enter our customized name.
The values for all the parameters as they appear in this guide assume that you have followed all previous instructions ***to the letter***. If you haven't, then the values shown here will not work.
Also, as already stated, copying and pasting from the PDF file into the virtual machine causes problems, so the safest approach is to copy the values from the **PVWA**, paste them into Notepad, and then copy them into the appropriate fields in the **Platform**.

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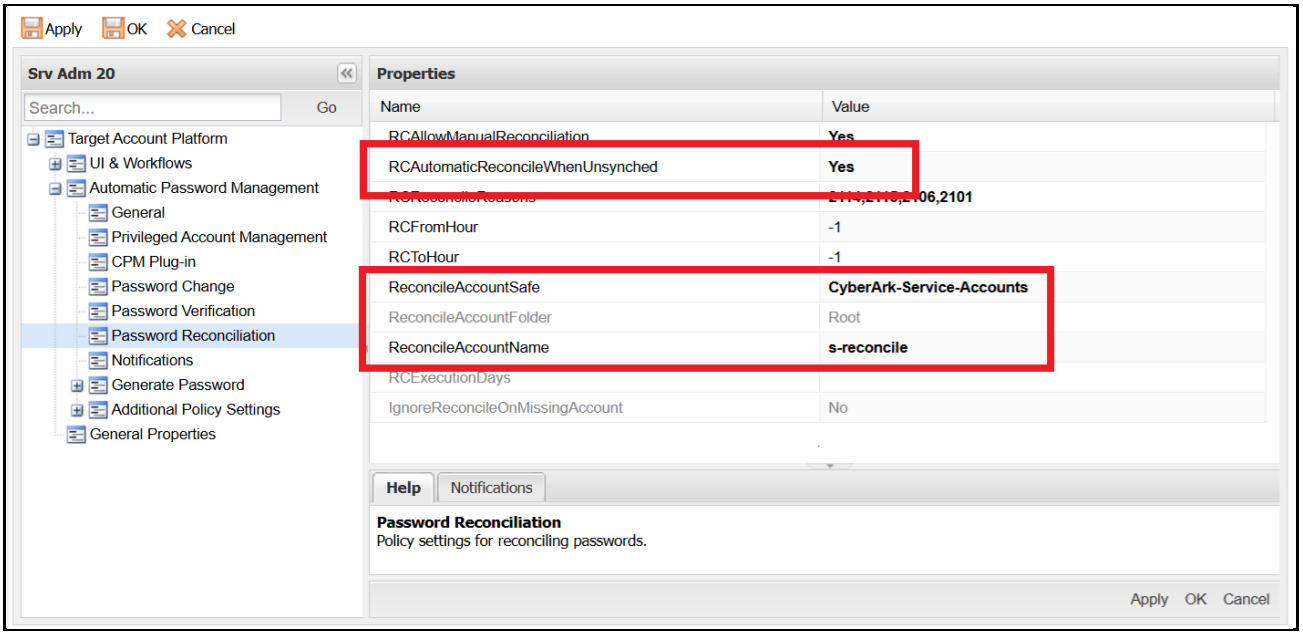

Note:	Think about what values would be appropriate for password length and complexity. Don't forget to add the relevant exception to the **Master Policy** to enable automatic password rotation every 20 days. Also, as we have now duplicated the *Windows Server Local Accounts*, you can deactivate this platform.

13. Press **Apply** and **OK** to close the platform.

Safe Manager Tasks

Once again, we are changing hats and are going to log in as a Safe Manager named *Tom*, who is responsible for the Windows servers team. In this part of the exercise, we will:

- Create a **Safe**
- Add members to the **Safe**
- Add an account

Create a Safe

Now we are going to create a **Safe** for our Windows server administrator accounts. To comply with data protection regulation, we are going to organize our **Safes** so that only US admins can access the passwords for US **Safes**.

1. Log in to the **PVWA** as the AD user *Tom* with the password *Cyberark1.*

2. Go to **POLICIES | Safes** and click **Create Safe**.

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3. Name the Safe ***P-BOS-WIN-S-LA-HR***. Leave the default values for the rest.

4. Add the AD group *Windows Admins* to the **Safe** using the **Connect only** permission preset. This will give the group *Use* and *List* permissions, but not *Retrieve*. We don't want our local administrators to view passwords. Also add the permissions *View Safe members* and *View audit log* in the section **Safe management and monitoring**. Notice that we now have a **Custom** permission set. Click **Create Safe**.

5. Now add the LDAP group **CyberArk Vault Admins**. Remove the permissions: *Use accounts* and *Retrieve accounts*. Add *Account Management* (which will add all the permissions under it). We will need this for a later exercise. You can also give them *View safe members* and *View audit log*. When you are ready, click **Add**.

Adding an Account

Here we will add a local administrator account for your target Windows server: ***acme-123.acme.corp***. Remember, we don't know what the password is, so you could put anything in the password fields (although they must match). We are still using the *Tom* account.

1. Go to the **ACCOUNTS** page, and press **Add Account**. Enter the following and press the **Add** button:

System type	*Windows*
Platform	*Svr Adm 20*
Safe	*P-BOS-WIN-S-LA-HR*
Address	*acme-123.acme.corp*
User Name	*administrator*
Password	*123*
Confirm Password	*123*
Logon To (optional)	*acme-123.acme.corp*

2. Press **Refresh**. Because the password for this account is incorrect, **the password verification will fail**. This will trigger the reconciliation process.

3. Press **Refresh** again and after a short time and you should receive a message saying that the account was successfully reconciled. The first time an account is reconciled it can take a little while, so be patient.

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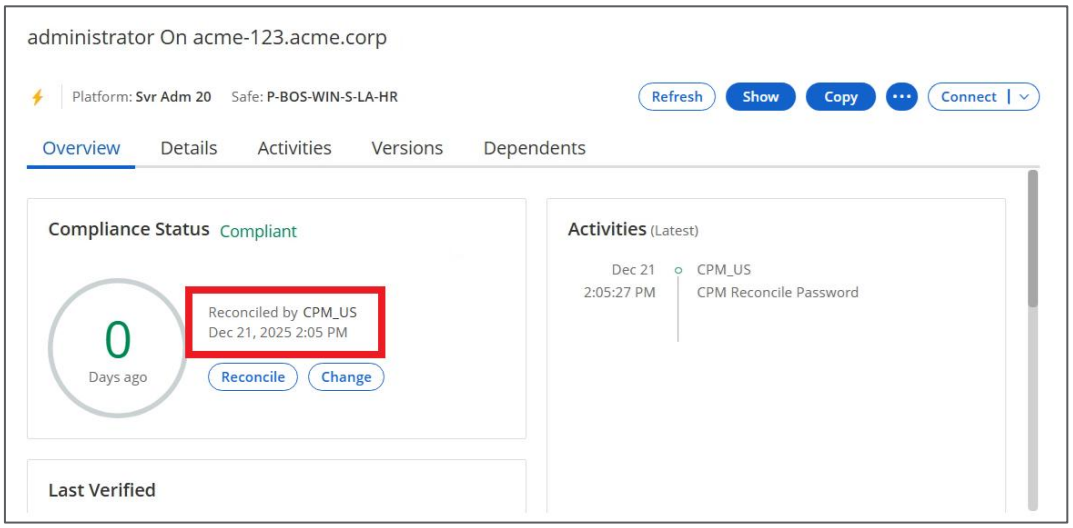
You may receive a message after the failed verification that the **CPM** failed to reconcile the password. You can ignore that message. If you wait a moment, you should see that the password has in fact been reconciled.



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Securing UNIX Accounts with SSH Keys

In this section, we will perform the tasks required to manage a Linux account that connects to its target server with a public-private key-pair.

- **Duplicate a Platform**
- **Verify the existing key-pair**
- **Add the account to PAM**

Note:	The plugin used by the **CPM** to rotate SSH keys uses both SSH and SFTP in order to perform its actions. The target server must therefore be configured to accept connections using both protocols.

Duplicating a Platform – Vault Administrator Task

As a first step, we need to prepare a **Platform** for Linux accounts that use SSH keys, because the log-in process with keys is very different from that with passwords.

1. Login to **PVWA** as *Mike* and go to **ADMINISTRATION | Platform Management**.
2. Locate the Platform ***NIX | Unix via SSH Keys** (make sure that you choose the **Unix via SSH Keys** platform, not the "Unix via SSH" platform).
3. Click on the three dots at the end of the line and select **Duplicate**.
4. Name your platform **Lin Keys 40**, provide a meaningful description, and click **Create**.
5. Select *Lin Keys 40* and select **Edit**.
6. In the **UI & Workflows** section, set **AutoChangeOnAdd** to *Yes*.
7. Under **UI & Workflows | Privileged Session Management**, modify the parameter **SessionRecorderSafe** to the value ***PSMR_LinKeys40.***
8. Go to **Automatic Password Management | General**.
    - Set *ImmediateInterval* to **1**
    - Set *AllowedSafes* to **-LIN-S-**
9. Press **Apply**.

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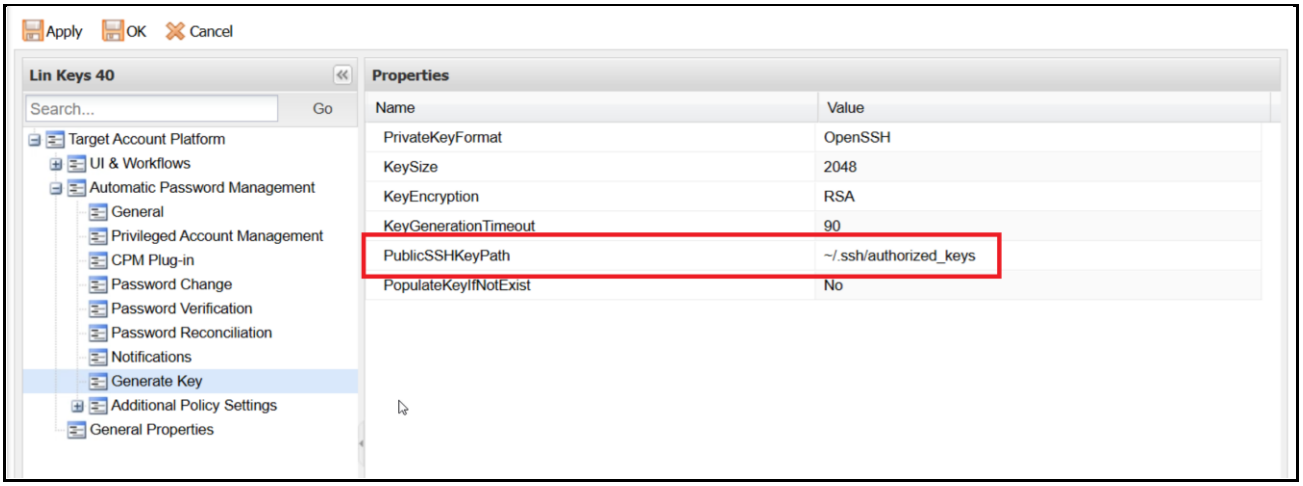
10. Now go to **Automatic Password Management** | **Password Change** and change the value of the parameter *PerformPeriodicChange* from **No** to **Yes**. This will enable the application of the Master Policy rule *Require password change every X days* to accounts managed by this platform.

11. Within the same window, go to **Password Verification** and change *VFPerformPeriodicVerification* from **No** to **Yes**. This will allow the password to be verified by the **CPM** automatically and without user intervention.

12. Click on **Generate Key**. Here we can see how our SSH key-pairs will be generated and where they will be stored on the target system: in the home directory of users in ***~.ssh/authorized_keys***. When you are ready, click on **OK** to save your changes and close the Platform.



Note:	Now that we have duplicated the *Unix via SSH Keys* platform, you can deactivate it. We don't need to add an exception for this platform because it aligns with the Master Policy.

Verify the existing key-pair

For the purposes of this exercise, we have already generated a key-pair for the account *root01b*, but let's verify that those elements are in place.

1. Log into the **PVWA** as *Paul* and launch a connection to lin-451.acme.corp with the account *root01b*.

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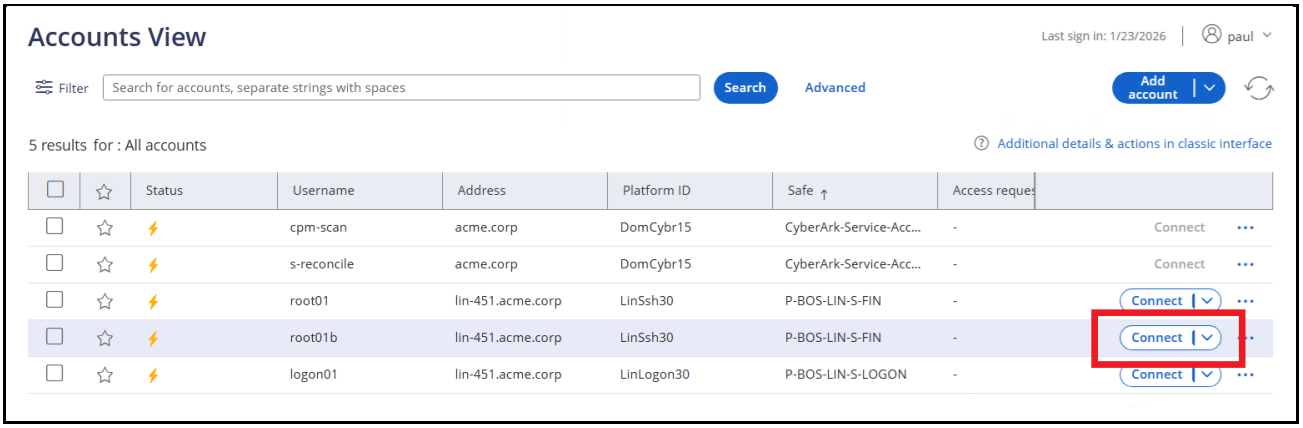
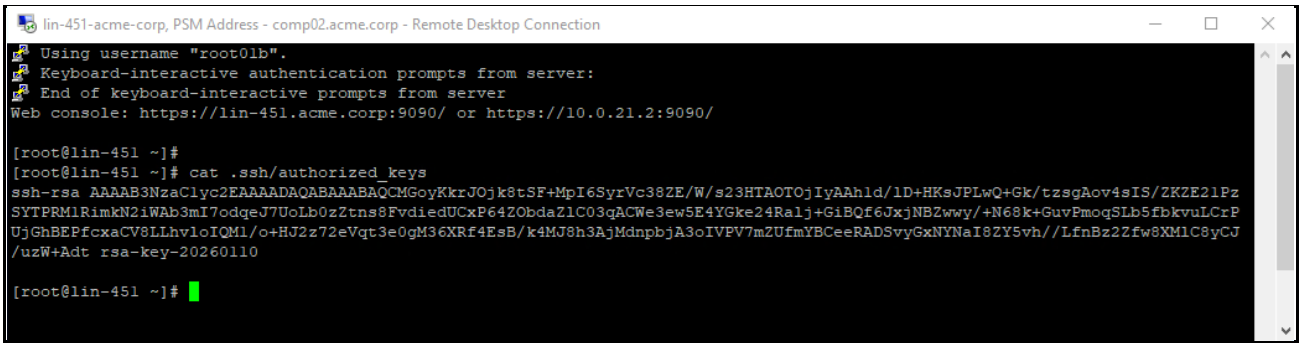


2. Execute the following command:

```
cat .ssh/authorized_keys
```

You should see the public key:



3. The private key can be found in *c:\CYBR_Files*. Open up Windows File Explorer and verify that the file *roo01b.ppk* is there.

Add an Account with an SSH key – Safe Manager Task

Now that we have the **Platform** and we know where the two parts of the key-pair are, we can add a new account. We can store accounts that use SSH keys in the same **Safes** as password accounts.

1. Log in to the **PVWA** as *Paul*.

2. Go to the **ACCOUNTS VIEW** page and click the **Add Account** button.

3. Add an account with the following properties. If you do not see the SSH Key configuration area, you may have duplicated the wrong platform.

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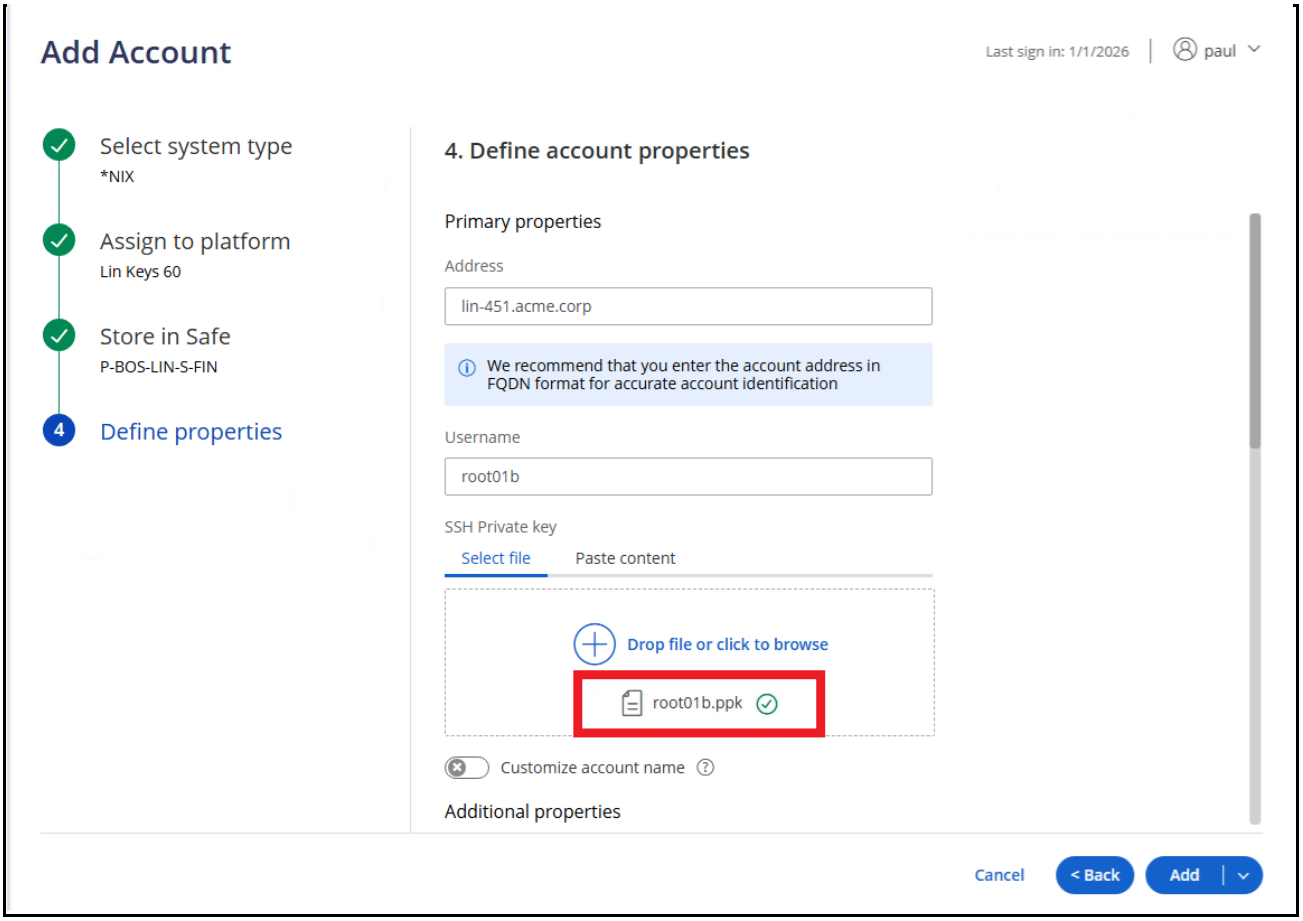
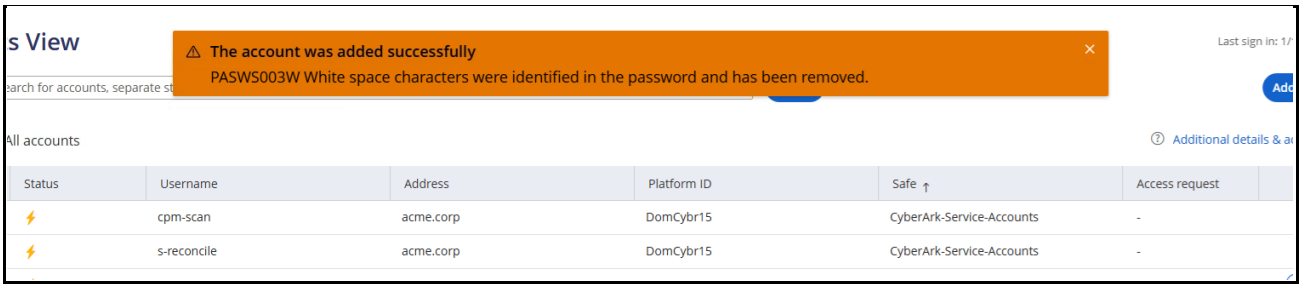
System Type:	**NIX*
Platform Name:	*Lin Keys 40*
Safe Name:	*P-BOS-LIN-S-FIN*
Address:	*lin-451.acme.corp*
Username	*root01b*
Private Key:	Browse to ***C:\CYBR_Files\root01b.ppk***



4. Click **Add**. You will receive a notification that the account has been added:



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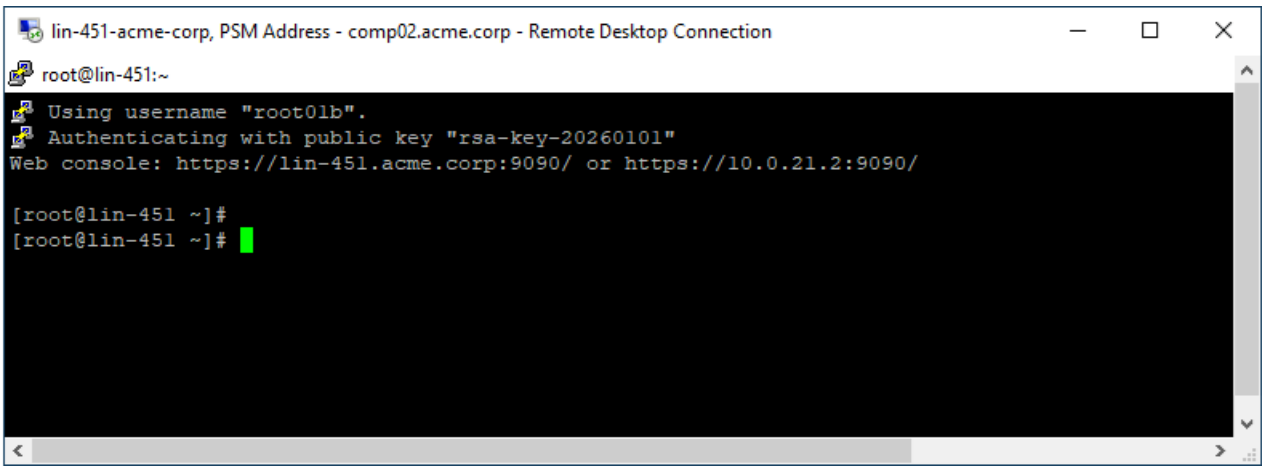
5. And because we set the **Platform** parameter **AutoChangeOnAdd** to *Yes*, the key-pair will be rotated automatically.

Note:	Key rotation is similar to a password change, except that the **CPM** is changing the ***public key*** in the *authorized_keys* file on the target server and updating the account in the **Vault** with the corresponding ***private key***.

6. Once the change completes, test that you can connect to the new account via the **PSM**.



Note:	Just because we have added an account with an SSH key-pair, that does not mean that the password for this account no longer works or is invalid. We have already onboarded the account with its password. The best way to manage this kind of scenario would be to create a dedicated Platform for root account passwords and **not** give the Linux Admins the permission to use accounts in this Safe. They would only have access to the accounts via the keys.

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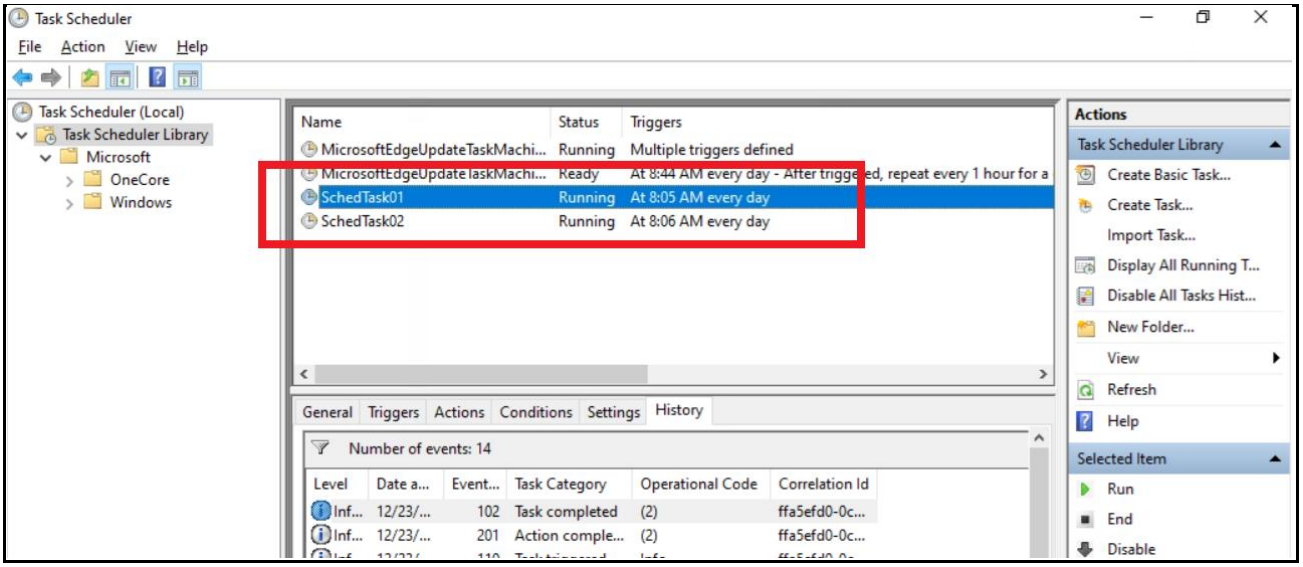
Dependents – Securing Service Accounts / Usages

In this section, we will look at dependent platforms and service account usages. Specifically, we will look at:

- Managing a Scheduled Task Usage
- Managing a Configuration File Usage

Managing a Scheduled Task Usage

There are two scheduled tasks configured on the Uplift virtual machine *acme-123.acme.corp*: **SchedTask01** and **SchedTask02**. Both tasks are configured to send emails to the email group **cva@acme.corp** (the group email for CyberArk Vault Admins) every time they are run. Mike can run these tasks remotely because he is a member of the local **Administrators** group on acme-123 (which is a privilege that should probably be removed). **SchedTask01** is configured to run with the local account *sendmail01,* while **SchedTask02** is configured to run with local account *sendmail02*.



We will deal with the second account in a later exercise. For the moment, let's concentrate on *sendmail01*. We will break down this process into a number of steps:

- Test the scheduled task to make sure it is working
- Add the *sendmail01* account to **CyberArk PAM** and change the password
- Test the scheduled task to make sure it is broken
- Add the scheduled task to the *sendmail01* account and run a password change

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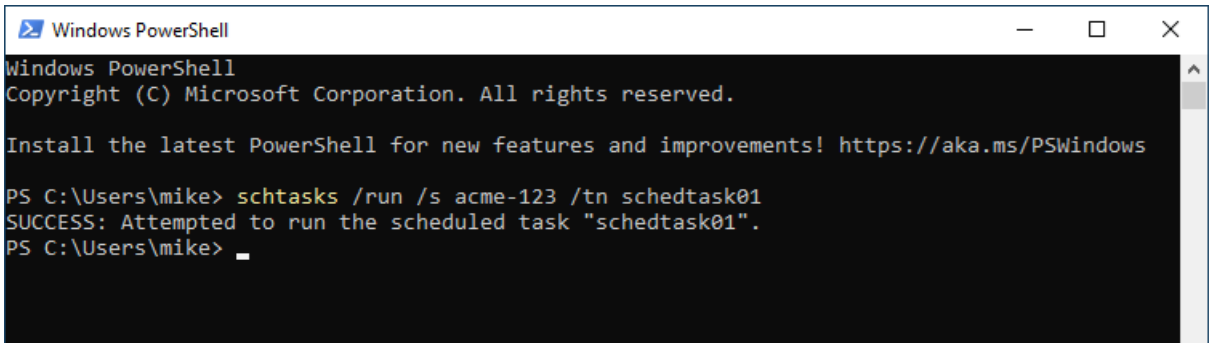
- Re-test SchedTask01 to make sure it is working again

Test the scheduled task

1. Let's start by testing the scheduled task. We need to run this as an administrator, so open a Powershell window as an administrator (there is a shortcut in the task bar). When prompted for credentials, you can use *dom* / *Cyberark1*. Then run the command below:

```
schtasks /run /s acme-123 /tn SchedTask01
```

2. You should receive the following message:



This does not mean that the task has worked. It only means that the command worked. To verify that the task has been executed, you need to check *Mike*'s email. Leave the terminal window open.

3. Launch Chrome and check *Mike's* email account. Launch a new browser tab, click on the short-cut **Acme WebMail** in the toolbar, and log in as *Mike* with the password *Cyberark1*.

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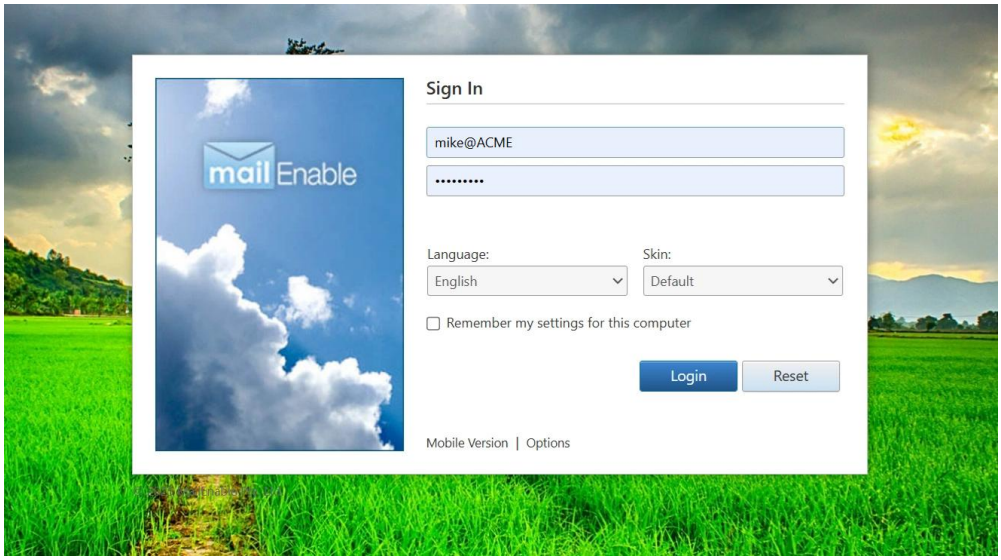
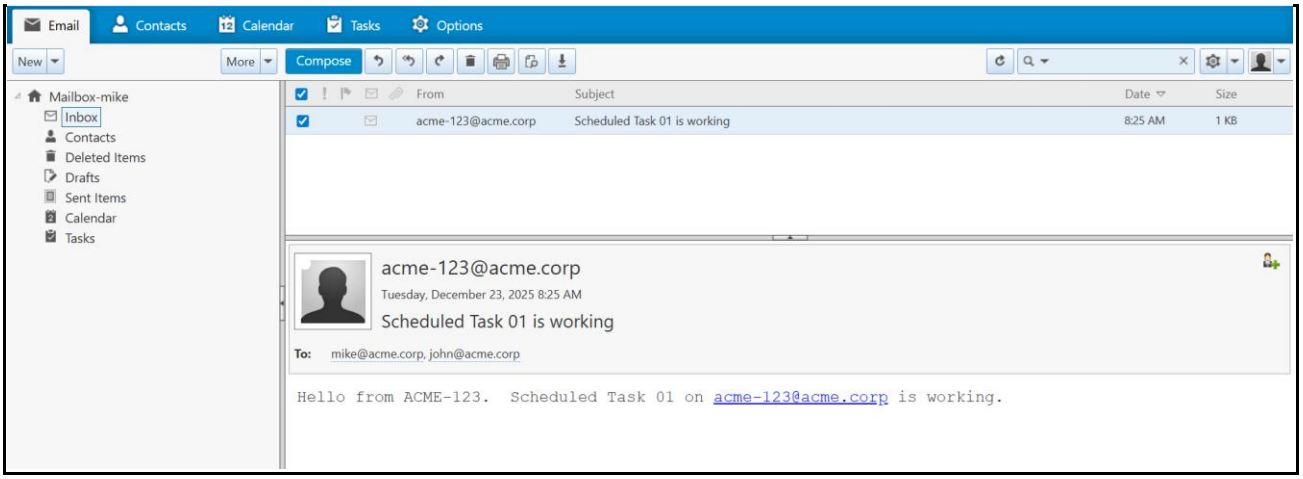


4. You should see that you have a new mail saying the scheduled task is working.



Leave the email tab open.

Add the sendmail01 account

1. Next, we will add the *sendmail01* account. Log in to the **PVWA** as *Tom* (the Safe owner) or *Mike* (who should have account management permissions on the **Safe**) and add the account using the properties below:

System type	*Windows*
Platform	*Svr Adm 20*
Safe	*P-BOS-WIN-S-LA-HR*

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Address	*acme-123.acme.corp*
Username	*sendmail01*
Password	*123*

This will prompt a reconciliation, because we have set the parameter *AutoChangeOnAdd* to ***Yes*** for this Platform and the password is not '123'.

You may see an error saying that reconcile has failed. Just wait a minute or two and then refresh. If for any reasons you get an error with automatic reconciliation, try to reconcile *sendmail01* manually.

Test the scheduled task again

1. Wait until the **CPM** has reconciled the password for *sendmail01* and run the command again.

```
schtasks /run /s acme-123 /tn SchedTask01
```

Because the *sendmail01* account password has been reconciled without accounting for the associated scheduled task, **the scheduled task will not run properly** (even though the return message says "SUCCESS"). You can confirm that the scheduled task **did not** complete properly by checking Mike's email account again. You will see that you do not have a new message referring to a scheduled task.

Leave the terminal window open.

Add the scheduled task to the sendmail01 account

1. Now, logged in to the **PVWA** as *Tom*, click on the *sendmail01* account and then the **Dependents** tab.

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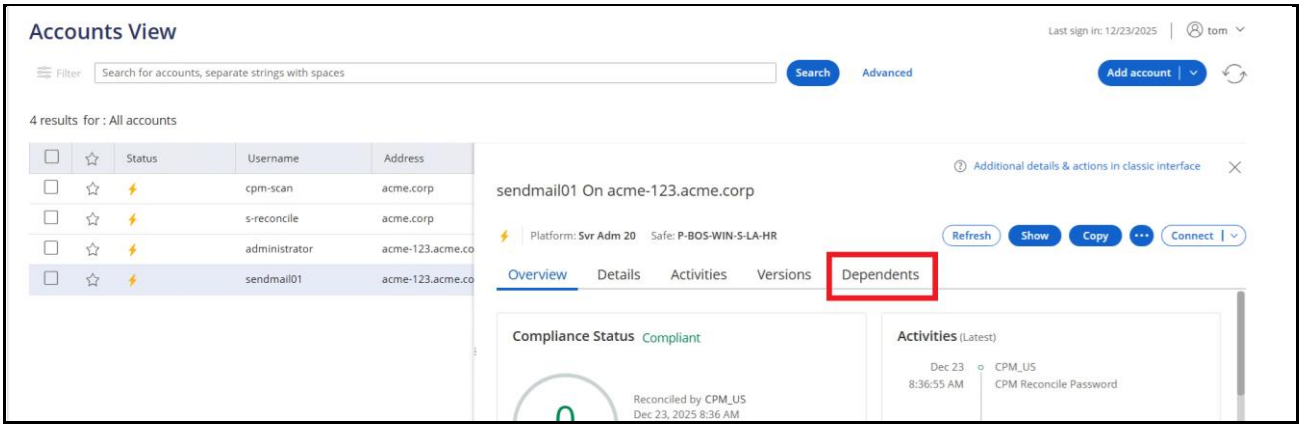
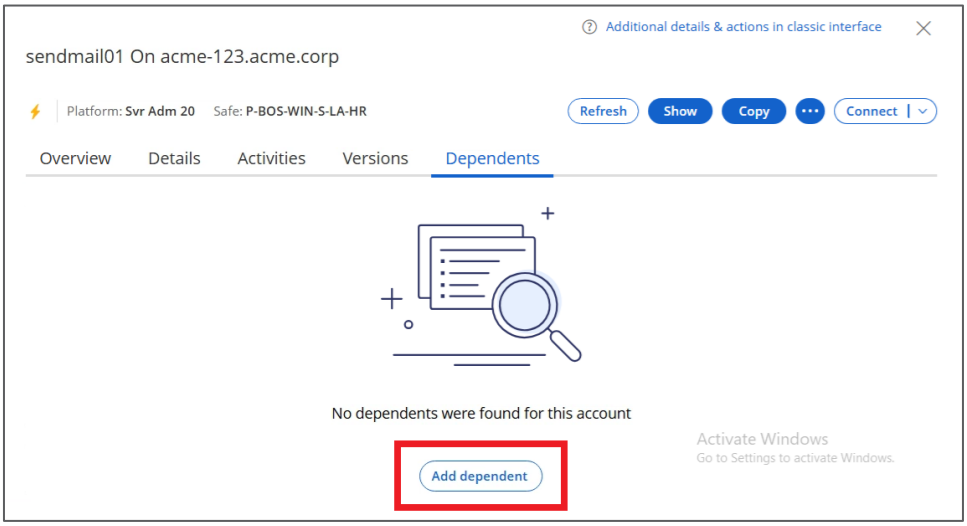


2. Click on the **Add dependent** button to add a new dependent account.



3. First we must choose the type of dependent account we want to add. In our case, we will use **Scheduled Task**, so click on that.

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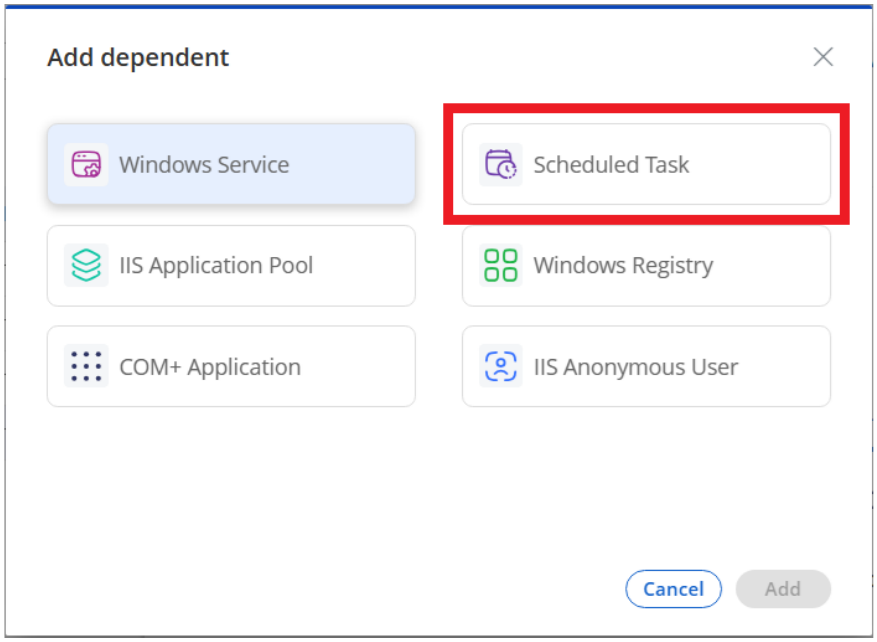
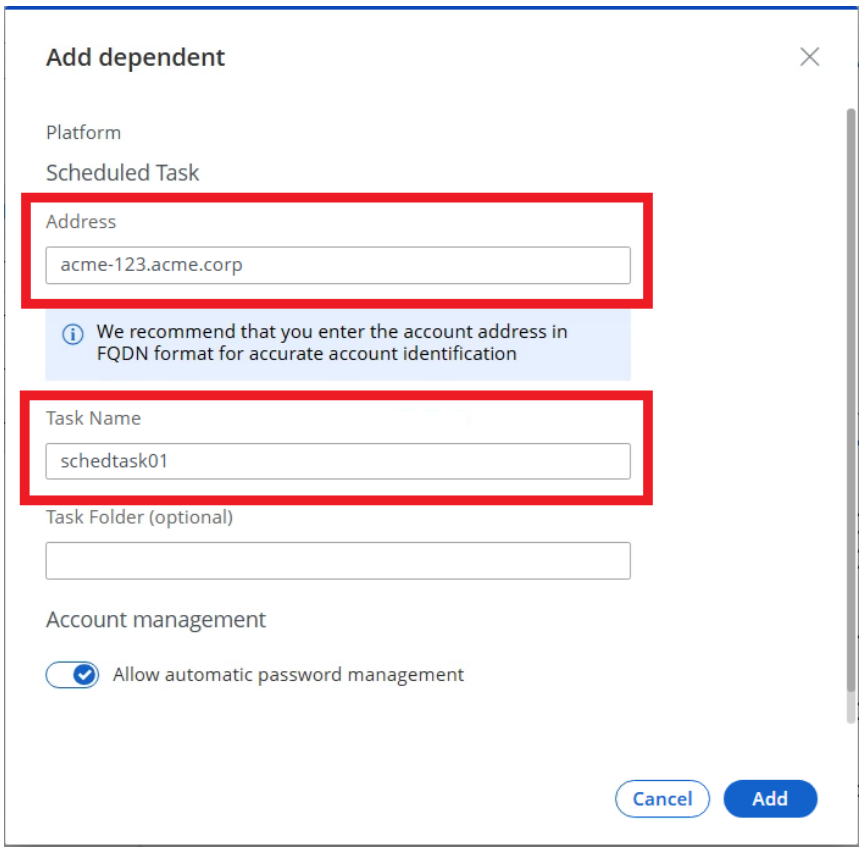


4. Enter **acme-123.acme.corp** in the *Address* field and **SchedTask01** in the *Task Name* field. We can ignore the other fields here. Click **Add**.



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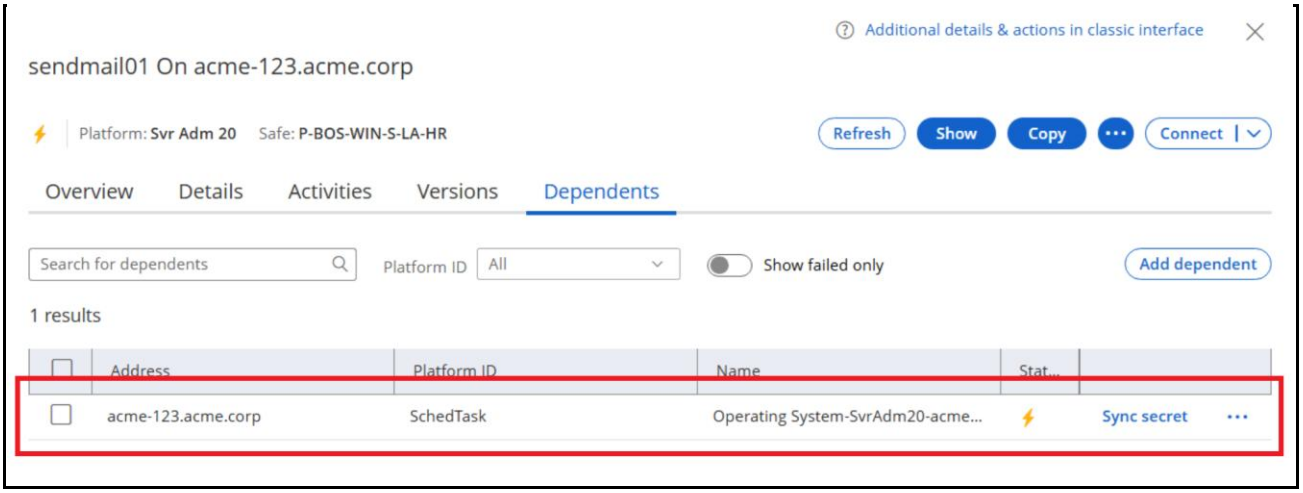
5. You will be able to see the new dependent account associated with the *sendmail01* account.

Associate a logon account with sendmail01

In previous versions of Windows, we were able to rotate the password for a dependent account directly with the account itself. However, it seems now that in order to change the password in the scheduled task, you must use another account to log on to the target system, so we will associate a logon account with *sendmail01*.

1. Select the *sendmail* account, click on the **Dependents** tab, and then click on the scheduled task.



2. In the new dialog, locate he **Linked Accounts** section and then click on the ellipsis to associate a **CPM Logon Account** and then click on **Link**.

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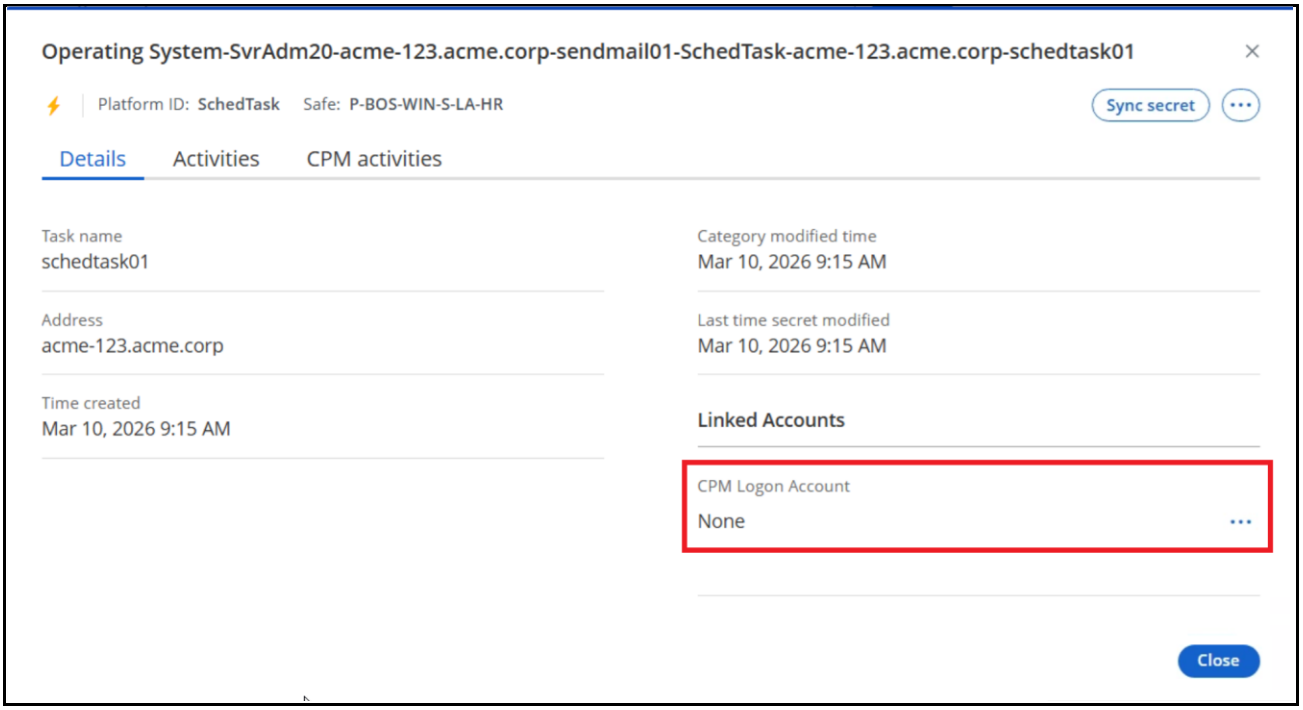
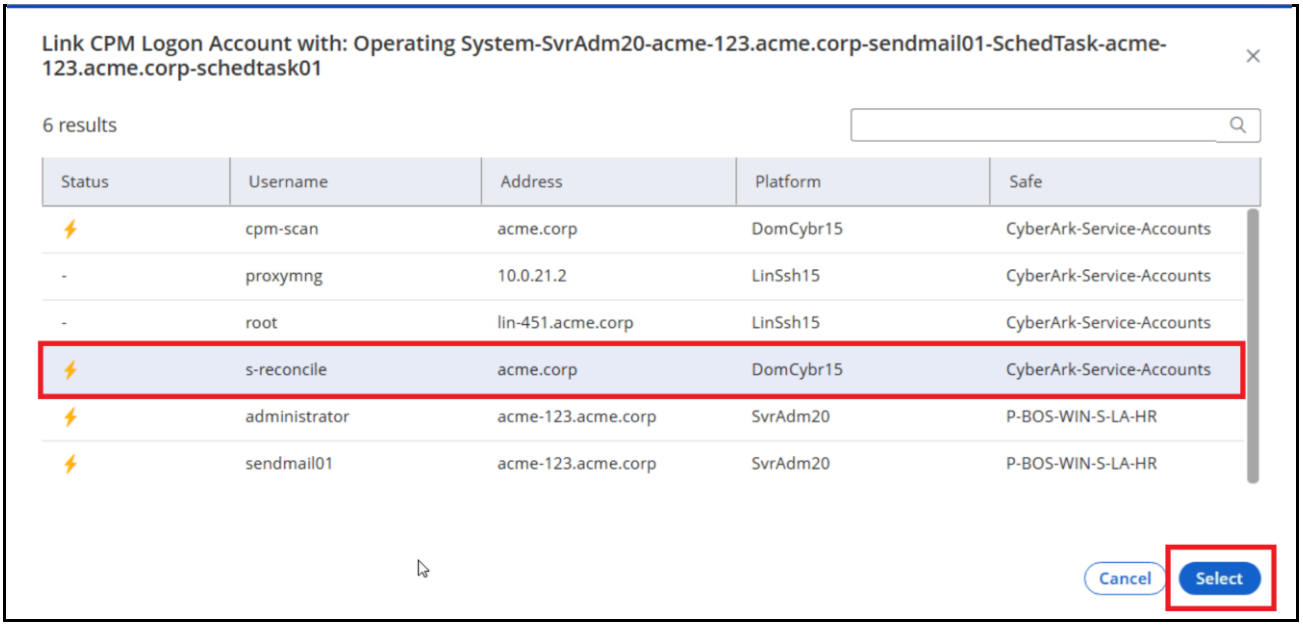


3. Locate the account you want to use as the logon account. We can use the reconcile account for this because it has the necessary permissions. Select it and then click the **Select** button. You can close the previous dialog box.



4. Now click on the **Sync secret** button. This will update the password in the scheduled task on the target server with the password that is currently stored in the **Vault**.

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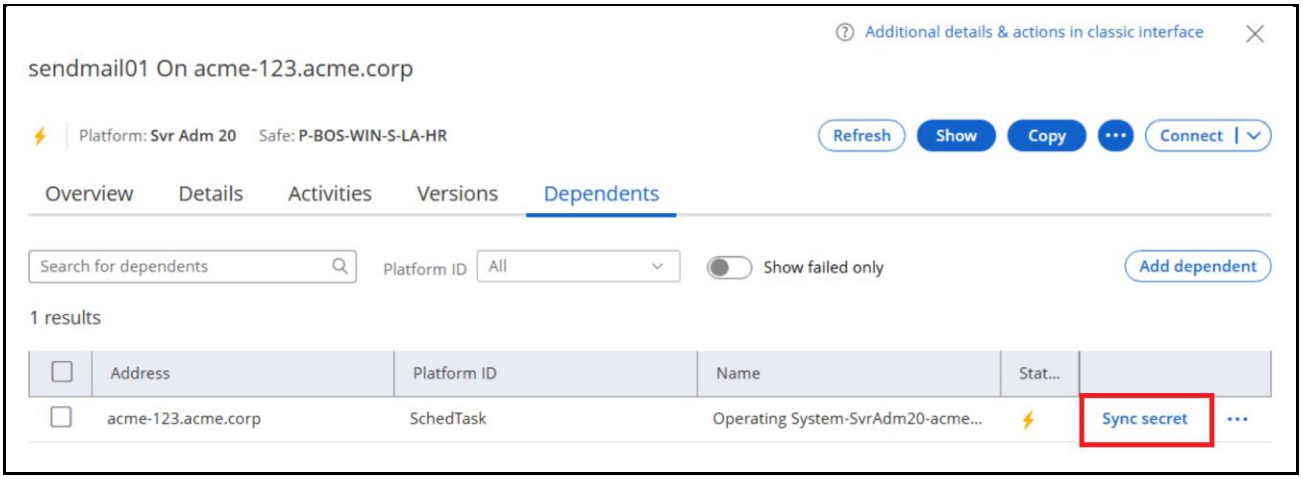
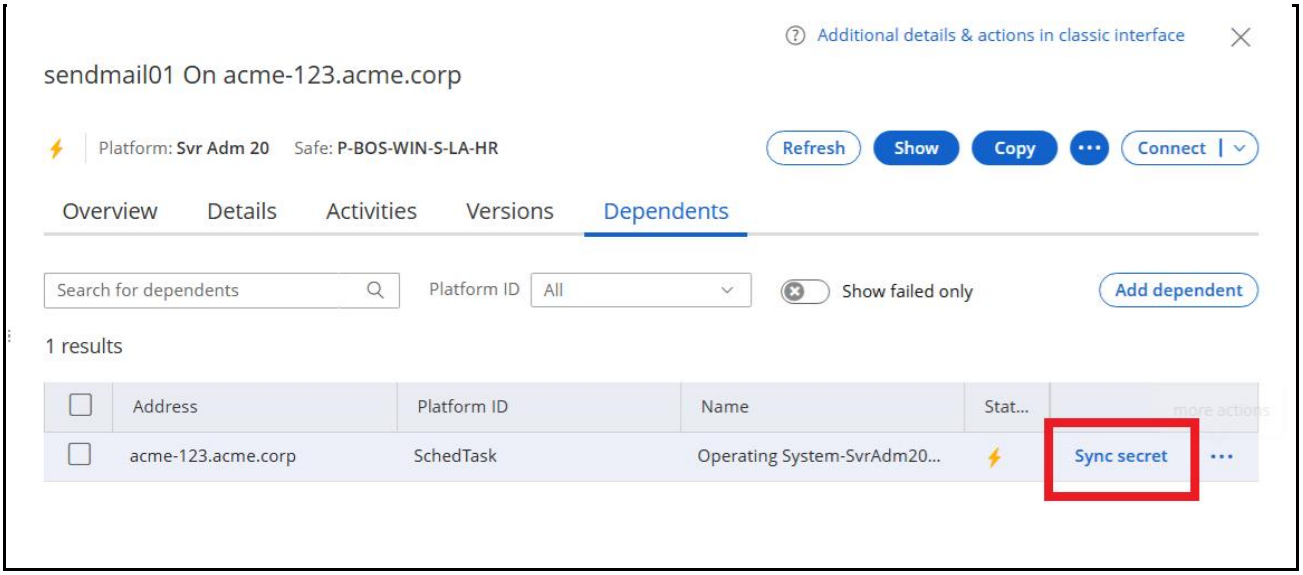


5. Now go back to the main **Accounts** view, select the *sendmail01* account and click on the **Sync secret** button and the CPM will synchronize the password in the scheduled task with the password for the primary account.



Note:	The scheduled task is associated with a different platform than the *sendmail01* account. If you click to change the password for the *sendmail01* account, the flag will be set for the scheduled task to be changed. The entire process could take up to 10 minutes to complete.

Re-test schedtask

1. Re-run the scheduled task from the terminal. Now check *Mike*'s email.

2. This time you should receive an email stating that the scheduled task is working.

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Note:	It is highly recommended to use the accounts discovery feature to detect, provision, and manage all Windows service accounts automatically. We will use the **Accounts Discovery** capability later to discover and onboard **SchedTask02** which is associated with *sendmail02*.

Managing a Configuration File Usage

In this exercise you will be configuring an existing account that has another kind of *dependency* (or *usage*). The idea is that we have an application (just imagine) running on the server **lin-451** that needs to connect to the Oracle database running on **acme-123** with the account *dba01*. In order to do that, the application needs the password for *dba01*. But if we change the password in Oracle and in the **Vault** without also updating the password on the application server, it will break the application's connection to Oracle.

There is a file named *app01.ini* on **lin-451.acme.corp** in the */var/opt/app* directory. In the section **[Server]** there is the **Password** parameter, the value of which is currently *Cyberark1*. We need to update this parameter whenever we change the password for the account *dba01*.

```
…
[Server]
Hostname=acme-123.acme.corp
Username=dba01
Password=Cyberark1

[Database]
Db=xepdb1
Port=1521
```

Note:	As with the SSH keys platform, the plugin used by the CPM to update passwords in INI files uses both SSH and SFTP in order to perform its actions. The target server must therefore be configured to accept connections using both protocols.

Configure Usages on the Oracle platform

We will begin by configuring the **Ora Dba 30** Platform so that the **CPM** will know it needs to search for usages when changing passwords for accounts managed by this Platform.

1. Login to the **PVWA** as *Mike* and navigate to the **ADMINISTRATION** tab and click **Platform Management**.

2. Select **Ora Dba 30** and press **Edit**.

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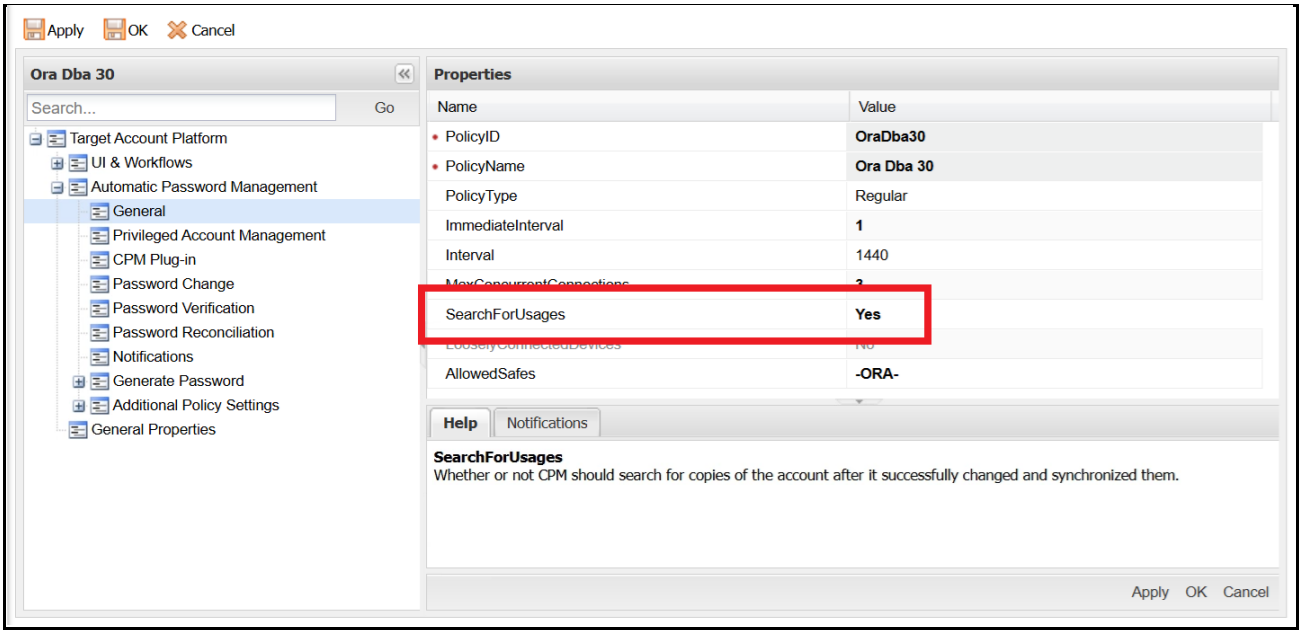
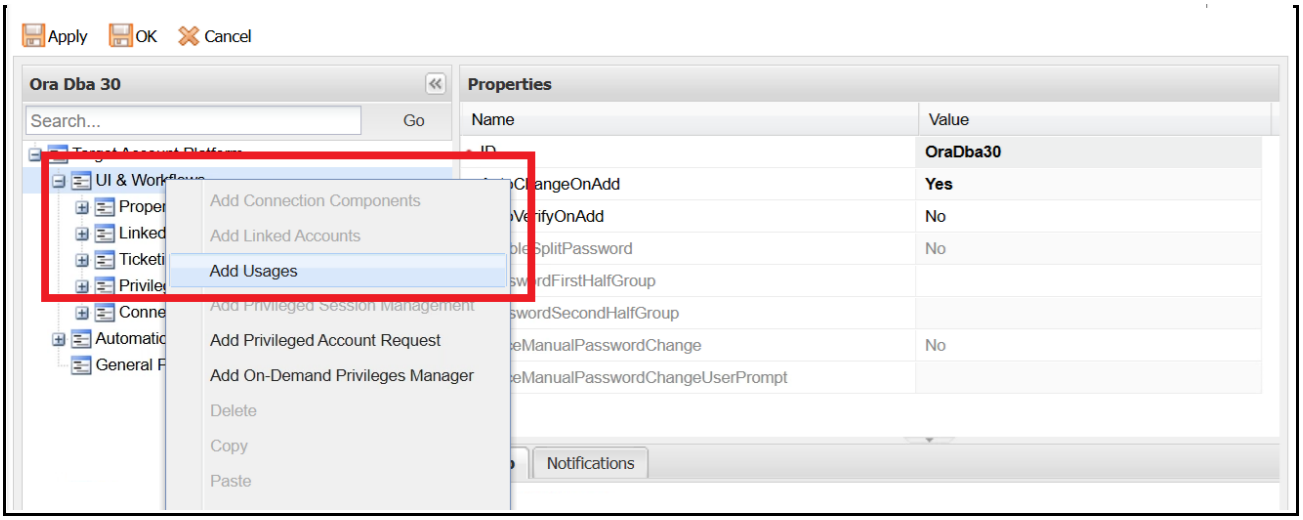
3. Go to **Automatic Password Management | General**, set *SearchForUsages* to *Yes* and press **Apply**.

Note:	The *SearchForUsages* parameter is set to **Yes** by default for Windows Platforms, but needs to be activated for others. This is another parameter that ought to be deactivated if it is not being used.



4. Now right-click on the node **UI & Workflows** and choose **Add Usages**.



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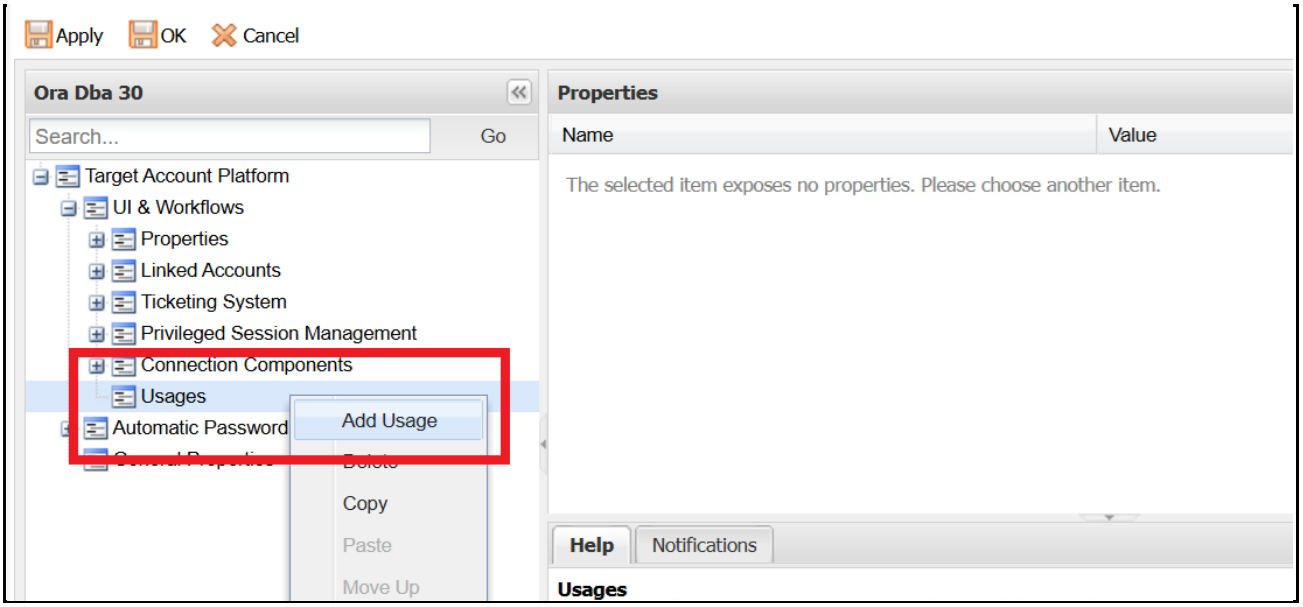
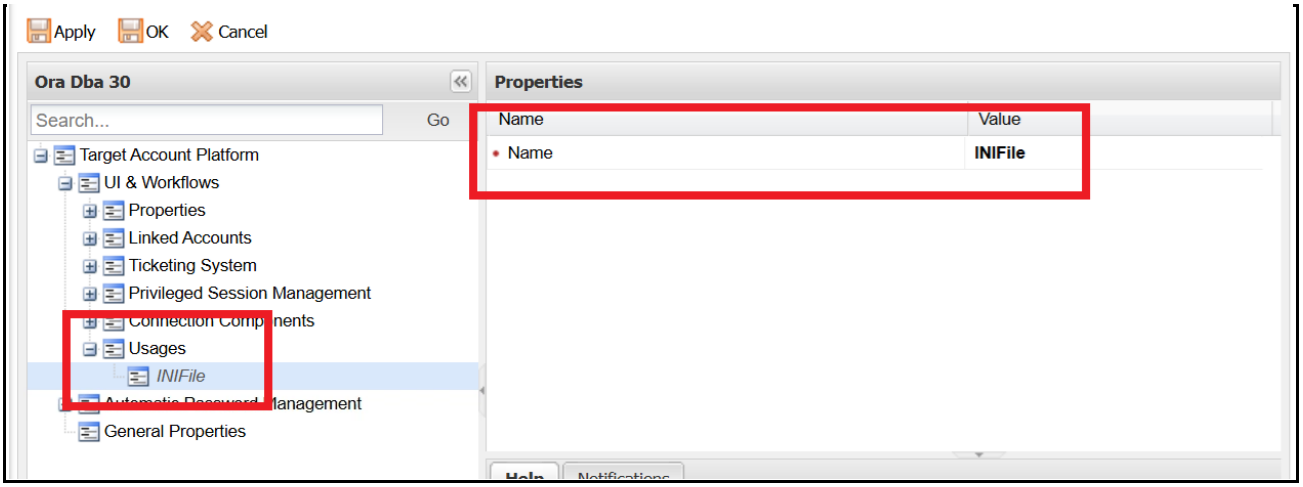
5. After selecting *Add Usages*, you will have a new 'Usages' entry at the end of the *UI & Workflows* section. Now we need to add a specific type of usage. Right click **Usages** and select **Add Usage**.



6. Enter *INIFile* as the **Value**. This value is a **case-sensitive** parameter and must be entered **exactly** as shown. This parameter is actually a pointer to the **ID** parameter of the dependent Platform.



7. Press and **OK** to save your changes and close the **Platform**.

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Create a Logon account

The account *dba01* is an Oracle database account and is therefore unable to log in to **lin-451** and modify a file. Consequently, we will need to create a logon account that will be used by the **CPM** to connect to the Linux target server and change the credentials stored in the *app01.ini* configuration file.

1. On your **Comp01** server, log in to the **PVWA** as *Paul* (this can also be performed as *Mike*).

2. In **ACCOUNTS**, press **Add Account** and enter the following:

System Type:	**NIX*
Platform Name:	*Lin Logon 30*
Store in Safe:	*P-BOS-LIN-S-LOGON*
Address:	*lin-451.acme.corp*
Username:	*app-account01*
Password:	*Cyberark1*
Confirm Password:	*Cyberark1*

3. Click on the newly created account and click on **Verify** or **Change** to confirm that the account has been created correctly.

Add the dependent account to the target account

The process here is similar to the one we used above for the scheduled task.

1. Log in to the **PVWA** as *Mike*, select *dba01*, click on the **Dependents** tab, and then click **Add dependent**.

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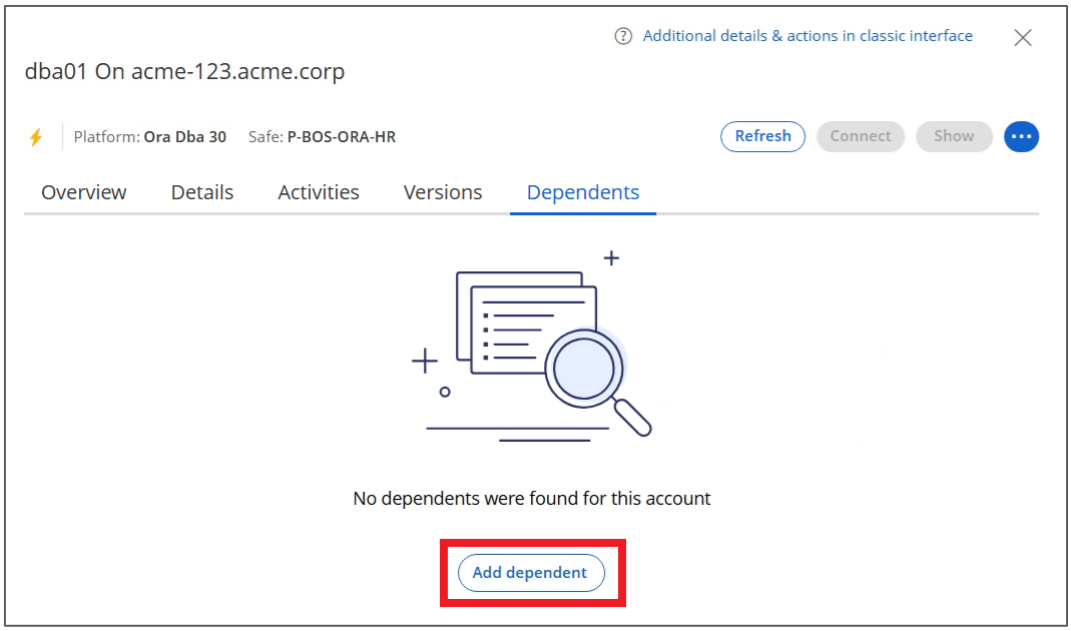
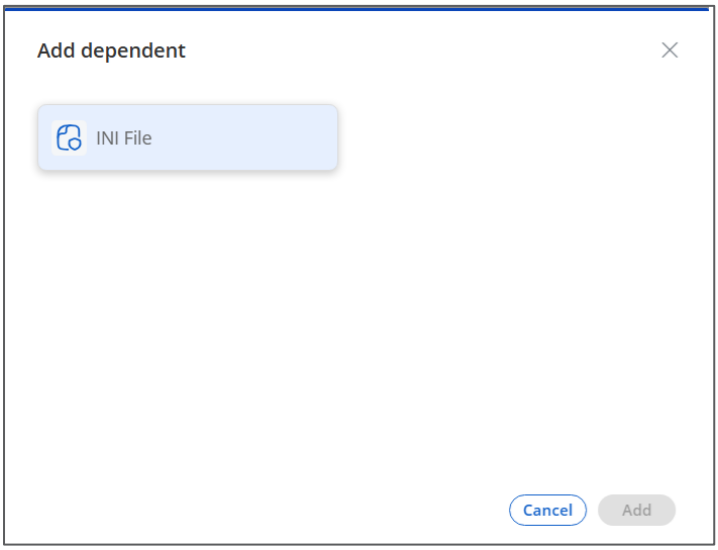


2. Notice that in this dialog, we are only seeing a single type of dependency: the **INI File**. That is because we only configure this one usage in the **Ora Dba 30** Platform. Click on **INI File** to configure the usage.



3. Here we need to provide all of the information that will enable the **CPM** to locate the ini file containing the password for the *dba01* account. Enter the following information:

Address	*lin-451.acme.corp*

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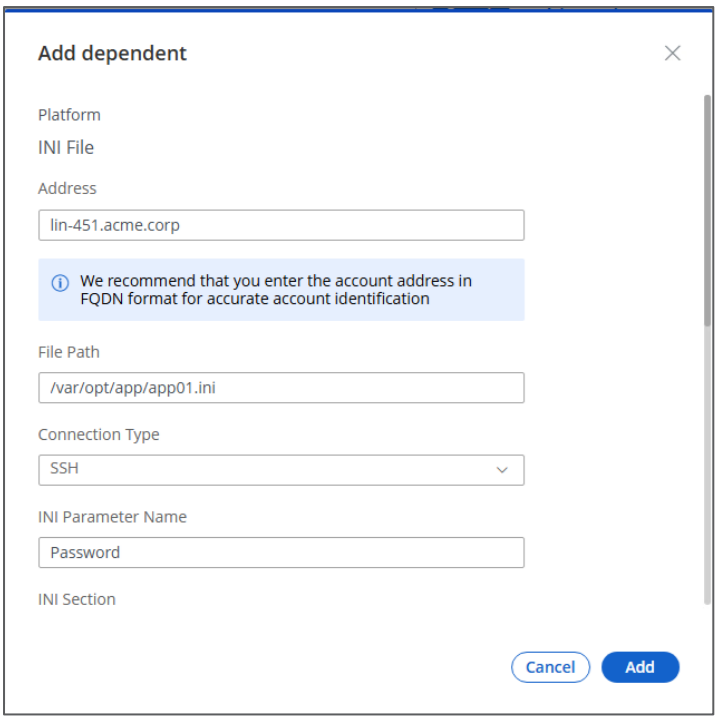
File path	*/var/opt/app/app01.ini*
Connection type	*SSH*
INI Parameter name	*Password*
INI Section	*Server*
Backup Password File	*No*



4. Once again, we will need to associate a logon account. Click on the scheduled task.

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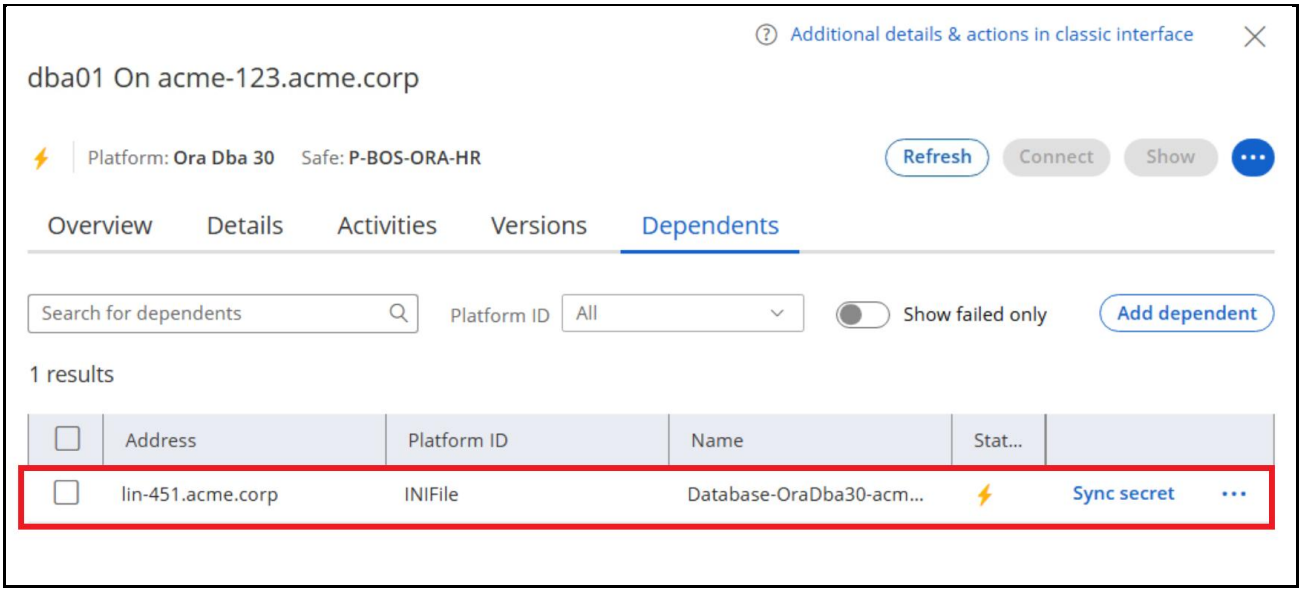
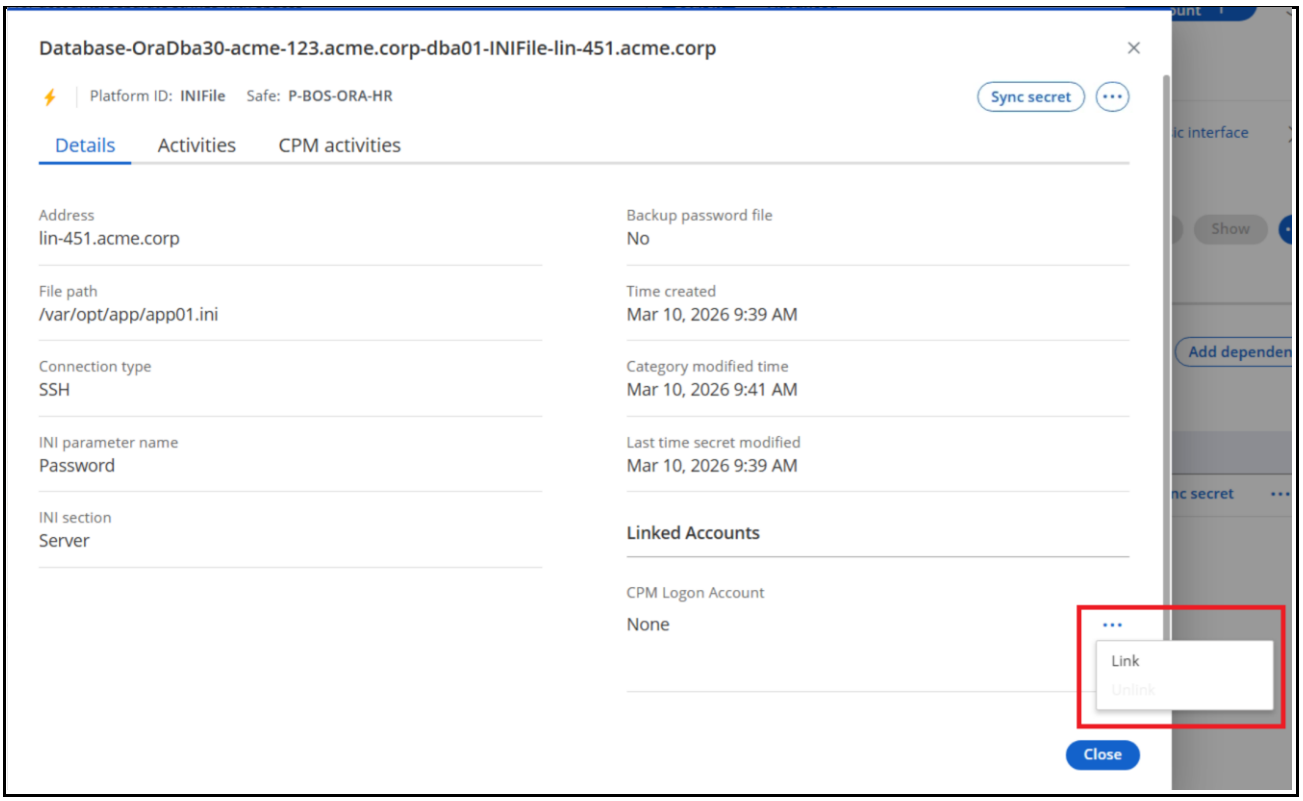


5. Under **Linked Accounts**, click on the ellipsis and then on **Link**.



6. Locate the account we created a moment ago – *app-account01*. Note that you can use the search box at the top to find the account. Select the account and then click **Select**. Then you can close the previous dialog box.

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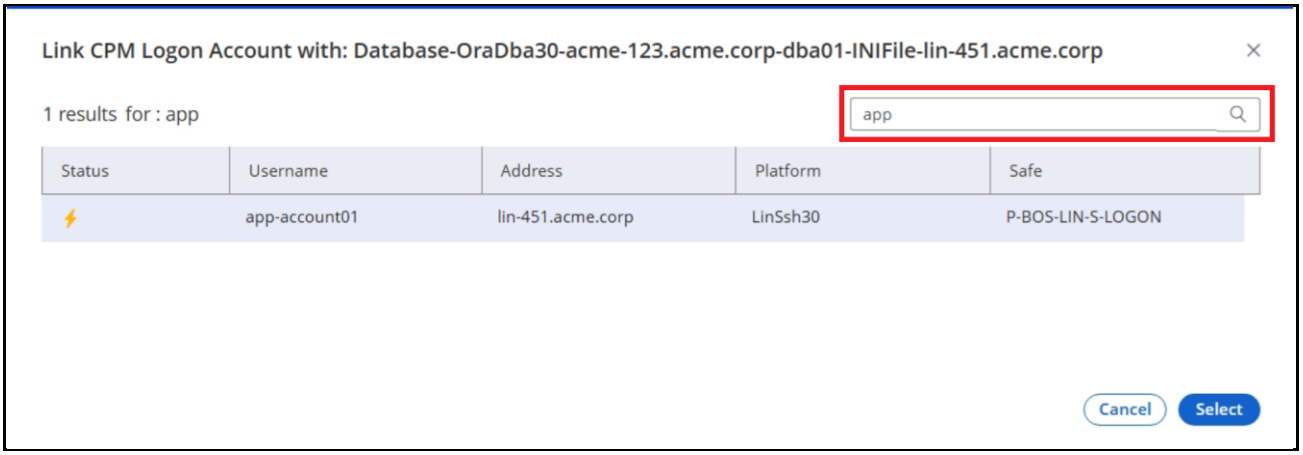
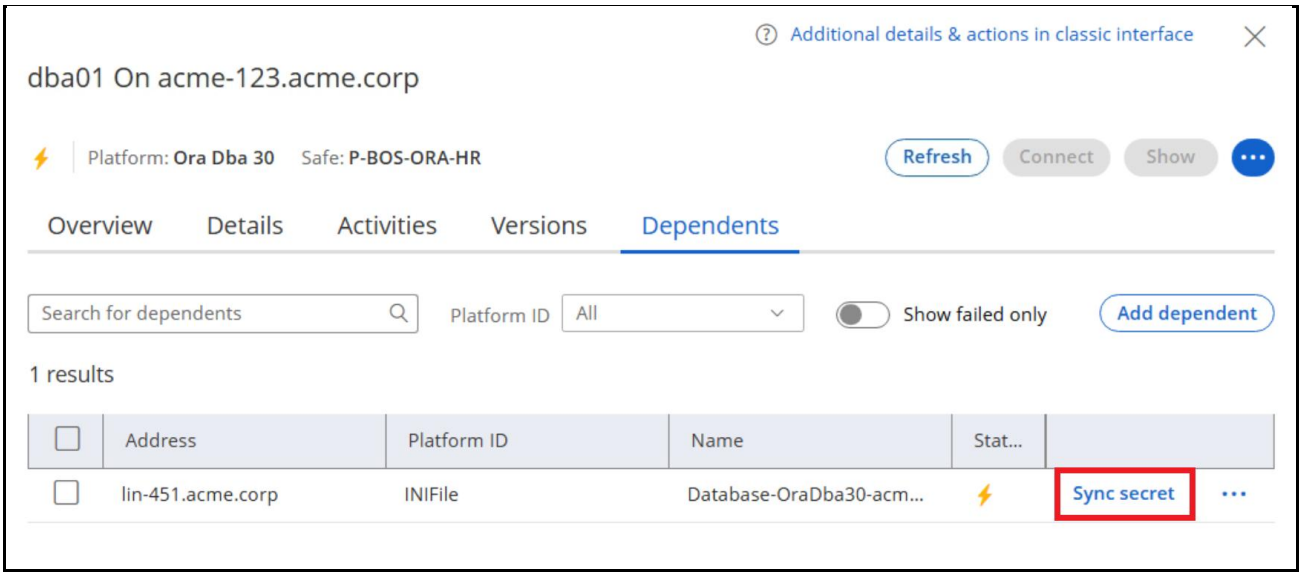


7. Then click on the **Sync secret** button to update the password in the ***ini*** file.



Reminder: The reason we are associating a logon account with the **Usage** is because the target account – *dba01* – is an Oracle account and so does not have permissions to login to the Linux server and change the password in the configuration file (*app01.ini*). The **CPM** will use the logon account (*app-account01*) to connect to the target Linux machine and change the password in the configuration file.

Verify the application password has been updated

Perform the following steps to verify the password *dba01* in the **Vault** matches the password in the *app01.ini* file.

1. First, log in to the **PVWA** as *Robert* and locate the *dba01* account. Select **Show** to see the password of *dba01*. Copy the password to Notepad.

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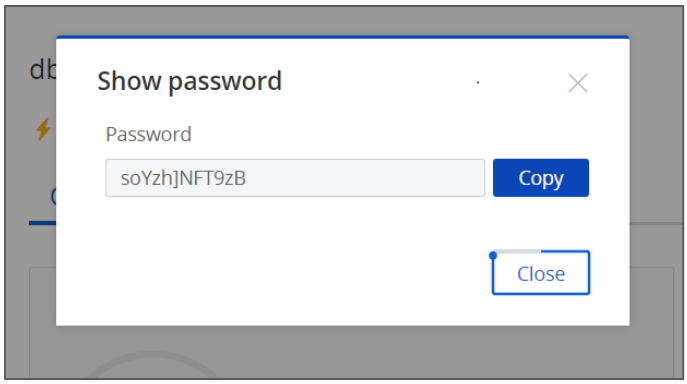
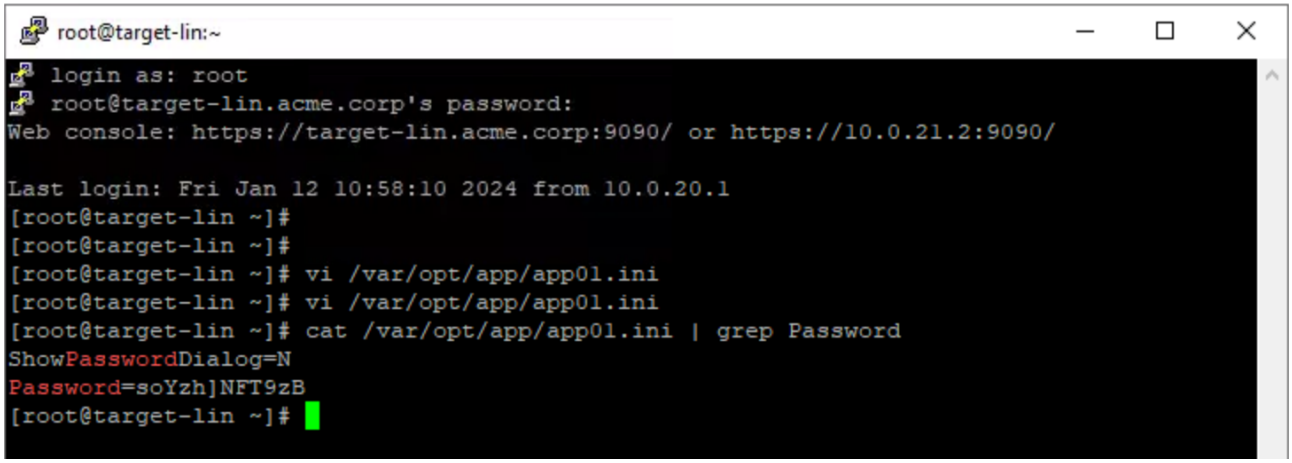


2. Now, log in to the **PVWA** as *Paul* and connect to **lin-451.acme.corp** with the *app-account01* account.

3. Enter the following:

```
cat /var/opt/app/app01.ini | grep Password
```

4. If everything was configured properly, you should see that the password in the file matches the new *dba01* password in the **Vault**.



5. You can close the Putty session by typing **exit** and hitting **Enter**.

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Privileged Access Workflows

In this section, we will configure the **Master Policy** for three **Privileged Access Workflows**:

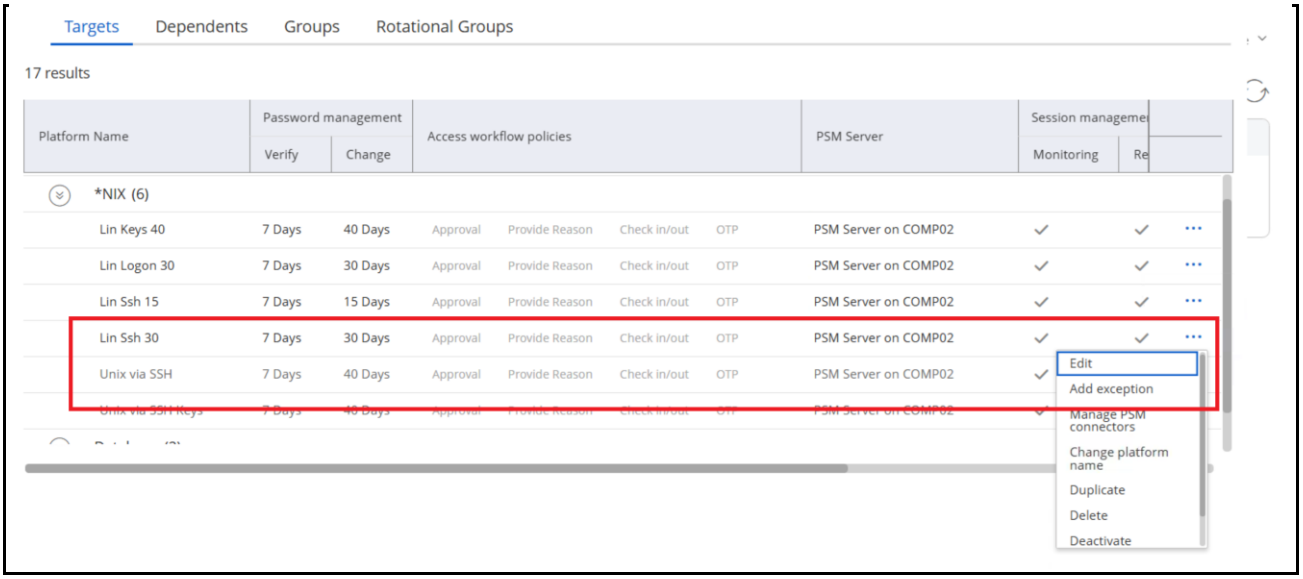
- Reason for access
- Dual control
- Exclusive passwords

Require users to specify reason for access

In this section we will test the *Require users to specify reason for access* workflow, as well as configure predefined reasons.

Activating the Policy

1. Log into the **PVWA** as *Mike* and go to **Administration | Platform Management**. Go to the Platform *Lin Ssh 30*, click on the ellipsis, and click **Add exception**.



2. Set *Require users to specify reason for access* to **Active**. Set *Allow users to specify reason for access* to **Inactive**. Click on **Save**.

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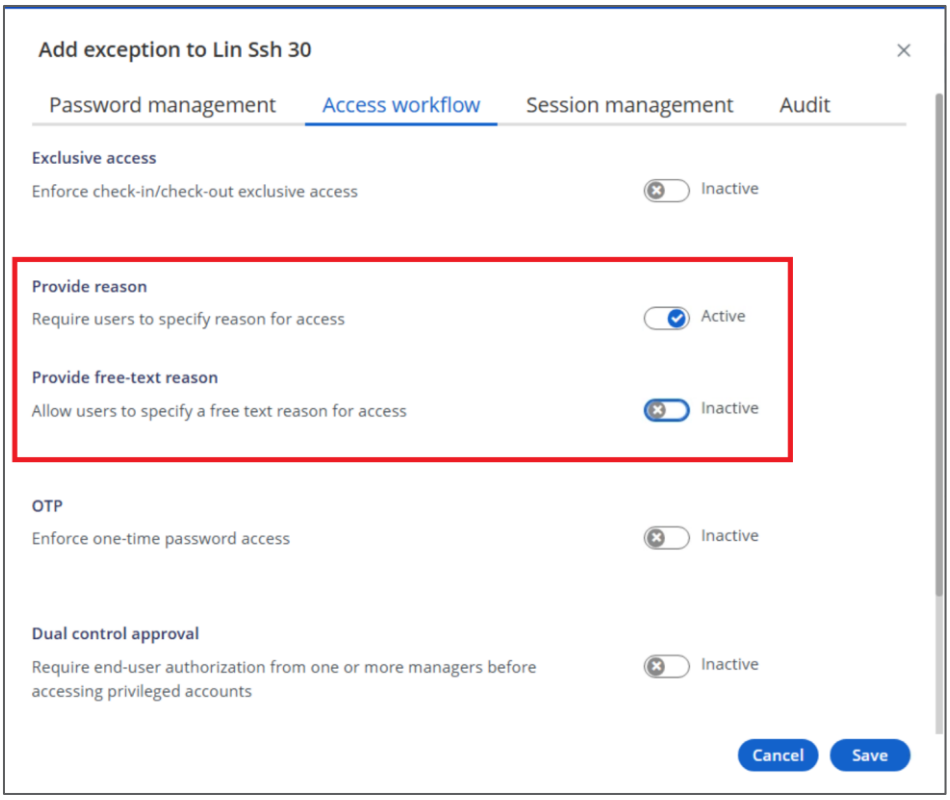

Add Predefined Reasons for Access

In this section, we will create a drop-down list of reasons that users will be required to select from when connecting to accounts managed by the Platform **Lin Ssh 30**.

1. Navigate to the **ADMINISTRATION** tab and click **Platform Management**.
2. Go back to the *Lin Ssh 30* platform, click on the ellipsis, and click on **Edit**.
3. Right-click on *UI & Workflows* and select *Add Privileged Account Request*.

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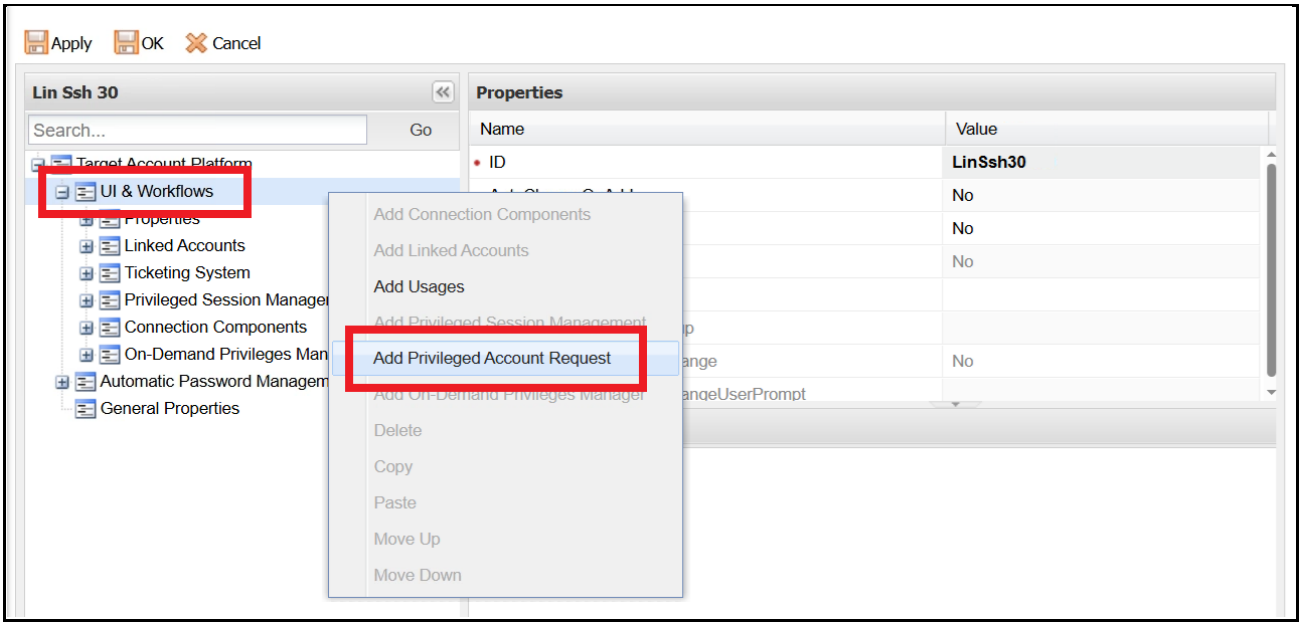
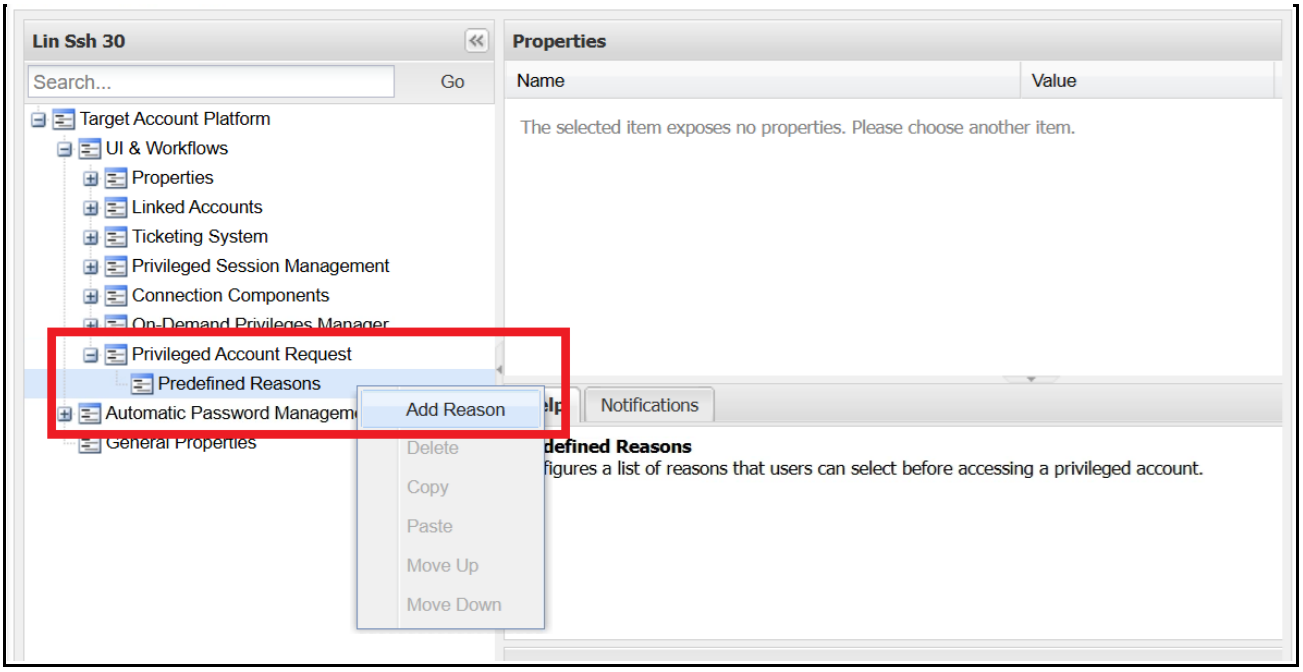


4. Expand *Privileged Account Request* and then right-click on *Predefined Reasons*. Select *Add Reason* to add predefined reasons.



5. Add the following predefined reasons in the **Value** parameter (you may also add your own if you wish).

   - **Approved change request**
   - **Incident**

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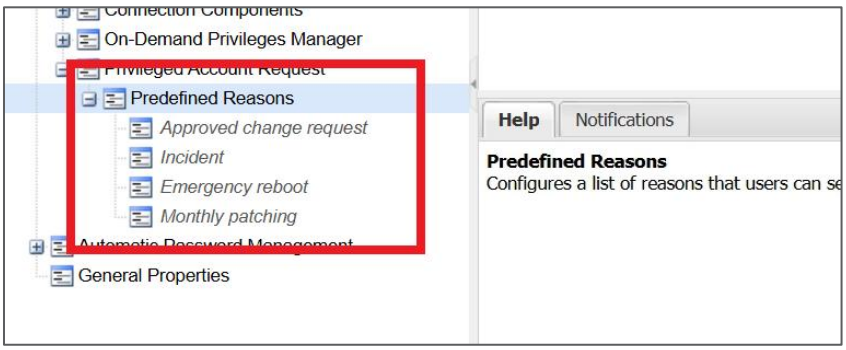
- **Emergency reboot**
- **Monthly patching**

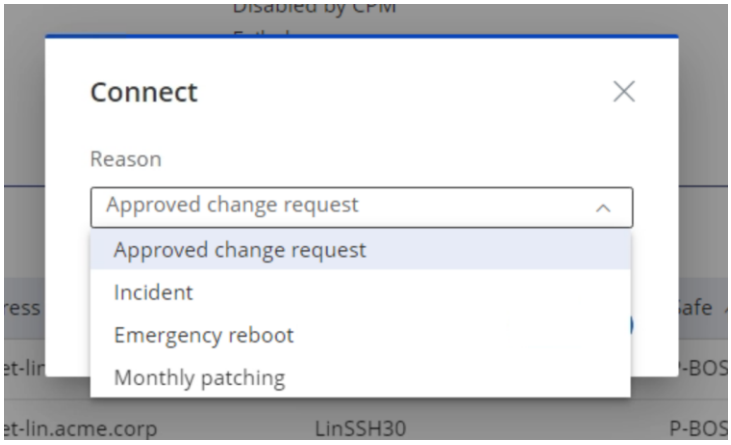


Hint:	A faster way to do this is to just add all four empty reasons at once and then go back and edit each one.

6. When you finish, click on **OK** to save and exit.

Testing Predefined Reasons for Access

1. Now, log into the **PVWA** as *Carlos* and select the *root01* account. Click on **Connect**. Select one of the predefined reasons, for example, *Incident*. Then click on **Connect** again to download the RDP file.



2. Click on the RDP file to connect to the target machine. We don't need to do anything, so you can disconnect from the target machine.

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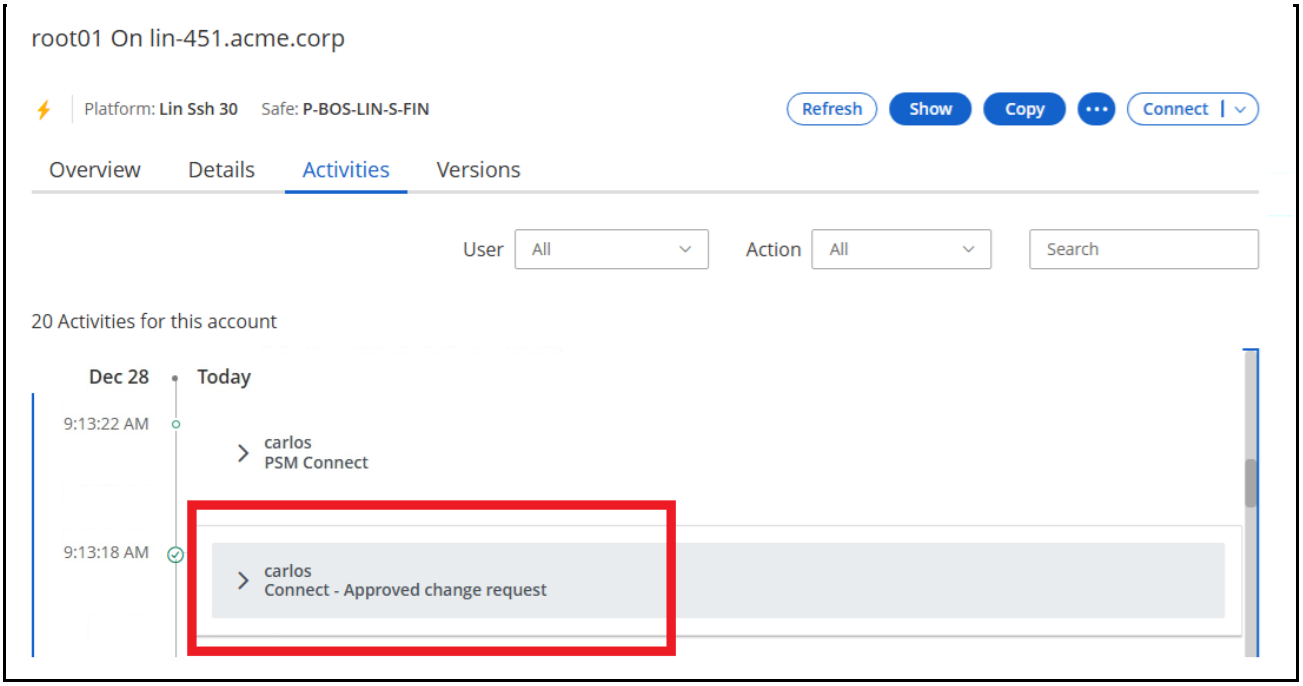
Carlos does not have audit rights on the **P-BOS-LIN-S-FIN** Safe, but if you log in as *Paul* and view the **Activities** tab, you can see that *Carlos's* actions have been recorded along with his selected reason.



Require dual control access approval

Dual control – requiring a manager to validate a request for access to certain accounts – is a 2-step process:

1. You must activate the policy *Require dual control password access approval*, either globally or by exception for a certain Platform (which is the usual case and what we will do).
2. Add an approver to a Safe, either a group or a user, with **at least** the *List Accounts* and *Authorize account requests* permissions.

This minimum configuration would give the manager/approver the right to validate the requests, but not the right to use the passwords to connect to target systems (they only have *List*, not *Use* or *Retrieve*).

Activating the Policy

This process is very similar to the previous one.

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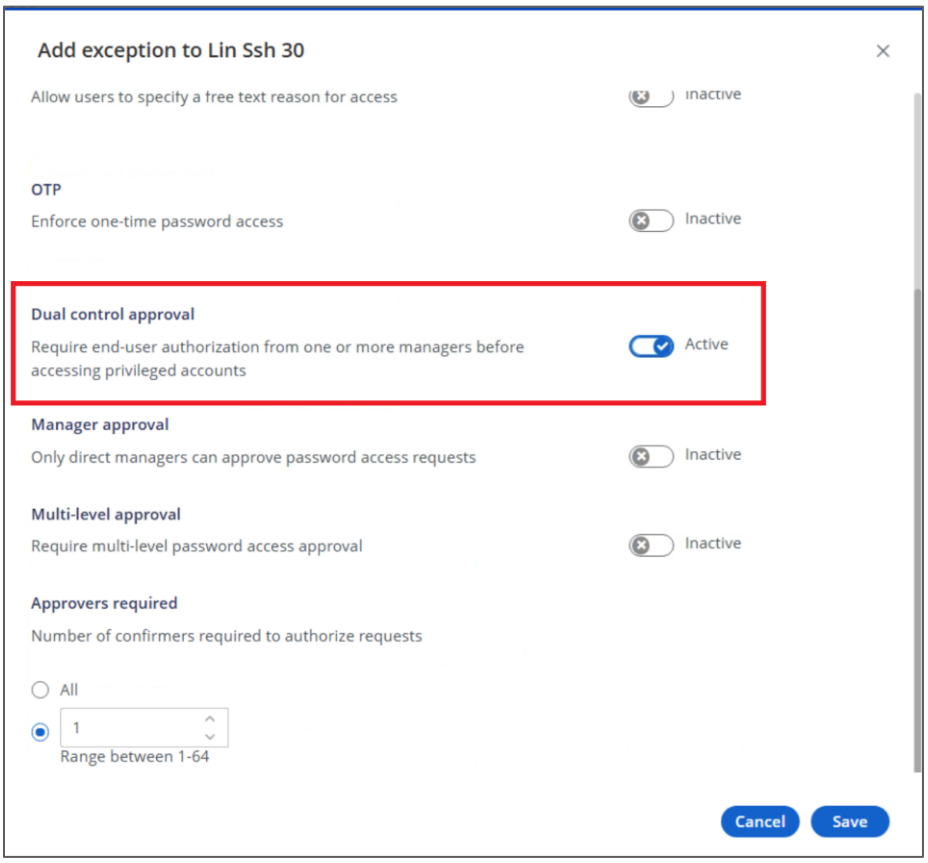
1. Log into the **PVWA** as *Mike* and go to **Administration | Platform Management**. Locate the *Lin Ssh 30* , click on the ellipsis, and click **Add exception**. Locate the option for **Dual control approval | Require end-user authorization…** and activate it. Click **Save** when you are done.



Adding an approver to a Safe

The workflow process is configured through Safe membership. Because *Paul* is the creator of the Safe, he is the Safe Owner and therefore has all permissions on the Safe, so he can approve requests. But what if *Paul* is out of the office? To make sure that there is always someone available to approve requests, we will add the AD group *ITManagers* to the Safe and give them the necessary permissions. They will be able to approve requests, but they will not be able to *Retrieve* the passwords or *Use* them.

1. Log on to the **PVWA** as *Paul* and go to **POLICIES | Safes**.

2. Select *P-BOS-LIN-S-FIN*, click on the **Members** tab, and click **Add members**.

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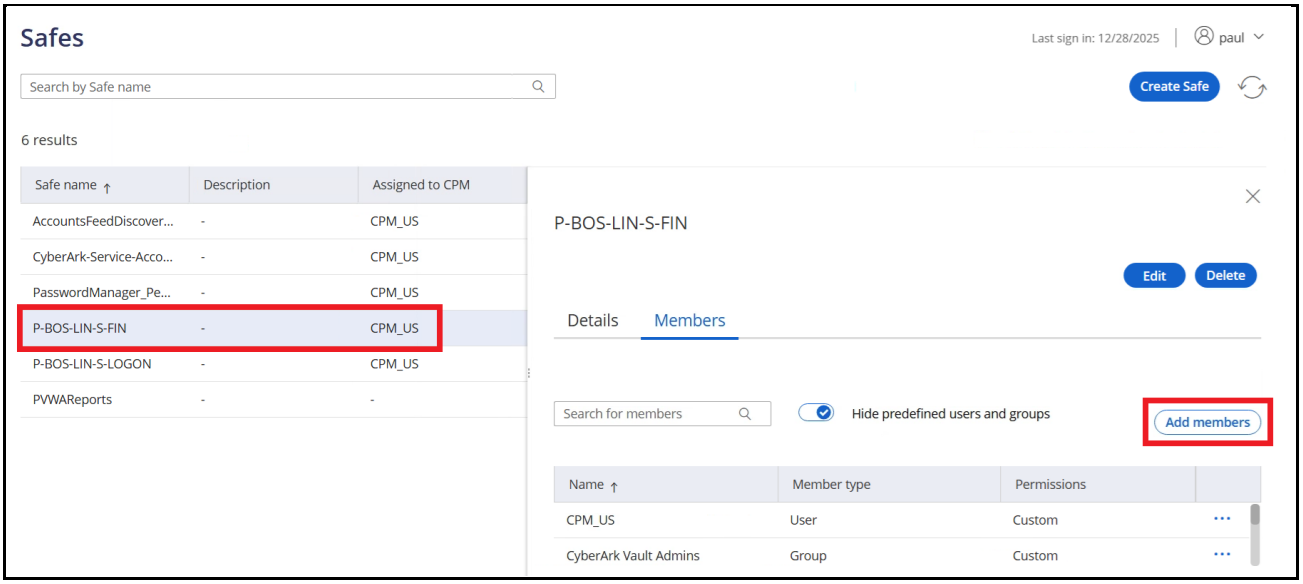
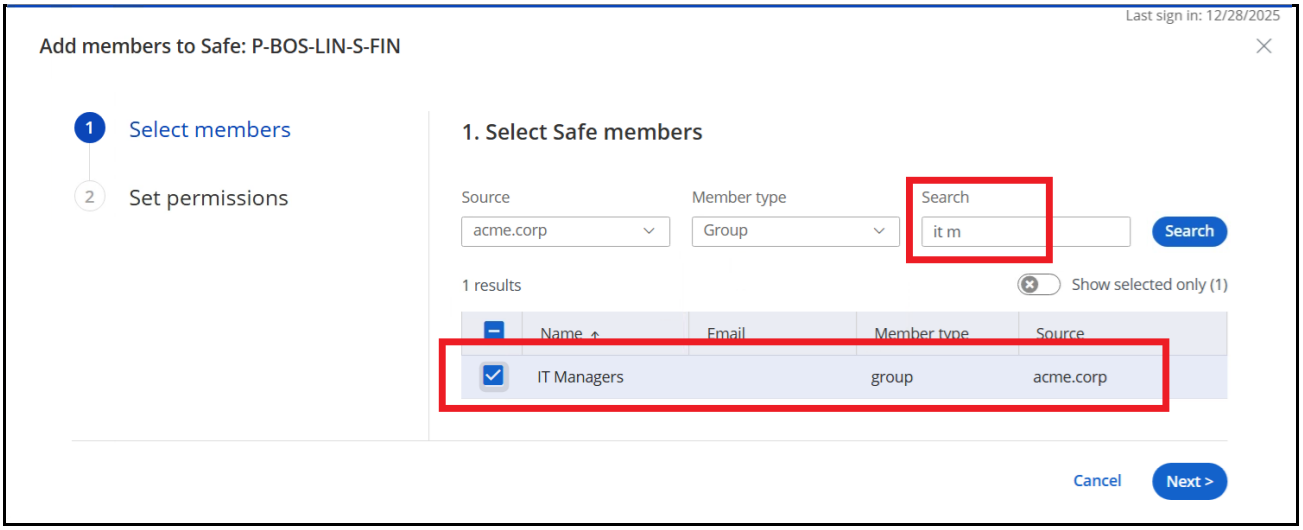


3. Select *acme.corp* in the **Source** field; in **Member type**, select *Group*; enter 'it m' in the **Search** field; and then press **Search**. Select the *IT Managers* group and click **Next**.



4. To set the Safe permissions, we can use the **Permission preset** *Approver*. This pre-configured permission set should include all the permissions necessary for the IT Managers to approve requests, notably the permission to confirm workflow requests, as shown below.

   Take a second to review the other permissions we are granting to IT Managers. Notice that we are not giving them the right to use these accounts, just list them. When you are ready, click **Add**. You can log out of the **PVWA**.

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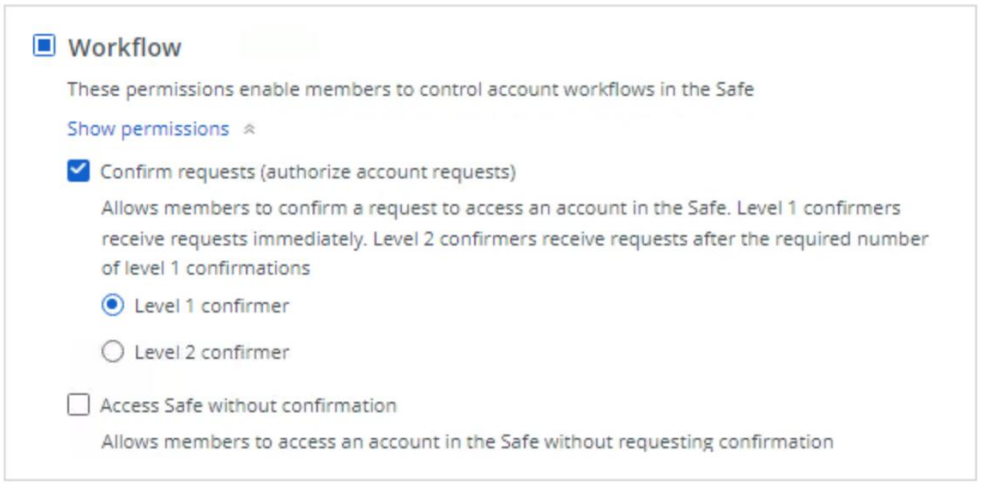
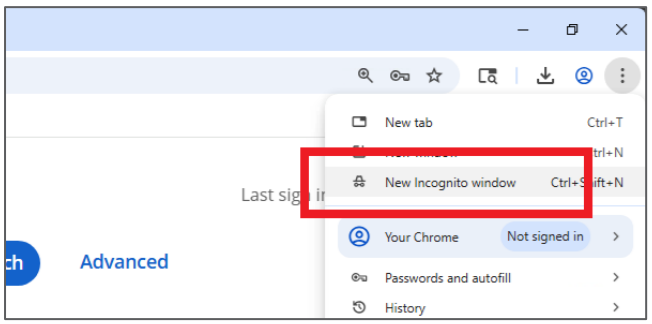


Testing Dual Control

Testing this workflow requires us to wear a number of hats and we'll do this using two instances of Chrome running – one regular and one in *incognito mode* – and log in to each one with a different user: one for *Carlos* and one for *Tom*.

- First we will connect as a user who has the right to use a password, but only with manager approval – *Carlos* – and request access to an account.
- Then, we will log in as *Tom*. Although *Tom* is the manager of the Windows team, he is also a member of the ITManagers AD group and can therefore approve any access requests, whether they be from Windows, Linux, or Oracle teams.
- We will see the request arrive for *Tom* and then approve the request.
- Finally, we will return to *Carlos*, find the approval notification, and access the target system.

1. Log into the **PVWA** as Carlos. Then, in the upper right-hand corner of the browser window, drop down the menu as shown and click on *New Incognito window*.



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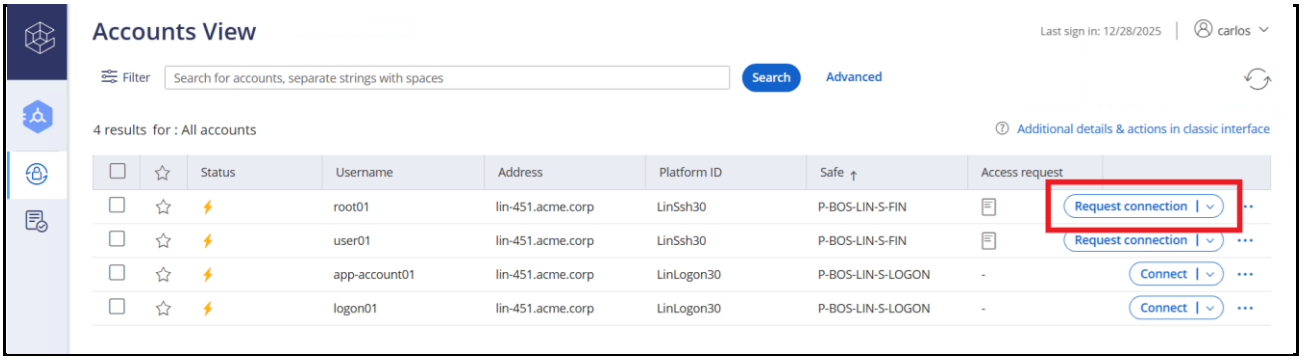
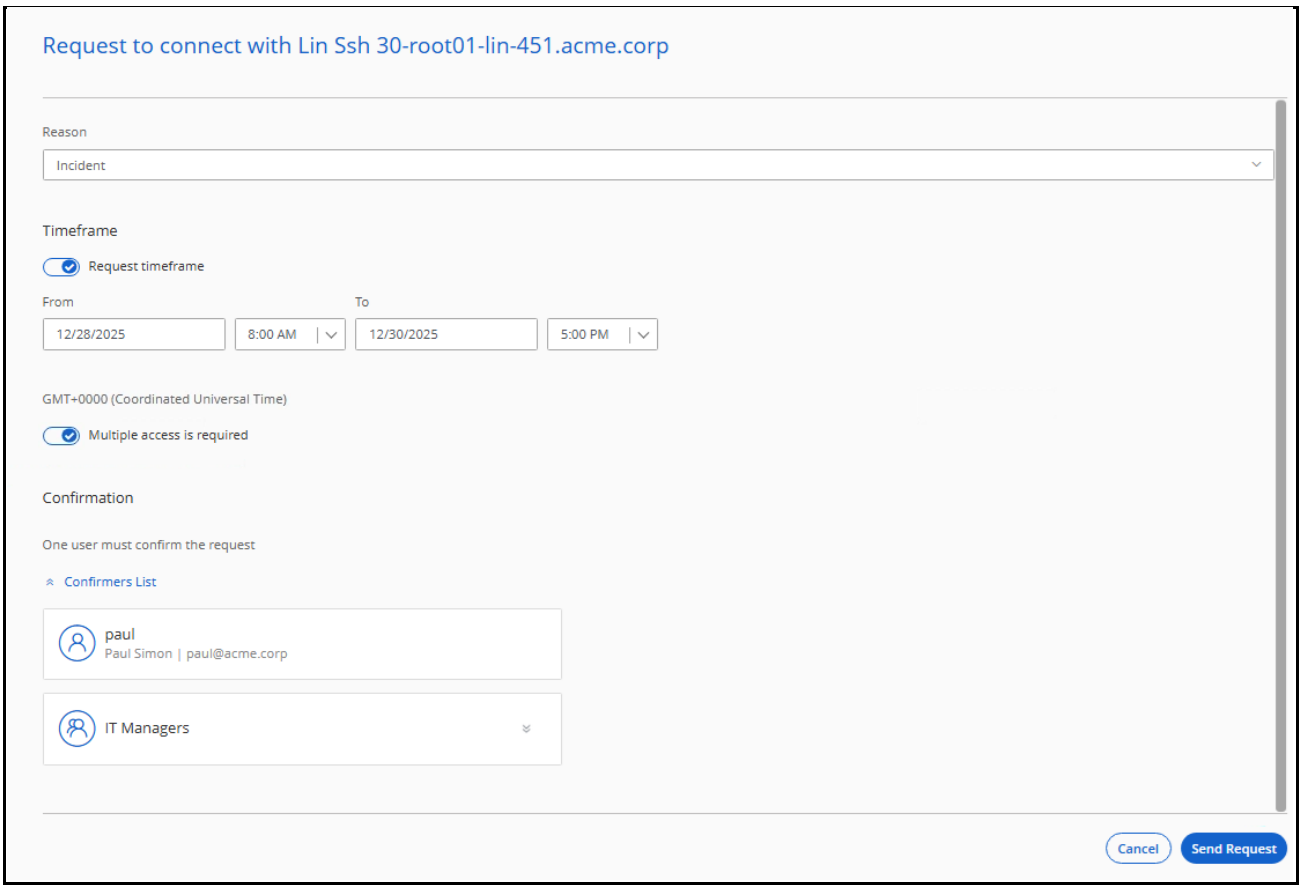
In the new incognito window, click on the **PVWA** shortcut in the bookmarks bar, and log in as *Tom*. Now we have two active browsers, one for each user.

2. Switch to Carlos' browser. Locate the *root01* account and click **Request connection.**



3. This opens the **Request to connect** dialog.



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Note:	It may take a moment for the PVWA to update the list of Confirmers. If you do not see **IT Managers** immediately, don't worry, just go ahead with the exercise.

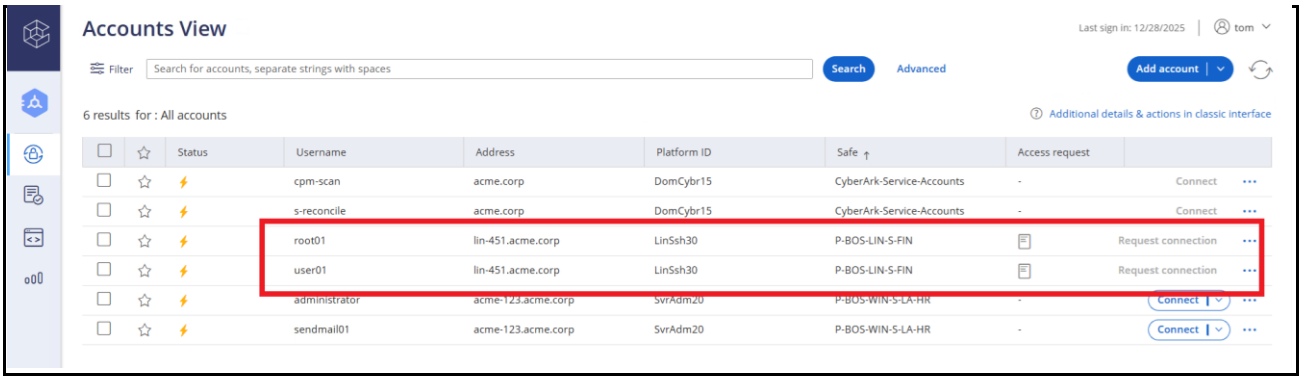
4. Select a **Reason** to access the account. Note that you are unable to enter free text and can only see the pre-defined reasons we configured in the previous exercise.

5. Activate the **Timeframe** and specify *FROM* the current date and hour in the morning *TO* the end of the last day of the class. Also activate *Multiple access is required*.

Note:	Make sure that the start time for the request is either in the past (for example, one half hour before) or not too far in the future so that you will be able to test the connection without having to wait too long.

6. Lastly, expand the *Confirmers List* at the bottom and note who can approve this request. You can also expand the group and see who the members are. When you are ready, click the **Send Request** button.

7. Switch back to *Tom*'s Chrome instance. If you are not already, log into the **PVWA** and filter the view for *All accounts*. Notice that *Tom* can now see both Linux and Windows accounts, but he is unable to use the Linux accounts, only approve Dual Control requests by members of the Linux team.



8. Go back to the **PVWA** as *Tom*, go to **Accounts**, and select **Incoming Requests**. You can see that *Carlos*' is the only pending request. Click on it to view the details.

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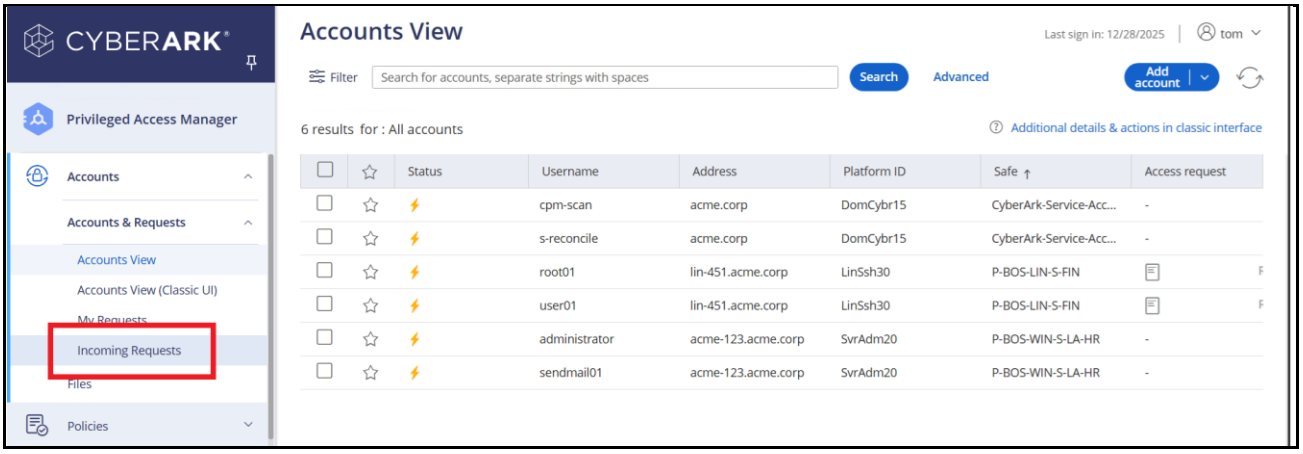
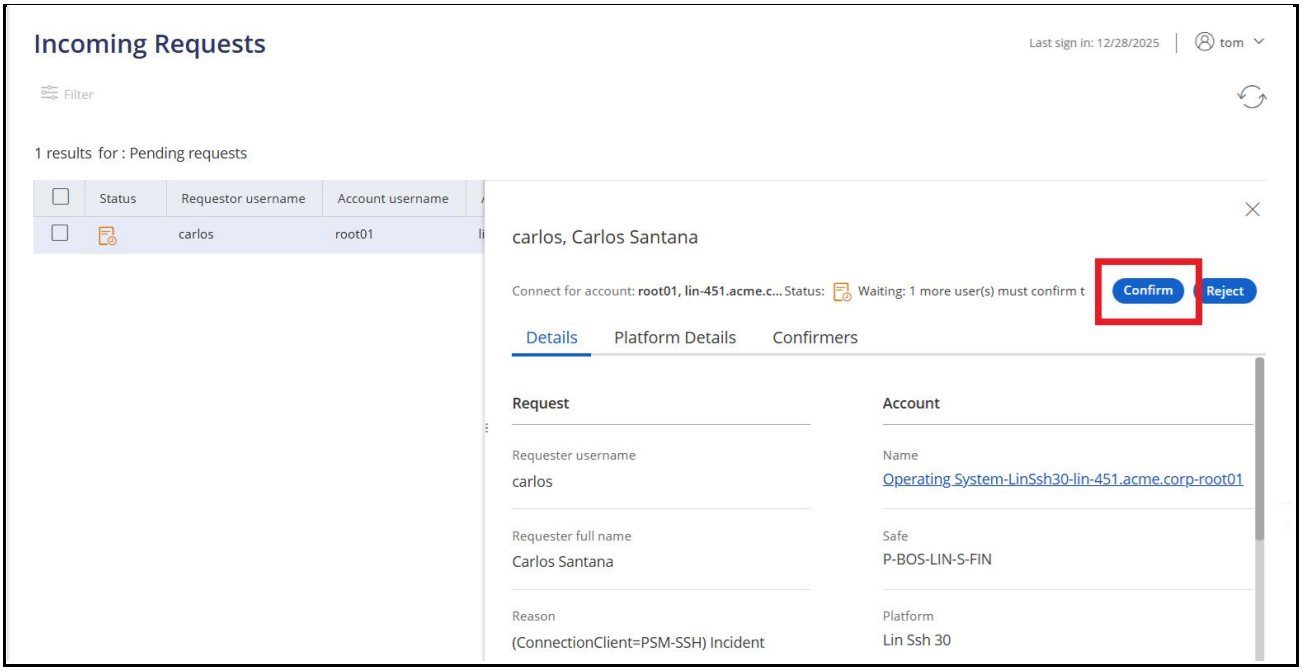


9. As an approver, it would be your role to ensure that this was a valid request to access a sensitive target machine. Press the **Confirm** button.



10. Optionally, enter a reason and press **Confirm**.

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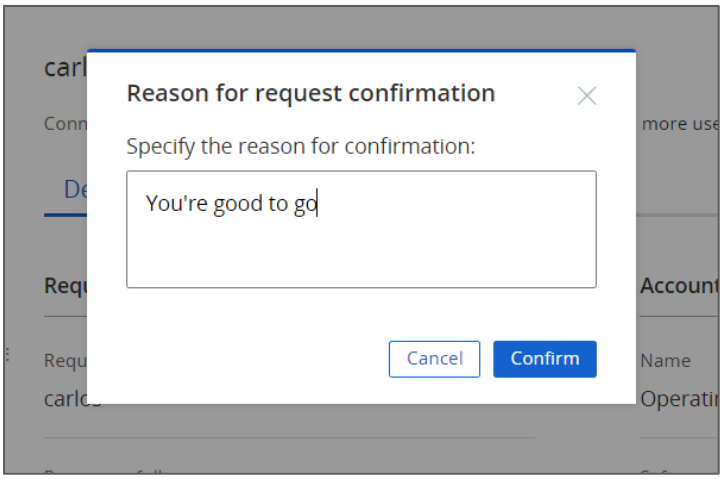
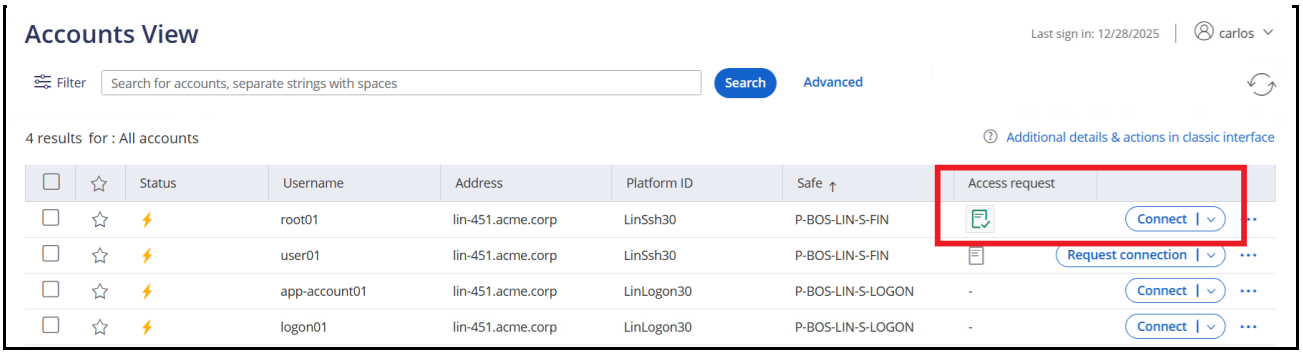


11. Sign out or close the browser to terminate *Tom's* session.

12. Browse to the email client and login as *Carlos*. You should receive an e-mail stating the request has been confirmed.

13. Login to the **PVWA** as *Carlos* (password *Cyberark1*), then go to the **Accounts View** page. Notice the **Status** of the request is now confirmed. You can now use the password and connect to the previously requested account.



14. Sign out of the *Carlos* session.

Exclusive Passwords with Automated Release and One-time Use

In this exercise, we will modify the policies for the **Platform** *Svr Adm 20*, which manages the Windows server local admin accounts, so that they have exclusive access with an automatic release based on the *Minimum Validity Period*. To achieve this, we will:

- Add a Master Policy exception for **Exclusive Passwords**
- Add a Master Policy exception for **One-time Passwords**

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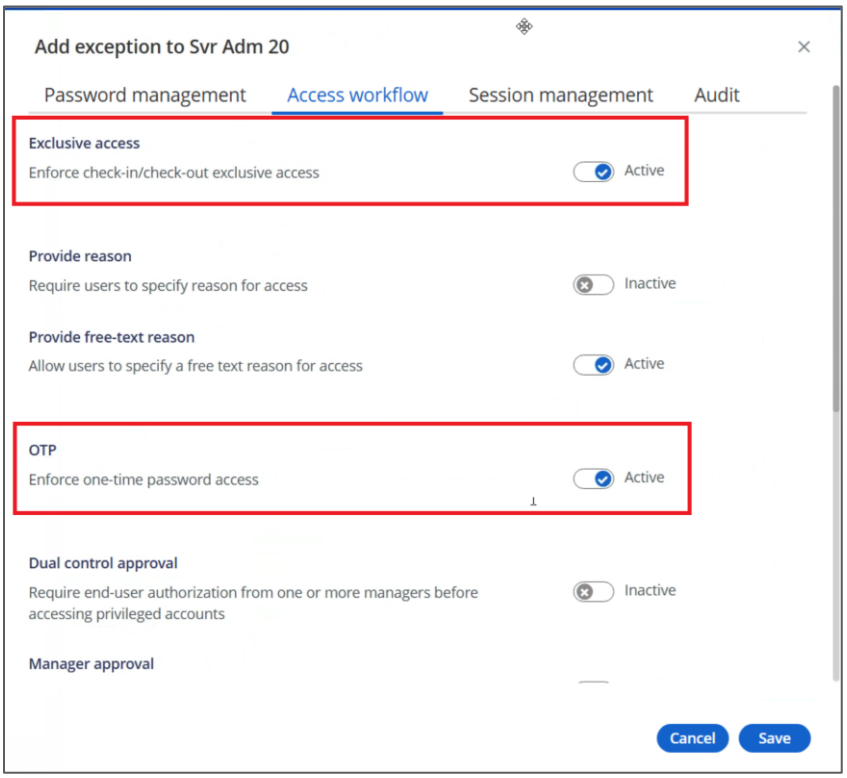
- Reduce the value of the parameter *MinValidityPeriod* for the purposes of testing

Adding Master Policy exceptions for Exclusive Access and One-time Passwords

1. Log into the **PVWA** as *Mike* and go to **Administration | Platform Management**. Locate ***Svr Adm 20***, click on the ellipsis, and click **Add exception**.

2. Click on the **Access workflow** tab and activate **Enforce check-in/check-out exclusive access** and **Enforce one-time password access**. Click **Save** when you are done.



Reducing the Minimum Validity Period (for testing only)

Note:	This next step is for testing/training purposes only and should not be used in a production environment.

We will set the *Minimum Validity Period* to **5 minutes**, so that we can see our results more quickly. The *MinValidityPeriod* parameter is configured in the Platform.

1. Go to **ADMINISTRATION | Platform Management**, select *Svr Adm 20*, and click **Edit**.

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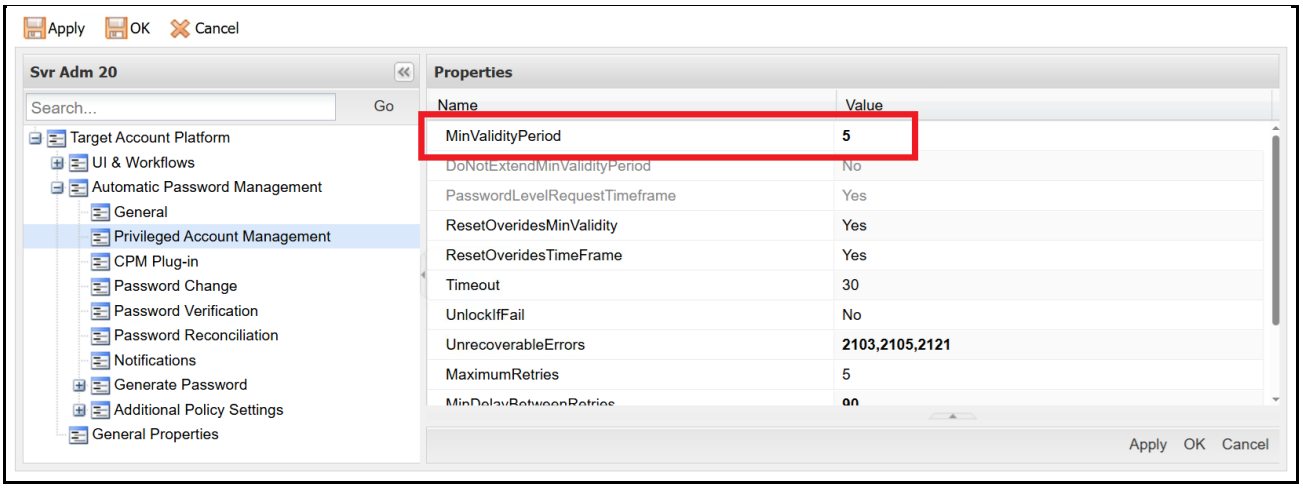
2. Go to **Automatic Password Managment | Privileged Account Management**.

3. Set *MinValidityPeriod* to **5**.



4. Press **Apply** and **OK** to close the Platform and then sign out of the **PVWA**.

Testing Exclusive Passwords

In this section, we will test our configuration of exclusive passwords with automatic release. We will use the users *Tom* and *John*. *Tom* is the Safe Manager (therefore its owner) and *John* is a member of the Active Directory group *Windows Admins*.

1. Login to the **PVWA** as the **LDAP** user *Tom* with the password *Cyberark1*.

2. Go to **ACCOUNTS**.

3. Click on the *administrator* for **acme-123.acme.corp**. At the end of the line, click on the three dots to drop down the actions menu and click the **Show** button. *Tom* has now checked out the password (even though he has not actually connected to the target machine).

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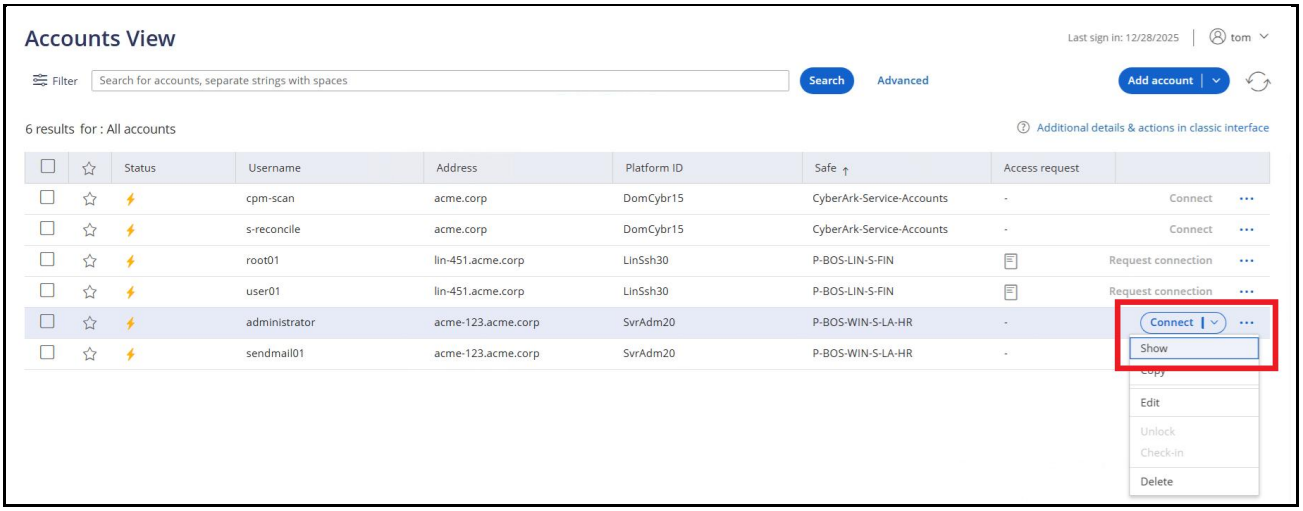
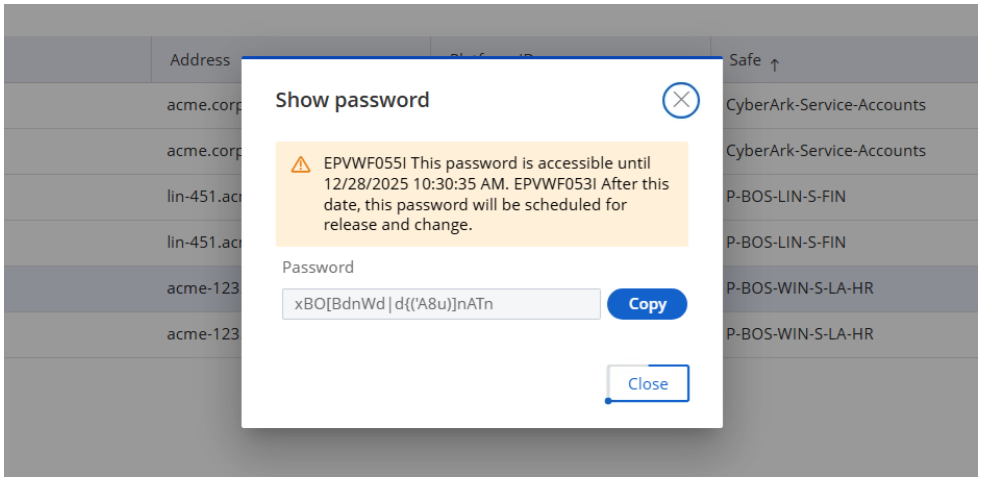


4. You should be able to see the password as well as a message stating the password is available for the next 5 minutes, after which it will be rotated.



5. Log out of the **PVWA** and log back in as *John*. You should notice a lock icon next to the *administrator* account.

Note:	Only *Tom* or a user who has the "Unlock Account" permissions on that Safe can release the account manually by using the "**Check-in**" option, however we will not use this option as we want to see the system release it automatically at the end of the Minimum Validity Period.

6. Hover over the lock icon, it should say "**The account is checked-out by tom**".

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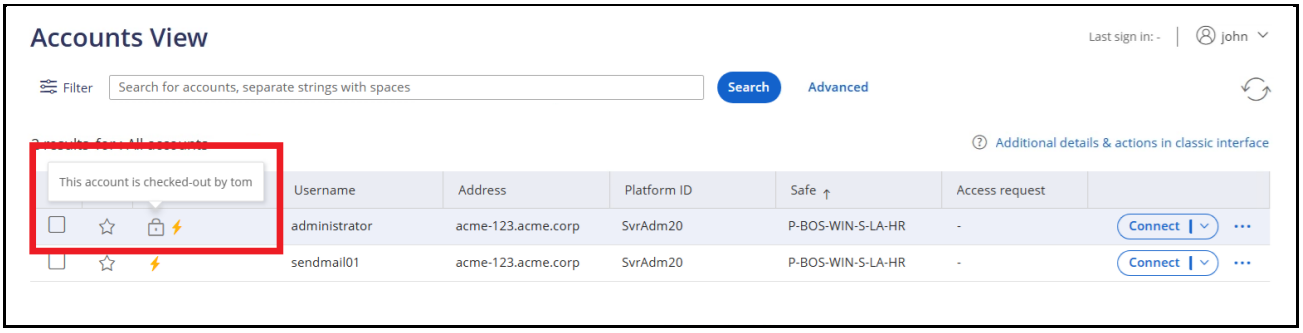
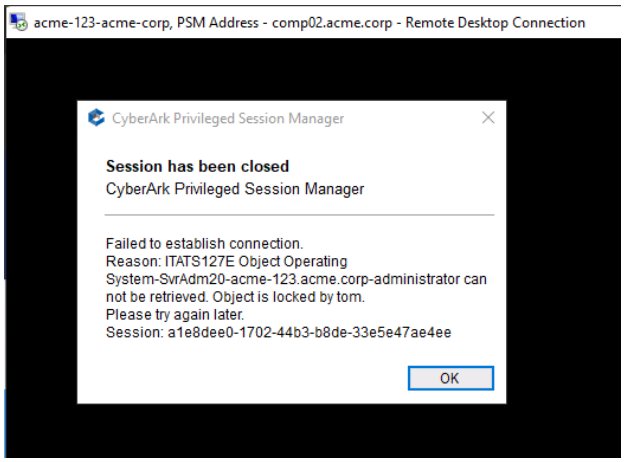


7. If you press **Connect**, you will be able to download the RDP file. However, if you click on the RDP file and attempt to launch a connection, you will receive an error message.



After several minutes (remember the minimum validity period was set to 5 minutes), the **CPM** will have changed the password and *John* will be able to access the account.

Hint:	If the account is not released after several minutes, run the *restart-services.bat* file and check again.

Automatic release by PSM

Starting with version 11.7, the **PSM** can also release an account locked by *exclusive access* upon closing the remote session. Perform the following steps to test automatic release by the **PSM**:

1. Login to the **PVWA** as *Mike* and navigate to **ADMINISTRATION | Platform Management**. select *Svr Adm 20* and click **Edit**.

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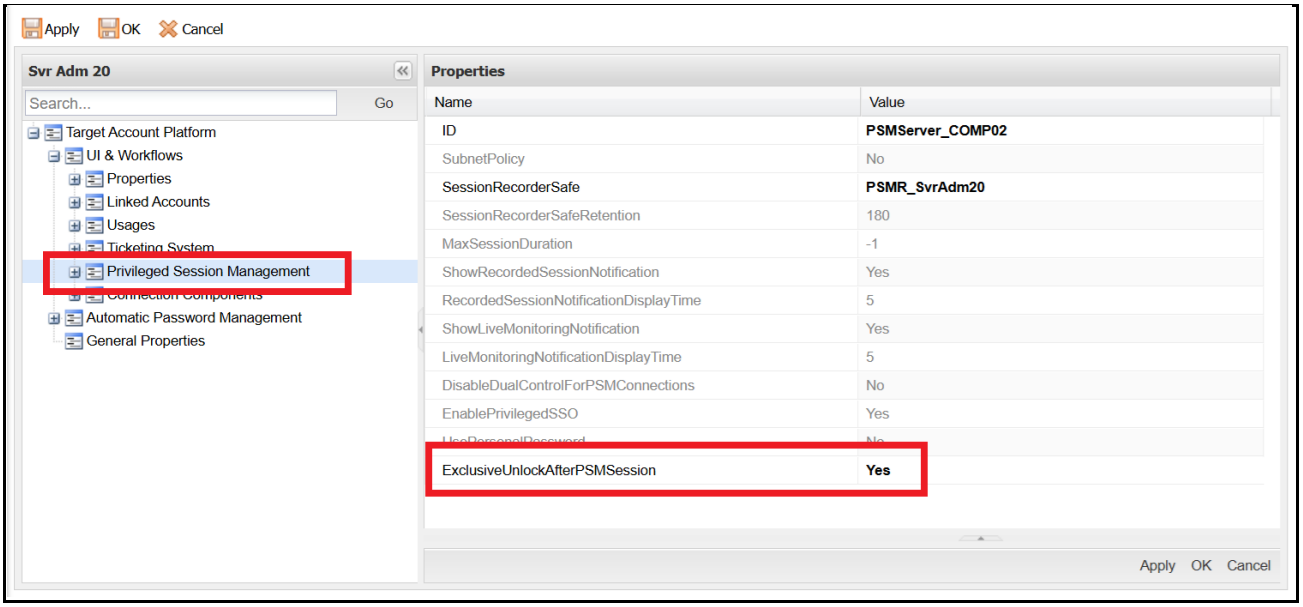
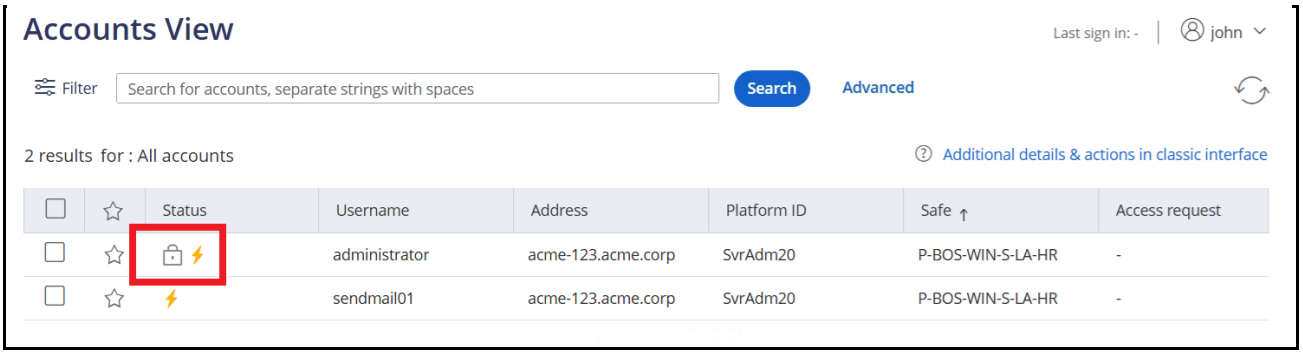
2. Navigate to **UI & Workflows | Privileged Session Management** and set ***ExclusiveUnlockAfterPSMSession*** to **Yes**.



3. Login to the **PVWA** as *John* and locate the *administrator* account. Click on **Connect**.

4. After the session to the target machine has been established, confirm the account is locked by *John*.



5. Now, disconnect from the target machine.

6. If everything has been configured correctly in the previous steps, the *administrator* should be unlocked immediately by the **PSM**. Then, after a few minutes, the account password will also be rotated by the **CPM** (thanks to the one-time password setting), which you can verify by looking at the time-stamp for the password change.

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Note:	The password change can take a little while, so continue to the next lesson and come back to verify it later.

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Onboarding Rules and Accounts Discovery

Up to now, we have been adding accounts one by one, which is fine for testing, but would be difficult to do if you are managing hundreds or thousands of servers and devices. In the following exercises we will see how to manage accounts in bulk. We will create an ***automatic onboarding rule*** that will place accounts responding to certain criteria directly into their appropriate destination **Safes**, associate them with a **Platform**, and will reconcile each account so that only **CyberArk PAM** knows what their passwords are.

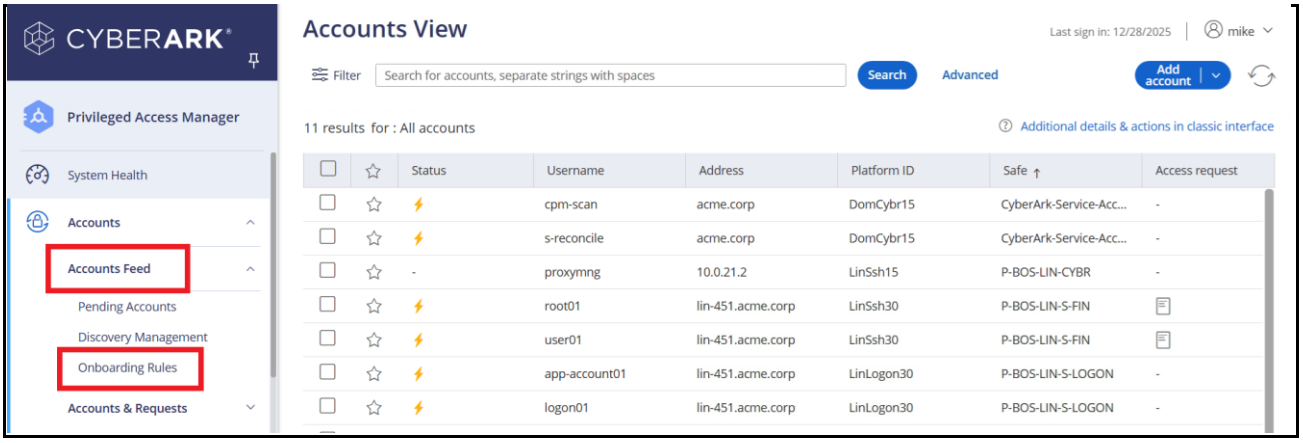
Note:	For this exercise, to make our lives easier we can just use the existing Platform **Svr Adm 20** and the existing Safe **P-BOS-WIN-S-LA-HR**. If you prefer, you can create a Platform and Safe dedicated to these accounts. It will not have an impact on later exercises.

Configure Automatic Onboarding Rules

In this section, we will configure an ***Onboarding Rule*** to add newly discovered accounts that meet certain criteria to the **Vault** without any human intervention.

1. Login to the **PVWA** as *Mike*.

2. Go to **Accounts | Accounts Feed | Onboarding Rules**.



3. Click on **Start creating a new onboarding rule**.

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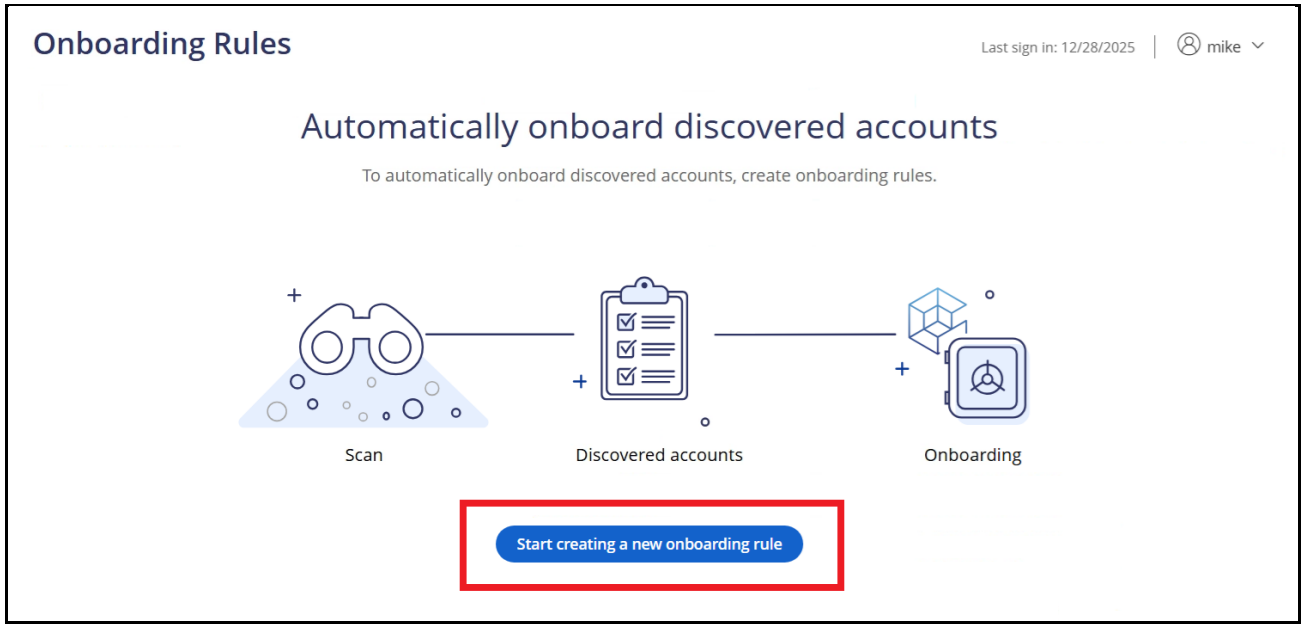
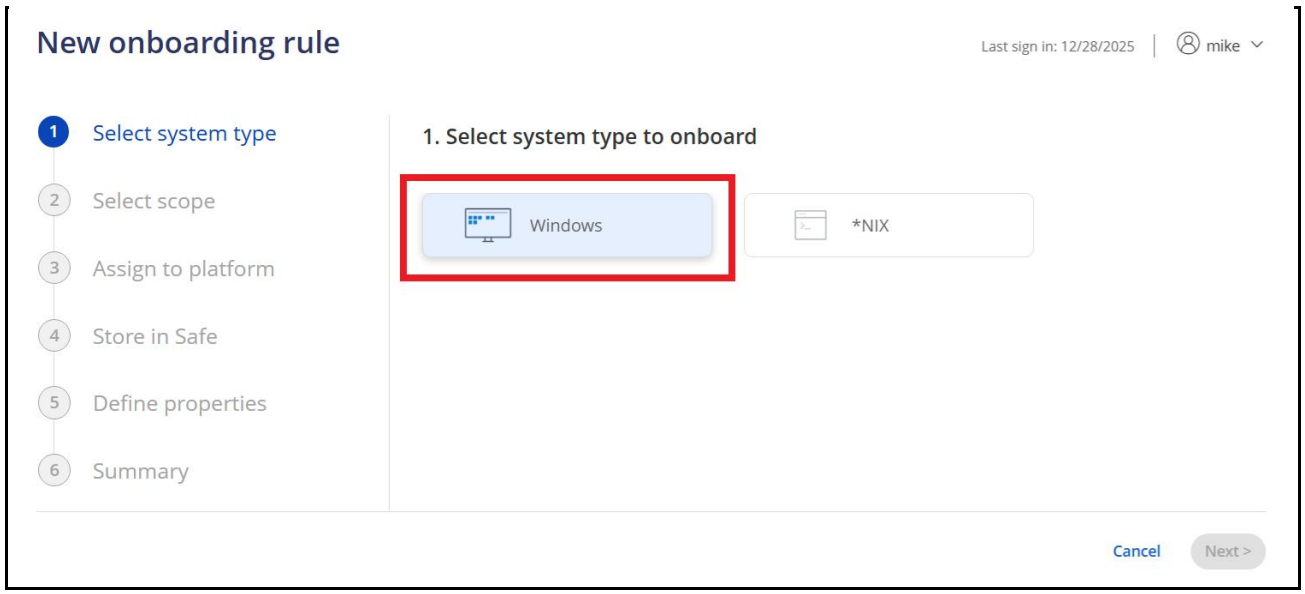


4. In **Select system type**, select **Windows**.



5. In **Select Scope** select the following:

Machine Type:	*Server*
Account Type:	*Local*
Account Category:	*Any*
Privileged Account Type:	*Any*

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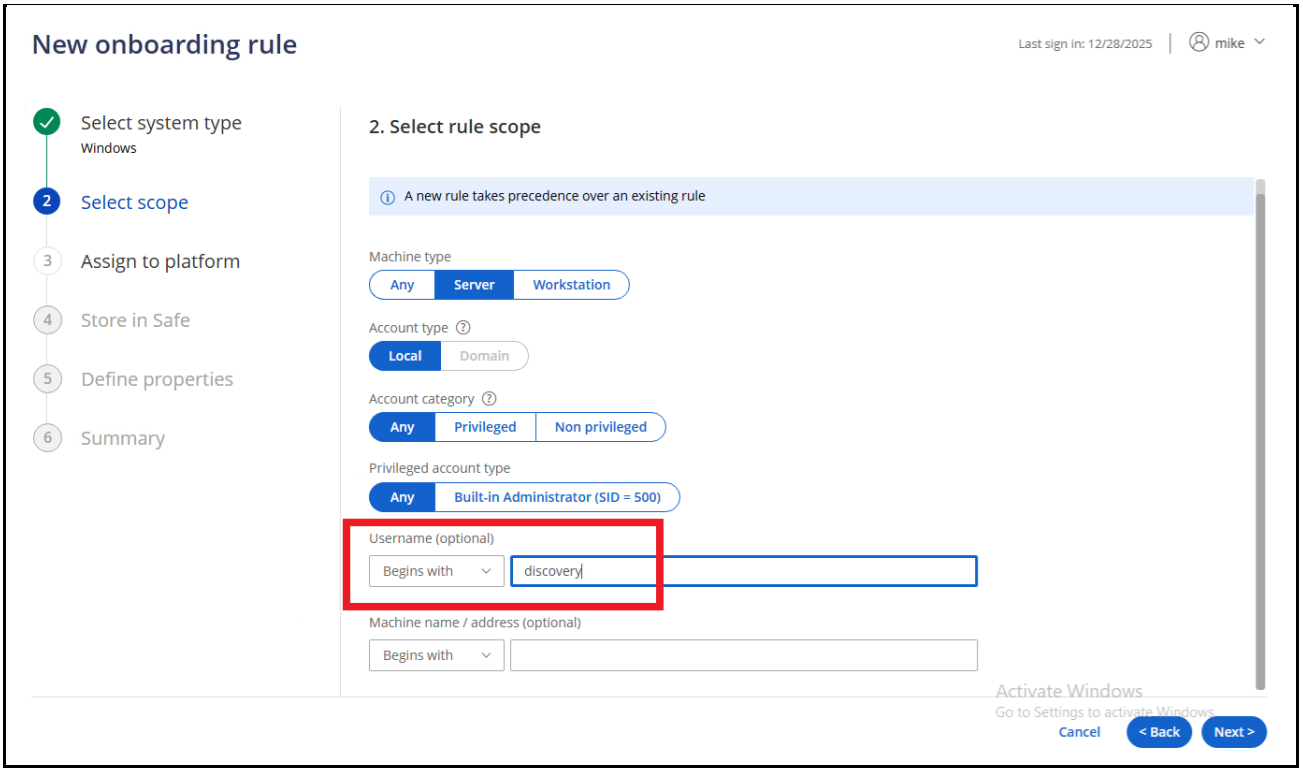
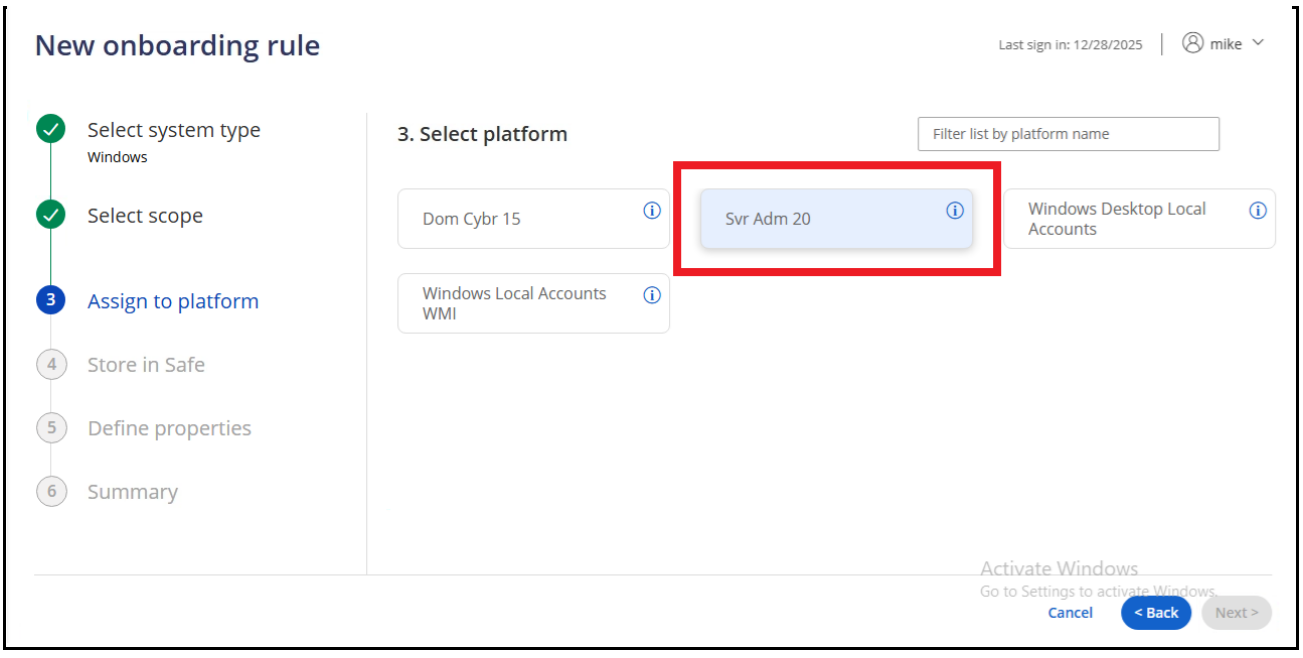
Username (begins with…):	*discovery*



6. Click **Next**.

7. In **Assign to platform** select *Svr Adm 20*.



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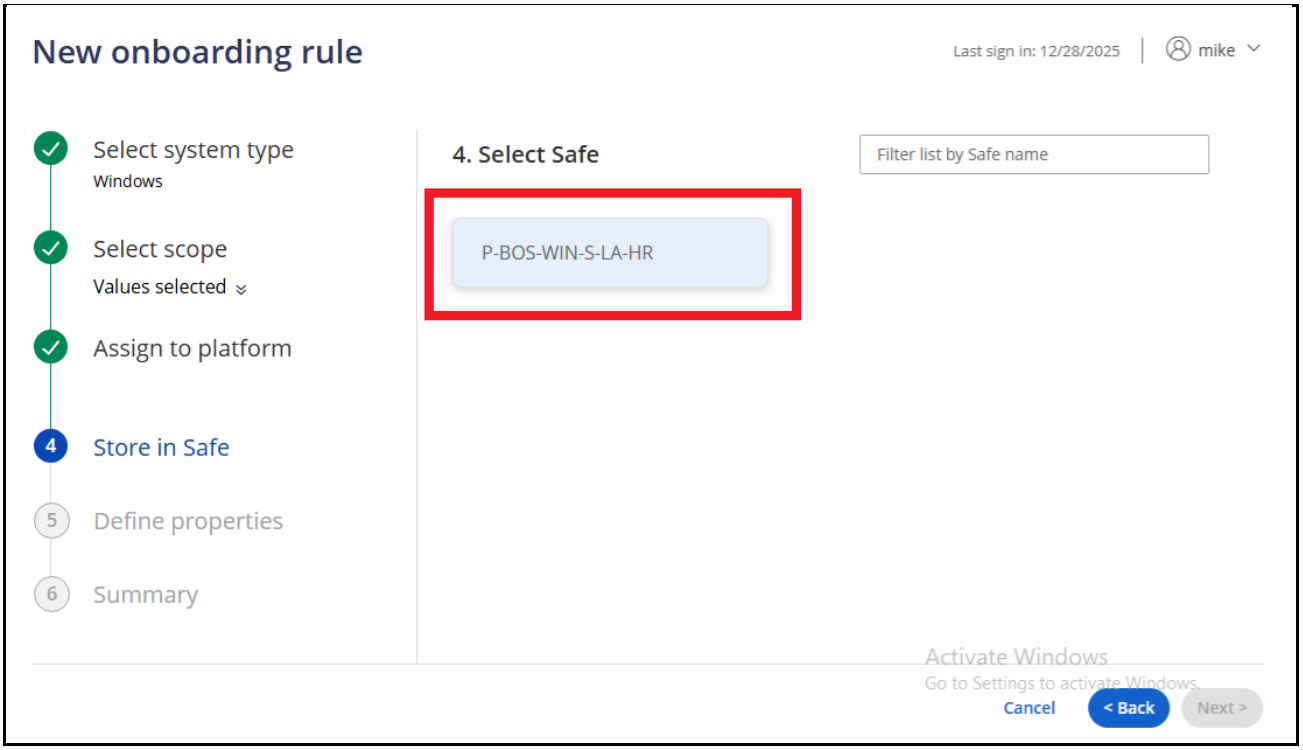
8. In **Store in Safe** select *P-BOS-WIN-S-LA-HR*.



9. In **Define rule** properties, enter an appropriate name and description, and click **Next**.

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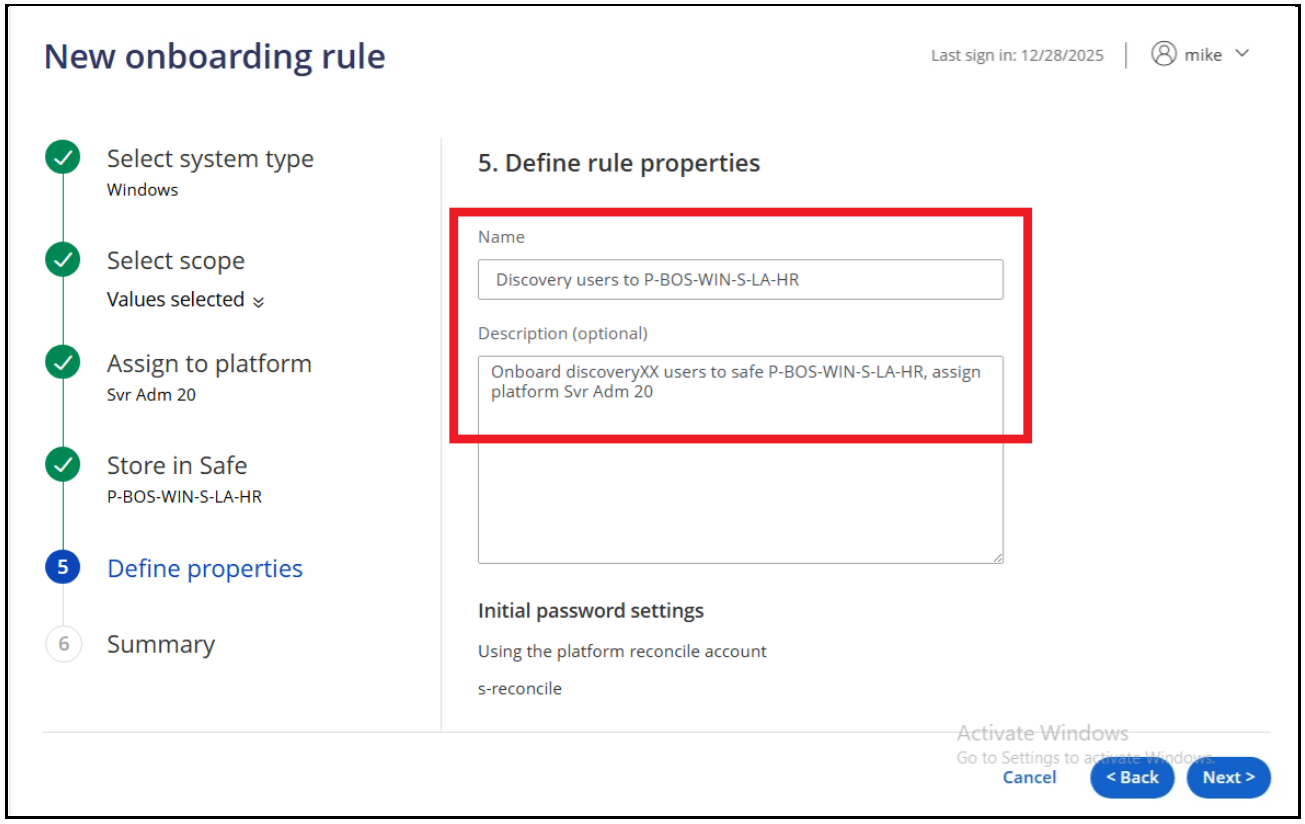

10. On the **Summary** page, review your rule and if everything seems to be in order, click on **Create rule**.

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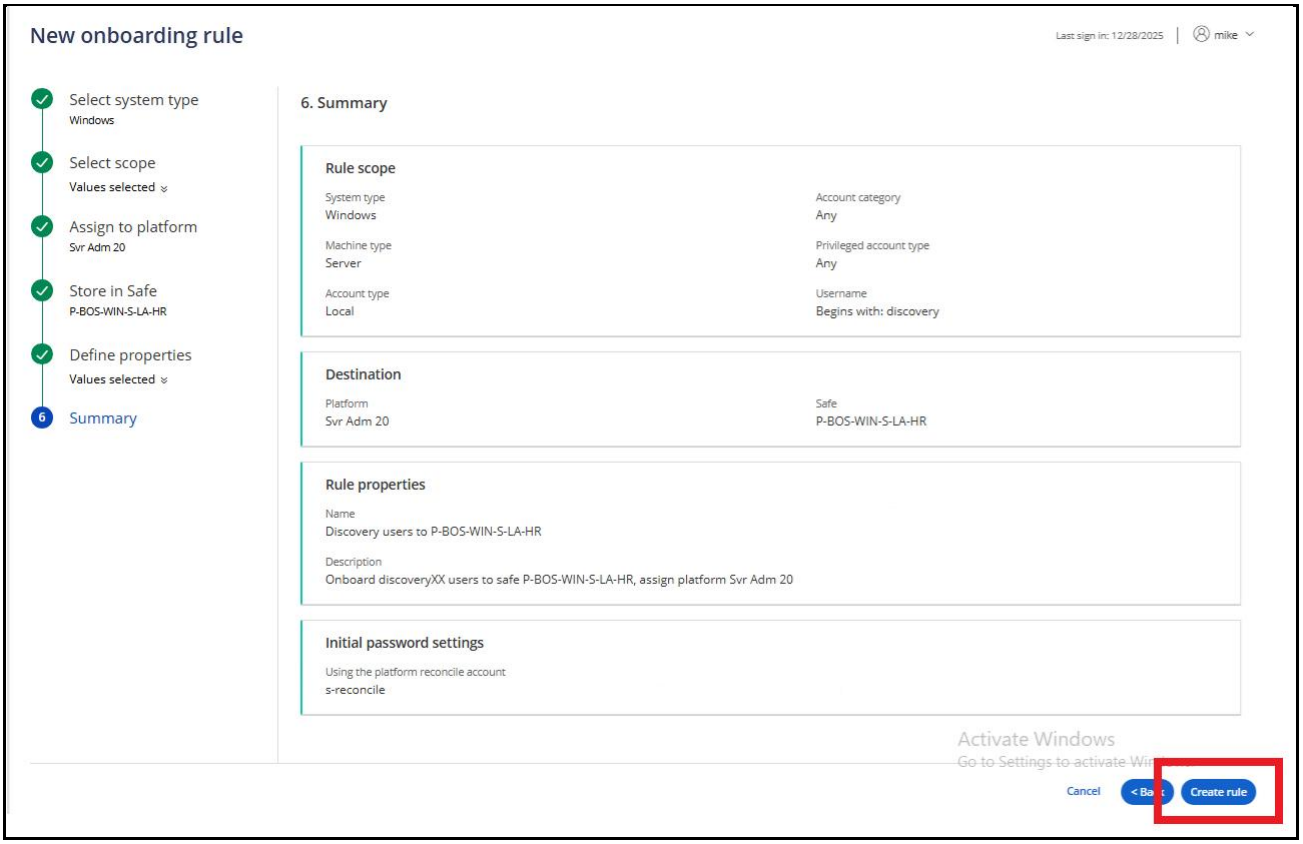

11. You will see a message informing you that the rule has been created.

Configure and Run Windows Accounts Discovery

The Accounts Discovery for Windows process requires an account to log in to the domain and scan the individual machines. We will use the *cpm-scan* account we created in the first exercise.

Note:	The account ***cpm-scan*** is an Active Directory account created specifically for the purpose of running Accounts Discovery scans. It is an unprivileged user at the domain level, but has been added to the local Administrators group on every server in our domain through a group policy.

1. Go to **Accounts | Accounts Feed | Discovery Management**. Click **Create discovery** to drop down the menu and select **Windows**.

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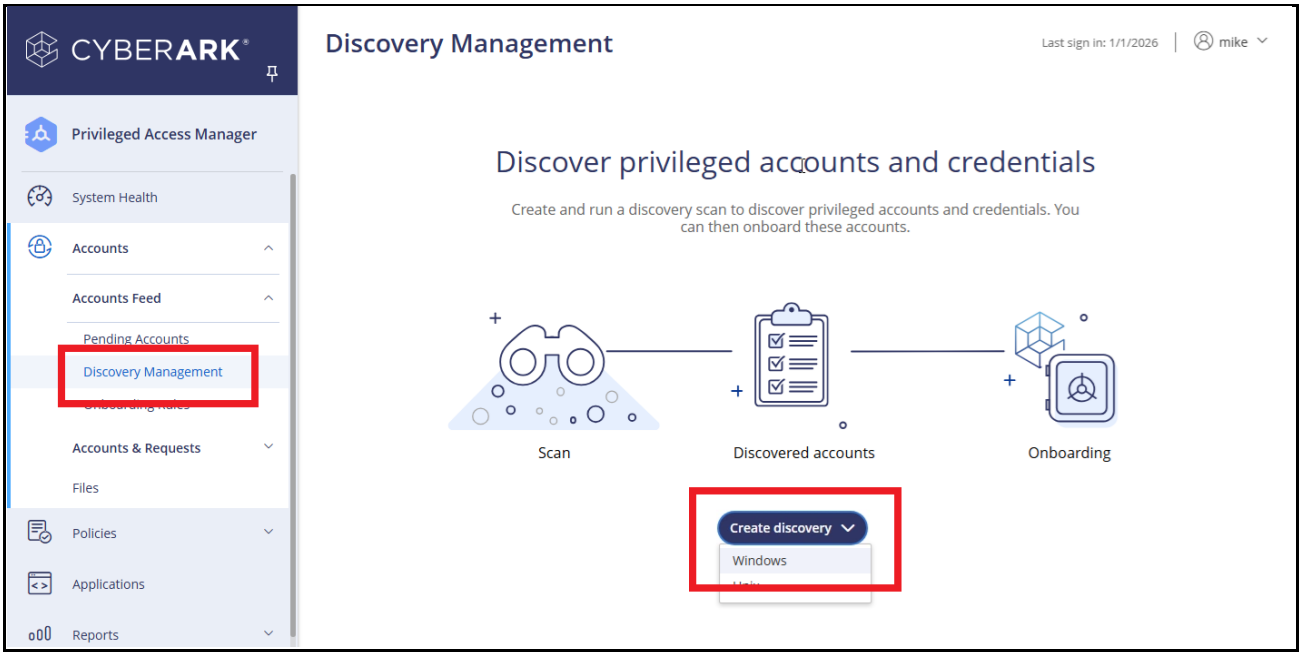
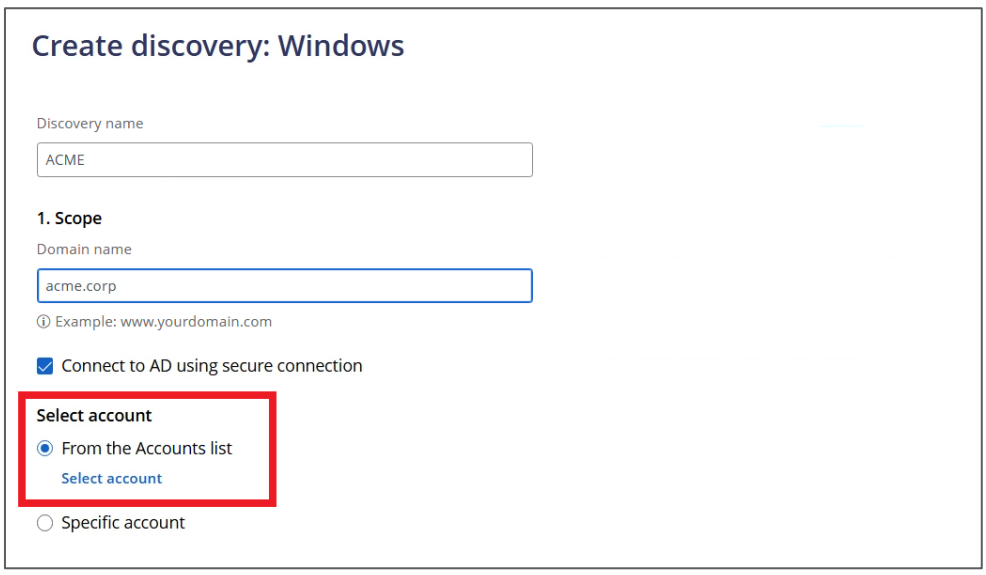


2. Because we have such a small domain, we can run a search on the entire Active Directory structure, so for the name enter **Acme**.

3. For the Domain name enter **acme.corp**.

4. Leave the option **Connect to AD using secure connection** checked.

5. Under the option **From the Accounts list**, click on **Select account**.



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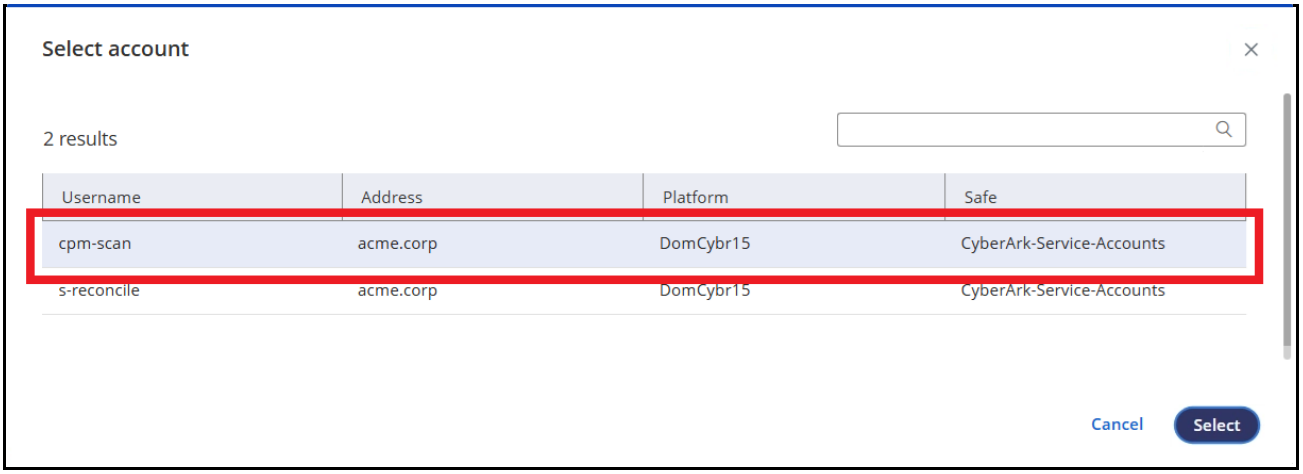
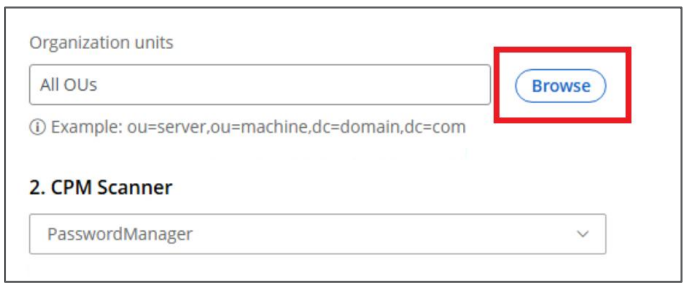
6. Select the account ***cpm-scan***. We have very few accounts for the moment, so locating the account is easy. In a more complex environment, you might need to use the search feature. Click **Select** when you are ready.



7. Scroll down to the section **Organization units**. Here we can select what part of the AD structure we want to search. Click on the **Browse** button.



8. Under **acme.corp**, select the node **IT**. This will include all the machines in our environment. Click **Select** when you are ready.

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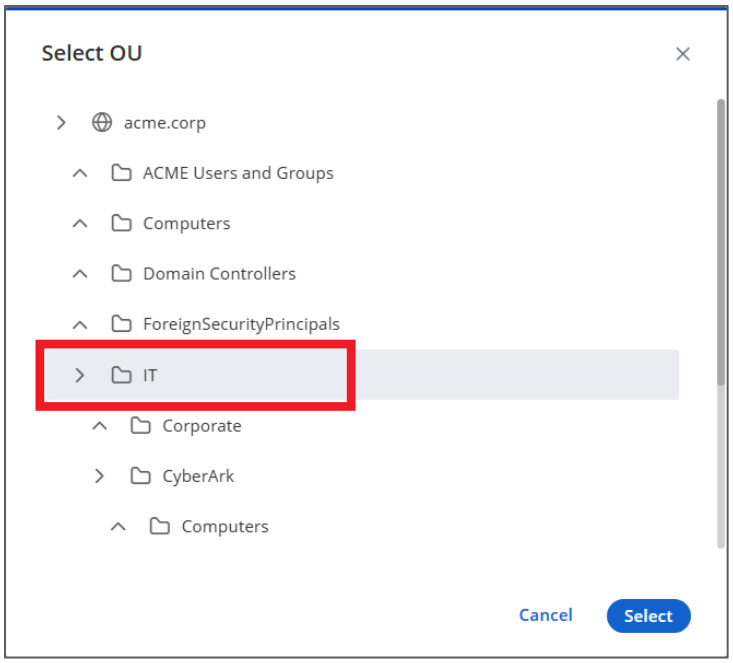
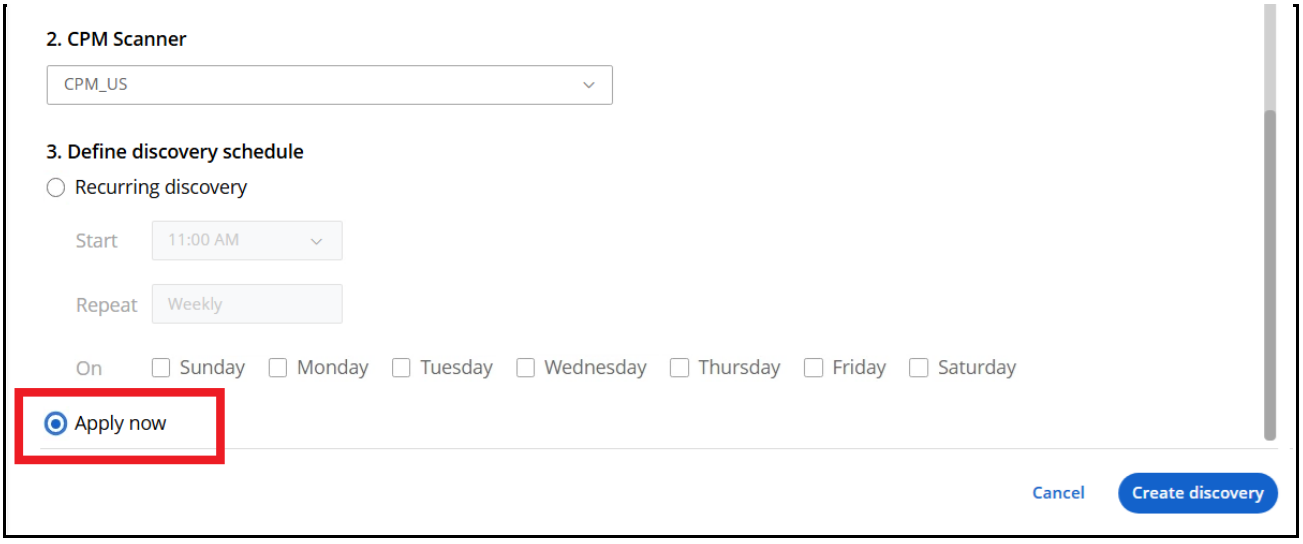


9. The last choices we need to make are the CPM to perform the scan, which for us is easy as we have only one and it is selected by default; and whether to perform a recurring discovery or to run it now. Select **Apply now** and then click **Create discovery**.



10. The new discovery is created and placed in a **Pending** list until the next time the **CPM Scanner** service connects to the **Vault** and sees that it has a new task to perform.

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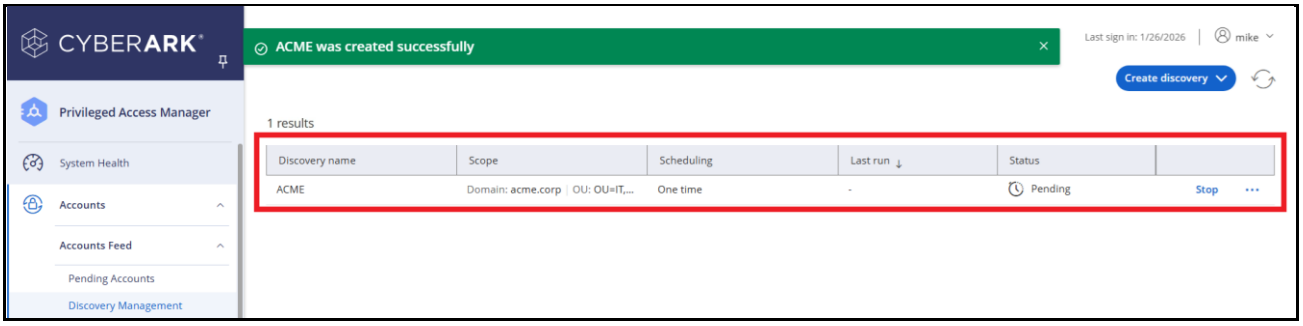
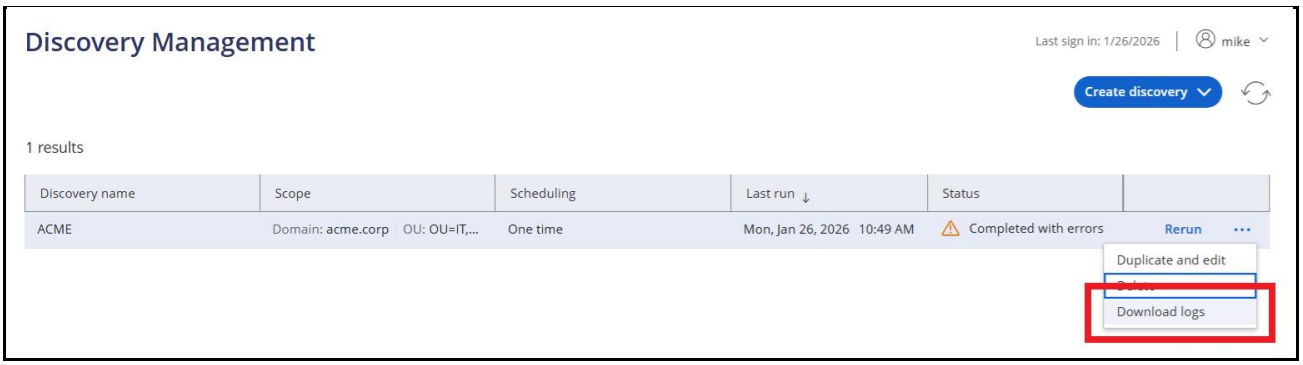


11. After a couple of minutes, refresh your screen. You may see that the discovery completed with errors. This does not necessarily mean that it failed. It simply means that it encountered some issues that may need to be resolved. Click on the ellipsis at the end of the line and select **Download logs**.



Review onboarded accounts

Let's see what accounts have been onboarded.

1. Click on the menu option **Pending Accounts** on the left. We can see that 52 new accounts have been onboarded. Most are from the various Windows servers in our environment – **acme-123**, **comp01**, and **comp02** – but there are some accounts from out **acme.corp** domain. What we cannot see here are the discovery accounts. This is normal. Navigate to the **Accounts View**.

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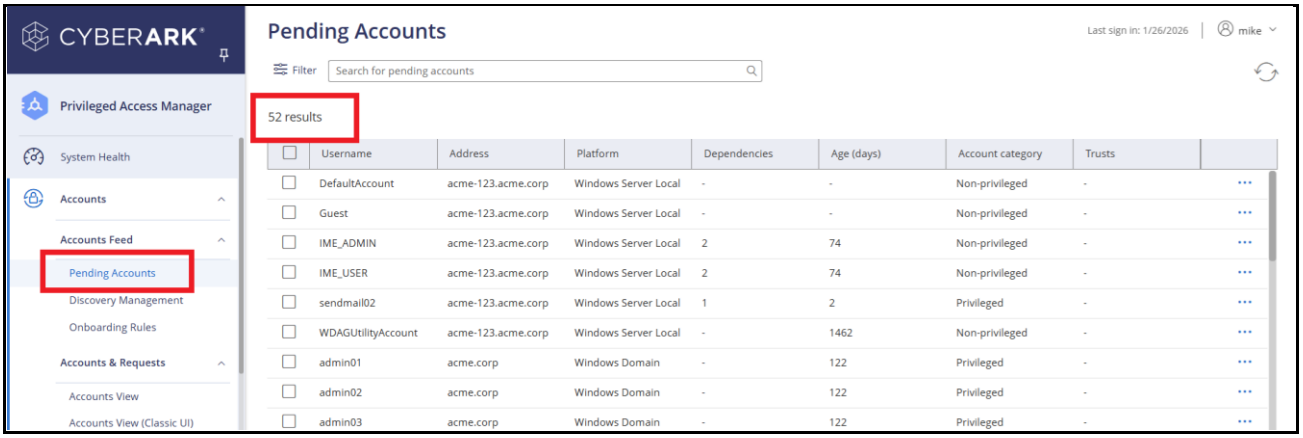
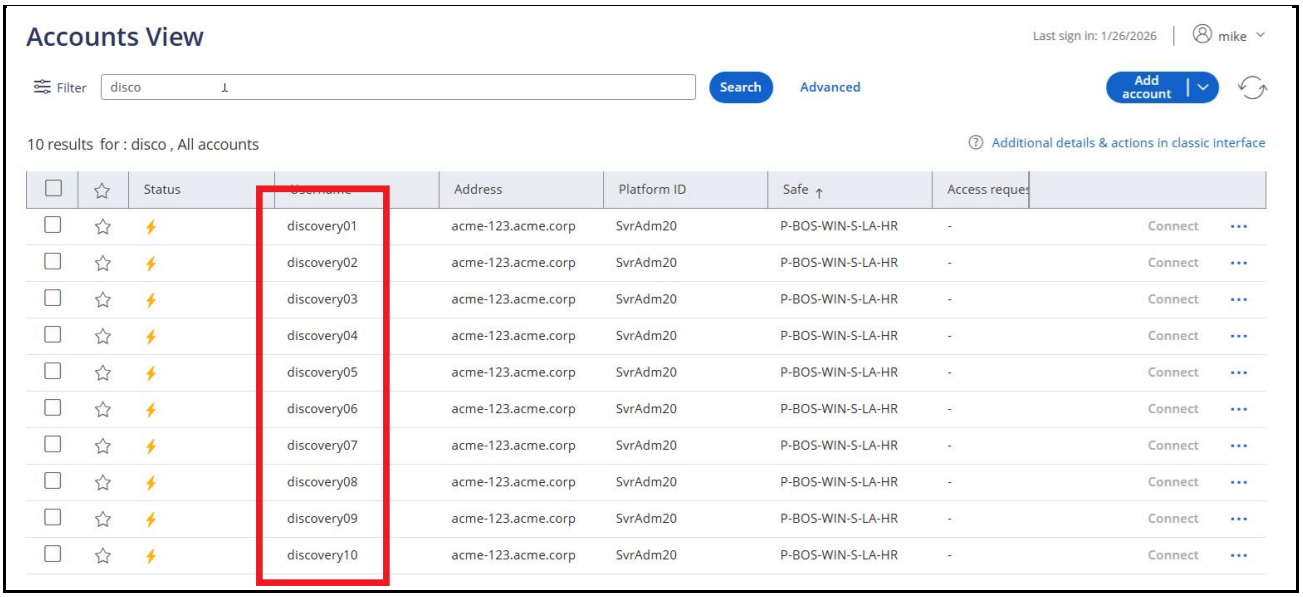


2. In the **Filter** at the top, enter the string 'disco' and hit **Enter** (or click **Search**). You will see that our Discovery and our Onboarding rules have brought the 10 accounts that meet the criteria for our search and rules. They have been associated with a Platform and placed in an appropriate Safe. Now click on one of the accounts to see the **Overview**.



3. Here we can see that the account was automatically reconciled during the onboarding process. All of our discovery accounts now have unique, complex passwords known only to the **CyberArk Vault**.

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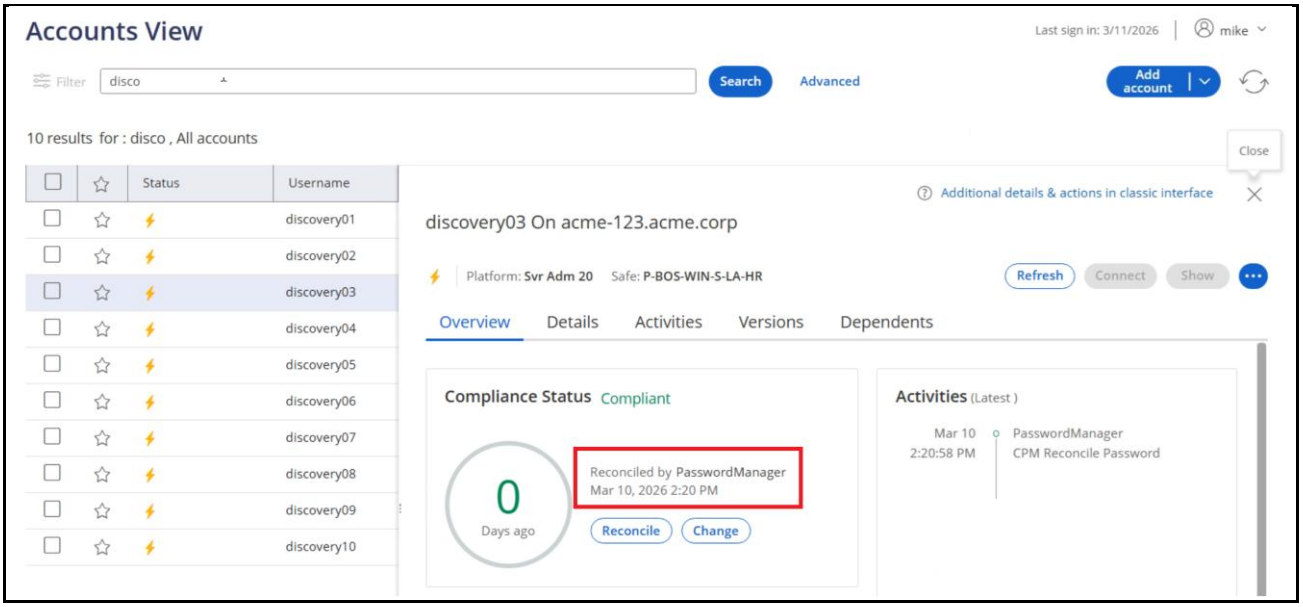
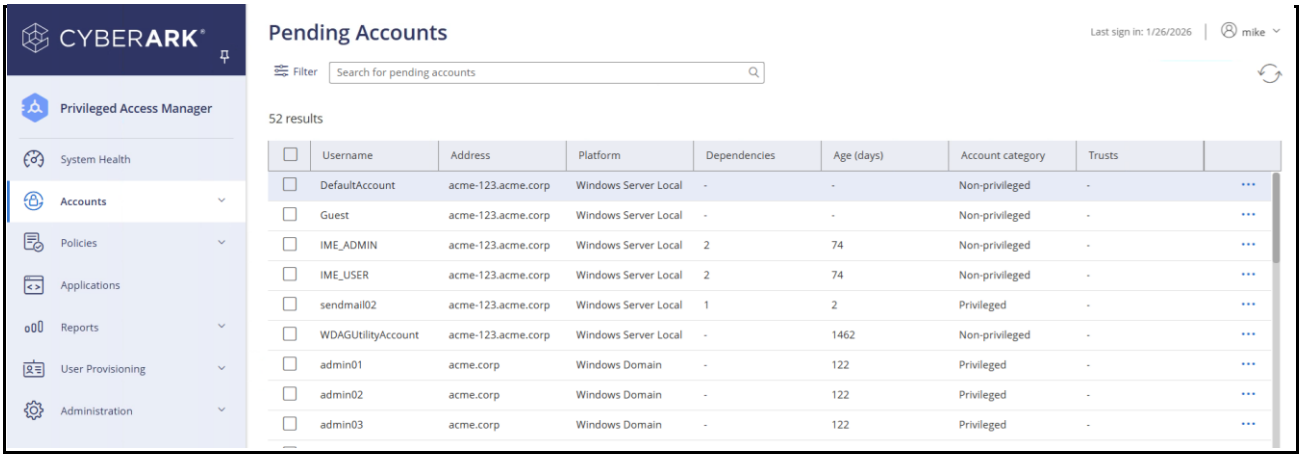


Manually onboard discovered accounts

Version 14 of **CyberArk PAM** updated the **Pending Accounts** interface to the new UI. The new interface provides easier access to **Pending Accounts**, enhanced filtering, a more efficient on-boarding process, and a new way to create **Safes** during onboarding. In this section, we will see how the new interface functions.



We will first manually onboard an account that was discovered but for which there was no automatic onboarding rule.

1. Go to the **Pending Accounts** list, enter *sendmail* in the **Filter** field, and hit **Enter**.

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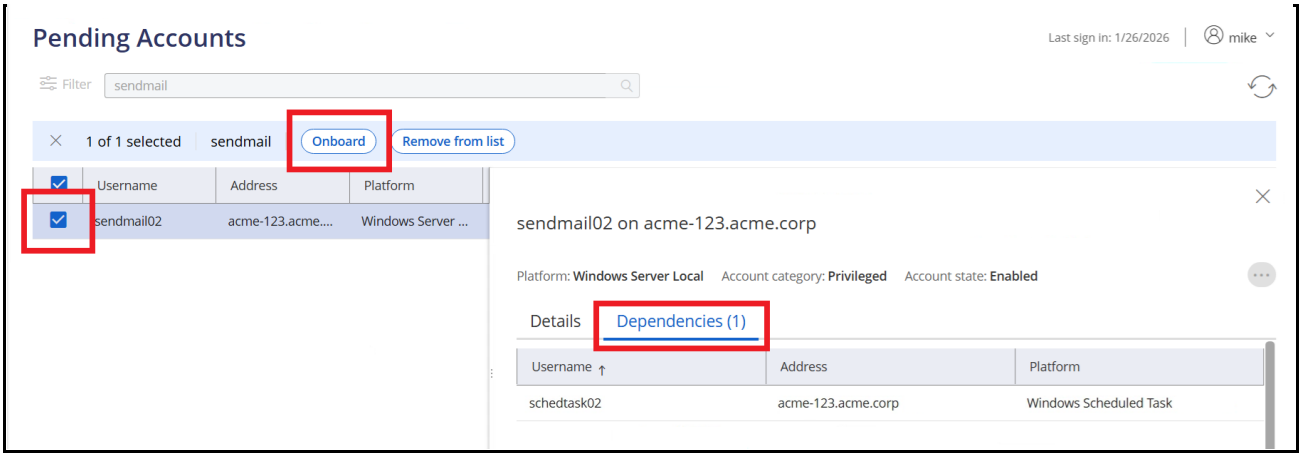
2. Tick the box next to the *sendmail* account. Now click on the **Dependencies** tab and you will see that the Discovery process has detected that the account *sendmail02* is also used in *schedtask02*. When you are ready, click on the **Onboard** button.



Note: One of the main benefits of discovery and onboarding is the ability to discover dependencies tied to Windows accounts. Unlike the earlier exercise, this time the dependency will be onboarded along with the target account, and the **CPM** will manage the dependency without any human intervention.

3. In the **Onboard Accounts** window, select the following:

Store in Safe	*P-BOS-WIN-S-LA-HR*
Assign platform	*Svr Adm 20*
Password	*Automatically reconcile password* (this will only be available if the assigned platform contains a reconcile account)

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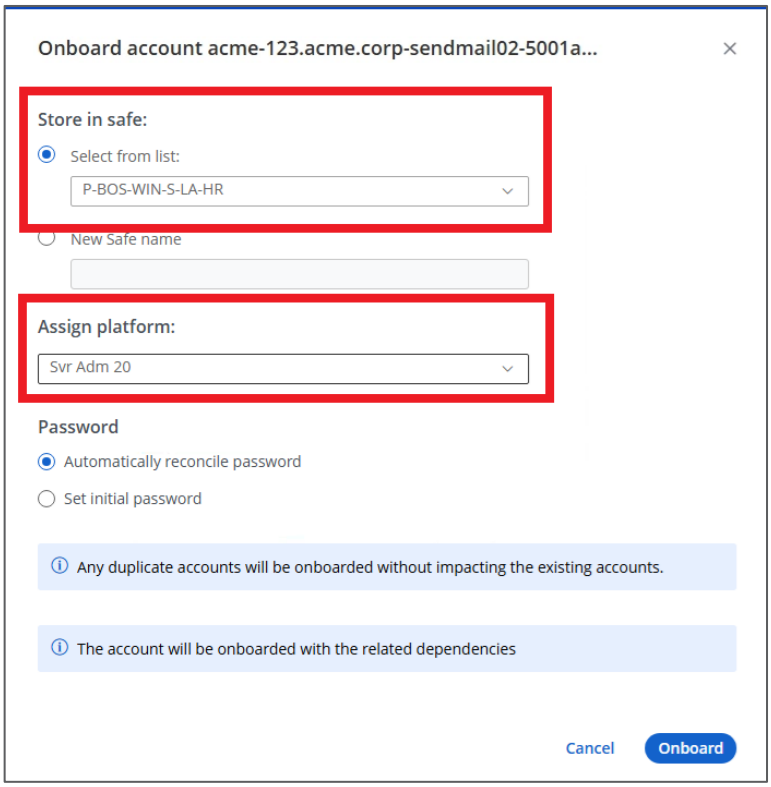
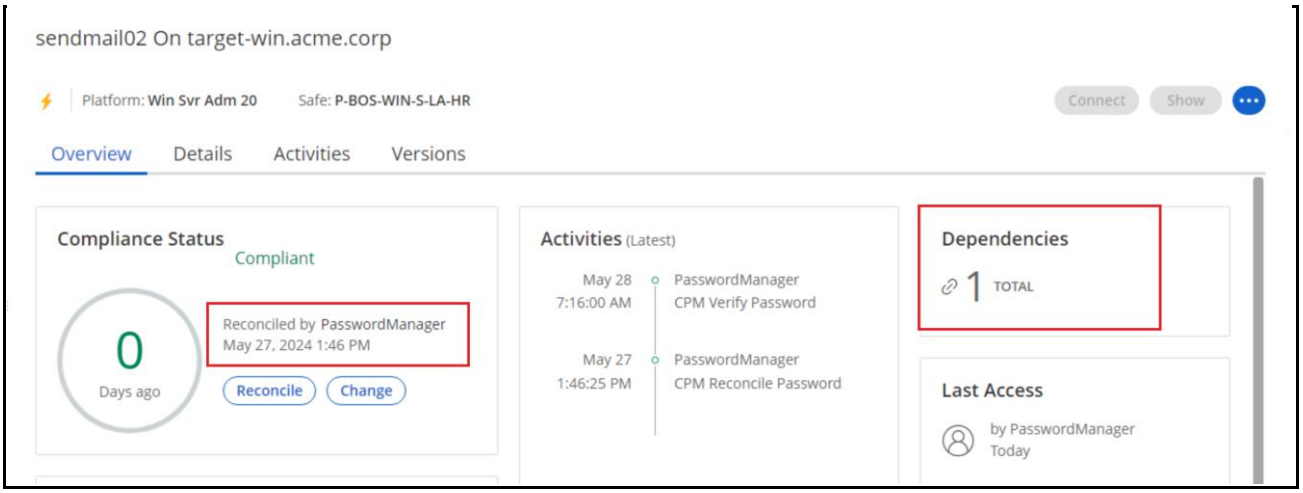


4. Press **Onboard**. You should receive a message saying "Successfully onboarded 1 account(s) and related dependencies.

5. Now go to the **Accounts View** and locate the newly onboarded account. Because the Platform was configured for automatic reconciliation, you should see that the account has been reconciled. Confirm that you can also see there is a dependency associated with the account.



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6. In order to change the password for the *sendmail02* account, you will need to associate a log on account for the scheduled task as we did for the first scheduled task account.

7. To confirm the scheduled task is also working, open a command line interface and input the following command.

```
schtasks /run /s acme-123 /tn SchedTask02
```

8. Now, login to the email client as *Mike* and verify that you received the email confirming *SchedTask02* is working.

Add Multiple Accounts from File

Frequently there is a need to upload many known accounts into **CyberArk PAM** from an existing repository. This is especially valuable during the early stages of implementing **CyberArk PAM**, for example when migrating from another solution, or when onboarding a new department into the **PAM** solution. We can also upload accounts with their logon accounts already associated with them. This requires careful configuration of the file to upload. We will see how to do this in this exercise.

In this section you will:

- Upload a CSV file with multiple root accounts and their logon accounts

1. Open the File explorer on your **comp01** server and go to *c:\Add-accounts*. Open the *root-accounts.csv* file. Make sure to select ***Comma*** in **Separator Options**.

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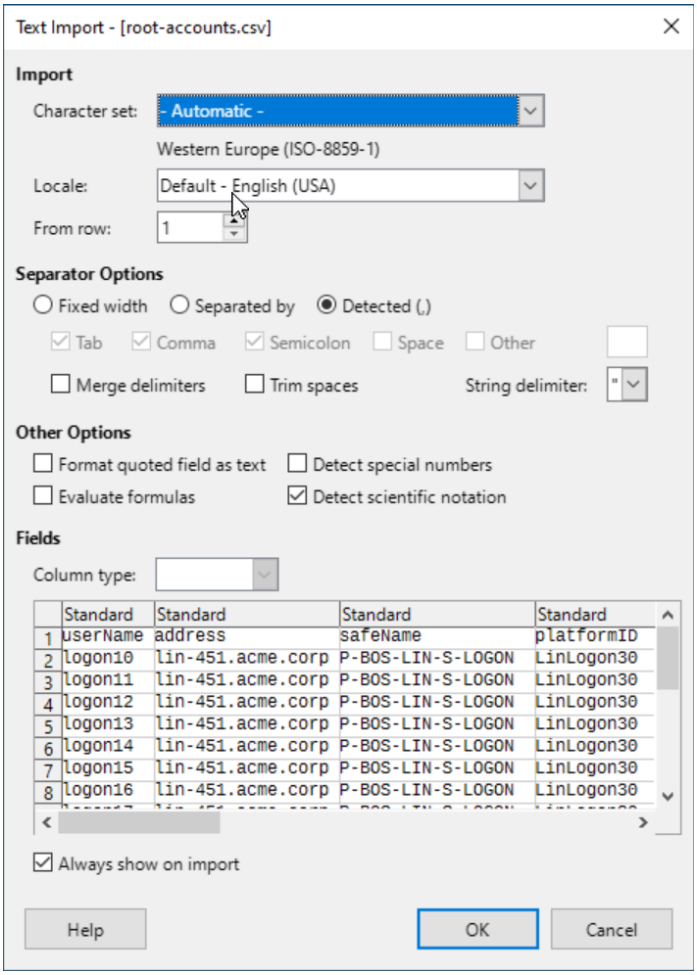

2. Review the file and the properties of the accounts we are about to upload to the **CyberArk PAM** solution. Note that the root accounts have three additional parameters with the logon account details. Based on our knowledge of **CyberArk PAM** and how it creates accounts, we can anticipate what the full name of these logon accounts will be. Using this file, we will create the logon accounts, then the root accounts

   We don't need to modify the file. You can close it.

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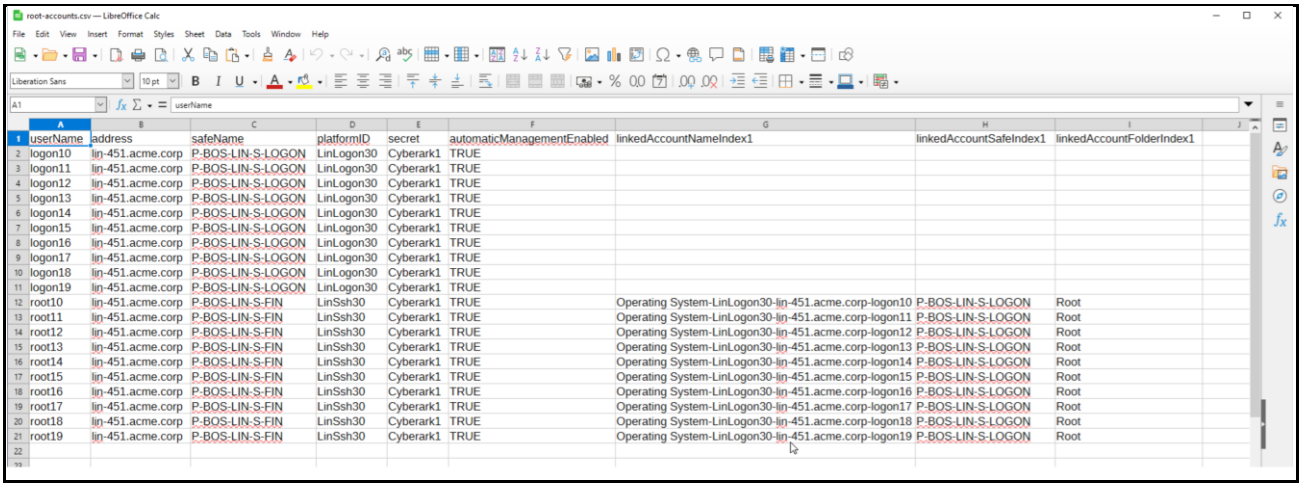
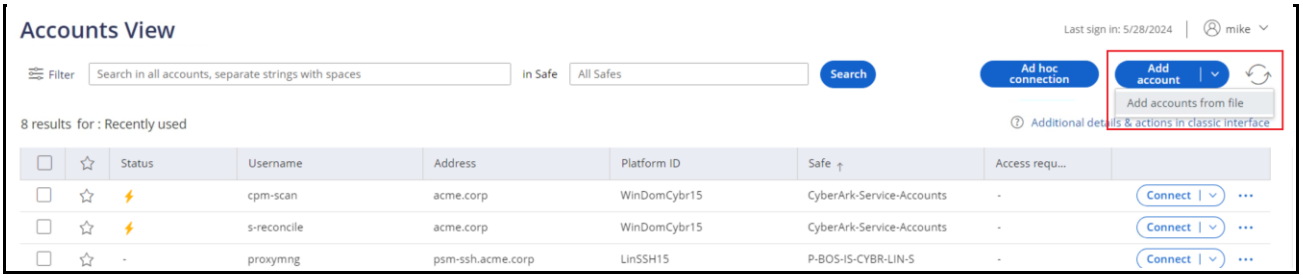


userName	address	safeName	platformID	secret	automaticManagementEnabled	linkedAccountNameIndex1	linkedAccountSafeIndex1	linkedAccountFolderIndex1
logon10	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon11	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon12	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon13	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon14	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon15	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon16	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon17	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon18	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
logon19	lin-451.acme.corp	P-BOS-LIN-S-LOGON	LinLogon30	Cyberark1	TRUE			
root10	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon10	P-BOS-LIN-S-LOGON	Root
root11	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon11	P-BOS-LIN-S-LOGON	Root
root12	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon12	P-BOS-LIN-S-LOGON	Root
root13	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon13	P-BOS-LIN-S-LOGON	Root
root14	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon14	P-BOS-LIN-S-LOGON	Root
root15	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon15	P-BOS-LIN-S-LOGON	Root
root16	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon16	P-BOS-LIN-S-LOGON	Root
root17	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon17	P-BOS-LIN-S-LOGON	Root
root18	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon18	P-BOS-LIN-S-LOGON	Root
root19	lin-451.acme.corp	P-BOS-LIN-S-FIN	LinSsh30	Cyberark1	TRUE	Operating System-LinLogon30-lin-451.acme.corp-logon19	P-BOS-LIN-S-LOGON	Root

3. Now, login to the **PVWA** as *Mike.*

Note:	We can perform this action as *Mike* because *Paul* gave the **CyberArk Vault Admins** group account management permissions on this Safe.

4. Go to **ACCOUNTS | Accounts View** and select **Add accounts from file**.



5. First, review the instructions on the page. Note you can also download a sample CSV file. When you are ready, click on **Drag and drop file or browse**. Navigate to *c:\Add-Accounts* and select the *root-accounts.csv* file. Review the page and click **Upload**.

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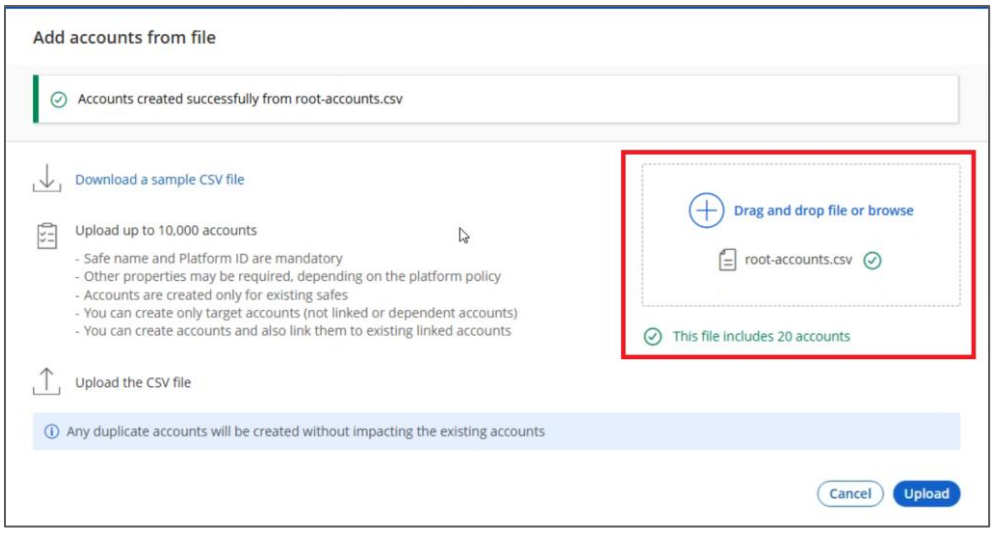
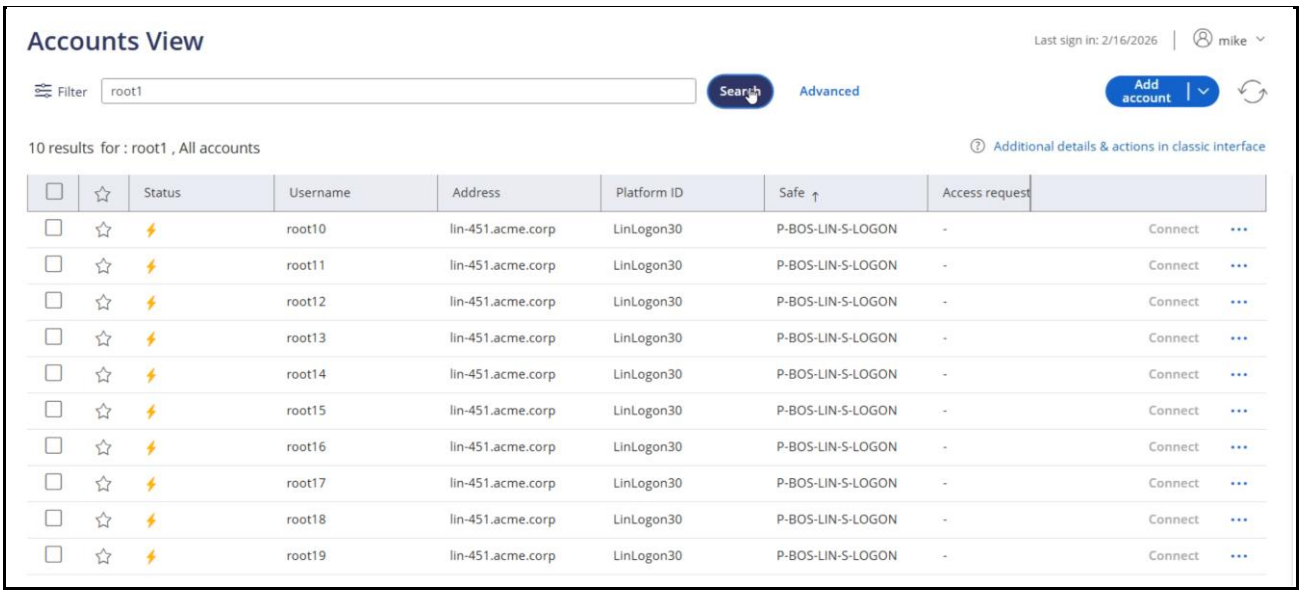


6. You should see a notification on your screen reporting the success of the action. Refresh the page. Search for *logon* and confirm the accounts were onboarded.



Changing the passwords on multiple accounts

Until recently, in order to change the password (or verify or reconcile) for multiple accounts, you needed to do it in the Classic UI. Let's see how to do it in the new interface.

We will first change the passwords for the logon accounts, then for the root accounts that use them. We don't want to change them all at once, because the CPM might try to change the accounts at the same time, which would result in errors.

1. First, go to the **Accounts View** and open the **Filter**. Filter the list for the following:

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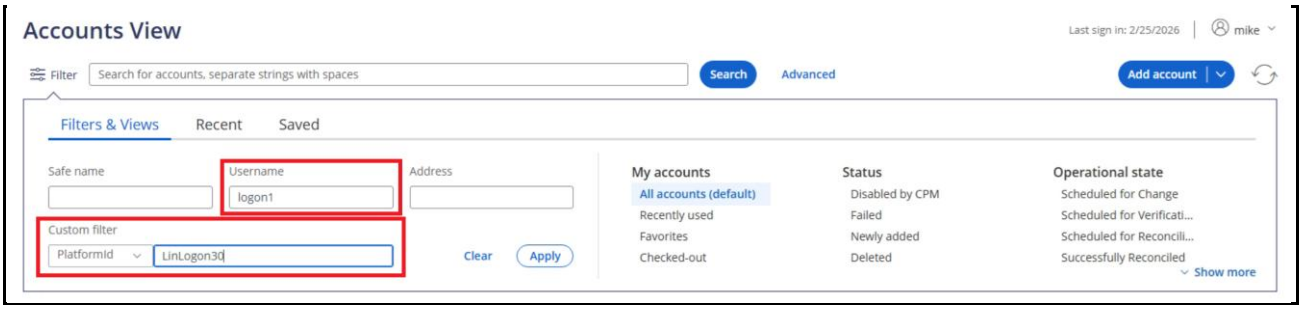
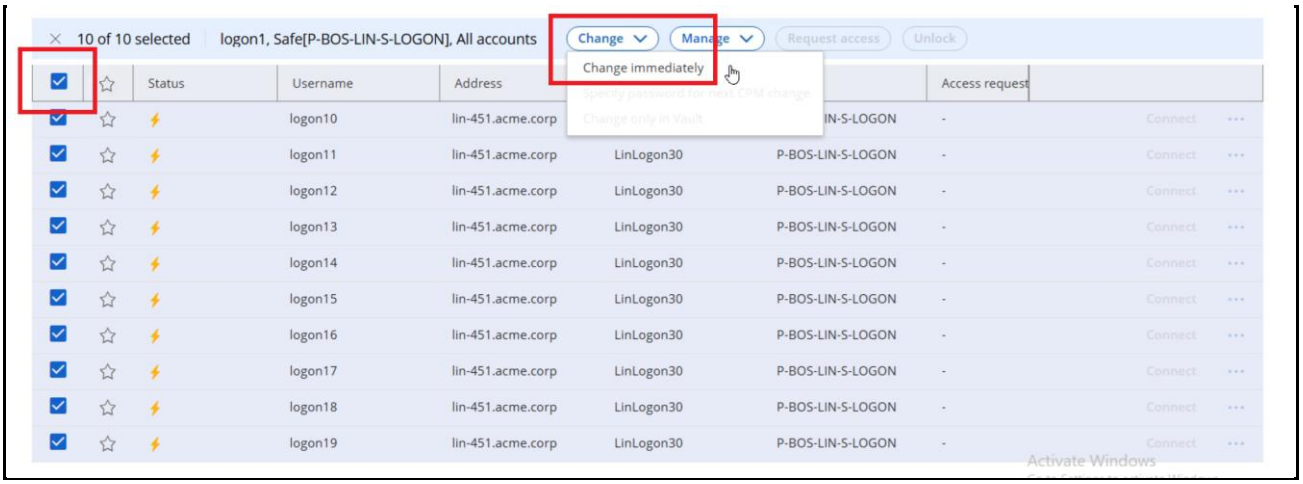
Username ***logon1***

And then under **Custom filter**, drop down the list and locate **Platformid**. Enter *LinLogon30* as shown in the image below. Click **Apply**.



2. Tick the box to select all the accounts shown and then drop down the **Change** button and select **Change immediately**. You will be prompted to confirm the action. Click the **Change** button.



3. After a few moments, you should see that all the passwords have been changed by **PasswordManager**.

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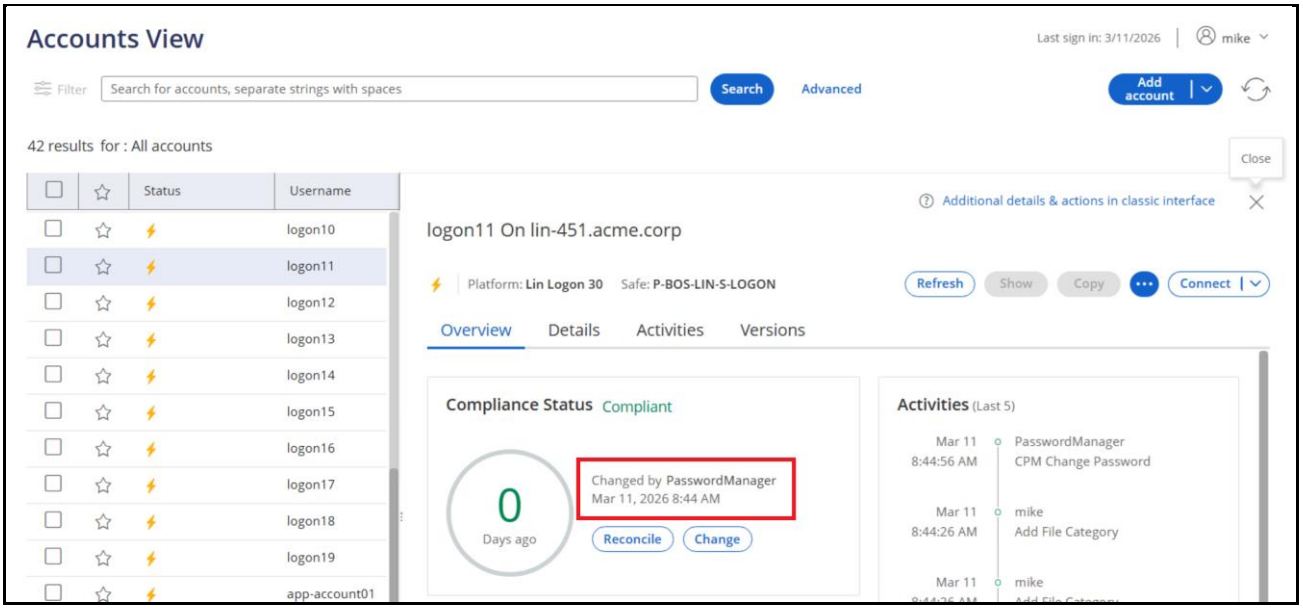

4. Once the all the logon accounts have been changed, perform the same action for the root accounts, this time filtering for **Username** *root1* and the **Platformid** *LinSsh30*.

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Privileged Session Management – Part 1

In this section, we will perform several tests to see the various privileged session management options that are available with **CyberArk PAM**.

First, we will disable the **PSM** globally and then activate it for specific platforms using exceptions. We will then perform tests to ensure that privileged session management is functioning properly using the various connection methods available:

- Privileged Session Manager (**PSM**) through the **PVWA**
- PSM for Windows
- PSM for SSH

Classic PSM

By ***classic PSM***, we mean connecting through the **PVWA** to the **PSM** running on a Windows server, which is what we have been doing so far, activating the **PSM** globally for all active Platforms. In this section, we will see how to activate the **PSM** using exceptions, which will give us greater control over to what and how our users connect.

Remove Privileged Access Workflows Exceptions

To simplify the **PSM** testing, we will first disable the **Privileged Access Workflows** that we modified in earlier exercises. That way, we will not have dual control or exclusive access interrupting our testing. We can perform this action using the classic interface. Just bear in mind that this interface will probably be removed in a not-too-distant future release.

Note:	Do **NOT** disable the Privileged Access Workflow ***Allow EPV transparent connections***.

1. Log in to the **PVWA** as *Mike* using LDAP authentication.
2. Go to **Administration | Platform Management** and go to the platform ***Lin Ssh 30***. Click on the ellipsis and select **Add exception**.
3. Go to the **Access workflow** tab and deactivate all the options **except** the last two:
   - ***Allow users to view passwords***
   - ***Allow EPV transparent connection***

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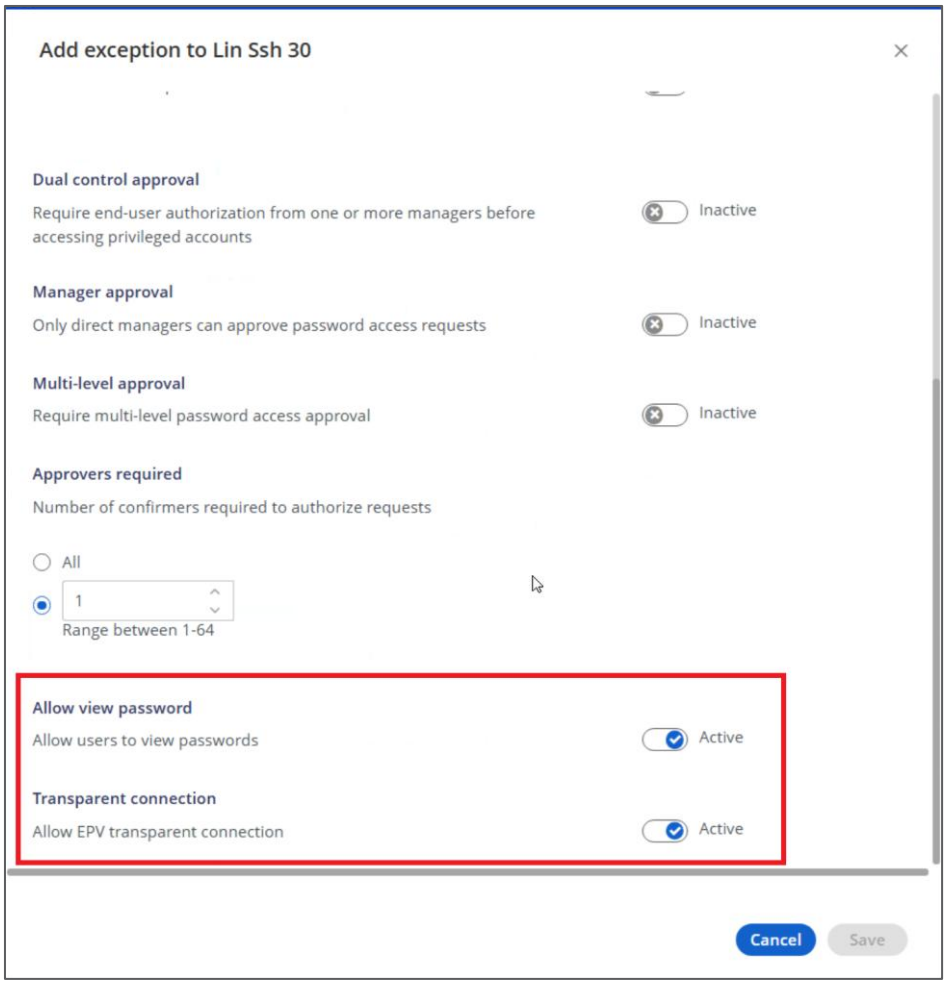

4. Remember to click the **Save** button.

5. Do the same thing for the Platform **Svr Adm 20**.

Disabling the PSM Globally

As we saw in one of our first exercises, the **PSM** is enabled through the **Master Policy**. The **PSM** can be enabled either globally for all platforms or disabled globally and only activated through exceptions, which is what we will test here.

The **Master Policy** can now be edited in the new interface, so we will see how to do that in this exercise.

1. Login to the **PVWA** as *Mike* using LDAP authentication.

2. Go to **Administration | Platform management**. Drop down **Master policy** and select **Edit master policy**.

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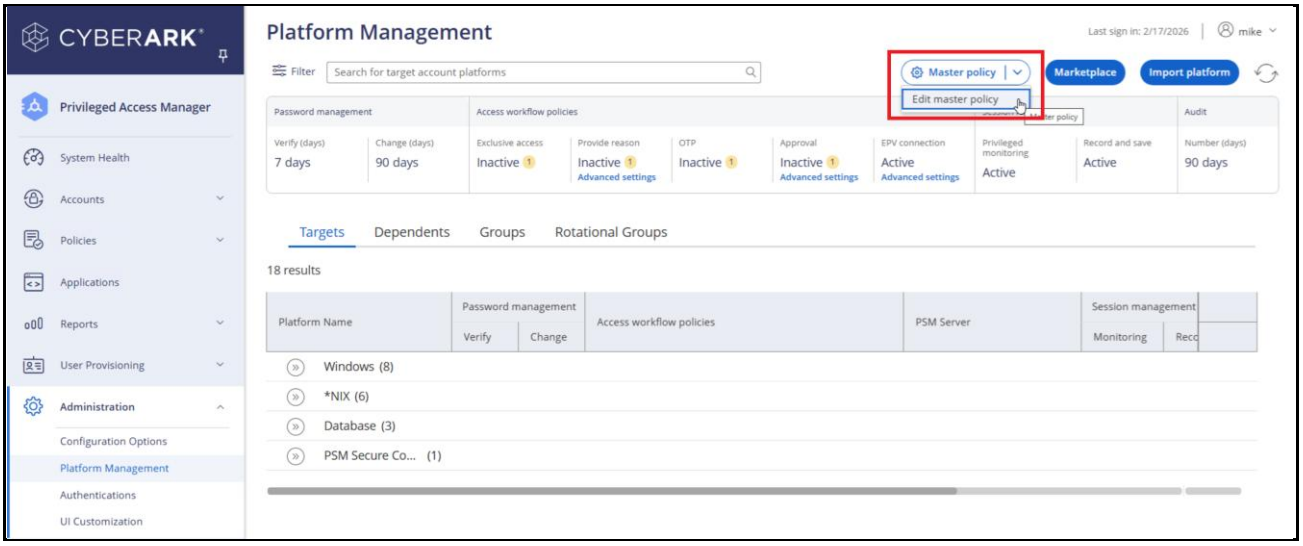
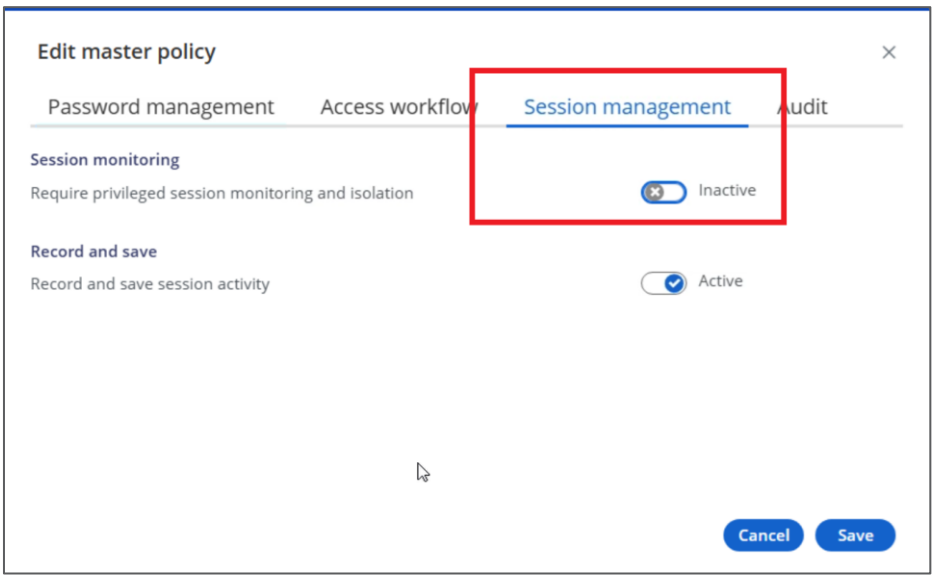


3. Click on the **Session Management** tab and then switch the toggle for **Session monitoring** to ***Inactive***. When you have finished, click **Save**.



4. Remain in the **Platform Management** section.

Activate the PSM by exceptions

This method enables users to connect via the **PSM** only to those accounts that are associated with Platforms that have been explicitly enabled by a Vault administrator. For instance, users will be able to connect to a target system using a *root* account that requires a logon account, but not directly via those logon accounts.

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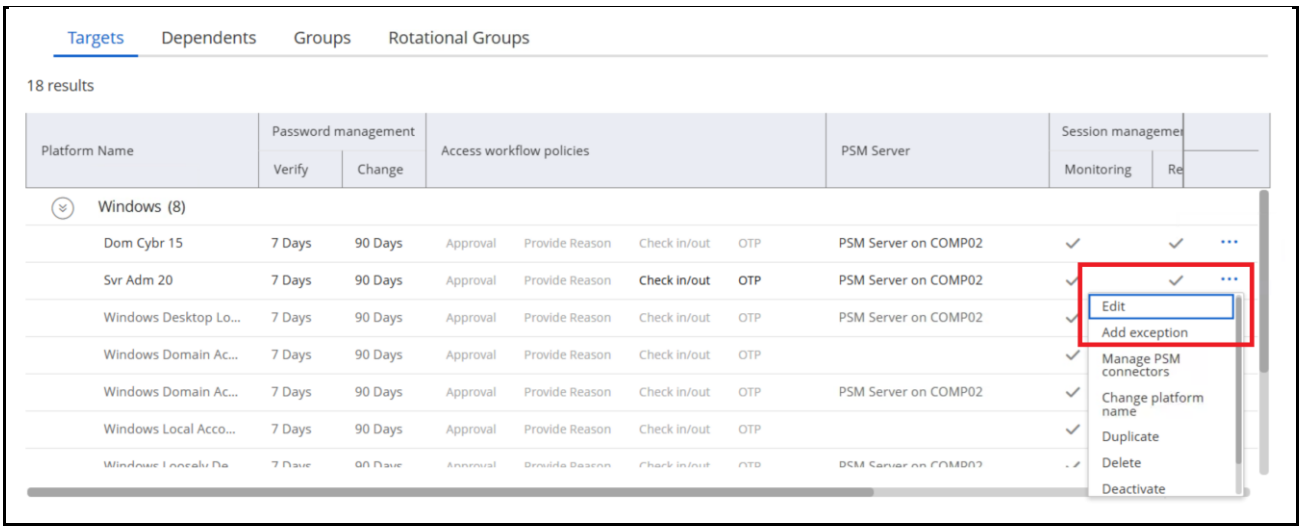
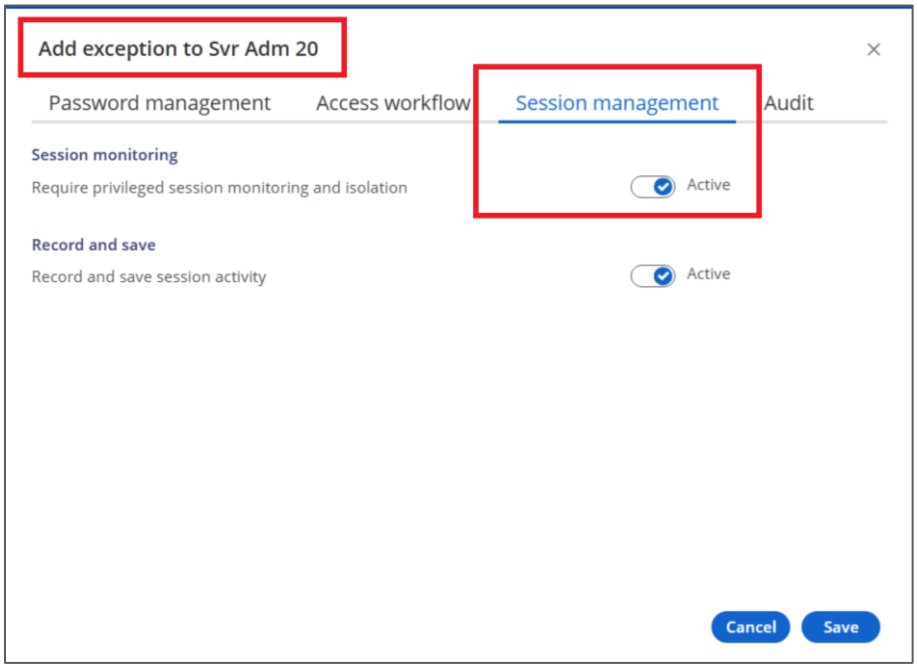
Adding Exceptions

Let's activate the **PSM** by exception for our Platforms. We will begin with the Platform for local admins on Windows servers.

1. Go to **Administration | Platform Management**. Expand the **Windows** section under **Targets**. Then click on the ellipsis at the end of the line for **Svr Adm 20** and select **Add exception**.



2. Click on the **Session management** tab and then activate the toggle for **Session monitoring**.



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Repeat the above steps to enable the **PSM** for the ***Lin Ssh 30*** and **Ora Dba 30** platforms.

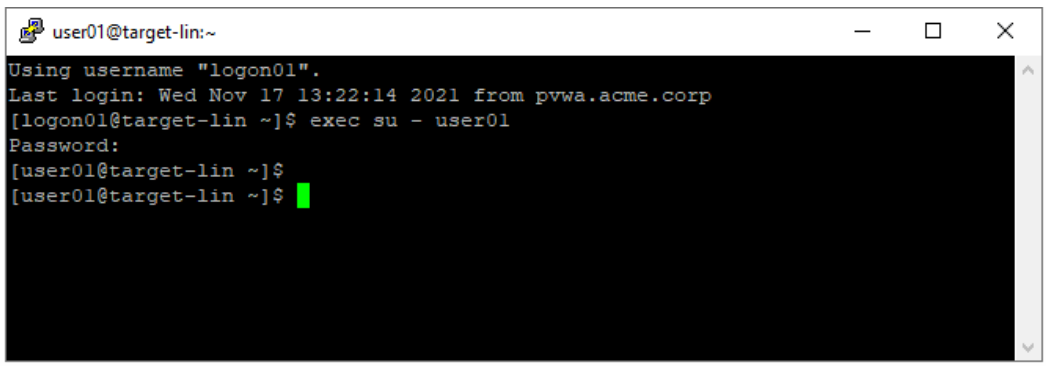
Connect with a Linux Account

In this first test, we will connect via the **PSM** exactly as we did previously, the only difference being that now the **PSM** is disabled by default and enabled by exception.

1. Login to the **PVWA** as *Carlos*, go to the **ACCOUNTS** page, and locate *root01*. Press the **Connect** button. As before, you will receive an RDP file. Click on it to launch the RDP session.

2. If everything was configured correctly, you should see a message that your session is being recorded.

3. And you will be connected to the target server.



4. Optionally, run some Linux commands.

```
mkdir test
rm -R test
```

5. Type **`exit`** to end the session.

Note: Notice that the connect button for the logon accounts are now greyed out.

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Connect with an Oracle Account

1. Log out of the **PVWA** and log back in as the user *Robert.*
2. In the **Account view**, find the account *dba01* and click the **Connect** button.
3. Once again, you will see a message stating that your session is being recorded.

Note:	If you receive a Remote Desktop Connect pop-up, "Your Remote Desktop Services session has ended", retry the connection component. You may have to connect a couple of times before seeing the message.

4. In a later exercise, we will be searching for activity in this session by the commands that were run. Execute the following commands so that we will have some content for later:

```
select * from employees;
```

You will see a list of employees with their IDs, their salaries, and home cities.

Now let's give one of them a raise.

```
UPDATE employees set employee_salary = '75000' where employee_id = 0003;
commit;
exit
```

5. Type **`exit`** to end the session.

HTML5 Gateway

In this section, we will see how to configure the **PSM HTML5 Gateway**, which enables us to tunnel sessions between end users and the **PSM** server using a secure WebSocket protocol (port 443). This eliminates the need to open an RDP connection from the end user's machine. The RDP session is delivered to the end user through a browser tab, rather than via an RDP window.

Enable the HTML5 Gateway

Note:	In this environment, the HTML5 Gateway has already been installed for you. It is running on **lin-451**, but it has not been enabled in the **PVWA**.

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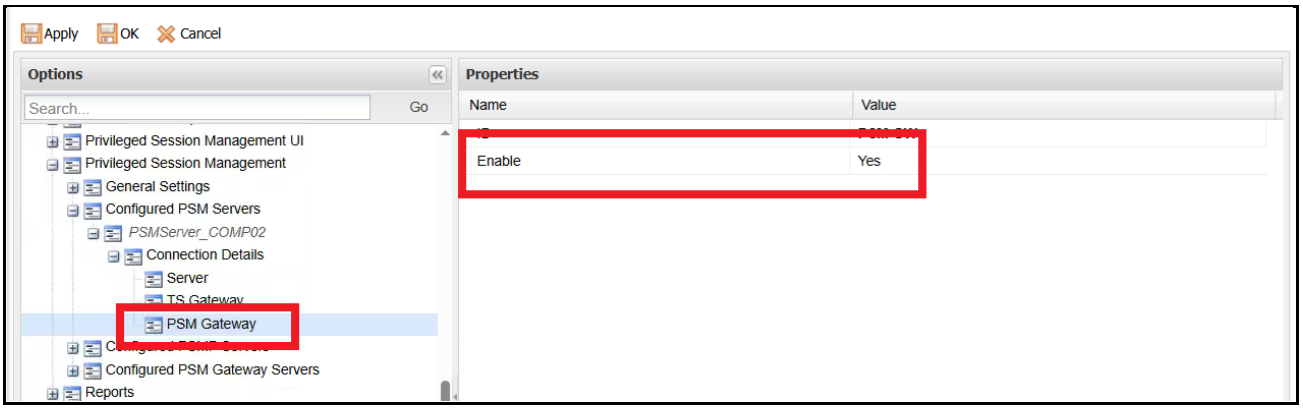
1. First, login to the **PVWA** as *Mike*, go to **ADMINISTRATION | Configuration Options | Options**.

2. Next, go to **Privileged Session Management | Configured PSM Servers | PSMServer_COMP02 | Connection Details | PSM Gateway**.

3. Set the *Enable* parameter to ***Yes*** and click the **Apply** button.



4. Click on **Apply** and then **OK** to save your changes.

Connect via HTML5 Gateway

In this section we will see how to connect to a target system through an RDP session delivered through your browser. We will also see how to transfer files from your local desktop to the remote system and the reverse.

1. Now log in as the user *John* and go back to the **ACCOUNTS** page and locate *administrator*.

2. Press the **Connect** button. This time, instead of downloading an RDP file, you will receive a pop-up asking whether you want to map your local drives (yes) and whether you want to *Connect using HTML5 GW* (also yes). Enable **Connect with your browser**. Optionally provide a reason for launching the connection and then click **Connect** to launch an HTML5 connection.

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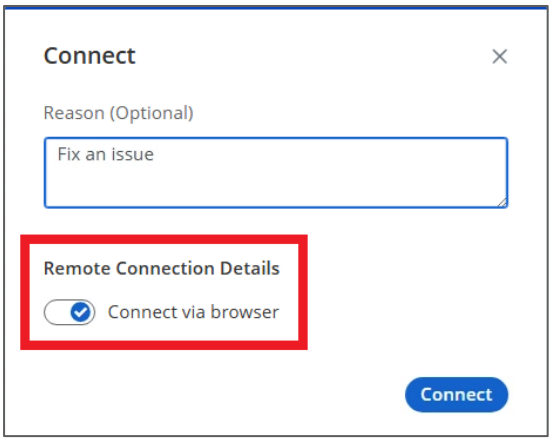
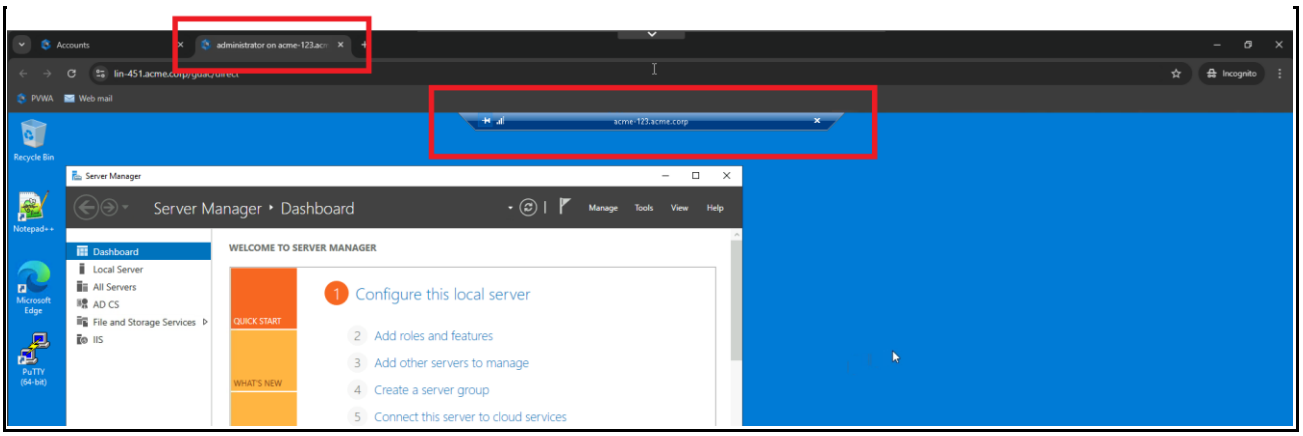


3. Press **Yes** to accept the host's RSA key, if asked.

Note:	The ability to toggle between RDP file and HTML5GW connections is defined at the **Connection Component** level. For your convenience, the functionality has been preconfigured for RDP and SSH connections in this lab. To enable this functionality for connection types other than RDP or SSH, go to ***Options | Connection Components | PSM-RDP | User Parameters*** and copy the ***AllowSelectHTML5*** parameter. Then paste it into a different connection component, for example: ***PSM-WinSCP***

4. A new tab opens in the browser and you can see the RDP toolbar at the top. Keep the HTML5 connection open.



5. You can close this session either by closing the tab or by signing out. Make sure that you don't accidentally power off the server.

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Transfer files via HTML5 GW – Optional exercise

In this section we will see how to copy a file from our workstation to the remote machine via the **HTML5 Gateway**.

Enable drive mapping

For this exercise to work, you must first enable the drive mapping feature.

1. Log into the **PVWA** as *Mike* and navigate to: **Administration | Configuration Options | Options | Connection Components | PSM-RDP | User Parameters**. Select the parameter **AllowMappingLocalDrives**.

2. Enter the following values:

DisplayName	*Map local drives?*
Value	*Yes*
Visible	*Yes*

When you are finished, click **OK** to save the changes.

Test drive mapping

1. Relaunch *John*'s connection with the *administrator* account and activate **Map local drives?**

2. Once connected, resize the Chrome browser window with the HTML5 connection in such a way that you can view the HTML5 session and the desktop for **COMP01**, as shown in the image below.

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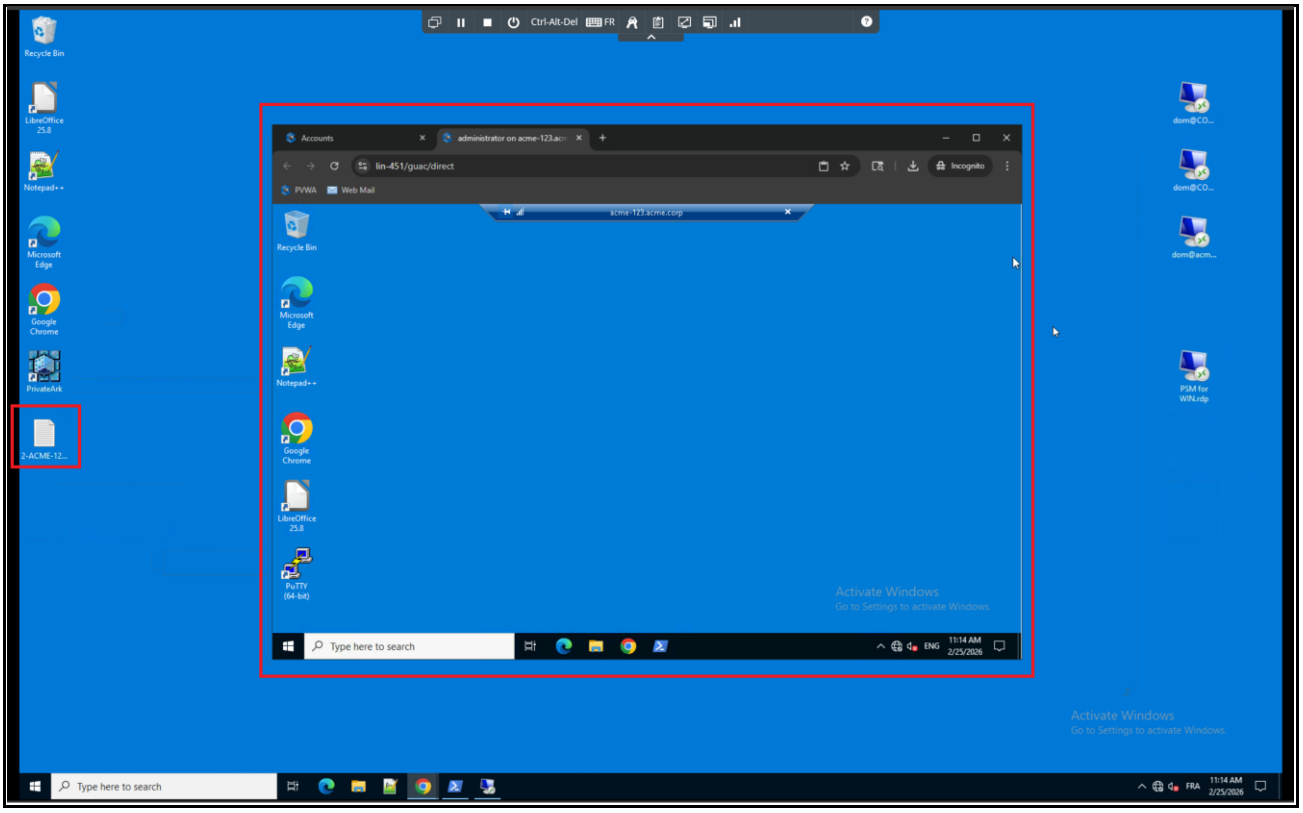
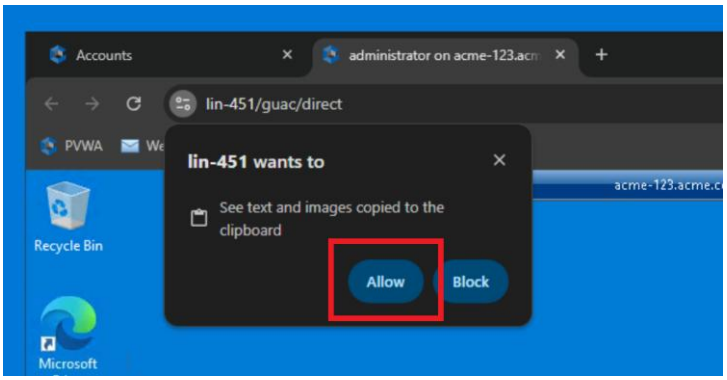


3. You will see a pop-up in Chrome asking you to allow the gateway server access to the clipboard. Click **Allow**.



4. On the **COMP01** desktop, you will find a file named *2-ACME-123.txt*. Drag and drop this file into the browser RDP window.

5. You should be able to see the following message stating that the file has been copied to the mapped drive **Z**, which you can view on the remote machine **ACME-123**. When the file transfer is finished, click on **Close**.

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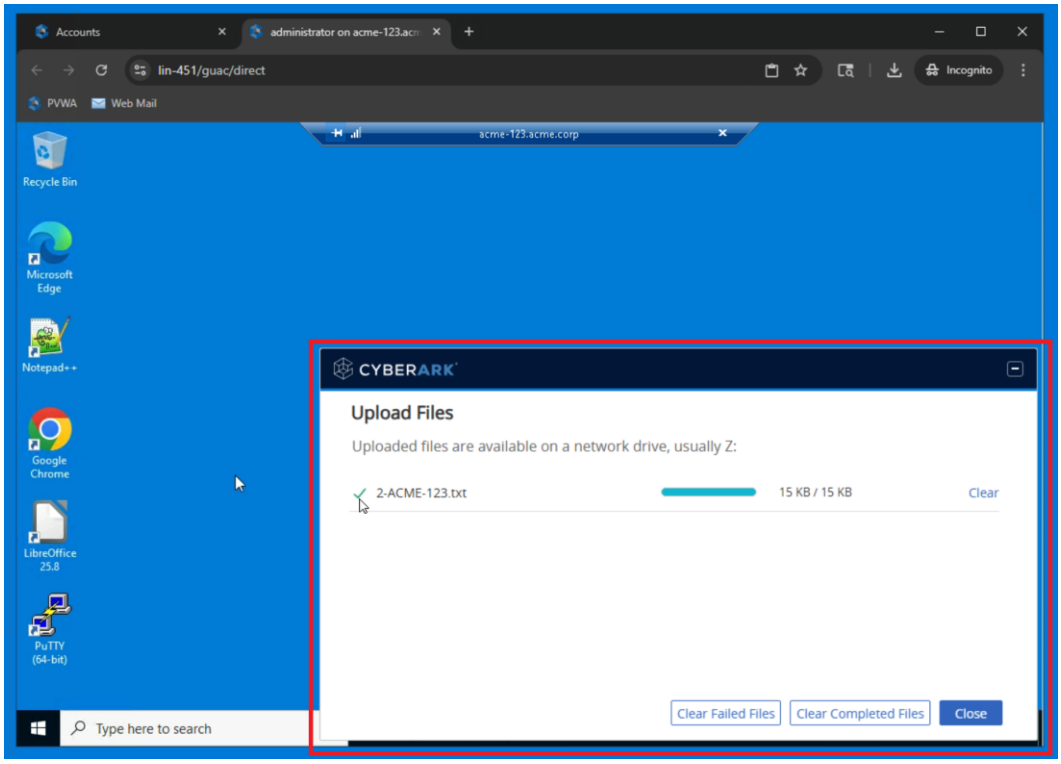
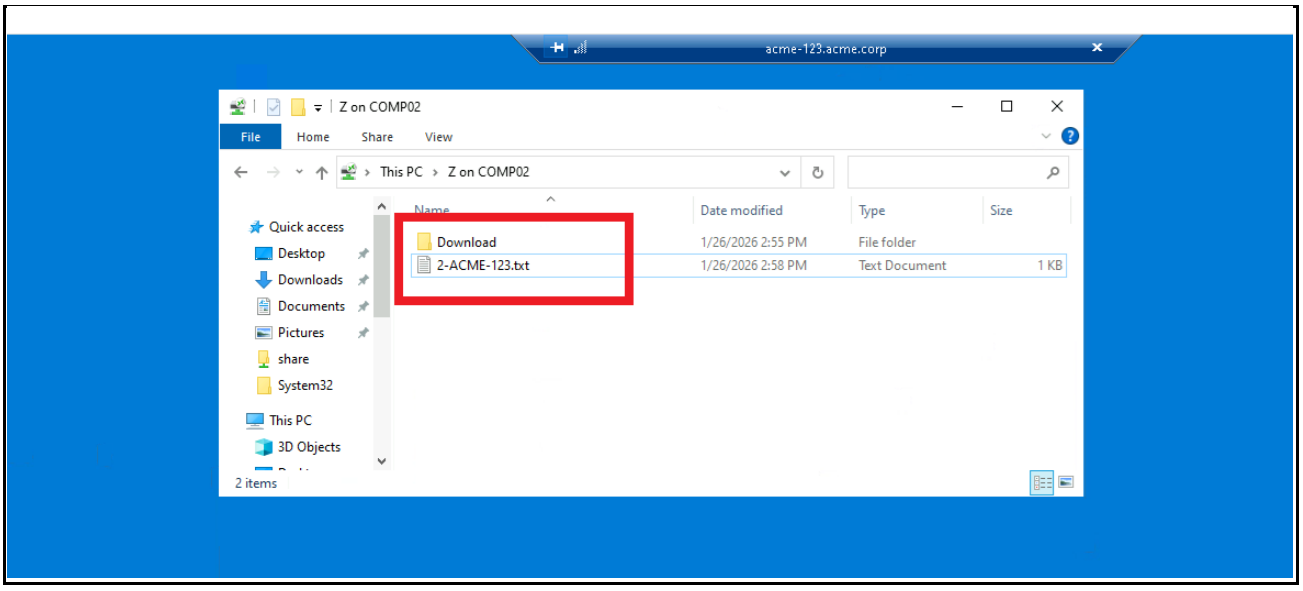


6. Next, inside the HTML5 session, open the file explorer, browse to the **z:** drive, and copy the file to the desktop on **ACME-123**.



Now we are going to copy a file in the other direction, from the remote machine back to our workstation.

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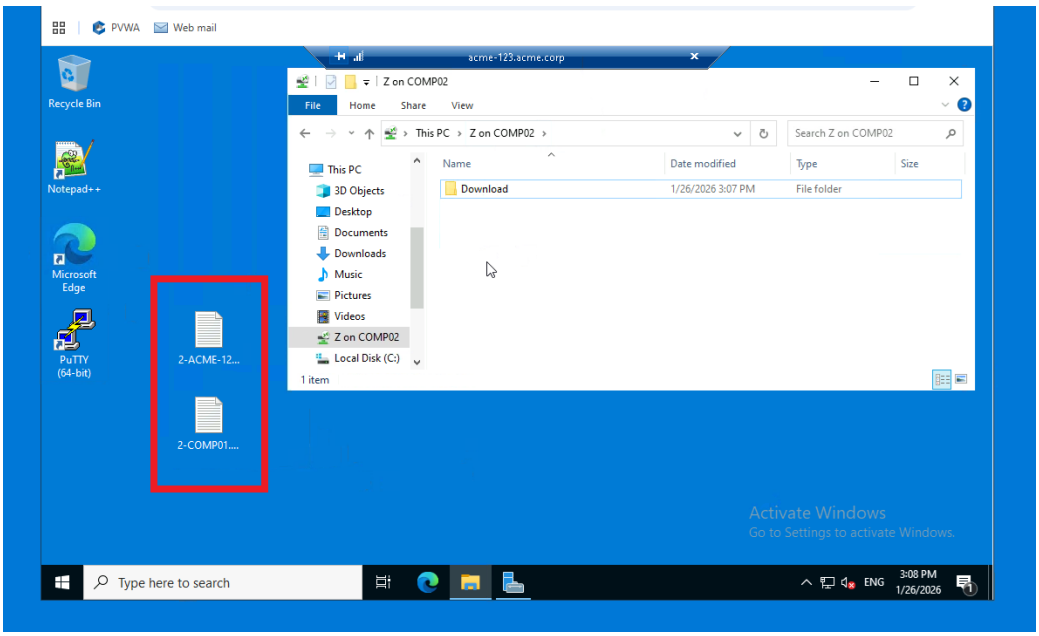
7. Still working in the browser RDP window (so on **ACME-123**), make a copy of the file named *2-ACME-123.txt* that is now on the Desktop of **ACME-123**, and name it *2-COMP01.txt*.



8. Next, open the *Download* directory **Z on COMP01**. Drag and drop the *2-COMP01.txt*. file that is on the desktop of the **ACME-123** into the *Download* directory.

Note:	It will look like nothing has happened, but if you dropped the file to be transferred into the mapped *Download* directory, it has been transferred to the *Downloads* directory on the machine from which you initiated the connection. That is the **COMP01** server.

9. When you are finished, disconnect from the target server.

Connect using PSM Ad-Hoc Connection

Next, you will configure a **PSM Ad-Hoc Connection** (previously known as **Secure Connect**), which allows you to launch a **PSM** connection using unmanaged accounts. We will:

- Duplicate the **Platform** *PSM Secure Connect*, giving it an appropriate name.
- Activate the new **Platform**
- Activate the **PSM** for this **Platform** using an exception
- Test the Ad Hoc Connection

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1. First, log into the **PVWA** as *Mike*, and go to **ADMINISTRATION | Platform Management**.

2. Locate the **Platform** *PSM Secure Connect* and duplicate it. Give it the name ***PSM Secure Connect US***.

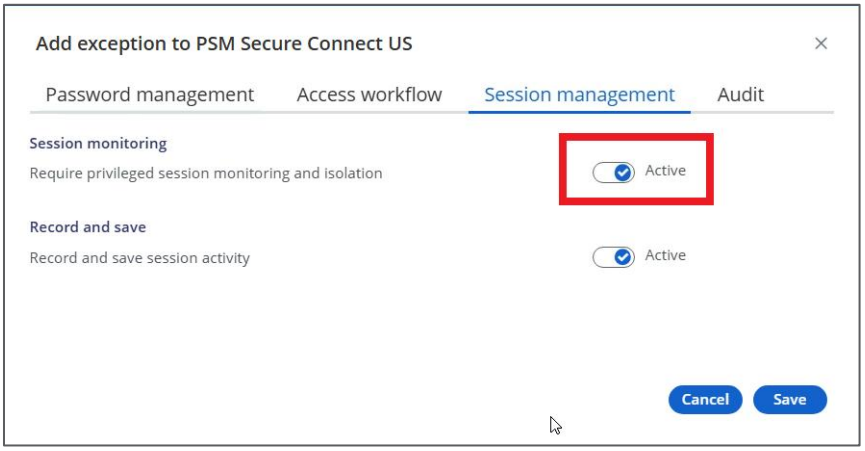
Hint:	*PSM Secure Connect* is at the bottom of the list.

3. Because we have duplicated an inactive Platform, we will need to **activate** the new **Platform**.

Note:	As a matter of best practice, you should configure a dedicated recording Safe for this Platform.

4. Add an exception to the Master Policy by clicking on the ellipsis at the end of the line for the new Platform and clicking **Add exception**.

5. Click on the **Session management** tab and set the toggle for **Require privileged session monitoring and isolation** to ***Active***. When you are ready, click **Save**.



We could wait for the **PSM** to refresh its policies, but that could take up to 20 minutes. We can move things along by restarting the service.

6. To restart the **PSM** Service, you will need to connect to the server **COMP02**. There is an RDP shortcut on *Mike*'s desktop for the user *dom* to connect to that machine. Once connected to **COMP02**, open the Windows Services app and restart the **CyberArk Privileged Session Manager** service.

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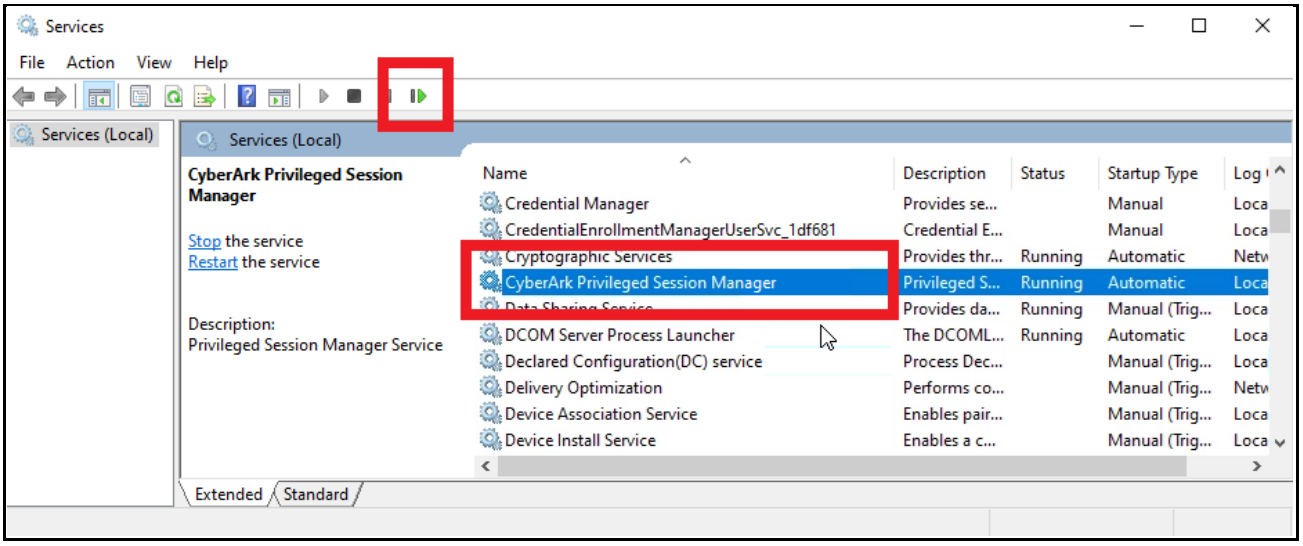
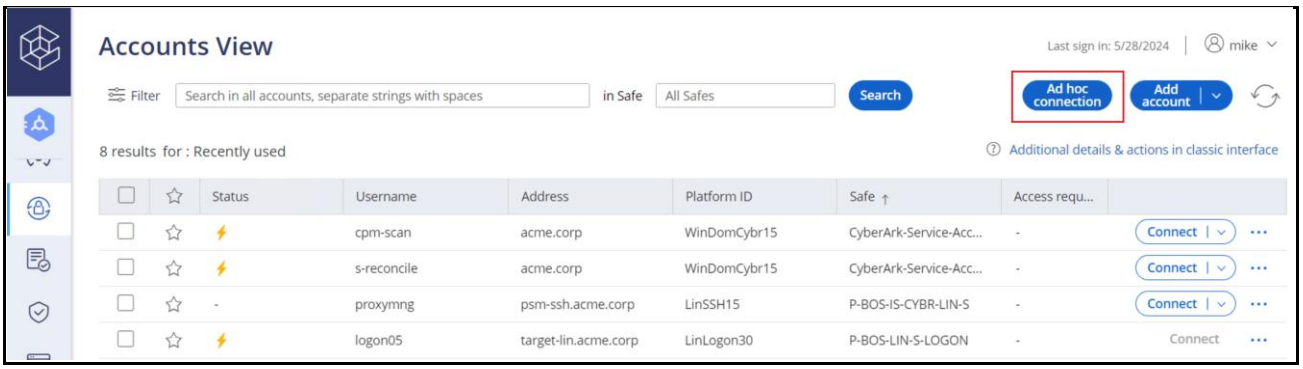


7. Now go to the **ACCOUNTS** page and click on **Ad hoc connection**.



8. Enter the following and press **Connect**:

Platform	*PSMSecureConnectUS*
Client:	*WinSCP*
Address:	10.0.21.2
User Name:	*linuxusr01*
Password:	*Cyberark1*
Map Local Drives:	*Checked* (scroll down)
Port	*22*

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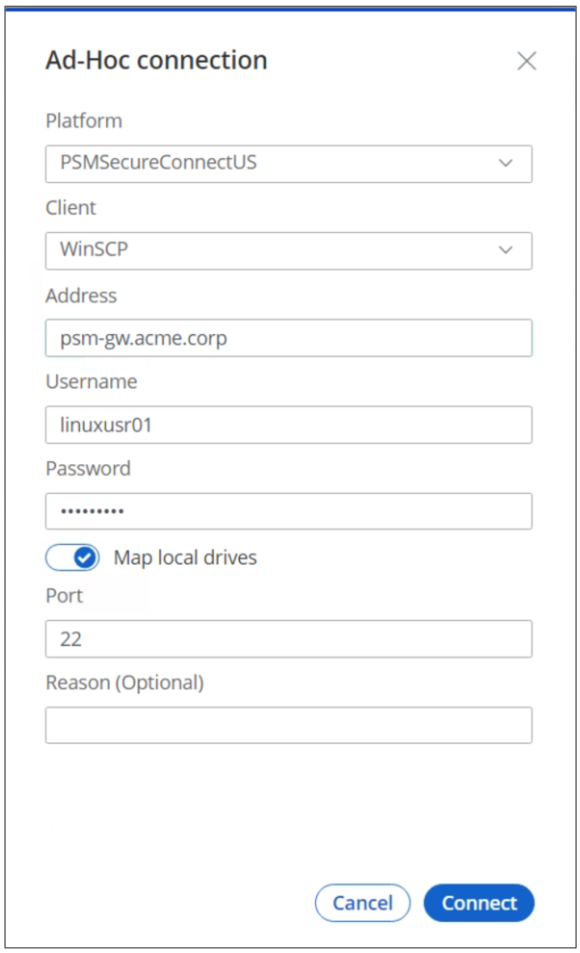
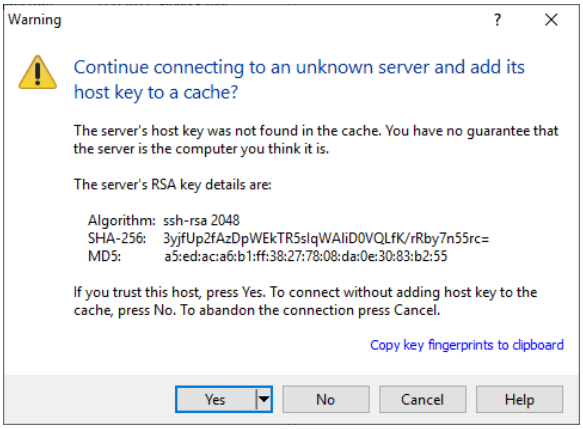


9. Press **Yes** to accept the host's RSA key.



Optional:	When you have connected to WinSCP, copy a file from the **PSM** server to the target machine. Suggestion: *C:\Add-Accounts\root-accounts.csv*

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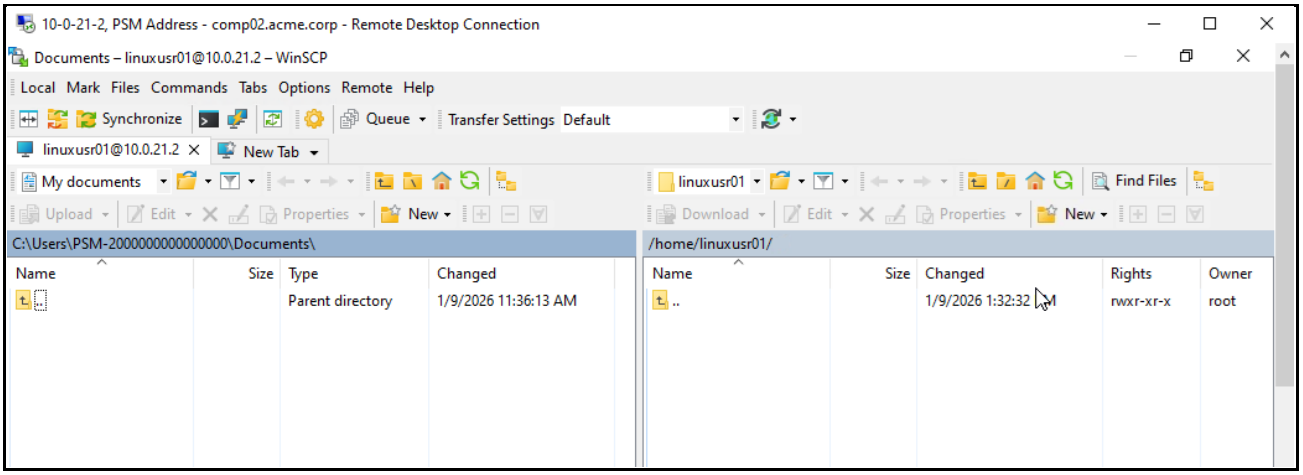

10. Press **F10** to exit and quit the application.

Privileged Session Manager for Windows

PSM for Windows (previously known as "PSM RDP Proxy") enables users to connect through **PSM** to any remote target securely with a standard remote desktop client application like *mstsc* or an RDP connection manager.

You can also use preconfigured RDP files. When using RDP files, you can configure a single RDP file to connect through the **PSM** without providing the target system details or configure separate RDP files that include the target system details in advance. In this exercise we will look at both options for using preconfigured RDP files.

Connect using RDP file

In the first example, we will use a preconfigured RDP file without providing the target system details in advance. In this way, we are able to bypass the **PVWA**, but not **CyberArk** security.

1. On the desktop of the **comp01** server, you will find an RDP file named *PSM for WIN.* Double click on the file. When prompted, click on **Connect**.

2. Accept to run the application.

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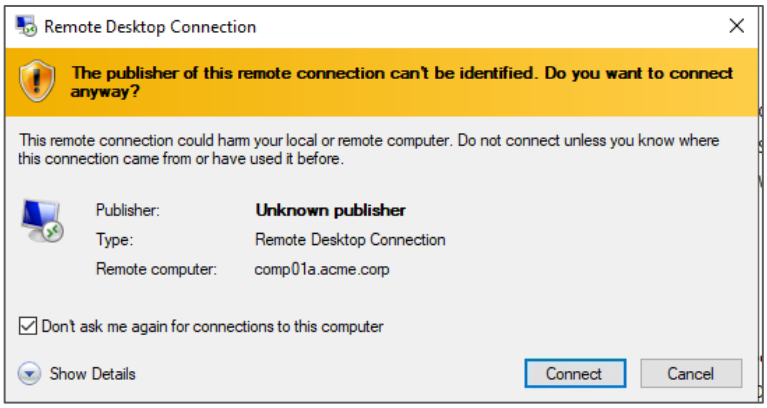
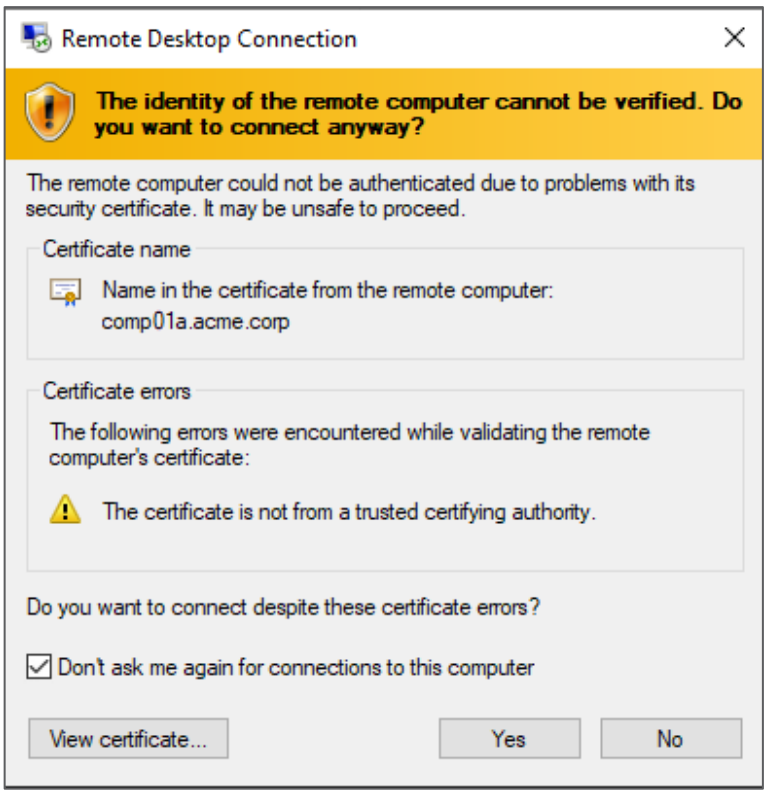


3. You may receive a certificate warning. Just click **Yes** to accept the certificate.



4. Enter the *John*'s domain credentials:

Vault username:	*John*
Password:	*Cyberark1*

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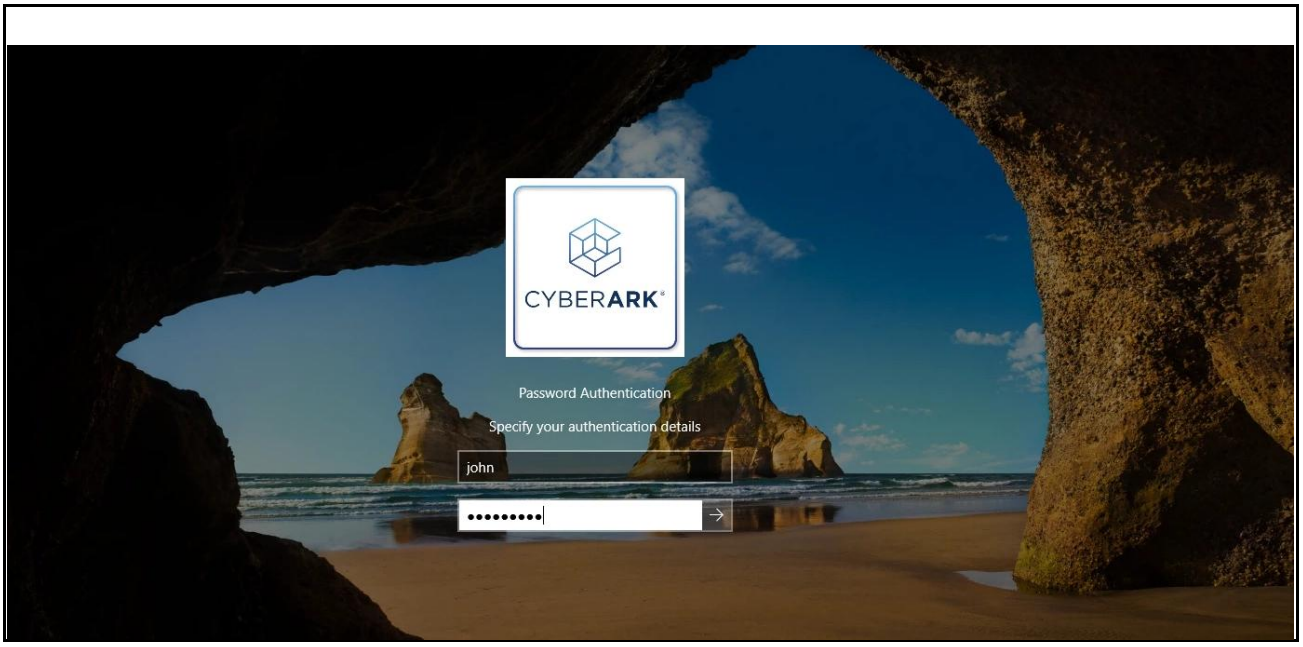

5. Next, input the target system details:

User Name:	*administrator*
Address:	*acme-123.acme.corp*

Note:	The account name and address information must be provided here exactly as it is shown in **CyberArk PAM**. Because this account was created with the FQDN as the address, you cannot connect with the IP address or the short domain name.

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6. Lastly, specify **PSM-RDP** as the connection type:

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7. Confirm you were able to connect to the target system as *administrator*. Then disconnect (or sign out) from the target system.

Connect using RDP file with the target system details

We can make our jobs even easier by including the target details directly in the RDP file.

1. Right-click to open the ***PSM for WIN*** RDP file for editing using Notepad++.

2. Scroll to the bottom of the file. Note the two different alternate shells in the file. One is commented out, the other is active.

3. Edit the RDP file as follows to switch the active shells. This will include the target system details in advance. We can also insert a line providing *John*'s user name so that on connection, *John* will only have to provide his LDAP password to connect. The lines should appear as below:

```
alternate shell:s:psm /u administrator /a acme-123.acme.corp /c PSM-RDP
username:s:john
# alternate shell:s:psm
```


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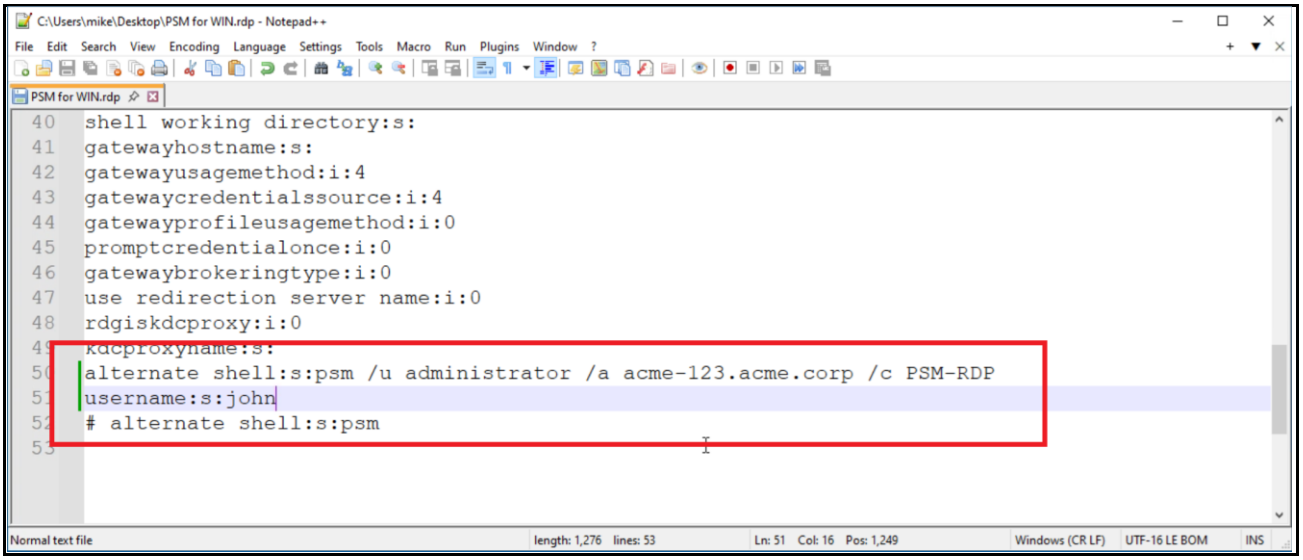

4. Save the file and exit Notepad++.

5. Double-click on the RDP file to launch the connection. If configured properly, you will be prompted only for the Vault user's LDAP password. After you authenticate as *John*, the connection to the target machine as *administrator* will be made automatically.

Note: You can use any RDP client application to connect to any target system via the **PSM**. When setting up your RDP client, make sure to input the following details:
- *PSM Address*
- *Vault username*
- *RDP Start Program setting*

For more details on configuring RDP clients, please review the online documentation.

Privileged Session Manager for SSH

PSM for SSH (previously known as **PSM SSH Proxy** or **PSMP**) is designed to provide a native Unix/Linux user experience, connecting to any SSH target.

1. Open up a Powershell window on the **comp01** server. There is a shortcut in the Task bar.

2. Use the following connection string to connect to the **lin-451** machine using the *root01* account where the **Vault** user is *Carlos*.

```
ssh carlos@root01@lin-451.acme.corp@10.0.21.2
```


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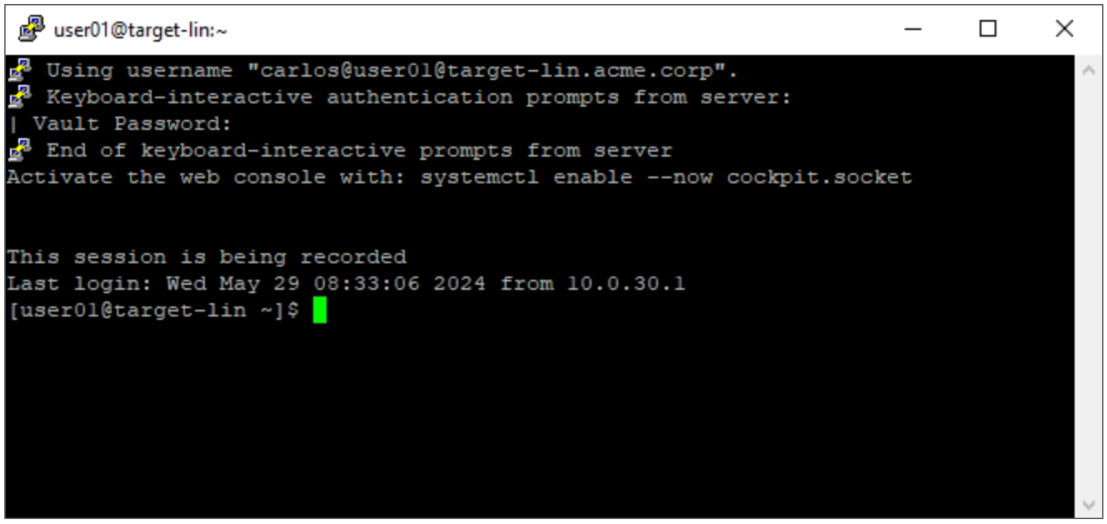
Hint:	To be able troubleshoot any connection issues easily, make sure you mark “Never” under “Close window on exit”.

3. Enter **yes** to accept the server’s host key when prompted.

4. When prompted for a password, enter the password for *Carlos* (password: *Cyberark1*).



5. Execute a few simple such as **`pwd`** and **`ls -al`** to generate some session activity. When you are done, enter **`exit`** and hit **Enter** to close the session.

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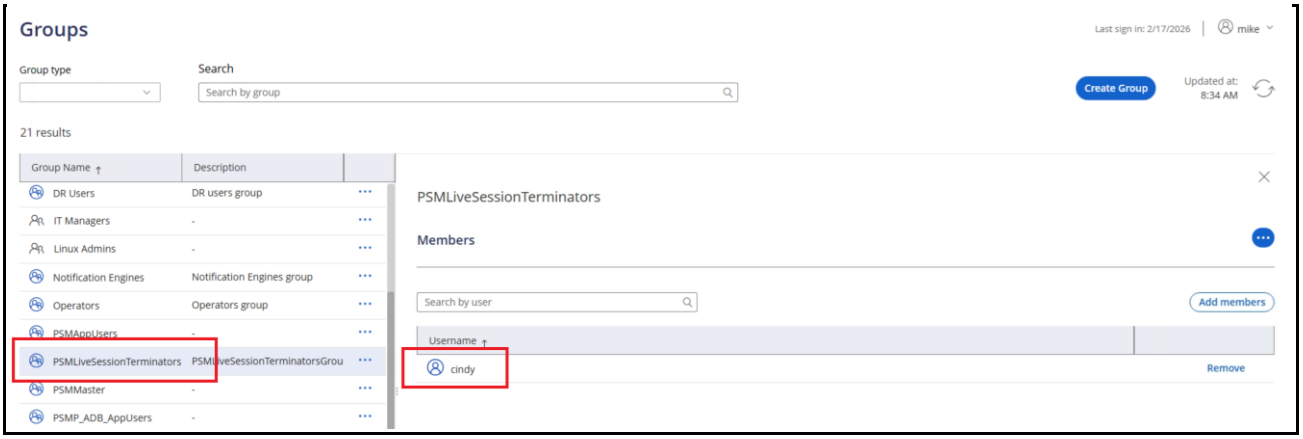
Privileged Session Management – Part 2

In this section, we are going to look at some of the audit information that was gathered by **CyberArk PAM** during our **PSM** testing. We will also be monitoring live sessions and test session termination and suspension. To do so, we will need to connect as a user who is a member of the Auditors group – *Cindy*.

PSM Session Terminators

As mentioned, we will be testing live monitoring, as well as session suspension and session termination. While all members of the *Auditors* group can monitor live sessions, not all members of the *Auditors* group have permissions to terminate or suspend sessions. Only users who are also members of the built-in *PSMLiveSessionTerminators* group have permissions to do so. For your convenience, *Cindy,* the ACME corporation auditor, has already been added to this group.



Monitor, Suspend, and Terminate Active Sessions

1. Launch a session to the **acme-123** server as *John* **using the RDP shortcut** on your Desktop, as we did in the earlier exercise. Leave the session open.

2. Log in to the **PVWA** via LDAP as *Cindy* and go to the **MONITORING** pane.

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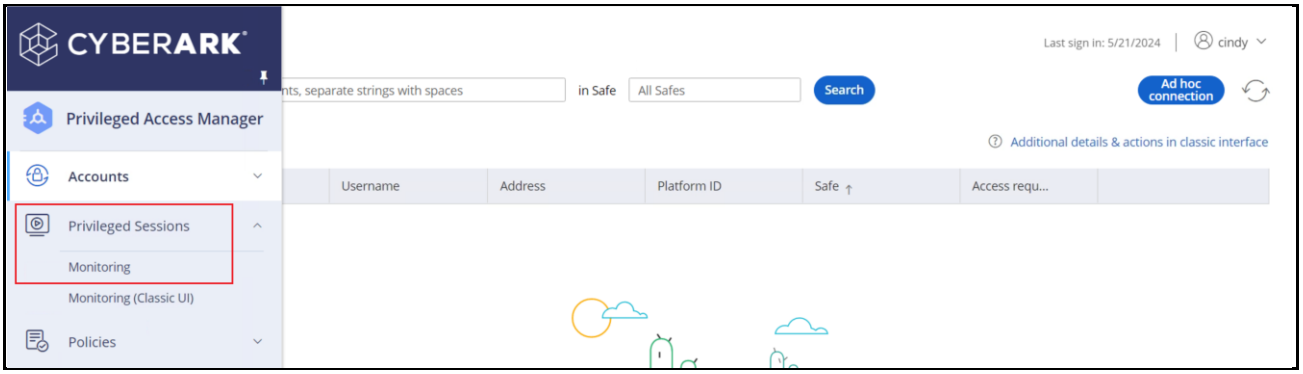
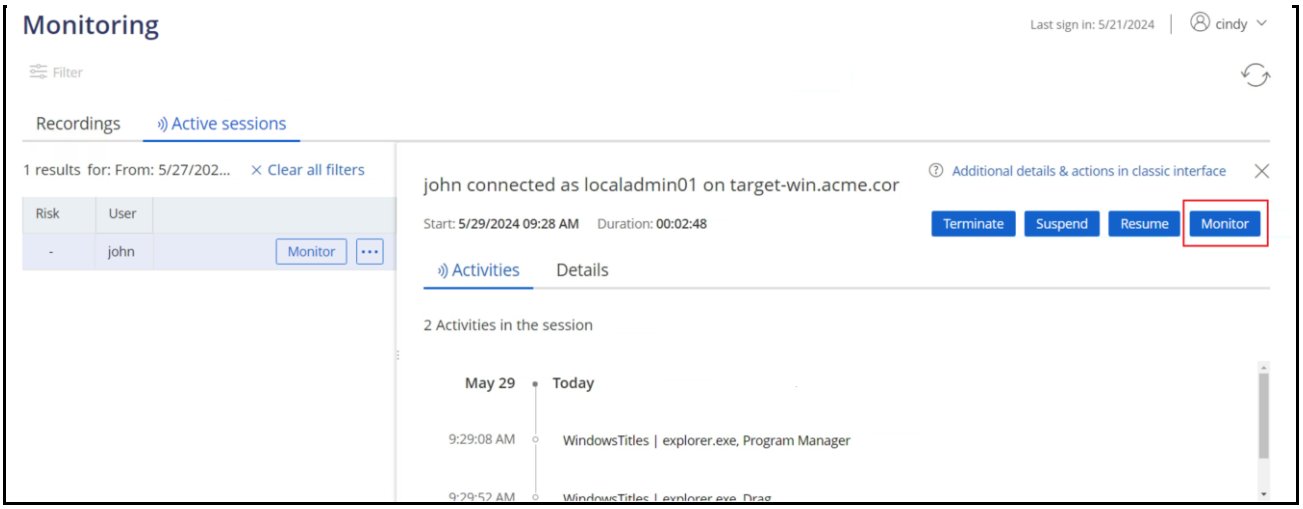


3. Go to **Active Sessions** and locate the session opened by *John* and click on **Monitor**. You should now be able to monitor *John's* session as it happens.



4. As *Cindy*, try to *Monitor*, *Suspend*, *Resume,* and ultimately *Terminate* the session.

Note:	When you click on **Terminate**, you will receive an RDP file like you would if you were establishing a connection. Clicking on this file will terminate the other user's session.

View Recordings

1. As *Cindy*, verify that you can see the recordings related to your earlier user sessions and try to play some of these recordings. Note that recordings related to **PSM for SSH** are presented in the classic UI.

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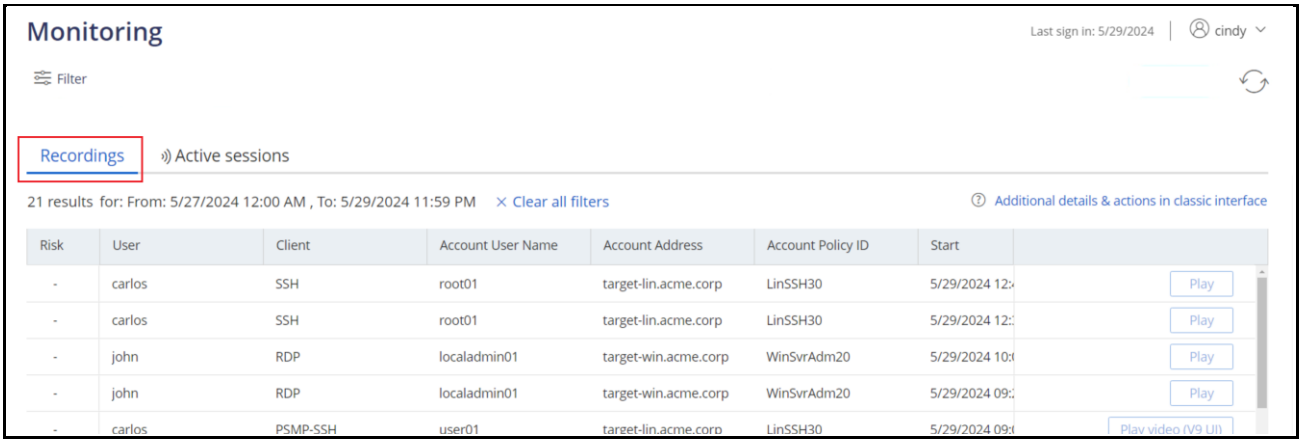
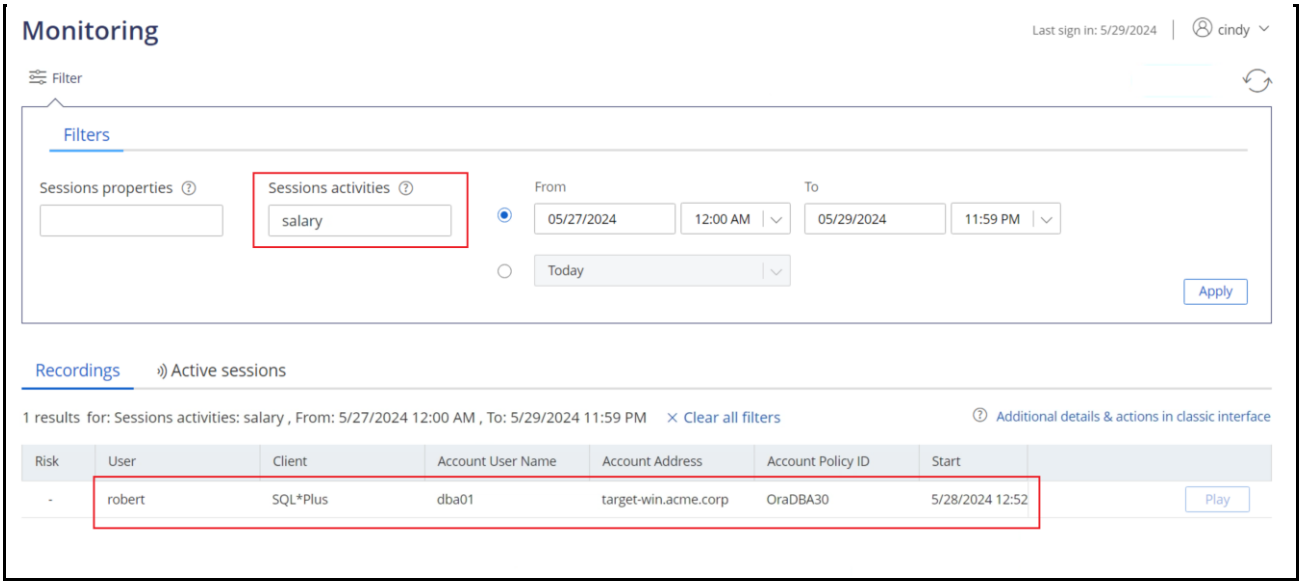


2. You can also search recordings by activities in a privileged session. For example, enter *salary* in the **Session activities** field and press **Apply**. Once you locate the SQL recording, click on **Play**.



3. Review the recording. Click on the session line for more detail and find the command "*select * from employees*". Note that the recording will now start at the command selected. Close the playback window when you are done.

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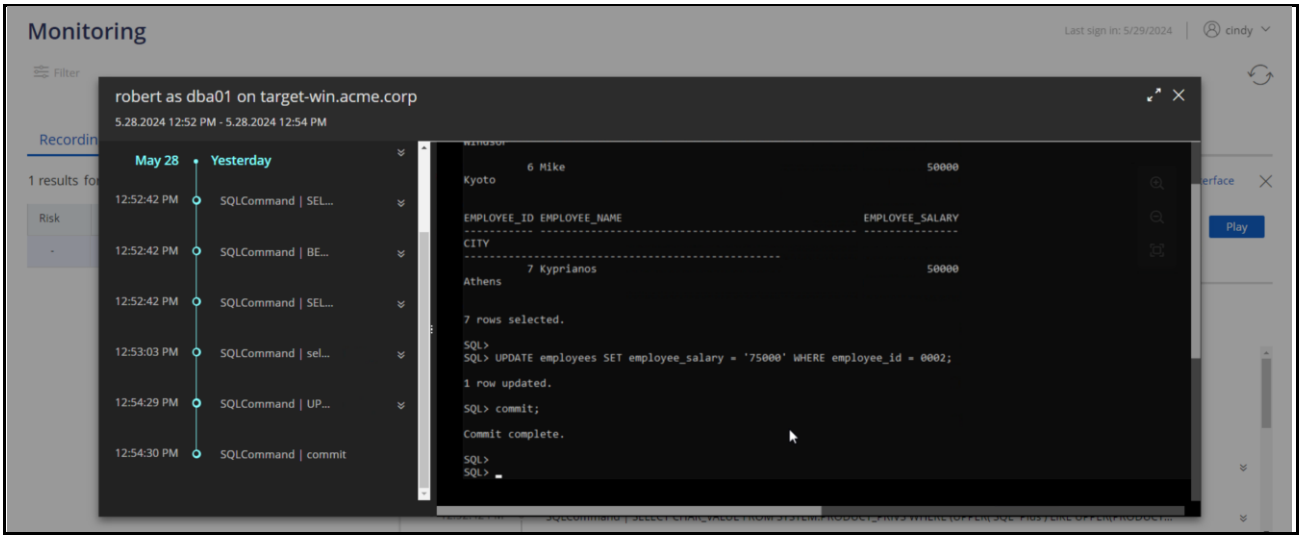

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Reports

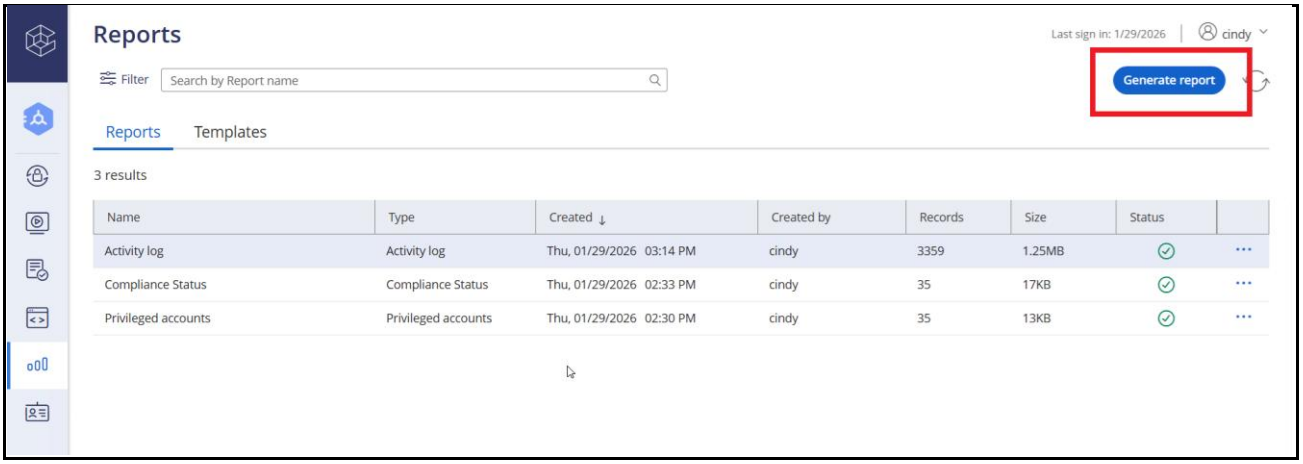
In this section we will look at how to generate reports.

Reports are now available in the modern interface (although you can still connect to the classic interface for the time being).

In keeping with **CyberArk**'s strict compartmentalization of information, users can only generate reports on **Safes** to which they have access. We will use *Cindy* to generate the reports, because as an auditor she has the permissions to view **Safe** members and to view the audit logs on all the **Safes**.

Generate a Privileged Accounts Inventory report

1. Login to the **PVWA** as *Cindy* and go to the **Reports** tab. Click on **Generate Report**.



The reports are divided into three groups:

- **Operational reports**
- **Audit/Compliance reports**
- **List reports**

2. Under **Operational** reports, select "Privileged Accounts Inventory" and click **Generate**.

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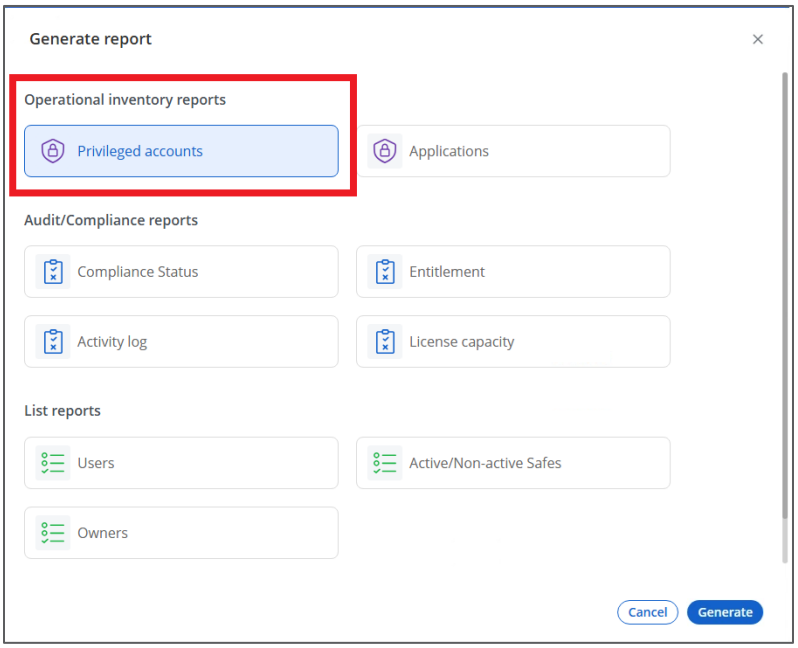
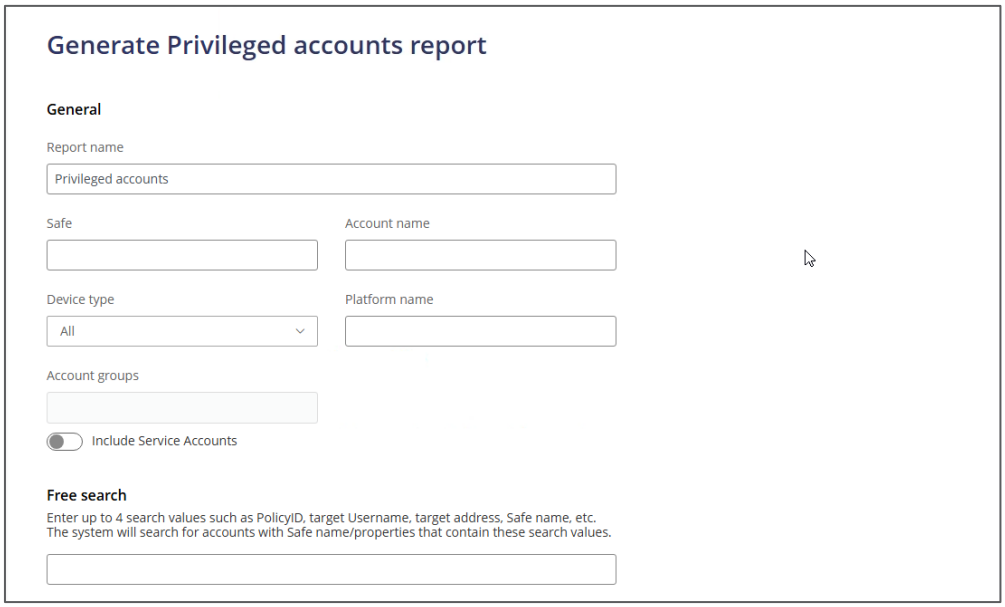


3. Review the options to filter the report, but keep the default values. There are options for filtering by general parameters, automatic management status, and activities. Scroll down to view more options.



4. Note that you can add subscribers to the report output. We are going to create a one time report (so that we don't have to wait for the output). Click on **One time report** then click the **Generate** button.

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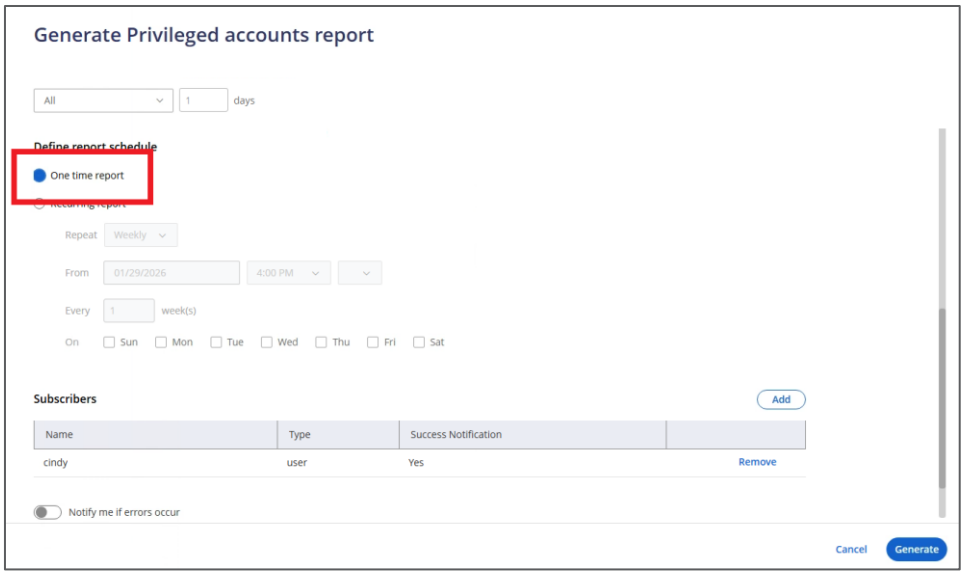
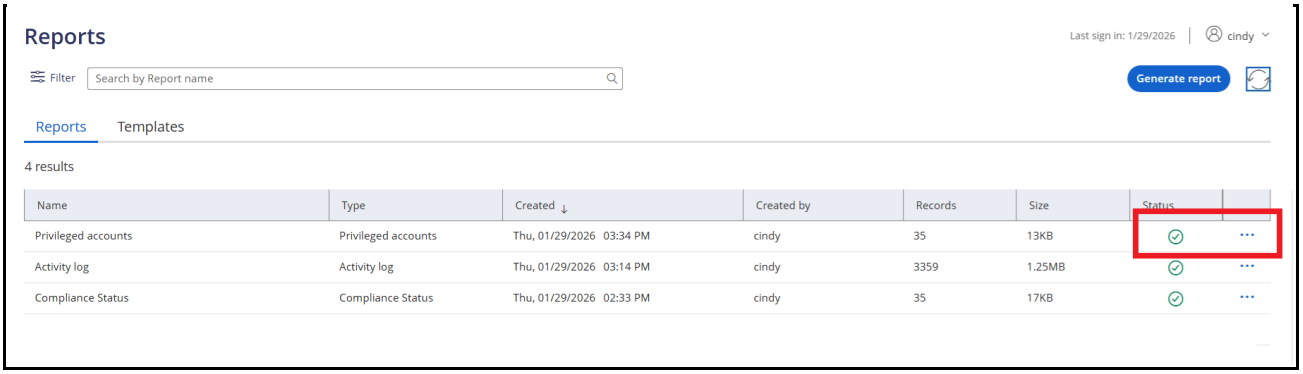


5. Select the refresh icon in the upper right-hand corner until the report status shows **Done**. Click on the ellipsis and select **Download CSV**.



6. When the download appears, open it with the default ***LibreOffice Calc***.

7. After going over the report, save the new report to the desktop of the **Comp01** server. If you are asked if you want to save the document in its current format, click **Use Text CSV Format.**

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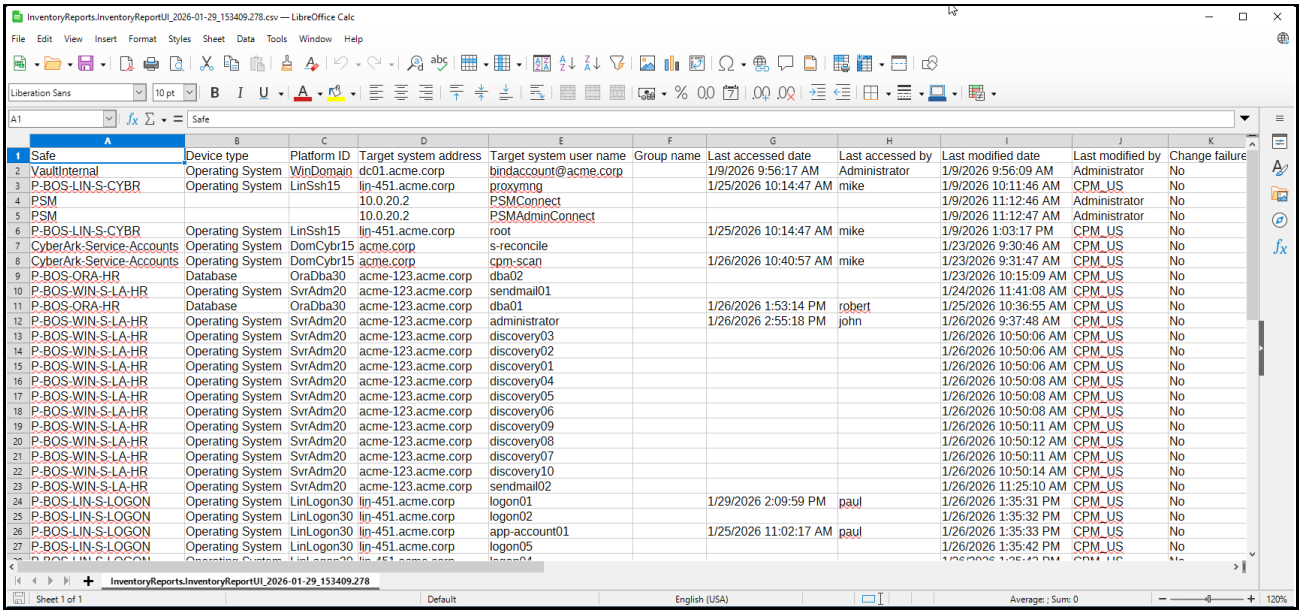

Safe	Device type	Platform ID	Target system address	Target system user name	Group name	Last accessed date	Last accessed by	Last modified date	Last modified by	Change failure
VaultInternal	Operating System	WinDomain	dc01.acme.corp	bindaccount@acme.corp		1/9/2026 9:56:17 AM	Administrator	1/9/2026 9:56:09 AM	Administrator	No
P-BOS-LIN-S-CYBR	Operating System	LinSsh15	lin-451.acme.corp	proxymng		1/25/2026 10:14:47 AM	mike	1/9/2026 10:11:46 AM	CPM_US	No
PSM			10.0.20.2	PSMConnect				1/9/2026 11:12:46 AM	Administrator	No
PSM			10.0.20.2	PSMAdminConnect				1/9/2026 11:12:47 AM	Administrator	No
P-BOS-LIN-S-CYBR	Operating System	LinSsh15	lin-451.acme.corp	root		1/25/2026 10:14:47 AM	mike	1/9/2026 1:03:17 PM	CPM_US	No
CyberArk-Service-Accounts	Operating System	DomCybr15	acme.corp	s-reconcile				1/23/2026 9:30:46 AM	CPM_US	No
CyberArk-Service-Accounts	Operating System	DomCybr15	acme.corp	cpm-scan		1/26/2026 10:40:57 AM	mike	1/23/2026 9:31:47 AM	CPM_US	No
P-BOS-ORA-HR	Database	OraDba30	acme-123.acme.corp	dba02				1/23/2026 10:15:09 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	sendmail01				1/24/2026 11:41:08 AM	CPM_US	No
P-BOS-ORA-HR	Database	OraDba30	acme-123.acme.corp	dba01		1/26/2026 1:53:14 PM	robert	1/25/2026 10:36:55 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	administrator		1/26/2026 2:55:18 PM	john	1/26/2026 9:37:48 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery03				1/26/2026 10:50:06 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery02				1/26/2026 10:50:06 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery01				1/26/2026 10:50:06 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery04				1/26/2026 10:50:08 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery05				1/26/2026 10:50:08 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery06				1/26/2026 10:50:08 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery09				1/26/2026 10:50:11 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery08				1/26/2026 10:50:12 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery07				1/26/2026 10:50:11 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	discovery10				1/26/2026 10:50:14 AM	CPM_US	No
P-BOS-WIN-S-LA-HR	Operating System	SvrAdm20	acme-123.acme.corp	sendmail02				1/26/2026 11:25:10 AM	CPM_US	No
P-BOS-LIN-S-LOGON	Operating System	LinLogon30	lin-451.acme.corp	logon01		1/29/2026 2:09:59 PM	paul	1/26/2026 1:35:31 PM	CPM_US	No
P-BOS-LIN-S-LOGON	Operating System	LinLogon30	lin-451.acme.corp	logon02				1/26/2026 1:35:32 PM	CPM_US	No
P-BOS-LIN-S-LOGON	Operating System	LinLogon30	lin-451.acme.corp	app-account01		1/25/2026 11:02:17 AM	paul	1/26/2026 1:35:33 PM	CPM_US	No
P-BOS-LIN-S-LOGON	Operating System	LinLogon30	lin-451.acme.corp	logon05				1/26/2026 1:35:42 PM	CPM_US	No

Note: Feel free to take some time to familiarize yourself with the different reports.

Extract data for reports using EVD

In this section we will use the **Export Vault Data** (EVD) utility to generate reports; or to be more precise, we will use the **EVD** to extract data from the **Vault** that can then be used to generate reports using a third-party tool.

The **EVD** utility exports data from the **Vault** to TXT or CSV files, from where they can be imported into third-party applications or databases. Each report is saved in a different file. Additional information about using **EVD** can be found in the online documentation.

Enable the Auditor user

This is another task that once had to be done through the **PrivateArk Client** but that now can be done in the PVWA.

1. Log in to the **PVWA** as *Mike*. In the side bar, click on **User Provisioning** | **Users**.

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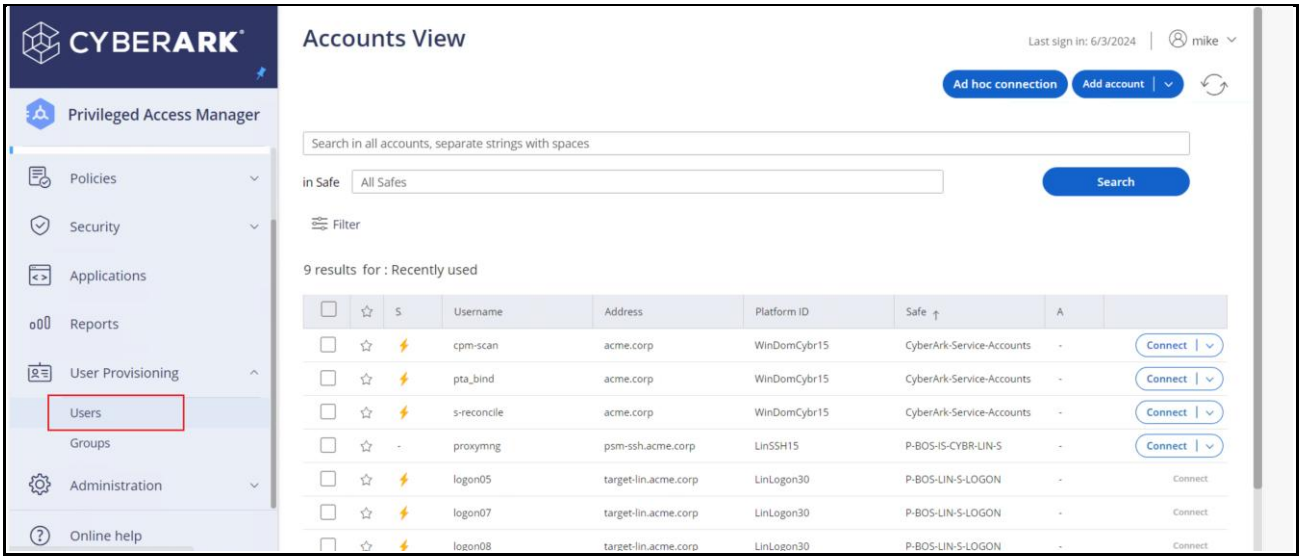
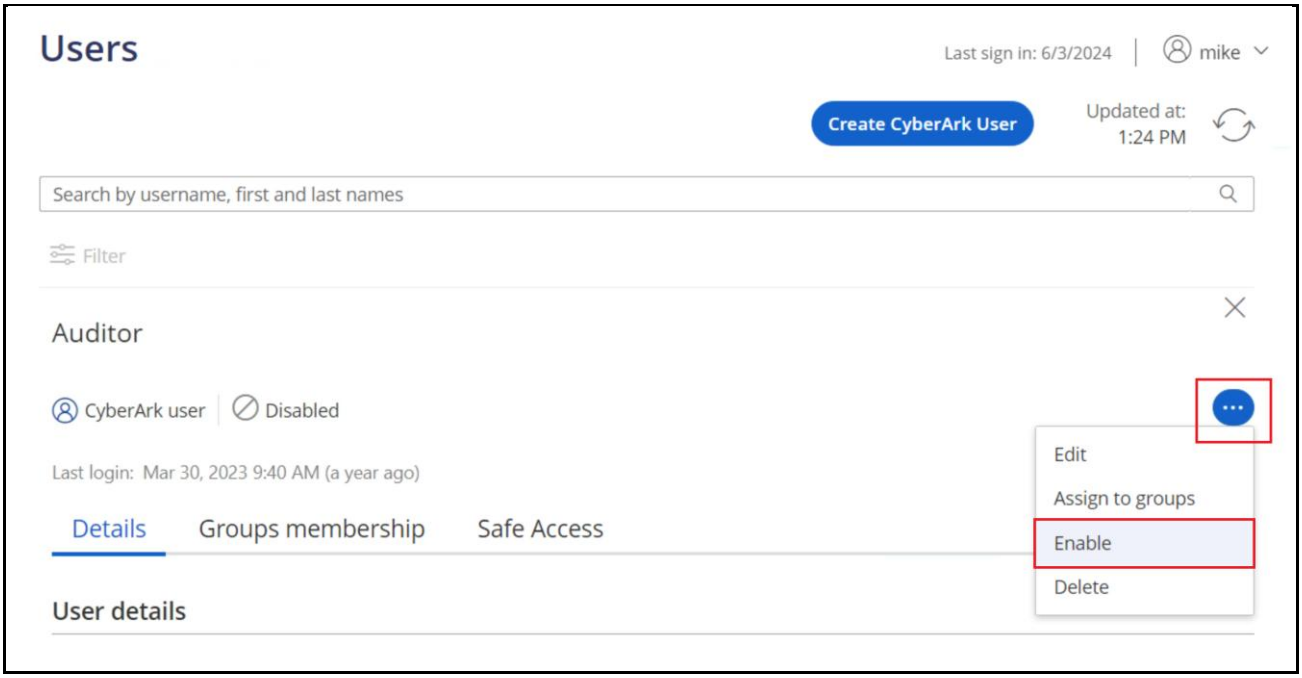


2. Click on the built-in *Auditor* user, then the ellipsis, and then on **Enable**.



3. Now click again on the ellipsis and select **Edit**. In the menu on the left, select **Authentication method**. Enter *Cyberark1* into the **Password** and **Confirm password** fields and then click **Save changes**.

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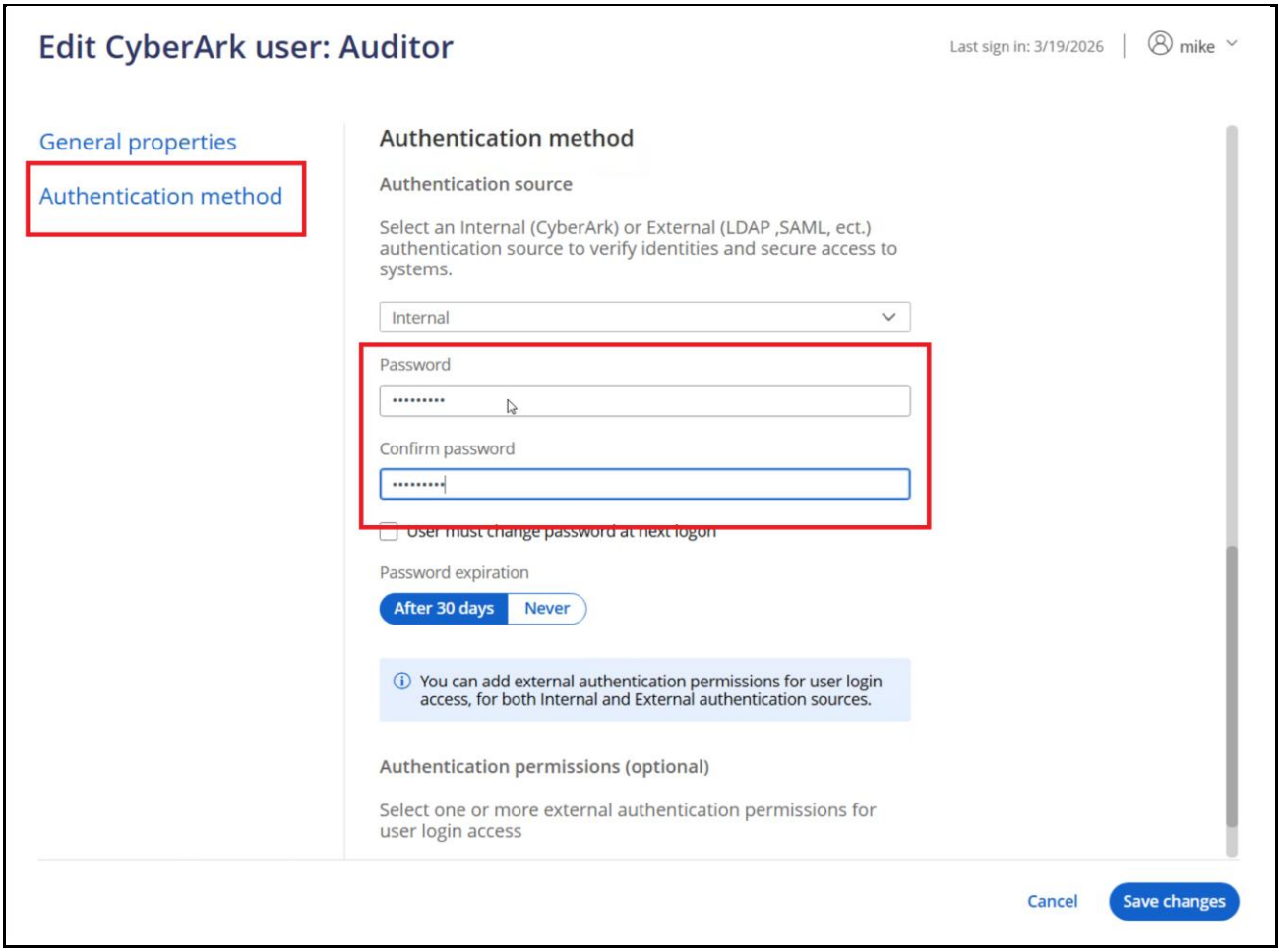

Create the Credential file

Note:	The commands in this exercise must be run as Administrator in an elevated Powershell window.

To ensure the security of your environment, it is essential to secure the credential file properly, in the same way you would protect any privileged account from abuse. To this end, the *CreateCredFile.exe* parameters allow you to harden the file so it can only be used by a specific component on a defined host system, as well as other restrictions you can choose. As there are many combinations of hardening rules possible, please consult with **CyberArk** for the best fit for your environment and document it internally.

1. Now using Windows File Explorer, go to *C:\ExportVaultData* and open the *Vault.ini* file using Notepad. Set the Vault IP address parameter to **10.0.10.1**. You may also change the **Vault** name to "Primary" or “Primary Vault” so that it aligns with the name used in **PrivateArk Client**, but it is not required.

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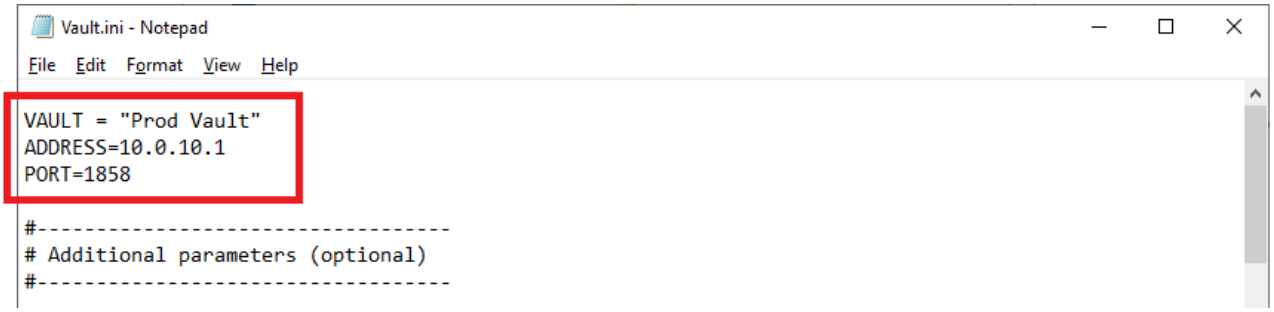

2. Save the *Vault.ini* file and close it.

3. Open a Powershell window as administrator. command prompt. Change directories to *c:\ExportVaultData\CreateCredFile* and run the following command to generate a credential file:

```
./CreateCredFile.exe auditor.cred Password /username Auditor /password Cyberark1
/IpAddress /Hostname /AppType EVD /EntropyFile
```

Note: You will find this command and the command to run the utility in a text file named *A1-EXPORT_VAULT_DATA_COMMANDS.txt* in the directory *c:\ ExportVaultData*.

There are multiple parameters possible in this command. You can find more information about them in the **CyberArk** documentation.

You can also invoke the **CreateCredFile** wizard using just the parameters of **executable** plus **outputfilename**, which steps you through the options available for securing a **CyberArk** credential file.

Note: Because the DR server is not running, the option “Disable wait for DR synchronization before allowing password change” will provoke an error in the Vault log: *Warning: CASTM057E Password has not yet been replicated to DR site(s). Password of user [Auditor] will not be changed.* This can safely be ignored.

Export Vault Data

Now that we have created a credential file, we are ready to export the data. We will export an activity log for the previous 4 days.

1. Change directories to *c:\ExportVaultData*:

```
cd ..
```


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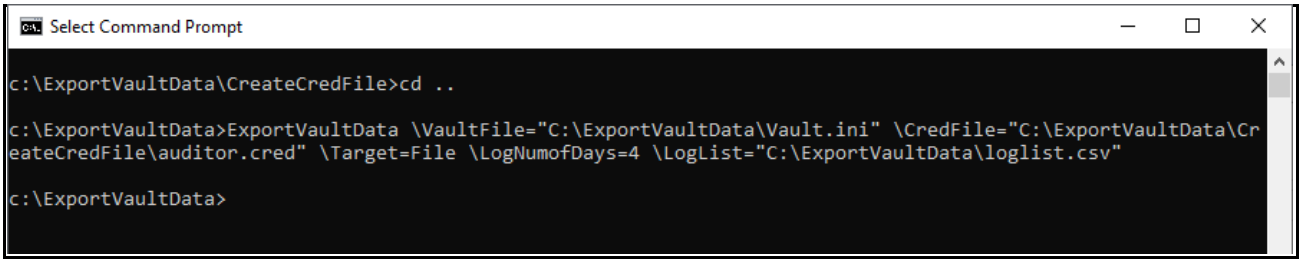
2. Now run the following command:

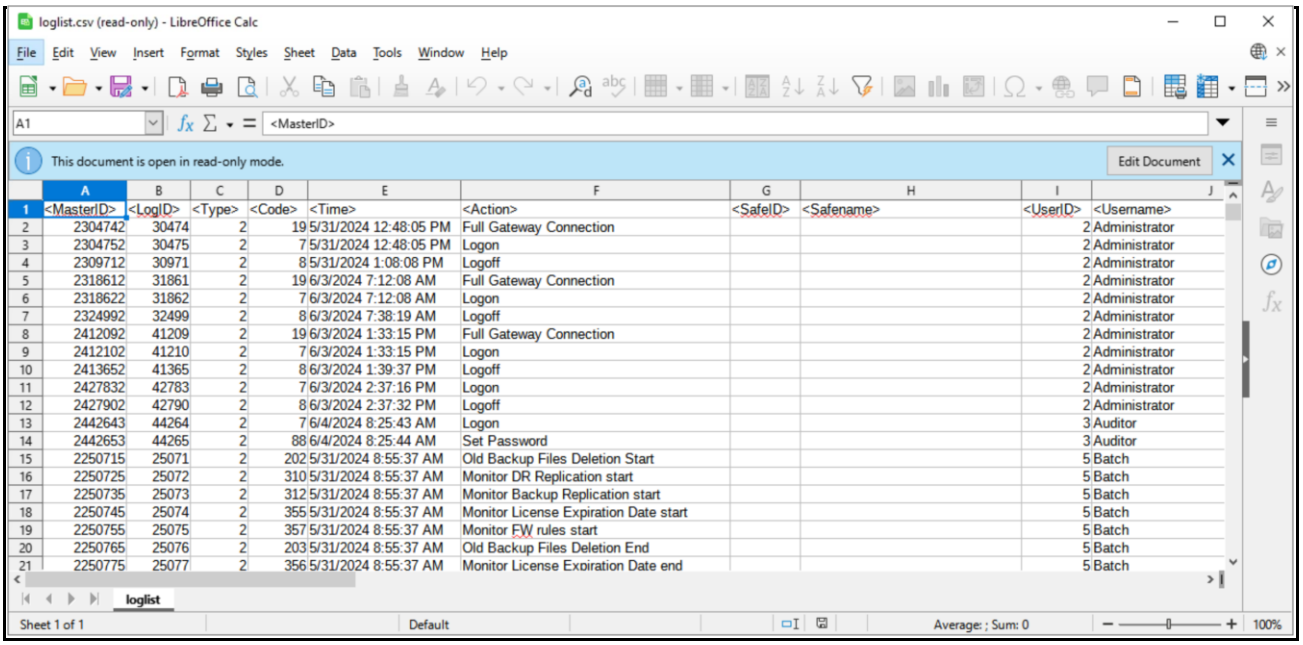
```
./ExportVaultData \VaultFile="C:\ExportVaultData\Vault.ini"
\CredFile="C:\ExportVaultData\CreateCredFile\auditor.cred" \Target=File
\LogNumOfDays=4 \LogList="C:\ExportVaultData\loglist.csv"
```

Note: You can find a copy of this command in the file *C:\ExportVaultData\A1-EXPORT_VAULT_DATA_COMMANDS.txt*



Note: The above example will create a log activity report for the **Vault** defined in the *Vault.ini* file in *C:\ExportVaultData*. The user who will access the **Vault** to generate this report is defined in the *auditor.cred* file in *C:\ExportVaultData*. The log activities report will be saved in a file called *loglist.csv*. The log is generated for the preceding 4 days.

3. A new file called *loglist.csv* was generated in the *C:\ExportVaultData* folder. Review the file using LibreOffice Calc to see the Activities log report generated by **EVD**.



	A	B	C	D	E	F	G	H	I	J
1	<MasterID>	<LogID>	<Type>	<Code>	<Time>	<Action>	<SafeID>	<Safename>	<UserID>	<Username>
2	2304742	30474	2	19	5/31/2024 12:48:05 PM	Full Gateway Connection			2	Administrator
3	2304752	30475	2	7	5/31/2024 12:48:05 PM	Logon			2	Administrator
4	2309712	30971	2	8	5/31/2024 1:08:08 PM	Logoff			2	Administrator
5	2318612	31861	2	19	6/3/2024 7:12:08 AM	Full Gateway Connection			2	Administrator
6	2318622	31862	2	7	6/3/2024 7:12:08 AM	Logon			2	Administrator
7	2324992	32499	2	8	6/3/2024 7:38:19 AM	Logoff			2	Administrator
8	2412092	41209	2	19	6/3/2024 1:33:15 PM	Full Gateway Connection			2	Administrator
9	2412102	41210	2	7	6/3/2024 1:33:15 PM	Logon			2	Administrator
10	2413652	41365	2	8	6/3/2024 1:39:37 PM	Logoff			2	Administrator
11	2427832	42783	2	7	6/3/2024 2:37:16 PM	Logon			2	Administrator
12	2427902	42790	2	8	6/3/2024 2:37:32 PM	Logoff			2	Administrator
13	2442643	44264	2	7	6/4/2024 8:25:43 AM	Logon			3	Auditor
14	2442653	44265	2	88	6/4/2024 8:25:44 AM	Set Password			3	Auditor
15	2250715	25071	2	202	5/31/2024 8:55:37 AM	Old Backup Files Deletion Start			5	Batch
16	2250725	25072	2	310	5/31/2024 8:55:37 AM	Monitor DR Replication start			5	Batch
17	2250735	25073	2	312	5/31/2024 8:55:37 AM	Monitor Backup Replication start			5	Batch
18	2250745	25074	2	355	5/31/2024 8:55:37 AM	Monitor License Expiration Date start			5	Batch
19	2250755	25075	2	357	5/31/2024 8:55:37 AM	Monitor FW rules start			5	Batch
20	2250765	25076	2	203	5/31/2024 8:55:37 AM	Old Backup Files Deletion End			5	Batch
21	2250775	25077	2	356	5/31/2024 8:55:37 AM	Monitor License Expiration Date end			5	Batch

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For more information about the different data export options available with the Export Vault Data utility, please refer to the **CyberArk PAM** documentation.

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Backup and Restore

In this section, you will use the **Replicator** utility to test backup and restoration of the **Vault** data. Like all the other components, the **Replicator** utility has already been installed in your environment by the implementation team. It has been installed on the server **comp01**, but to configure and run it, you need to use an administrator account such as *dom*.

Enable the users Backup and Operator

We will be using two **CyberArk** built-in users. The first user is *Backup*, which has permissions to backup all Safes. We will use *Backup* to execute the back up of all Safes. The second user is *Operator*, which has authority to restore all Safes. We will use *Operator* to restore a Safe. These two users are disabled by default, so for this exercise you will need to enable both in your environment in the same way we did for the *Auditor* user in the previous exercise.

Log in to the **PVWA** as *Mike* and enable the users ***Backup*** and ***Operator***. Set both users' passwords to *Cyberark1*.

Create a test Safe and an Account

To test back up and restore, we will create a test Safe and Account, run a replication, delete the Safe, and then restore it from the backup.

Log in as *Mike* and create the following Safe and Account:

Safe name:		*TEST_DELETE*
Account information:	***Platform***	*Lin Ssh 15*
	Name:	*root10b*
	Address	*lin-451.acme.corp*
	Password:	*Cyberark1*

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Configure the CyberArk Replicator Utility

Configure the *Vault.ini* file

1. Connect to the server **comp01a** via RDP using the account *dom* / *Cyberark1* (there is a shortcut on the desktop). Once connected, open Windows File Explorer and go to *C:\Program Files (x86)\PrivateArk\Replicate*.

2. Open the *Vault.ini* file, enter "Primary Vault" for the **VAULT** parameter (although this is not mandatory), and enter the IP address of your **Vault** server in the **ADDRESS** parameter: *10.0.10.1*:

```
VAULT = "Primary Vault"
ADDRESS=10.0.10.1
PORT=1858
```

3. Save and close the file.

Locate the output directory – *tsparm.ini*

In the same directory, open the file *tsparm.ini* and note the output location of the backup.

Create the credential file – *backup.cred*

This process is almost identical to the credential file generation that we performed in the preceding exercise. In this case, the *CreateCredFile.exe* is in the same directory as *Vault.ini*.

1. Open a Powershell window (you do not need to do this as an administrator because *Mike* has been given access) and change directories to the *Replicate* folder:

```
cd "c:\Program Files (x86)\PrivateArk\Replicate"
```


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2. In this example, we will lock down our credential file by adding additional parameters such as the full path to the executable that is allowed to use the credential file and DPAPI machine protection. Run the following:

```
./CreateCredFile.exe backup.cred Password /username backup /password Cyberark1
/ExePath "C:\Program Files (x86)\PrivateArk\Replicate\PAReplicate.exe"
/IpAddress /Hostname /AppType CABACKUP /EntropyFile /DpapiMachineProtection
```

Note:	You can also find a copy of those commands in the *A1-COMMANDS.txt* file in the *Replicate* home folder.

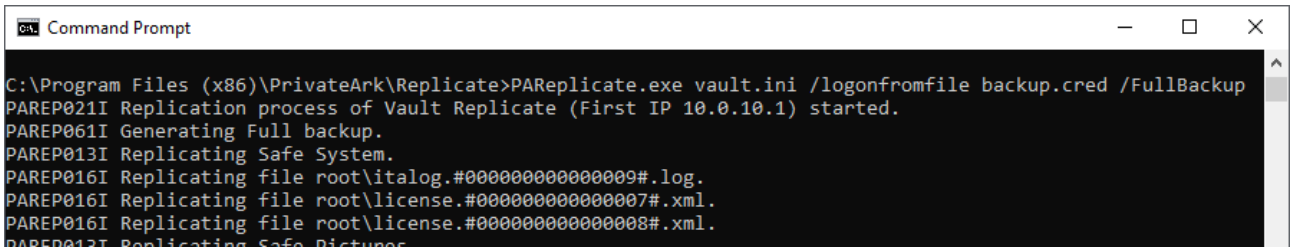
Run a Backup

Note:	**Make sure** that you have enabled the ***Backup*** and ***Operator*** **users** (and set the passwords to ***Cyberark1***) ***prior*** to running the backup command. These users can be found under the System branch. The process is identical to enabling the *Auditor* user performed earlier.

To perform a backup, run the following command from the *Replicate* folder:

```
./PAReplicate.exe vault.ini /logonfromfile backup.cred /FullBackup
```

If the backup is successful, you should see several messages indicating that files are being replicated with a final message stating that the replication process has ended. If everything worked properly, proceed to the next steps. If not, verify the configuration information and try again.



Delete A Safe

In this exercise, we will delete a **Safe** that we will later restore from the backup made in the previous exercise.

1. Login to the **PVWA** as *Mike* and go to **POLICIES | Safes**.

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2. Locate the **Safe** TEST_DELETE, click the ellipsis, and then the **Delete** button.

3. You will receive a prompt asking you to confirm deletion. Press **Delete** to confirm that you would like to delete the **Safe** and its contents.

4. Lastly, you will receive a message stating that the **Safe** cannot be deleted due to Safe retention rules. The **Safe** has not been permanently deleted, but it has been removed from usage. Click **Close**.

5. To confirm that the contents of the *TEST_DELETE* **Safe** have been removed, go to the **Accounts** page.

6. Enter *root10b* in the search box and press the **Search** button.

7. The account that you created earlier should not appear.

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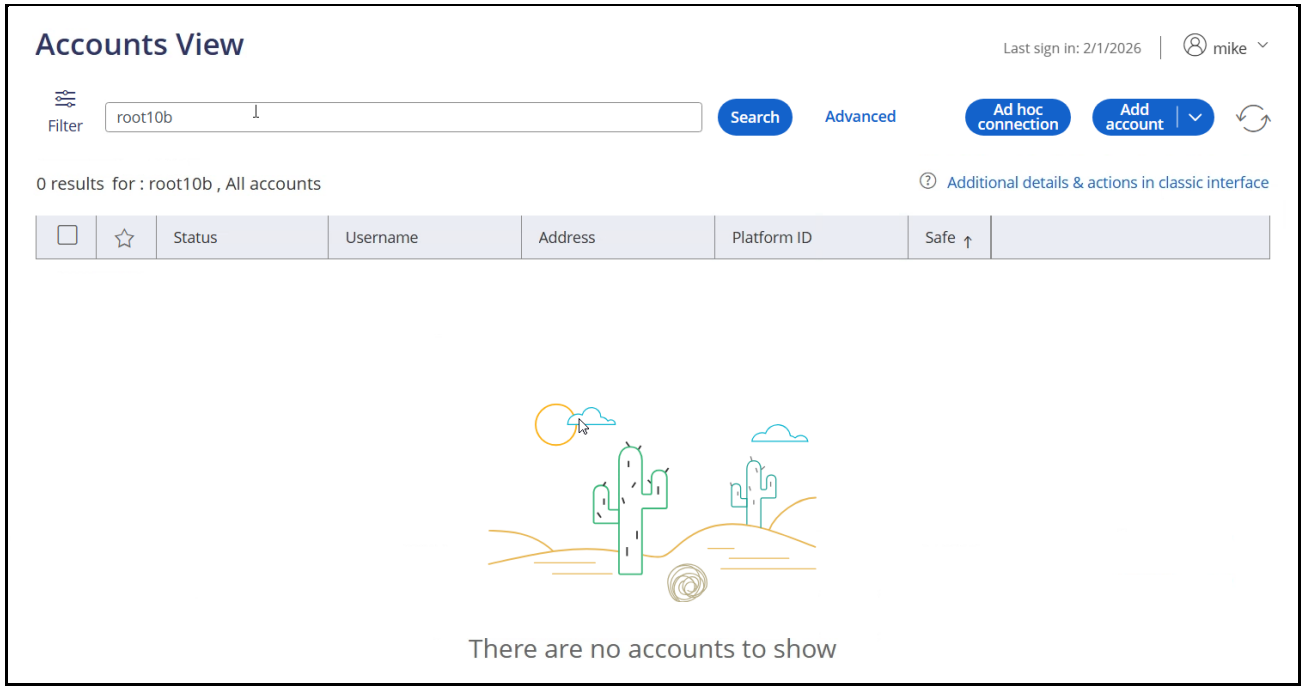

Run a Restore

In this section, we will restore the *TEST_DELETE* Safe from the backup we performed earlier. Because the *TEST_DELETE* Safe has not been permanently deleted, we will restore the contents of the Safe to a new Safe named *TEST_RESTORE*.

Note:	Make sure you have enabled the *Operator* user (and set the password to *Cyberark1*) prior to running the restore command. The process is identical to enabling the *Auditor* user performed earlier.
Note:	The *Operator* user is located under the **System** branch.

1. Go back to the command prompt, making sure you are still in the *Replicate* directory, and run the following command:

```
./PARestore.exe vault.ini operator /RestoreSafe TEST_DELETE /TargetSafe
TEST_RESTORE
```

You will be prompted for the password for the *Operator* user, which should be *Cyberark1*.

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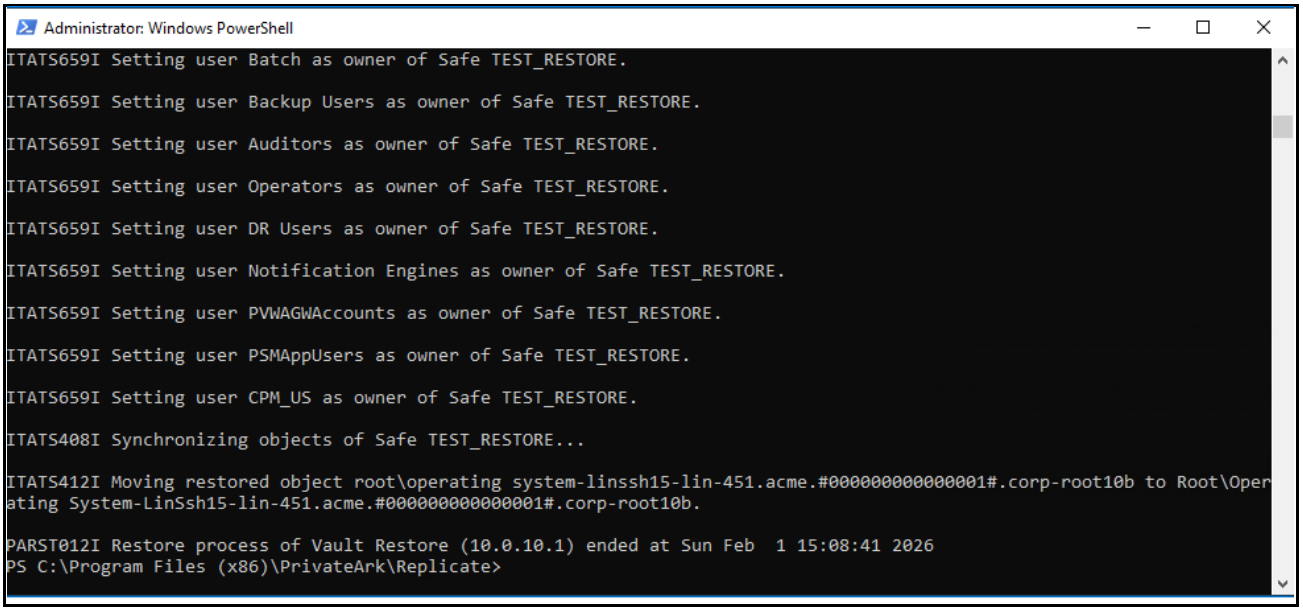

2. Once you see the message that the restore has been completed, go back to the **PVWA** (as *Mike)* and search for *root10b* again. You should now see the account residing in Safe **TEST_RESTORE**.

Note:	The Target Safe (**TEST_RESTORE**) is the name of the restored **Safe**. The restore process will not overwrite an existing **Safe** (remember, the original has been removed, but not yet deleted) so we must create a new one. Also, it may take a moment for the restored **Safe** to appear, so be patient.

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Disaster Recovery

In this section we will test the Disaster Recovery (DR) procedures for automatic failover and manual failback. We will perform this in two phases – **Failover** and **Failback** – and each phase will be broken down into multiple steps for, we hope, clarity.

The first phase will include the following steps:

1) First, we will enable the DR user.
2) Then, start the VM **drvault01**.
3) Then, we will configure the **Disaster Recovery module** on the **DR** server to perform an automatic failover in case the **Primary Vault** is no longer reachable.
4) We will execute a full replication from the **Primary Vault** to the **DR Vault**.
5) We will perform an automatic failover from the **Primary Vault** to the **DR Vault**. As part of the test, we will also confirm that our end users can still access critical systems via **CyberArk**, without any human intervention.

Once the failover is completed, we will return our systems to their original state by performing what we call a Failback.

1) We will set the **Primary Vault** to act as DR and replicate all data back from the **DR Vault** to the **Primary Vault**.
2) We will then perform a manual failback from the **DR Vault** to the **Primary Vault**
3) Lastly, we will set the **DR Vault** back to DR mode and confirm our end users are still able to connect to critical systems via **CyberArk**.

The steps below have already been performed by the implementation team:

- The **PrivateArk Server**, **PrivateArk Client**, and **Disaster Recovery** module have all been installed on both your **vault01** and DR servers by the implementation team.
- A second **DR** user called "*DR_Failback*" was manually created by the implementation team during the deployment of the **Primary Vault** for the purpose of supporting the failback procedure from the DR site back to the primary site.

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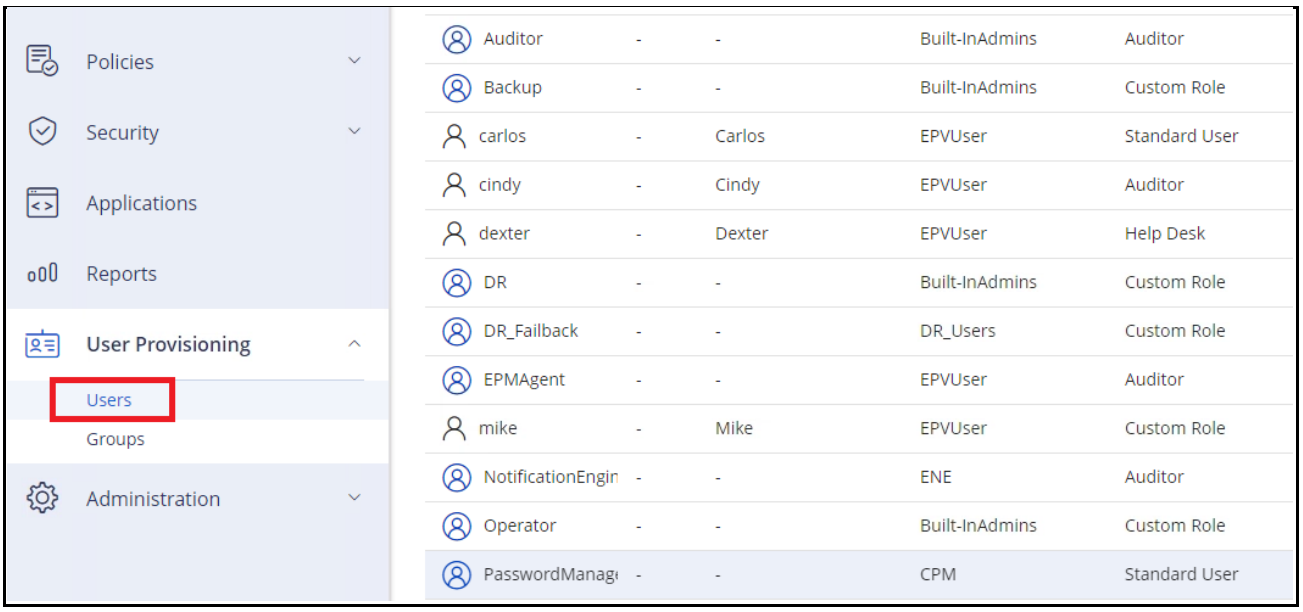
Failover

Step 1: Enable DR user

IMPORTANT:	Do **NOT** change the password for the **DR** user. This user has already authenticated to the **Vault** during the initial implementation and the password for the user has been rotated.

As noted above, the implementation team has already installed the **PrivateArk Server**, **PrivateArk Client** and **Disaster Recovery** service on the **DR** server. However, to avoid an unwanted automatic failover during the first days of the course, automatic failover was disabled, and the DR user deactivated. We are now going to enable the DR user on the **Primary Vault** and Automatic Failover to the **DR Vault**.

1. Launch Chrome to open the **PVWA** page. Log in as *Mike*. Go to the **User Provisioning** tab and then **Users**.



2. Locate the *DR* user, click on the ellipsis and **Enable**.

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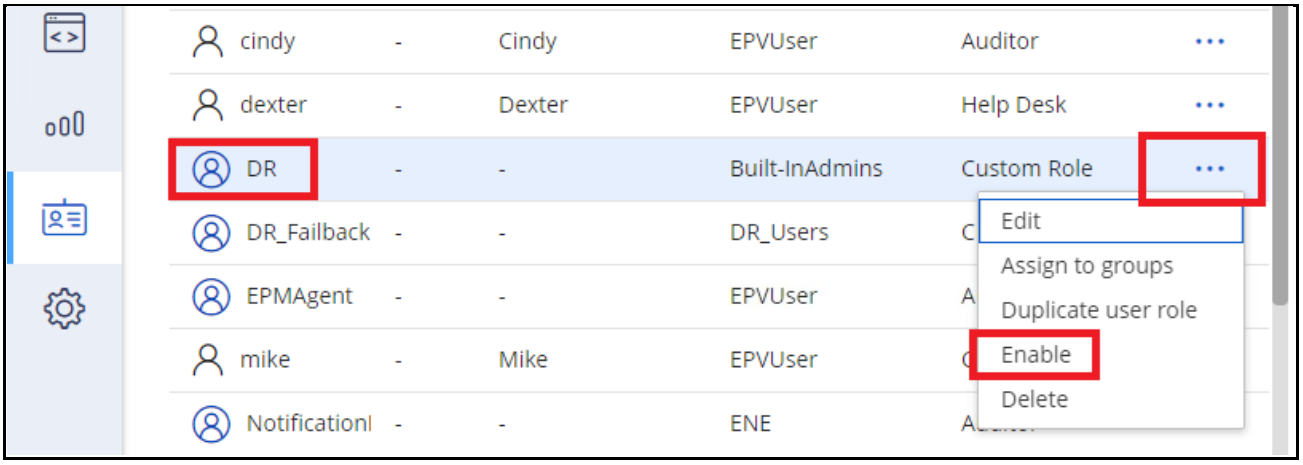
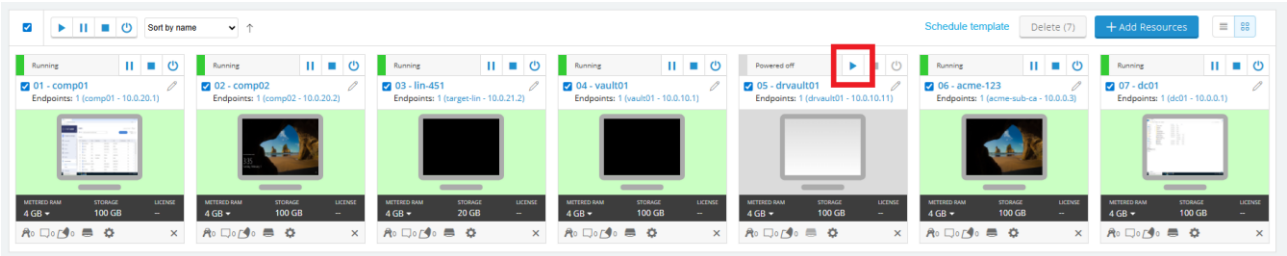


Remember! Do **NOT** reset the *DR* user's password.

Step 2: Start the DR Vault

1. Next, power on the **DR** server. Remember, it will take a moment for the machine to start.



Step 3: Activate Automatic Failover

1. Sign into Windows on the **drvault01** server as *Administrator* with the password *Cyberark1*.

2. Open the file explorer and navigate to *C:\Program Files (x86)\PrivateArk\PADR\Conf.* Double click on the *padr.ini* file to edit it with Notepad (there is a short-cut to the *PADR* folder on the desktop).

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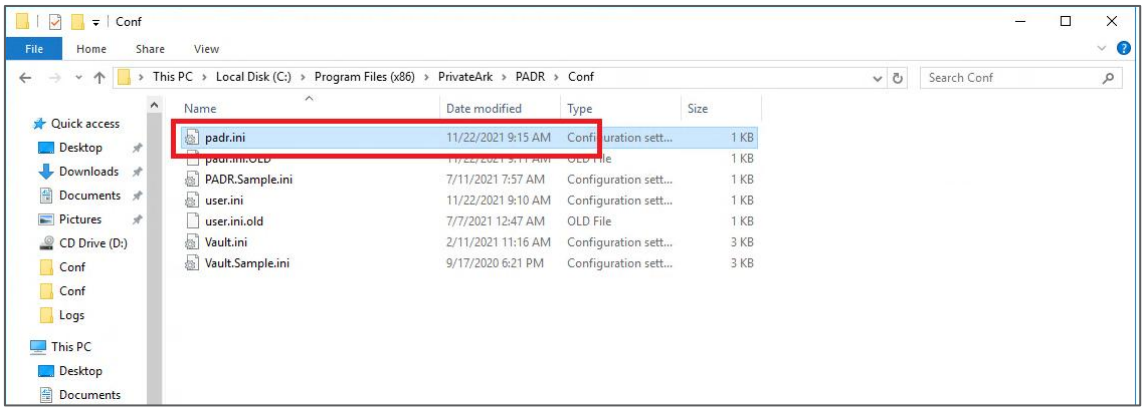
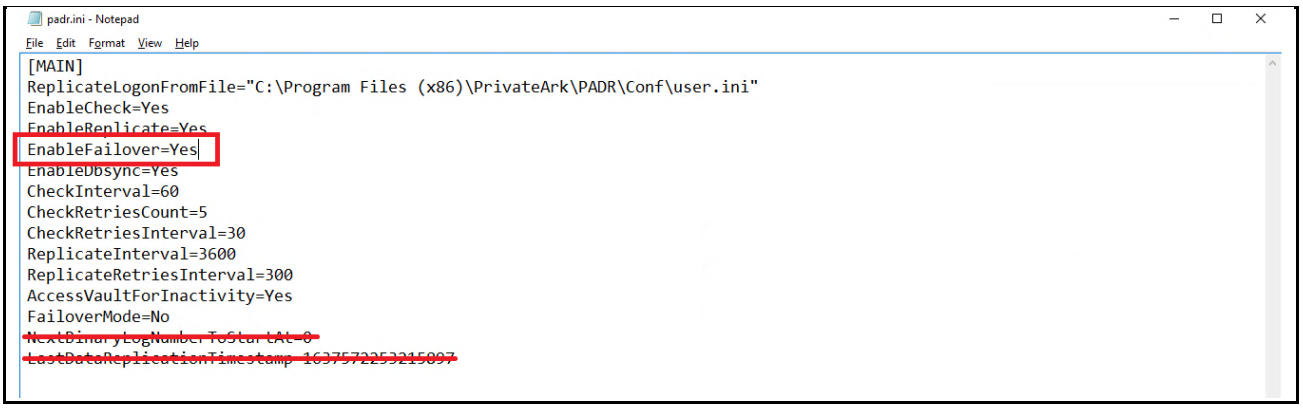


3. Verify that the *EnableFailover* setting is set to **Yes** and delete the last two lines of the file (if present). This will trigger a full replication when we restart the Disaster Recovery service, ensuring we have the most up-to-date data.



Note:	Notice *FailoverMode* is currently set to **No**. Do NOT change this setting. It will automatically change later when we test the failover process.

4. Save the file and exit Notepad.

Step 4: Execute a full replication to the DR Vault

In this step, we will simply re-start the **CyberArk Disaster Recovery** service. In so doing, the service will read the changes we made to the file *PADR.ini*.

1. On the **DR** server, open the Windows Services applet. There is a shortcut in the task bar.

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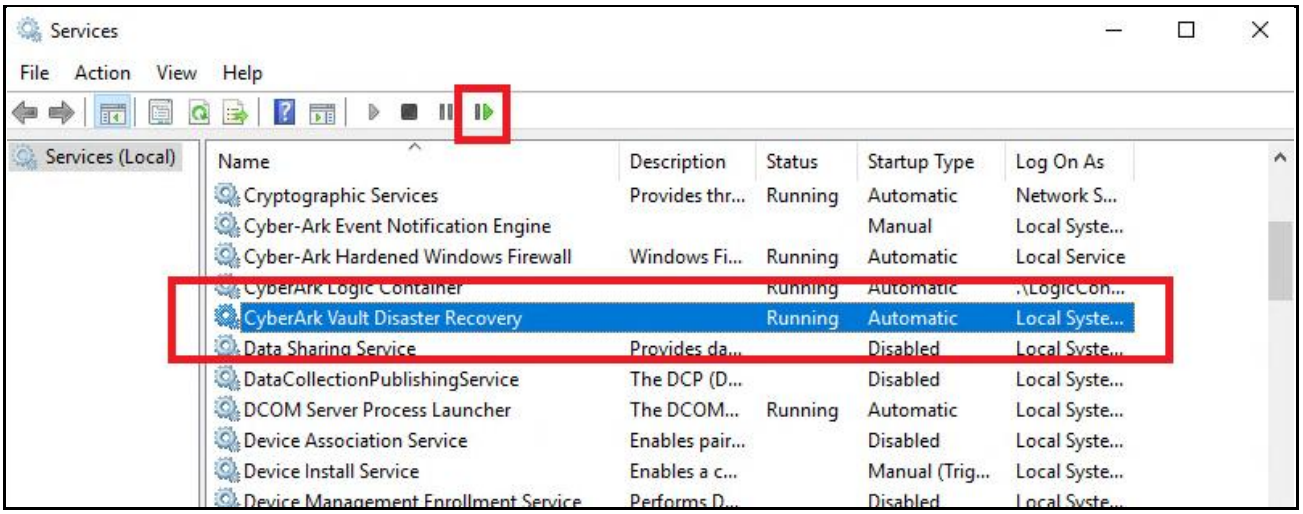
2. Restart the **CyberArk Vault Disaster Recovery** service.



3. Now go to the desktop. Right click on the ***padr_log.ps1*** file located on the desktop and select **Run with PowerShell**.

Note:	The above script will run a tail on the *padr.log* file located in *C:\Program Files (x86)\PrivateArk\PADR\logs* folder. The tail will allow you to monitor the actions performed by the Disaster Recovery service in real time. If you are prompted to allow running the script, select **Yes**.

4. Confirm the Disaster Recovery module has completed the replication of data from the **Primary Vault**. You should see entries with informational codes **PAREP013I Replicating Safe** and at the end:

   - **PADR0010I Replicate ended**
   - **PADR0099I Metadata Replication is running successfully**.

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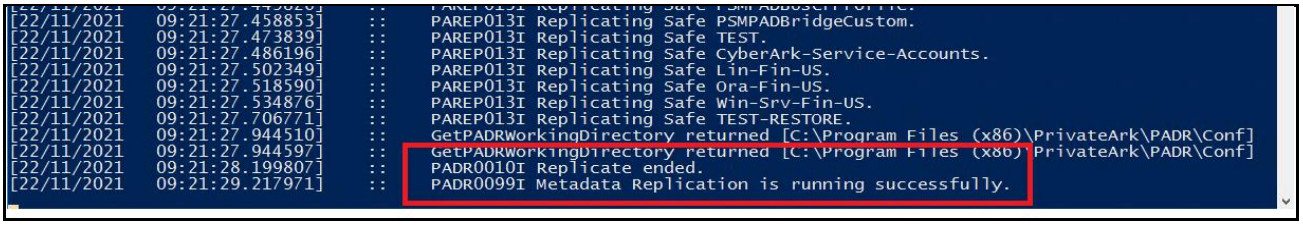
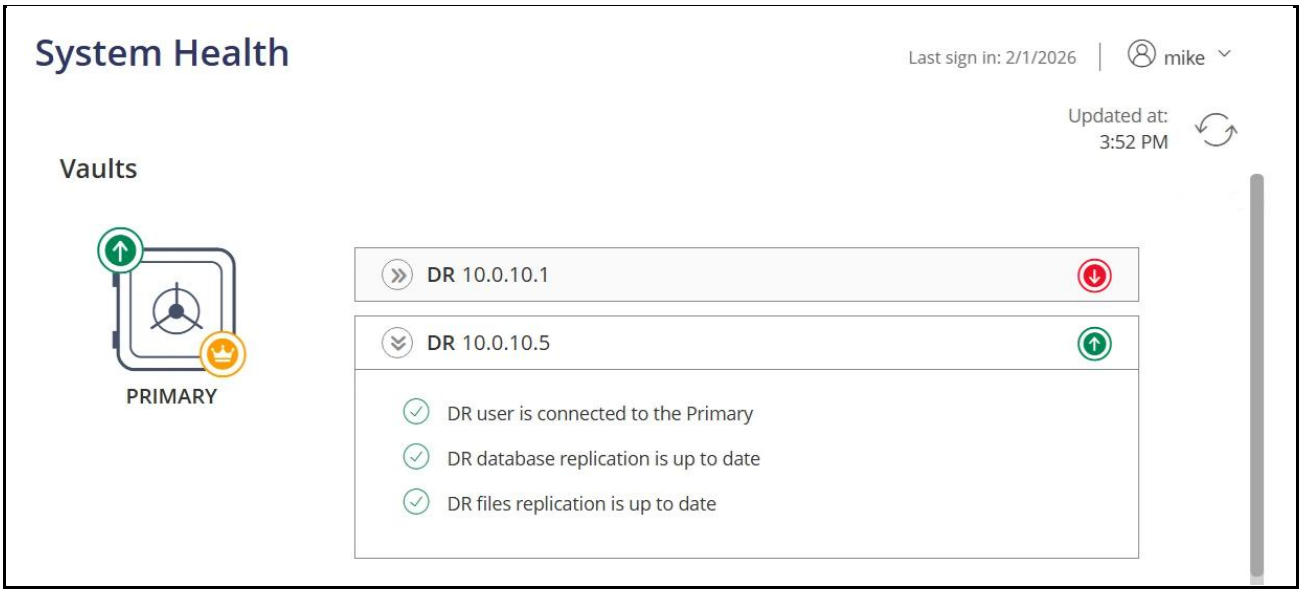


Note:	Keep the tail running for the remainder of the exercise.

5. Back on the **comp01** server and login to the **PVWA** as *Mike*.

6. Navigate to **System Health** to review the status of the DR Vault. Note that the Primary Vault is online and the **DR** on **10.0.10.5** is functioning normally.



Note:	The **System Health** is also aware that there is a Disaster Recovery component on **10.0.10.1**, that is our Primary Vault, but that it is currently offline. This is normal.

Step 5: Test Automatic Failover

Now, we will execute an automatic failover test by stopping the **PrivateArk Server** service on the **Primary Vault**. Perform this action using the **Remote Control Client** as described in an earlier exercise using the command “stop vault”.

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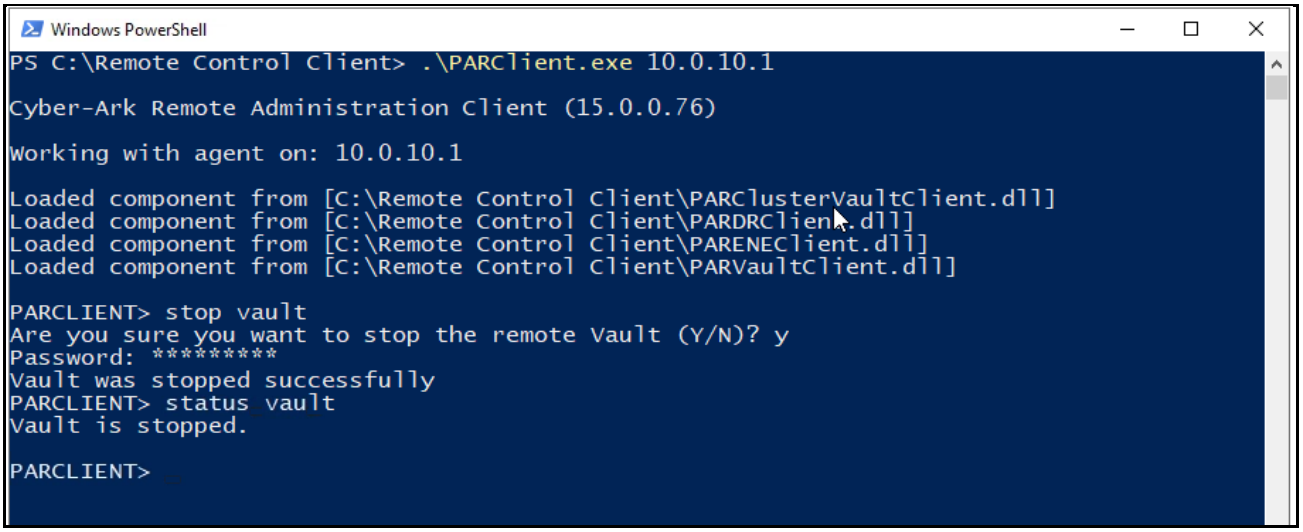
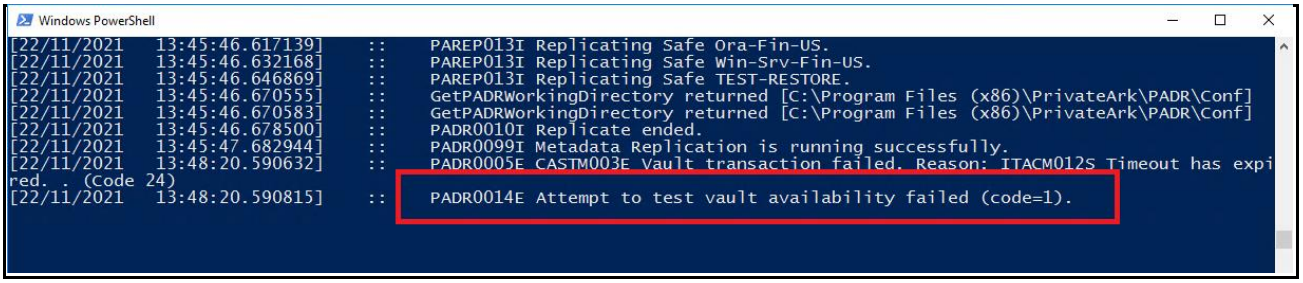


7. Once the **Primary Vault** has stopped, return to the console of the **DR** Server.

8. Monitor the the tail on the *padr.log* file. You should see messages stating that the Disaster Recovery service is unable to reach the **Primary Vault**.



Note: If you are not seeing new entries in the log file after a few minutes, press **Enter**. If you are still not seeing new entries, close the PowerShell window and run the script again.

9. After 5 failures, the **DR Vault** will go into failover mode (this is the default setting). Check the *padr.log* and review the sequence of events.

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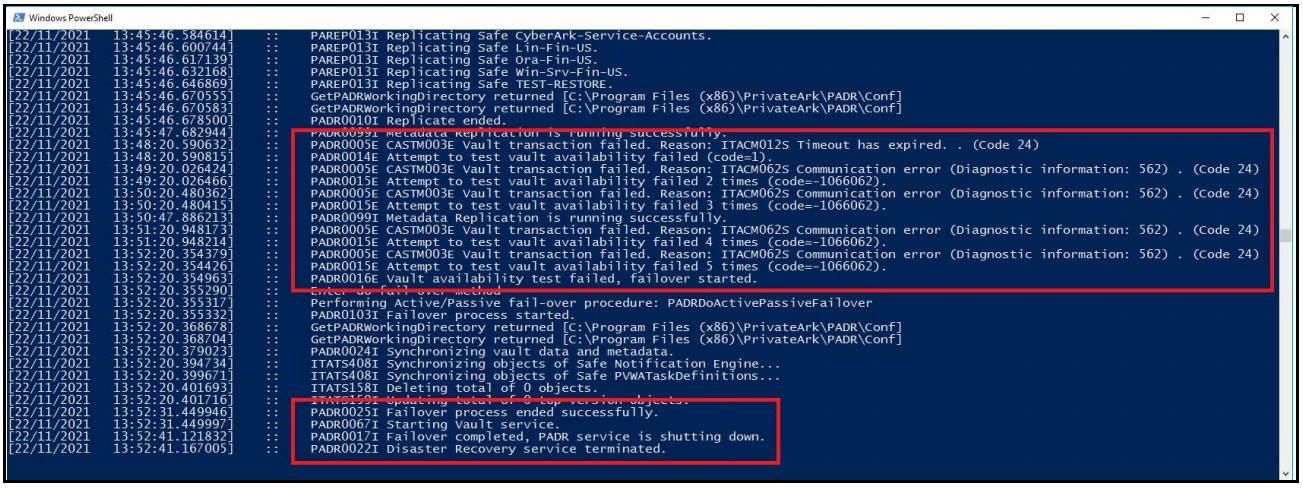
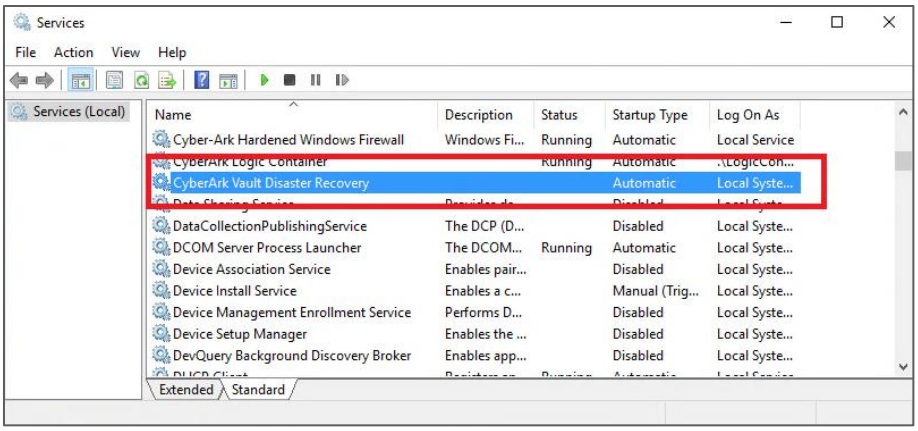


Note:	The entire process should take around 5 minutes.

Confirm Automatic Failover on the DR Vault

1. On the **DR** server (**10.0.10.5**), open the Windows Services applet and confirm the **CyberArk Vault Disaster Recovery** service has terminated.



2. Confirm the **PrivateArk Server** service is now running on the **DR** server (10.0.10.5).

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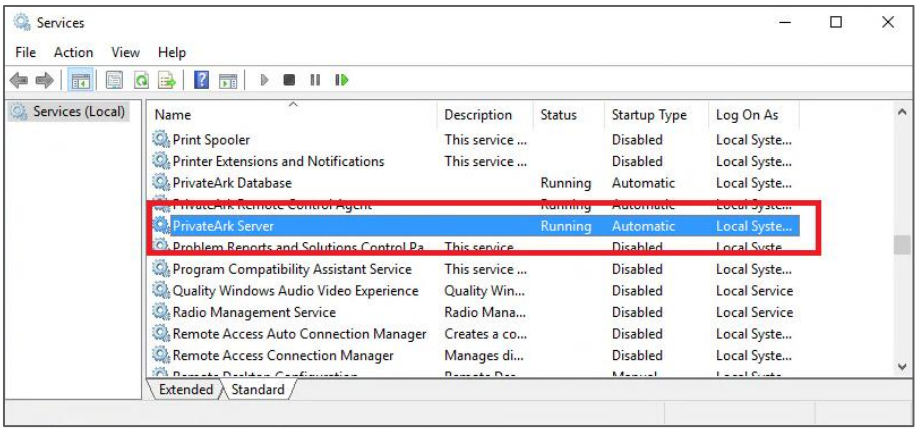
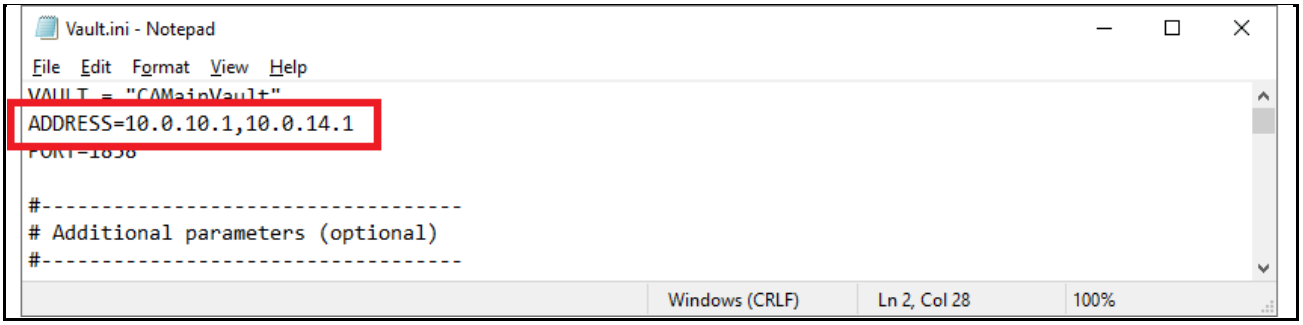


Confirm Automatic Failover of PVWA and PSM

In this section we will confirm that our end users (like *John*) can still access critical systems via **CyberArk**, even though the **Primary Vault** is offline, without human intervention.

Note the implementation team has already configured the **PVWA** and **PSM** to automatically failover to the **DR Vault** when the **Primary Vault** is no longer available. To support automatic failover, the *Vault.ini* file for both services has been configured with the IP addresses of both the **Primary Vault** and the **DR Vault** separated by a comma. The **CPM** has **not** been configured for automatic failover, nor should it be.

Here you can see the configuration of the **PVWA** *Vault.ini* file:



1. To confirm that both the **PVWA** and **PSM** automatic failover was successful, return to the console of the **comp01** server.

2. Open Chrome and verify that you can still login to the **PVWA** as *John*, even though the **Primary Vault** is offline.

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3. Now, verify you can launch a secure session to the target Windows machine using the *administrator* account via **PSM**. If everything worked as expected, *John* should still be able to access the target server via **CyberArk**, without any human intervention.

Note:	You may need to try to launch the connection via **PSM** a couple of time before it works, as it may take a few minutes before the **PSM** fails over to the **DR Vault**.

Failback

Now let's say that the problem with our old Primary Vault – **vault01** – has been resolved and we want to return it to its role as the Primary Vault. During the time of the failover, however, the DR Vault was acting as the Primary and recording the activities of users (although no passwords have been changed because we did not failover the **CPM**). So, before we failback to the **Primary Vault** we must first make sure we replicate all the latest data from the **DR Vault** (which served as the active **Vault** for the duration of resolving the incident). In this section we will use the **Disaster Recovery** module on **vault01** to replicate data back from the **DR Vault** to the **Primary Vault**.

Note:	The implementation team has already installed the **Disaster Recovery** module on **vault01**, and manually created a separate DR user for the purpose of performing replication from the **DR Vault** back to the **Primary Vault**. The new user is called *DR_Failback*, and has been made a member of the built-in group *DR_Users*. The user was assigned the following **Vault** authorizations: *Backup All Safes* and *Restore All Safes*.

Step 1: Execute a full replication back to the Primary Vault

1. First, we need to enable the *DR_Failback* user for the **DR Vault**. You can do this either in the **PrivateArk Client** or the **PVWA**.

2. We have already seen how to enable a user in the **PVWA**. To do it from the **PrivateArk Client** (which until recently was the only way it could be done), from **Comp01**, open the **PrivateArk Client**. Double-click on the **DR Vault** shortcut and login as *Administrator.* To enable the user *DR_Failback* by going to **Tools** | **Administrative Tools** | **Users and Groups**. Scroll down the list to the **System** location. Click on *DR_Failback*, then **Update**, then uncheck the box for **Disable user**.

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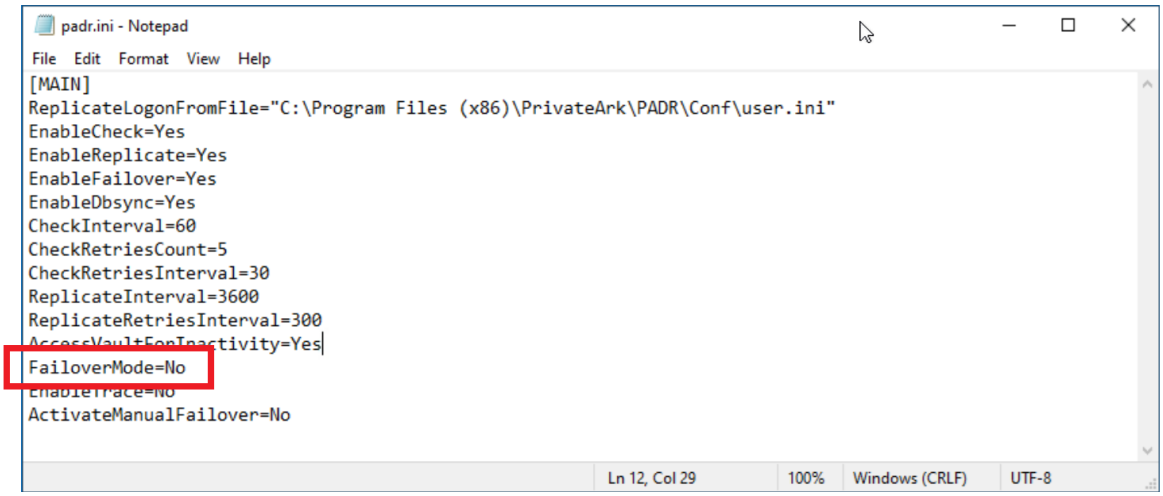
Note:	The process is identical to enabling the *Auditor* user performed earlier. Do **NOT** change the password as the **DR_Failback** user has already authenticated to the **Vault** during initial implementation, and the password for the user has already been rotated. If you change the password to *Cyberark1*, you will need to create a new cred file as well.

3. Next, we to configure Disaster Recovery on the **Primary Vault**, so connect to the VM **VAULT01 (10.0.10.1)**.

4. Open the file explorer and navigate to *C:\Program Files (x86)\PrivateArk\PADR\Conf.* Double click on the *padr.ini* file to edit it with Notepad.

5. Make the following changes to the *padr.ini* file on **vault01**:

   - Set *FailoverMode* to **No** if it is not set yet.
   - If the parameters ***NextBinaryLogNumberToStartAt=-1*** and ***LastDataReplicationTimestamp*** appear in the file, delete them.



Note:	The above changes will trigger the Disaster Recovery module on the Primary Vault to perform a full replication of the data from the DR Vault once the service is restarted.

6. Save the file and close it.

7. Start the ***CyberArk Vault Disaster Recovery*** service on the **Primary Vault**.

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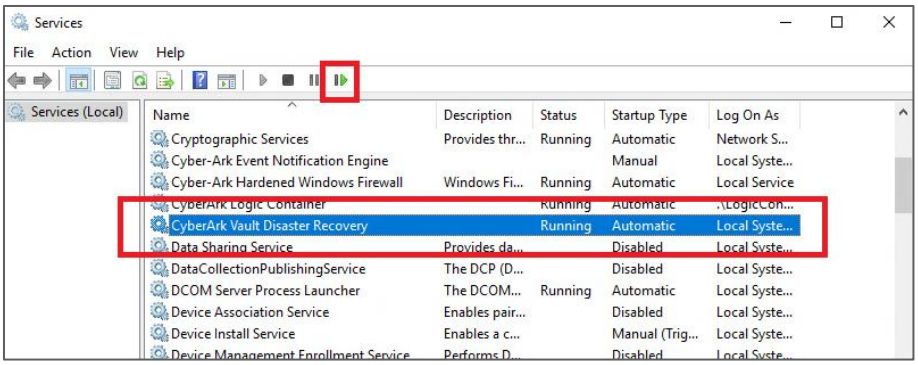
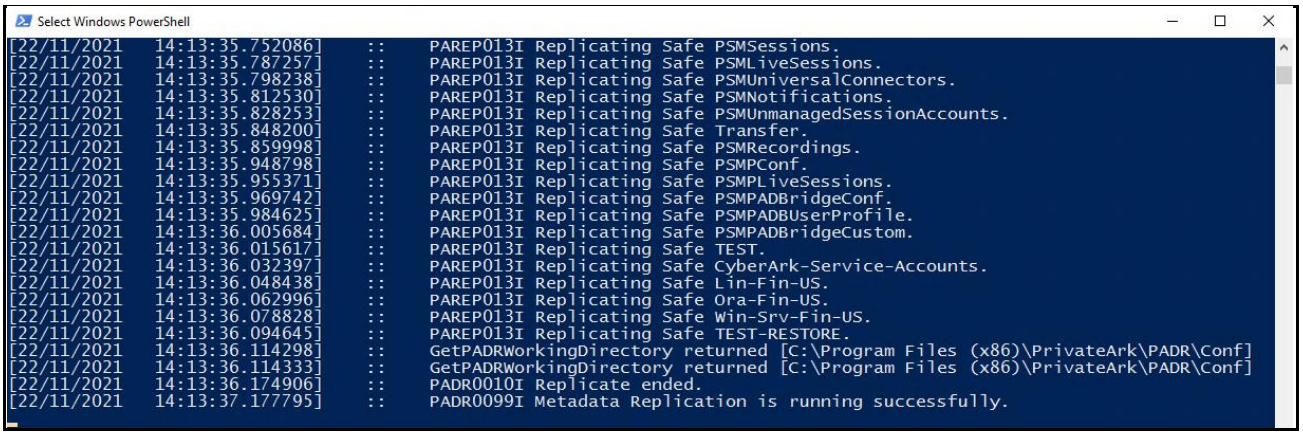


8. Once the Disaster Recovery service is running, Right click on the *padr_log.ps1* file located on the desktop of the **vault01** and select **Run with PowerShell**.

Note:	If you are prompted to allow running the script, select **Yes**.

9. Monitor the tail of the *padr_log.ps1* to verify that the **Primary Vault** has replicated all the changes from the **DR Vault**.



10. On the **comp01** server, login to the **PVWA** as *Mike*. Navigate to **SYSTEM HEALTH** to review the current system health. Note that Vault **10.0.10.1** is now the **DR**, while the old DR – **10.0.10.5** – is showing as disconnected, which it is with respect to its former role as the DR Vault.

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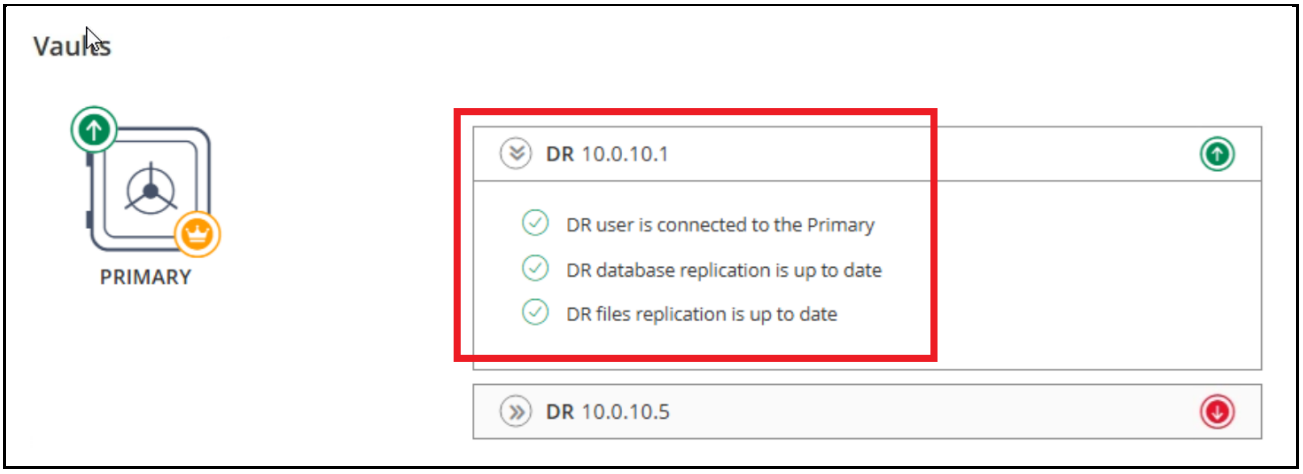


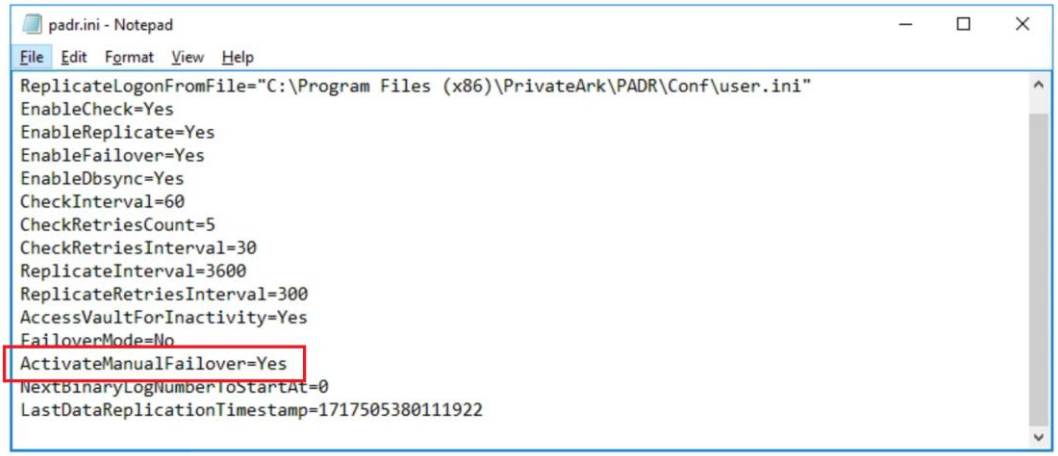
Step 2: Stop the PrivateArk Server service on drvault01

To prevent there being two Vaults active at a given moment, first stop the PrivateArk Server service on **drvault01** (10.0.10.5).

Step 3: Perfom a manual failback procedure

Now we can proceed with performing a manual failback from the **DR Vault** to the **Primary Vault**. The failback procedure will be performed using a **Manual Failover**.

1. Make sure you are working on **vault01** (10.0.10.1).
2. Open the file explorer and navigate to *C:\Program Files (x86)\PrivateArk\PADR\Conf.* Double click on the *padr.ini* file to edit it with Notepad.
3. Set *ActivateManualFailover* to **Yes**.



4. Save the file and close it.

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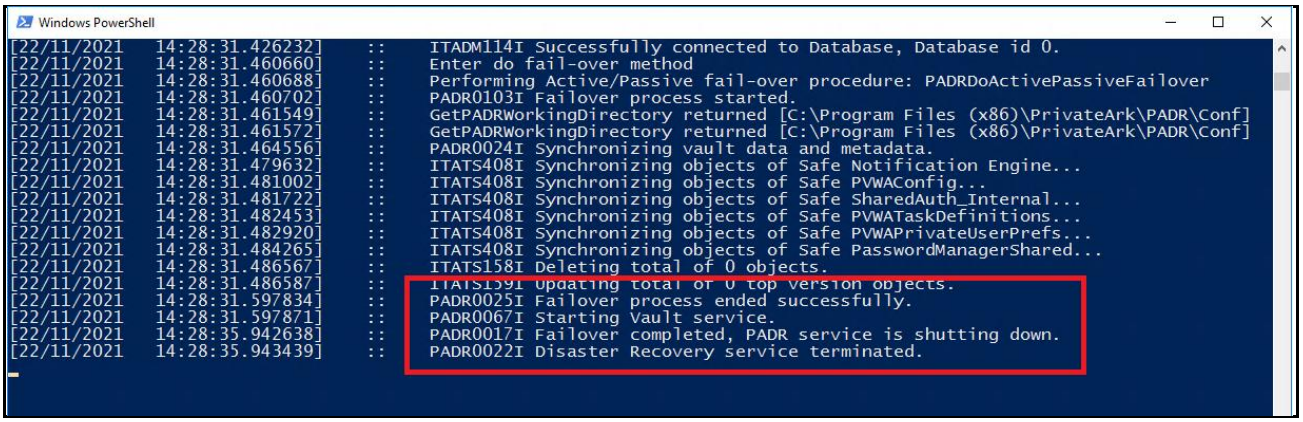
5. Start the **CyberArk Vault Disaster Recovery** service on **vault01** (10.0.10.1). The service should start and stop immediately (because of the *ActivateManualFailover* setting). Then the **PrivateArk Server** service should start.

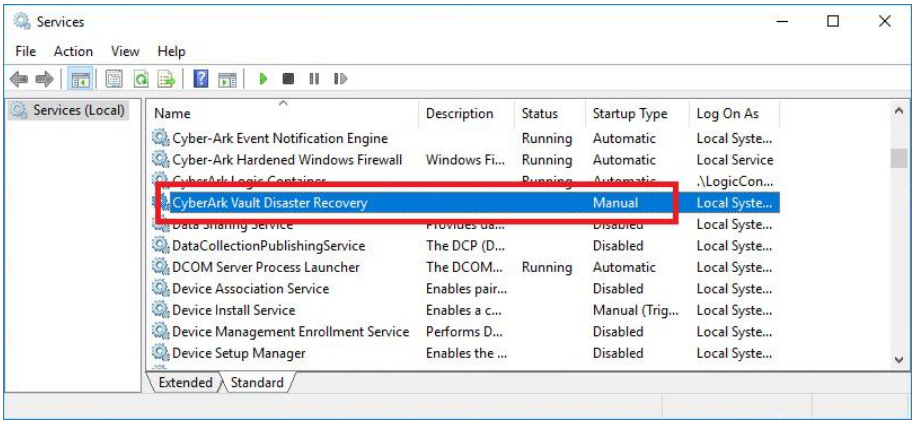
Important: The above steps are critical for a successful failback from the DR Vault to the Primary Vault. Reverting to the Primary Vault without first performing a proper failover can result in data inconsistencies.

Confirm Manual Failover on the Primary Vault

1. Monitor the tail running on the *padr.log* file on **vault01** (10.0.10.1). Confirm you can see the messages stating that the Failover process ended successfully, that the Vault service is starting, and that the *Disaster Recovery* service has terminated.



2. Verify that the the **CyberArk Vault Disaster Recovery** service has terminated on **vault01** (10.0.10.1).



3. Verify that the **PrivateArk Server** service has started successfully on **vault01** (10.0.10.1).

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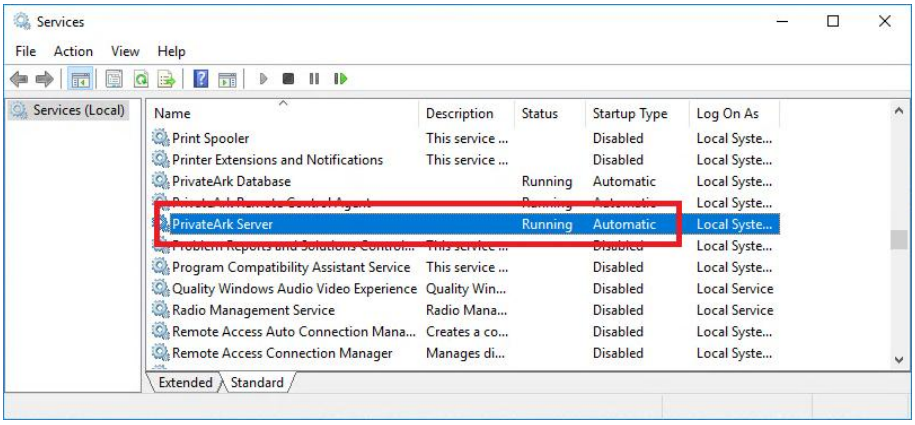


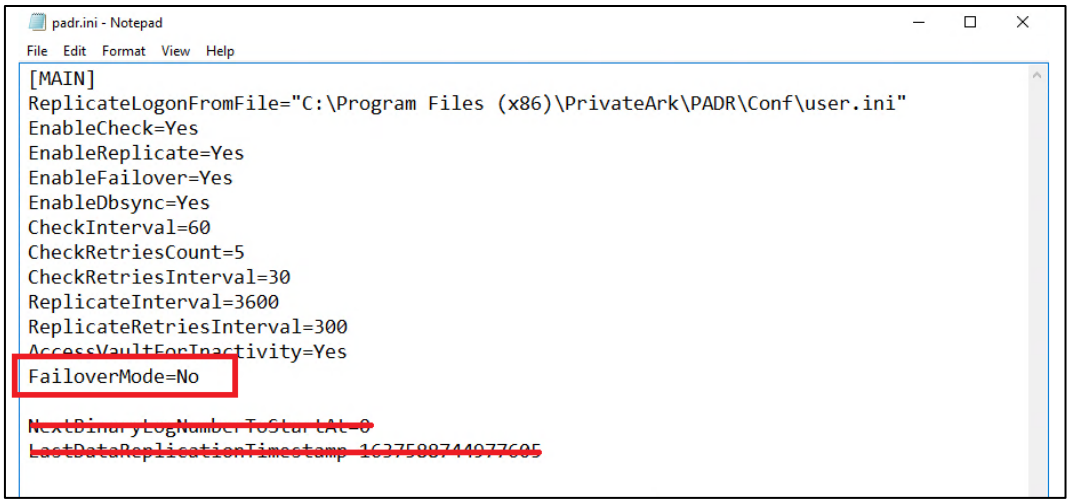
Step 4: Set the DR server back to DR mode

In the last section of this exercise, we will set the **DR** server back to DR mode.

1. Return to the console of **DR** (10.0.10.5).

2. On the **DR** server, edit the *padr.ini* file and make the following changes:

   - Set *FailoverMode* to **No.**
   - Delete the last two lines (log number and timestamp of the last successful replication) in the file.
   - Save and exit the file.



Note:	Click **Yes** to stop **the Cyber-Ark Event Notification Engine** service as well.

3. Then, **start** the **CyberArk Vault Disaster Recovery** service.

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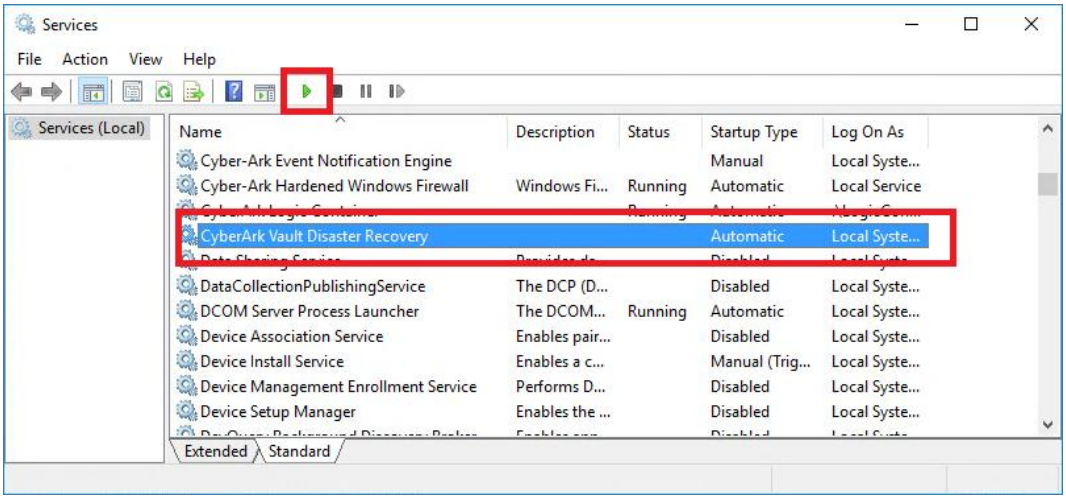
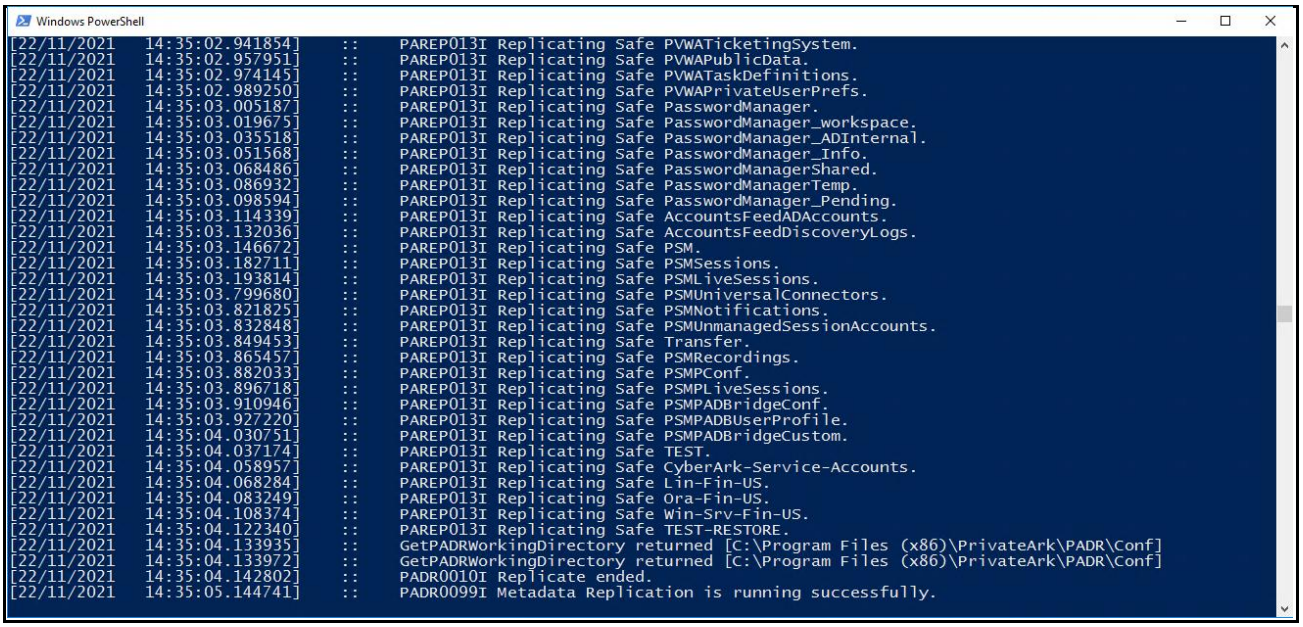


4. Check the tail running on the *padr.log* file on the **DR** server (10.0.10.5) and confirm that a full replication process started and that the replication (from the **Primary Vault** to the **DR Vault**) has ended succesfuly.



Confirm Automatic Failover for PVWA and PSM

In this step we will confirm that our end users can still access critical systems via **CyberArk**.

1. Login to the **PVWA** as *John* and launch a secure connection to the target Windows machine using the account *administrator*. If everything works as expected, *John* should be able to launch the secure connection without any human intervention.

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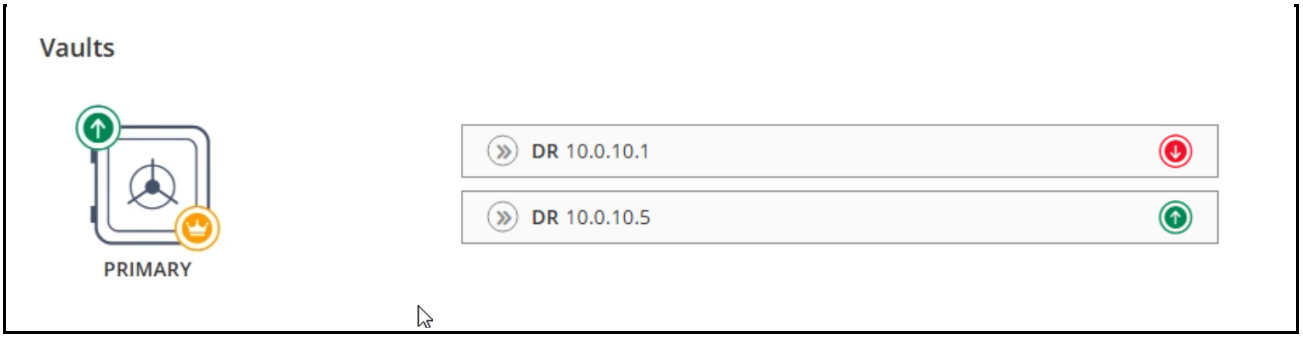
2. Lastly, login to the **PVWA** as *Mike* and navigate to **SYSTEM HEALTH**. Confirm server 10.0.10.1 once again acts as **PRIMARY** and server 10.0.10.5 acts as **DR**. Confirm all other components are connected.



Note:	It may take a little longer for the **PSM for SSH** service to failover, but eventually it should failover to the functioning Vault.

3. Because we don't want to failover again accidentally, which could happen with our virtual machines during a shutdown or a startup, it is important to disable both the **DR** and the **DR_Failback** users (using either the **PVWA** or the **PrivateArk Client**) until the time you want to work again on the disaster recovery exercise.

4. You can also shut down the DR Vault if you are not using it.

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Optional Exercises

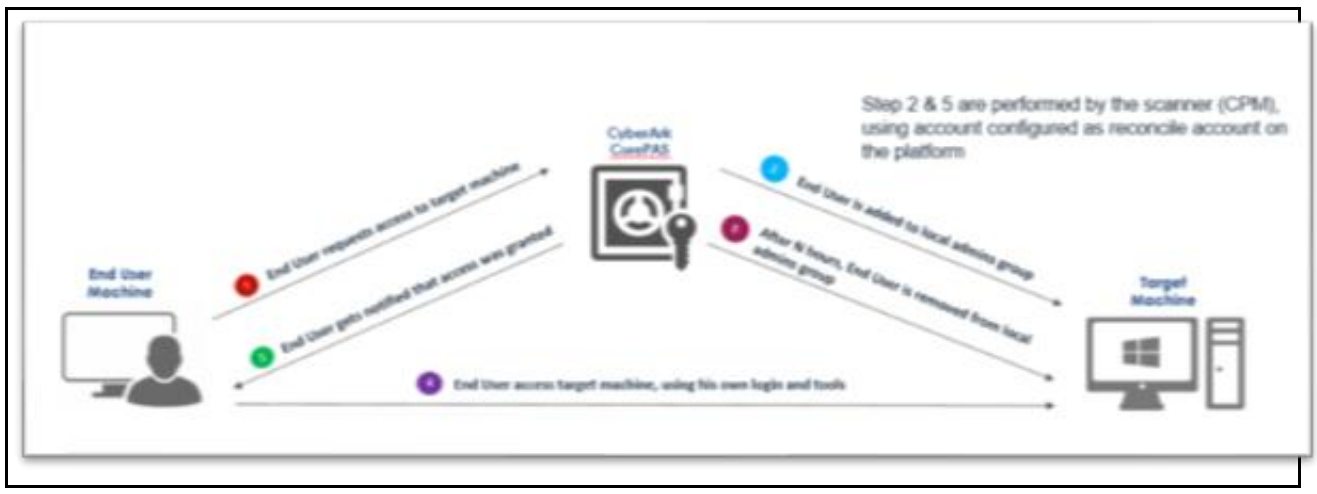
Just-in-Time (JIT) Access

A major step in the **Privilege Access Management** program is to secure the Windows local administrators. This is essential to reduce the risk of lateral movement. **CyberArk** enables securing local administrator credentials, as well as using **PSM** to access these accounts.

There are cases, however, where managing the local administrator passwords is not possible at the initial stage of deployment, whether because of objection from the IT users, or other reasons. Just-in-Time (JIT) access allows you to gain control over local administrator security without inconveniencing administrative users. It can be used as an intermediate step towards full implementation of Vaulting the local administrator accounts. You can grant Windows admins on-demand, ad-hoc privileged access to Windows targets, for a predefined number of hours (4 by default).

During this time, domain users can request access to a system as a local administrator. If authorized, the system temporarily adds the logged-on Windows users into the target system's local administrator group, without the need to manage the credentials of the local administrator on that target. This allows for a frictionless and lightweight solution that enables your organization to introduce privileged controls and help establish habitual security, before moving into a robust PAM program.

The workflow, as shown in the following diagram, starts when an end user requests access to a designated target machine and then is added to the local administrator group. The end user is notified that they have been granted access (or not), and once granted, is able to access the target machine using their own login for 4 hours (by default). After this period, the user is automatically removed from the local admin group.



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Set up the JIT Access Platform

In this exercise, you will set up Just-in-Time access for the Windows admin user (*John*), allowing him to be added to the local admin group on the **acme-123** machine for 4 hours.

1. Log into the **PVWA** as *Mike*.

2. Go to **ADMINISTRATION** | **Platform Management** and duplicate the *Svr Adm 20* Platform to a new platform called *Svr JIT*. You may add a description stating accounts associated with this platform are not managed by the **CPM**.

3. **Edit** the new platform and set the following parameters to **No**.

    - **UI & Workflows**
        - AutoChangeOnAdd
        - AutoVerifyOnAdd
    - **Automatic Password Management | Password Change**
        - AllowManualChange
        - PerformPeriodicChange
    - **Automatic Password Management | Password Verification**
        - VFAllowManualVerification
        - VFPerformPeriodicVerification
    - **Automatic Password Management | Password Reconciliation**
        - RCAllowManualReconciliation
        - RCAutomaticReconcileWhenUnsynched

4. Go to **UI & Workflows | Properties | Required**. Delete the *Username* property from *Required*, and add a new property called *Username* under **Optional**. For **DisplayName**, enter *User name*.

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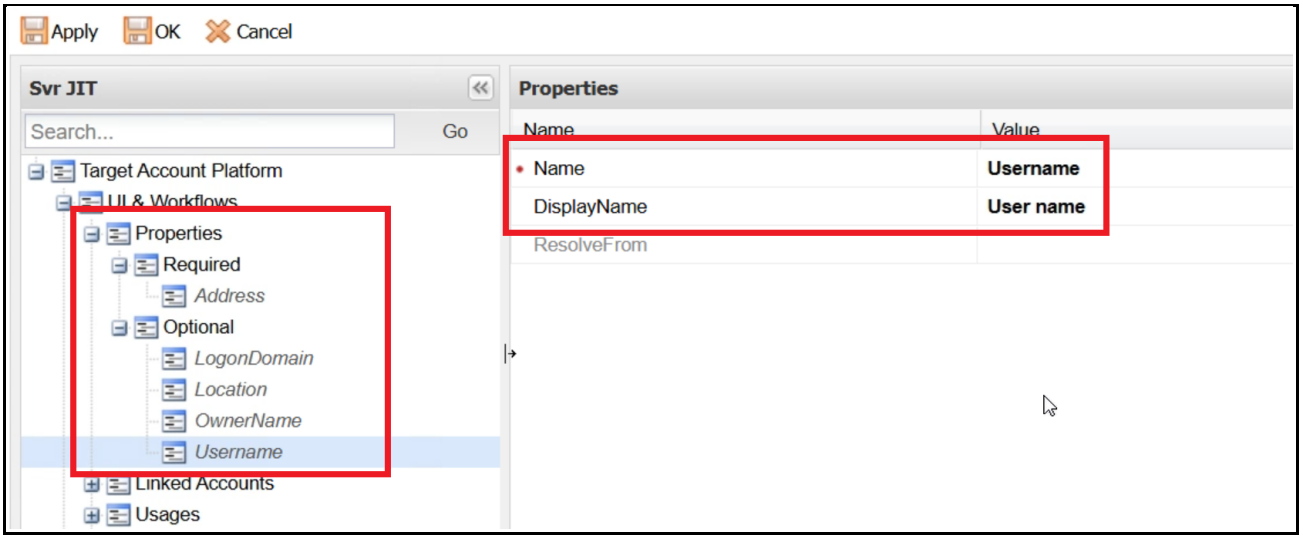
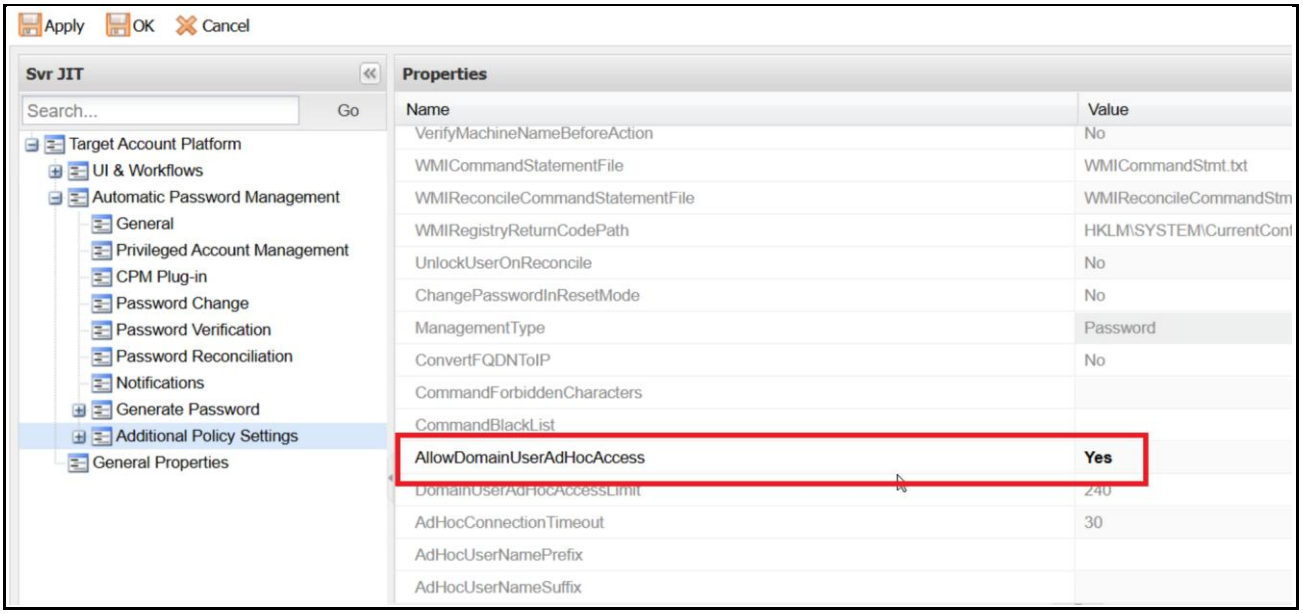


5. Then go to **UI & Workflows | Privileged Session Management** and set the **SessionRecorderSafe** parameter to: ***PSMR_SvrJIT***.

6. Next go to **Automatic Password Management | Add Additional Policy Settings**. Set **AllowDomainUserAdHocAccess** to ***Yes***. It is near the bottom of the list.



7. You will see a pop-up about the **AllowedSafes** parameter. Just click **Yes**.

8. Click **OK** to save your changes and close the **Platform**.

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Note:	Under **Additional Policy Settings**, you can also set the time, in minutes, after which a user is automatically removed from the Administrators group on the target machine. By default, the parameter is set to 240 minutes (4 hours). For JIT access, a domain account that has been configured as a reconcile account should be associated with the platform. In our case, this has already been defined in the base platform we duplicated: *Svr Adm 20.* For security best practice, you need to limit the Safes that are required for ad hoc access, by setting the ***AllowedSafes*** parameter with a regular expression that lists the Safes that this platform can be applied to. This too has already been defined in the base platform we duplicated: *Svr Adm 20.*

Add the Local Administrator Account

1. As either *Mike* or *Tom*, go to the **Accounts** view and click on **Add Account**. Add an account with the following parameters. Note the different password.

System Type:	*Windows*
Assign to Platform:	*Srv JIT*
Store in Safe:	*P-BOS-WIN-S-LA-HR*
Address:	*comp02.acme.corp*
User Name:	*Administrator*
Password:	*CyberArk1999-**
Confirm Password:	*CyberArk1999-**
Logon To (optional):	*comp02.acme.corp*

Test Just-in-Time Access

1. First, open MSTSC.

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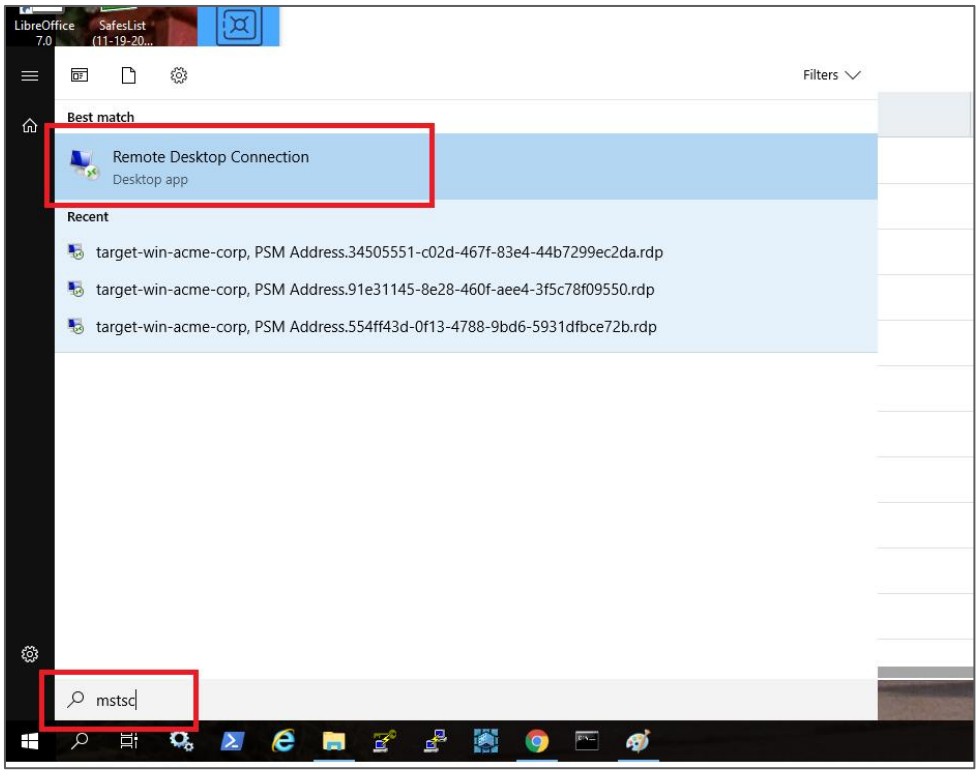
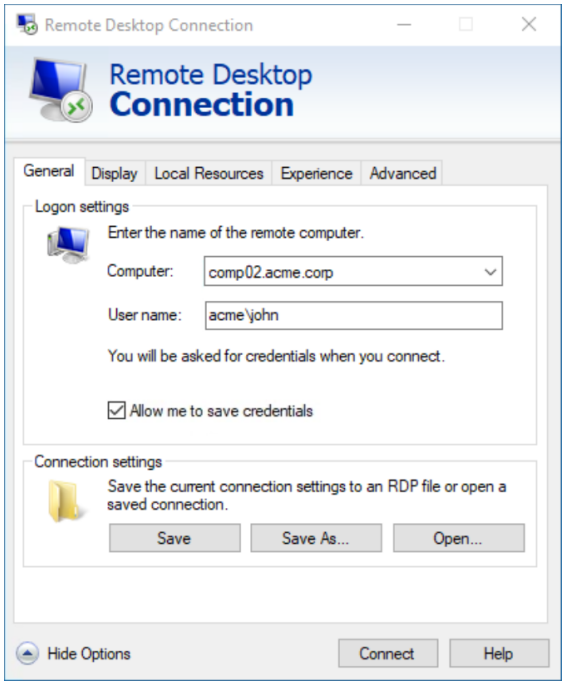


2. Attempt to connect to **comp02.acme.corp** as *acme\John*.



3. You should receive an error stating that *John* is not authorized for remote login:

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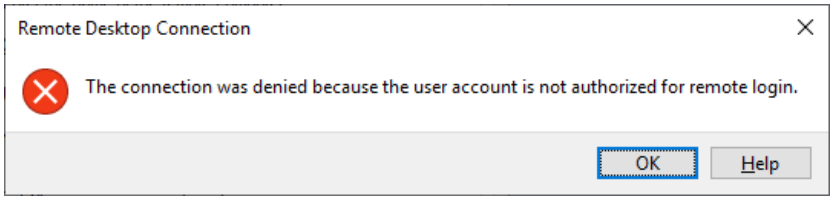
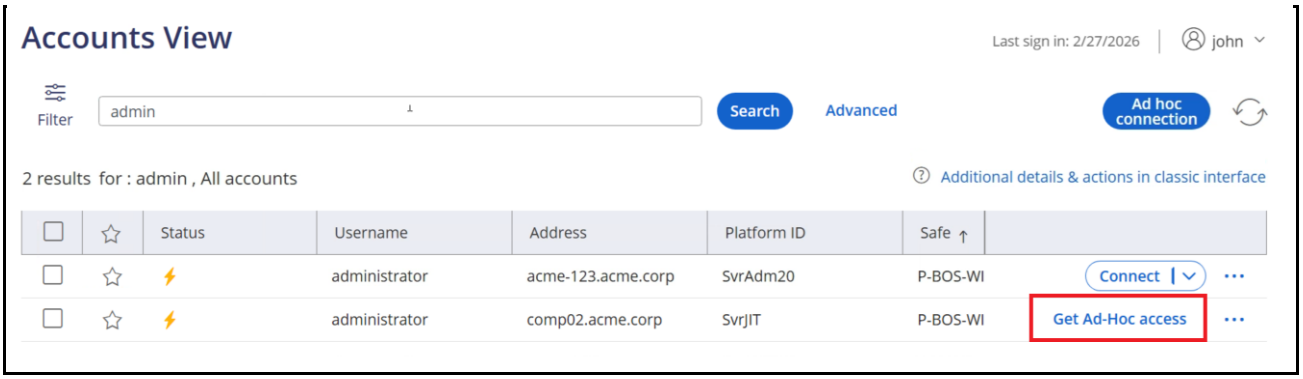


4. Now, login to the **PVWA** as *John*. Search for the Target Windows local *Administrator* account and click on ***Get Ad-Hoc Access***.



5. If you configured everything successfully, you should receive a notification saying you've been granted admin access for 4 hours.

6. Now try to launch another RDP connection to the server **comp02.acme.corp** as *acme\John*. You should be able to login this time.

7. After successfully connecting to the server, go to **Computer Management | Local Users and Groups | Groups** and open the local Administrators group. Verify that *acme\John* was added to the group.

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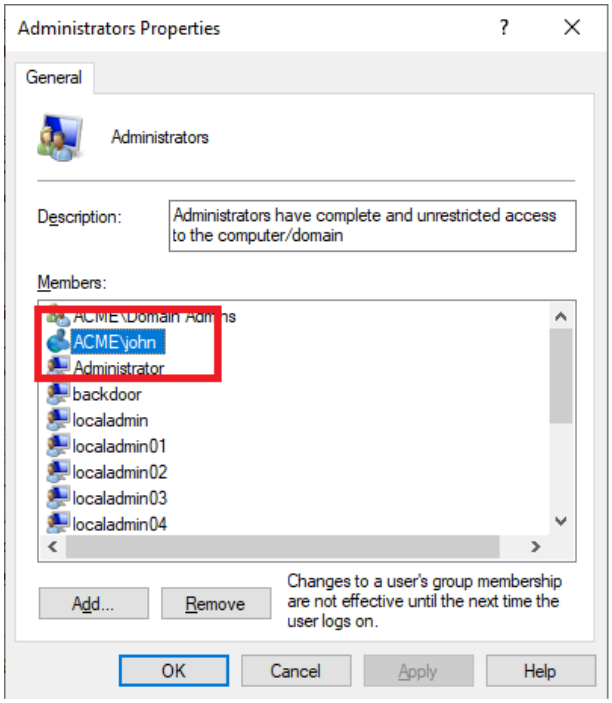

8. You can disconnect from the server.

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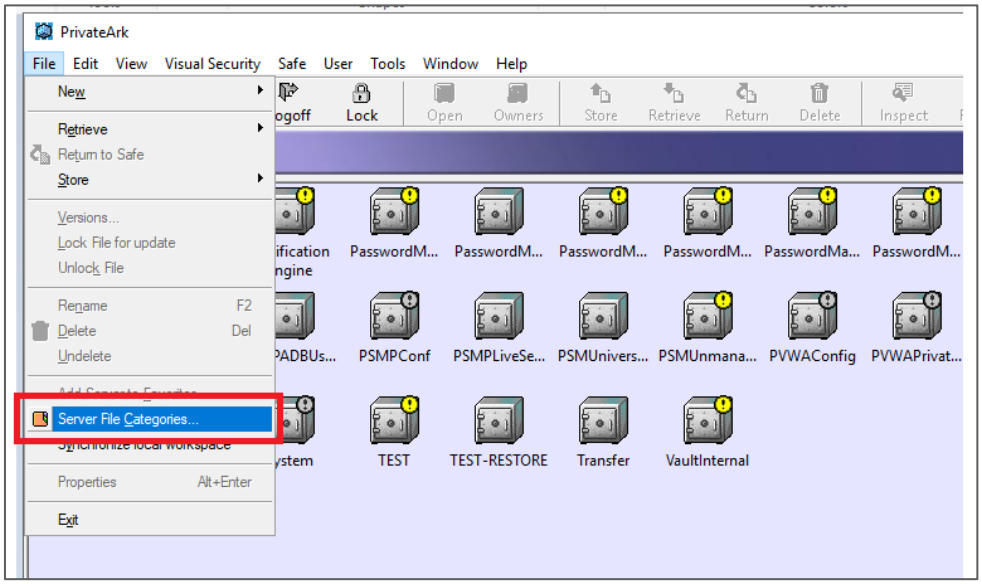
Custom File Categories

File category is the **CyberArk** term for the properties or fields available on accounts (Address, User Name, etc.). This section will detail the steps required to create and use custom file categories, allowing you to categorize accounts based your organization's requirements.

In this final exercise, we will create a custom file category called *BusinessUnit* and provide a list of possible choices: *International, Retail,* and *Corporate*. We will then modify our Oracle platform so that when users add new accounts, they will be required to associate the new account with one of these business units. Finally, we will make the new parameter searchable within the **PVWA** and, of course, we will test what we have done.

Creating the Custom File Category

1. On the **Comp01** server, from the **PrivateArk Client**, log onto the **Primary Vault** as *Mike* and go to **File | Server File Categories.**



2. Press the **New…** button.

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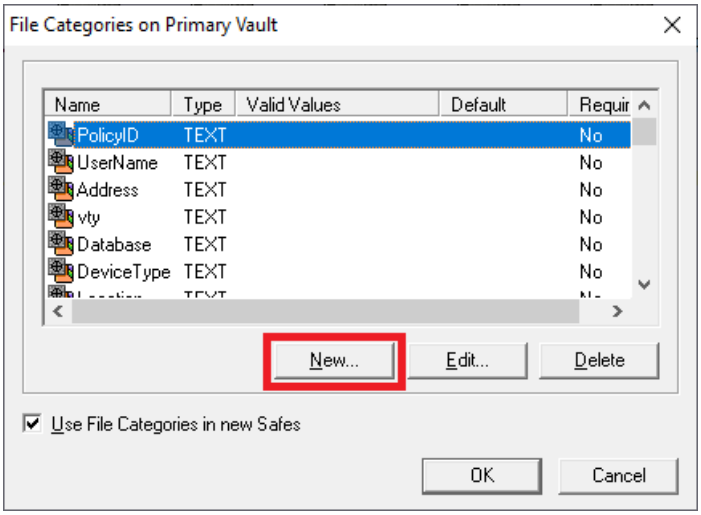
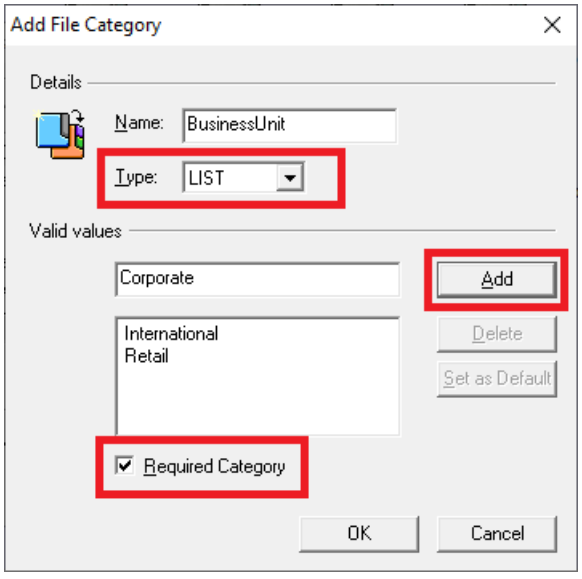


3. In the *Add File Category* window, check the box for *Required Category*. Then enter the information below. To enter the list values, type an entry into the **Valid values** field and then click **Add**. When you are finished, click **OK** to save the changes.

Name: *BusinessUnit*

Type: *List*

Valid values: *International, Retail,* and *Corporate*



4. Log out of the **PrivateArk Client**.

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Adding the Custom File Category to the Platform

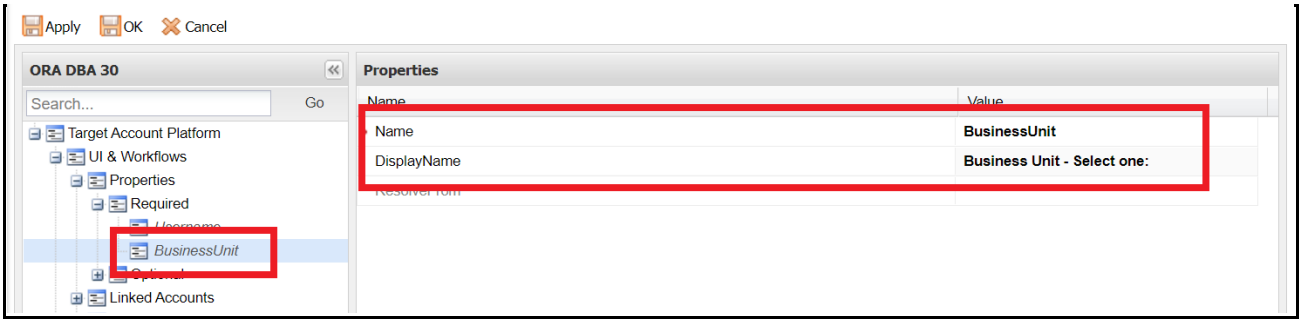
Now we'll make the new *BusinessUnit* **File Category** a required field for accounts assigned to the *ORA DBA 30* platform.

1. Log into the **PVWA** as *Mike* and go to the **ADMINISTRATION** tab and click **Platform Management**.

2. Highlight *Ora Dba 30* and press **Edit**.

3. Go to **UI & Workflows | Properties | Required.** Right-click and select **Add Property** from the context menu.

4. Enter *BusinessUnit* in the **Name** field and then *Business Unit – Select one* in the *DisplayName* field.



5. Press **Apply** and **OK**. This will make the parameter *BusinessUnit* a required field on any accounts attached to the *ORA DBA 30* policy, but will display a more helpful name when they need to use it.

Making the File Category Searchable

Now we will make the new *BusinessUnit* file category searchable.

1. Go to **ADMINISTRATION | Configuration Options | Options**.

2. Right-click on **Search Properties** and select **Add Property**.

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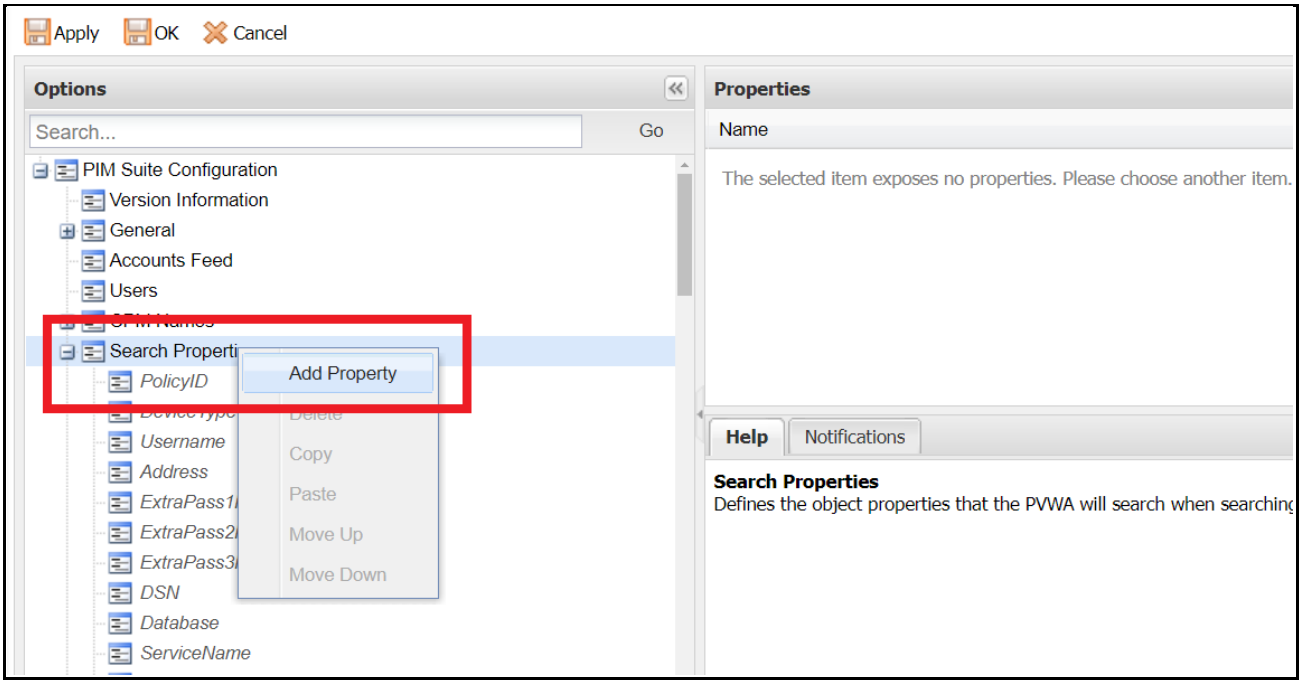
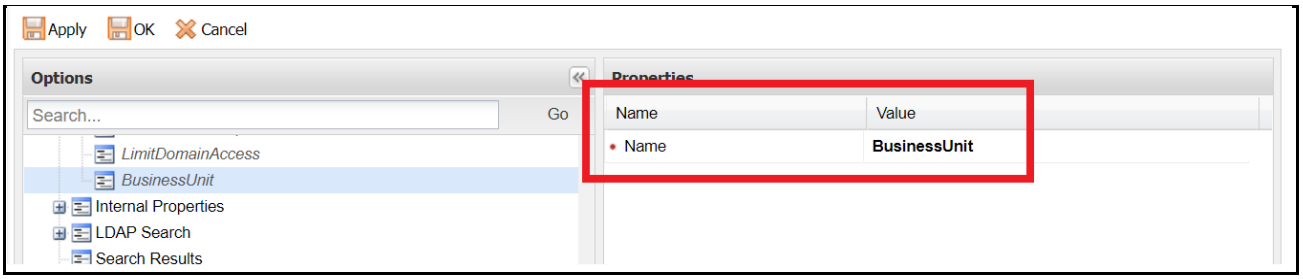


3. Enter *BusinessUnit* in the **Name** field and press **Apply** and **OK**. This will allow the new file category to be searchable.



4. Sign out of the **PVWA** session.

Testing the New File Category

1. Login to the **PVWA** as *Robert*.

2. Locate the account *dba01*, click on the ellipsis button, and select **Edit**.

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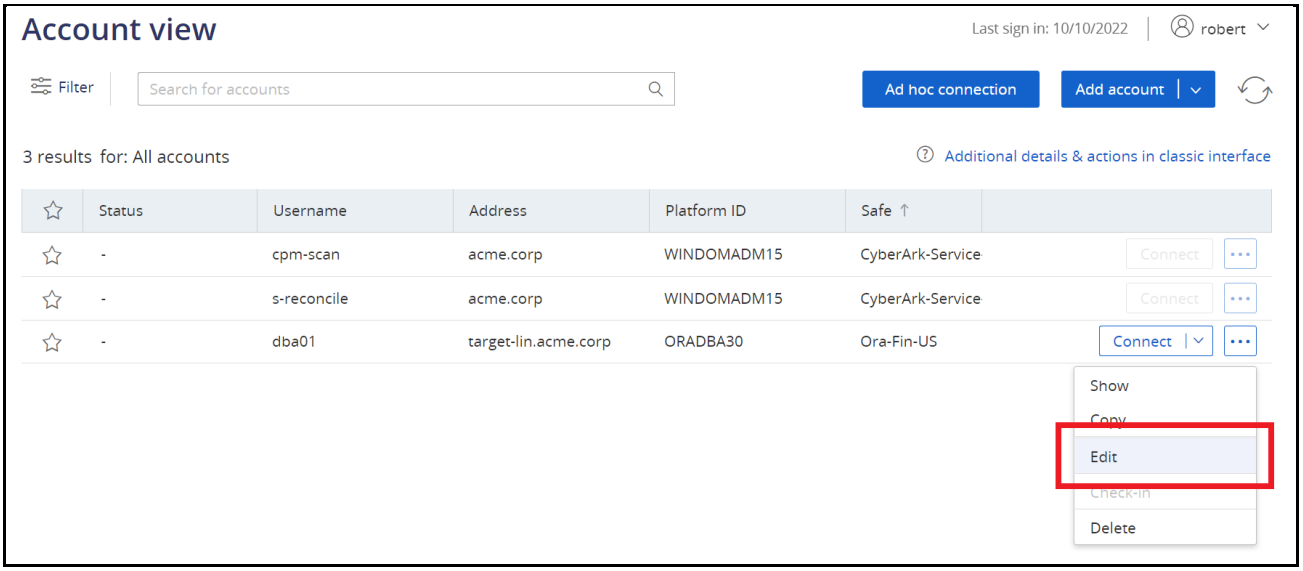
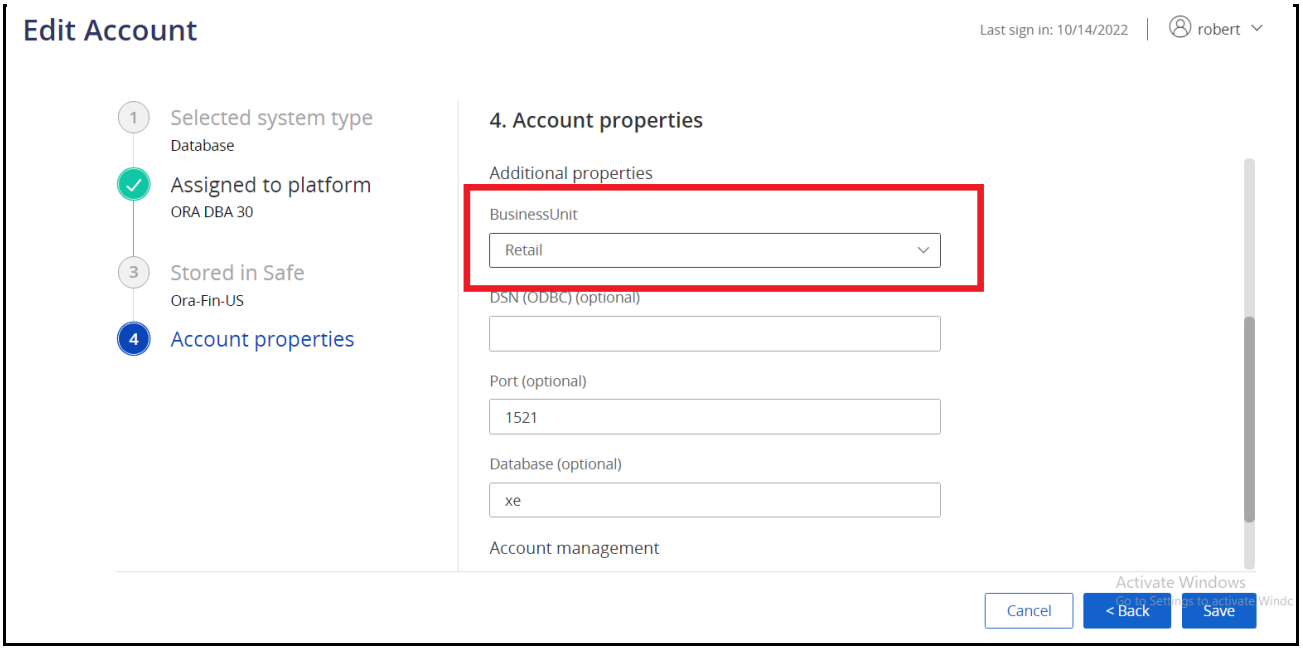


3. Under **Additional** Properties, you will see your new display name: *Business Unit – Select* One. Select *Retail* and press **Save**.



4. Enter *retail* in the **Search** field on the **ACCOUNTS** tab and press **Enter**.

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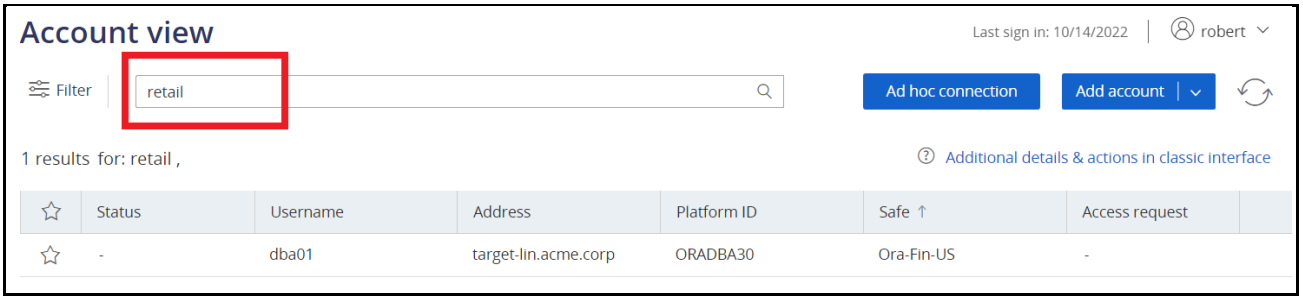

5. The account *dba01* should be returned based on the new searchable file category.

The end

And that is the end of this **CyberArk Privileged Access Manager Administration** training.

We hope to see you again soon in one of our other trainings.

https://training.cyberark.com/learn

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Appendix

Pre-defined Vault Users

The Batch user is used by the Vault for internal maintenance processes and is completely inaccessible to users.

The Notification Engine User is used by the Event Notification Engine. This is the component responsible for sending email alerts from the Vault. The Notification Engine User inherits all the permissions it needs from the Notification Engines group, which exists for the unlikely (but not impossible) chance that there is more than one notification engine.

The DR user is used by the Disaster Recovery component, which is capable of creating an offsite replica of the Vault. It gets its permission from the DR Users Group, which again exists in case there are more than one DR Vault. This is actually a common occurrence. However, since these are not always used, this account is disabled by the installer and would need to be enabled in order to setup the DR component.

The Backup user is used by the Replicate component which is used to backup Vault data to unhardened server where it can be backed up to tape. It gets its permission from the Backup Users Group, which again exists in case there are more than one Backup servers. Since these are not always used, this account is disabled by the installer and would need to be enabled in order to setup the Backup component.

The users we've looked at so far are used by components and so their passwords are managed by those components. Even if the account is enabled and used, no user will know the password and be able to use them to access data. The next set of the accounts CAN be used by real people. These accounts might be themselves used as built in administrators or may serve as template for admin roles you'll wish to create in the Vault.

The operator user, by virtue of being a member of the Operators group has the ability to change properties of safes, delete safes, and restore safes from a backup. It is the counterpart to the backup user. It establishes a role that acts as a custodian of the physical data on the Vault without having an access to read that data. The operator user is disabled by the installer.

The auditor users, by virtue of being a member of the auditors group has the ability to see what accounts exist in each safe--though no access to see the password—as well as the ability to view the auditor records of each safe. It establishes a role that acts as an internal auditor for the organization. The auditor user is disabled by the installer.

That leaves only two accounts that are both enabled and accessible to users.

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First the *administrator* user, by virtue of being a member of the Vault Admins group, has full access to all Safes that were created by **CyberArk** installer. This means a member of that group would have full access to configure **CyberArk** applications, but no access to see, or even be aware of the existence of, user passwords. This establishes a role of application owner, but allows us to prevent administrators from having access to user data.

Finally, the *Master* user is the true super-user of the Vault. It has all Vault authorizations and all Safe authorizations on all Safes. Its access to the Vault cannot be altered, even by the *Master* user itself. This account exists for emergency recovery purposes, and is, by design, difficult to use.

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